

GAZBOT V7 — the Friday report

Week ending **Fri 2026-08-21** · Mon 17 – Fri 21 August 2026 · published 2026-08-26 ·
MNQ-led, MGC captured · PAPER · 50 plays · 10 sections

★★ READ THIS BEFORE ANYTHING ELSE — HOW THE OPEN POSITION ENDED

This report covers the week ending Friday 2026-08-21 and is being published on 2026-08-26, so the biggest single item in it has already resolved and you need the ending before you read the rest. The day rider was long 4 MNQU6 from 29,491.625, opened 2026-08-21T13:13:06Z, and it went through the CME halt into the weekend unmanaged. **You closed it by hand on Monday 2026-08-24 at 11:59:39Z, at 29,224.00, for – \$2,149.44.** (That is the booked figure and it is the one every other section uses. The fills average 29,491.625 where the trades row stores 29,491.93; priced off the fills the same close is –\$2,147.00, \$2.44 apart. The row is left as it is rather than quietly re-derived — but do not re-compute the P&L from its entry price, because that price disagrees with its own fills.) That single position is larger than everything else in this report put together, and it is not in any of the totals below — it closed on the 24th, and this report's books stop on the 21st.

★ **Not one of the automatic paths closed it, and the reason is two separate faults in a row.** The IB gateway entered a partial wedge at about 19:57Z on the Friday, and the rider's own hard-flat, the watchdog, the EOD flatten and the reconciler all share one `ib.connectAsync`, so all four died on the same call. The EOD flatten's timer is **Mon-Fri**, so a Friday failure gets no weekend catch-up at all. `gazbot7-weekend-flatten.timer` — built for exactly this — then fired **119 times** between 22:00Z and 23:59Z on the Sunday. **Every one failed: 105 of them on the wedge, and then, once the 23:45Z gateway close had healed the wedge, the last 14 on a PermissionError writing root-owned data/day_rider.env — which happens BEFORE the claim write and BEFORE the page.** So it never placed an order and never told anyone. Fixing either fault alone still gets you nothing: the file-mode fix buys 14 attempts that would all have hit the wedge, and the gateway fix buys 105 that would all have hit the file mode.

★ **AND THE TIMER IS STILL ARMED.** It is written to self-disable once the venue reads flat, but it dies before it ever gets that far, so it never disabled itself: `systemctl list-timers` has it next elapsing **Sunday 2026-08-30 22:00Z**, where it will re-arm the rider and write a bare-stamp claim — which means claim everything — against whatever happens to be open that night. It has to be deleted, and that is SATURDAY #1.

★★ IF YOU READ ONLY THIS — THREE THINGS, IN ORDER

The card says SATURDAY and MONDAY and both of those days have been and gone. The plays were written for the weekend of 2026-08-22 and the Monday of 08-24; you are reading this on 08-26. So the windows below are re-labelled against the publication date rather than the week they were drafted in, and anything overtaken by events says so on its own row.

1 — Before Sunday, delete the weekend-flatten timer. `systemctl disable --now gazbot7-weekend-flatten.timer`. This is the only play on the card with a hard deadline: `systemctl list-timers` has it next elapsing **Sunday 2026-08-30 22:00Z**, and what it does when it fires is re-arm the rider and write a bare-stamp claim — which means claim everything — against whatever is open that night. It has fired 119 times and placed zero orders and sent zero pages, so it will not tell you either way. SATURDAY #1.

2 — Delete the expired carve-out comments from the top of data/gate_switches.env. Nothing prunes them, and `switch_notes()` splices the top 45 comment lines into every router prompt labelled ACTIVE. ★ **RE-READ AT PUBLICATION, BY IMPORT, AND IT HAS MOVED — which is worse than the version Part 2.6 §5 describes.** A sixth carve-out was written on **2026-08-24T13:46Z** by the open-hour watcher, and prepending it pushed the 2026-08-09 “abs_veto_short IS ARMED BY DEFAULT” line past the 45-line cap: `switch_notes()` now returns five blocks ending mid-sentence inside the 11 August one, and the dangerous line is silently no longer spliced. Nobody closed it — it was evicted, and the eviction is invisible from the file. So the header is an unmanaged FIFO as well as a set of comments that never expire, and both halves need the same remedy: delete the lines. SATURDAY #4.

3 — Make the tournament standdown a FILE that every writer checks. It is currently a variable inside one process, and `open_hour_watch.py` is a second writer to the same switch file with off→on authority and no awareness of it. It armed a gate straight through the standdown at 14:03Z on the Friday and cost \$0 only because the router's next tick caught it. ★ **AND IT HAS HAPPENED AGAIN SINCE THE REPORT'S WINDOW CLOSED.** On **2026-08-24 at 13:46:32Z** the watcher auto-armed `abs_veto_short` through the same standing standdown a second time (APPLIED {'abs_veto_short': 'on'} [OPEN-HOUR

WATCHER, automatic]); the router re-benched it at 13:50:22Z. That is **3m50s of exposure against Friday's 2m15s**, \$0 again because the tape produced no signal, and **zero pages** both times. Part 2.6 §4 argues this from one incident; it is two, the second one longer, and the only guard on either is a 5-minute poll. SATURDAY #3.

What has been overtaken: SATURDAY #2 (the rider stop) is re-argued from scratch below and against 231 sessions rather than one week — read that row before acting on it, because it re-opens a decision you took on 08-20. MONDAY #1's bench of `abs_veto_short` has already happened: the switch file has read off since 08-24 13:50Z. And `exhaustion_short` also reads off, which the Rev1 card did not know — see HOLD #6.

★ THE HONEST HEADLINE — a small red week, and the loss is not where you would look for it

The desk booked **−\$174.00** across 43 clean legs (49 contracts — the day rider's legs are 1–3 lots each, so a row count is not a lot count), **3 of the five days green**. Three sessions made \$1,183.50 between them and two lost −\$1,357.50, and essentially all of that loss is **one day-rider position on Thursday** that ran for six hours with no stop in the water. The tournament — the eight base gates filling six live slots that most of this report is about — traded only 36 legs all week for −\$4.50, because four of its six slots were stood down and never fired at all.

The three things worth carrying into Saturday. One: the standdown was right. Firing all six gates mechanically through this week's own tape loses **−\$1,905.36** on 117 fires (Movement 2), so the four silent slots are the best-performing thing on the desk. Two: your hands made \$578.50 on 16 claims and the machine's own profit exits made almost nothing — but the study that tried to automate your hand comes back **−\$4 out-of-sample**, so the answer is not to copy it. Three: the week's real damage was

done by things that are not strategies at all — a session-roll bug, an off-tape fill, a missing stop and a wedged gateway. ★ **And the sharpest version of that last one:** the four lots are a missing-stop incident that a wedge then made unfixable, not a wedge incident. Every stop width tested — all seven, from $0.5\times$ to $3\times$ ATR — closes that position within nineteen minutes of entry, six and a half hours before the gateway stopped answering. It eventually cost $-\$2,149.44$, which is more than the rest of the week's book combined; the stop would have cost a fraction of it.

There is no single honest total, so here are all of them, with what separates each from the next.

WHAT YOU ARE LOOKING AT	LEGS	NET \$	WHY IT DIFFERS FROM THE ROW ABOVE
Raw trades rows, 08-17 → 08-21	44	$-\$255.50$	everything the desk booked, flags and all — the CASH book
Clean ledger (data_quality IS NULL)	43	$-\$174.00$	drops 1 flagged lot worth $-\$81.50$ — this is the STRATEGY book and it is the number to judge a gate on
The day rider alone	7	$-\$169.50$	the second desk, on the same IB account. It made the week's best trades and its single worst one
The tournament alone (eight base gates, six live slots)	36	$-\$4.50$	what the gates this report is mostly about actually did — on two slots, because the other four were stood down
... plus the position that was still open on Friday night (4 lots long from 29,491.625, closed by hand 2026-08-24 at 29,224.00)	4	$-\$2,149.44$	not in any row above — on Friday because the trades table only enumerates what CLOSED, and now because it closed on the 24th, outside this report's window. The week's real economic position is $-\$2,404.94$.

THE ONE FLAGGED LOT, NAMED

It is a single day-rider lot from Friday morning, flagged BADFILL:stale_price_30pt_outside_book_20260821, worth $-\$81.50$. It filled at 29388.25, about 26 points below a 30-minute tape low, roughly 92 ms after its 3-lot sibling filled on-tape. **A BADFILL flag marks a defective fill PRICE, not a phantom trade** — the money left the account, and IB's own realizedPNL of $-\$149.38$ for Friday reconciles to the

FLAGGED figure, not the clean one. So: clean book for judging a gate, cash book for counting cash, and the venue position line above both.

DAY	DESK NET (CLEAN)
2026-08-17	\$473.00
2026-08-18	\$195.50
2026-08-19	\$515.00
2026-08-20	-\$1,288.50
2026-08-21	-\$69.00



THE SCORECARD — what survived the week

Lead with what survived the skeptic, never with the biggest number. Ten results carried real weight out of this week; here is where each one landed and the test that decided it.

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
The desk is better off with four of its six slots switched off	SURVIVED — the week's best-evidenced result All six gates fired mechanically in their exact live config across the whole of this week's captured tape, every fire counted rather than only the ones near a big move: — \$1,905.36 on 117 fires, and —\$517.79 of that is the four slots that were stood down. ★ REV3 — quote the whole-week figure, not the run-window one. —\$1,905.36 is CENSUS-INDEPENDENT: it fires all six gates across every second of the week's tape and never looks at the run list. The run-window arm does depend on it, and the list moved — Movement 1's frozen census is the 08-25 re-freeze at 68 runs / 62 sat out / \$9,610, while the —\$1,333.85 was measured on the superseded 08-22 freeze at 70 / 65 / \$9,987. Re-cut onto the runs that survive, it is —\$976.71 on 21 fires across 16 of 53 runs. Same sign, same instruction, smaller number — and 9 of the new list's 62 runs were never simulated at all, so that is a re-cut, not a re-replay. Strip every filter off and they lose more, not less. The base-rate control is what makes this different from the last three idle-gate labs, which measured only windows already known to contain a run. The honest limit: one week, mostly stood down. It licenses leaving them benched, not retiring them — and ★ REV2 puts a bootstrap round the headline: the whole-week total's 95% interval straddles zero, so —\$1,905.36 is a point estimate, not a	Movement 2

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
	measured saving. What IS statistically solid is the comparison Rev1 got backwards: the ten minutes before a big run are 3.6× worse per fire than the base rate (12.9× once the overlapping fires are removed), 95% CI excluding zero on both readings. Rev1 called those two rates “close”; they are not, and the difference strengthens the bench case rather than blurring it.	
“abs_veto_55s is the promotion candidate, +\$332 this week”	WITHDRAWN — \$308.50 of it is untradeable The permanent Asia bench (00:00-07:00Z) sits DOWNSTREAM of where a shadow arm records its fire, and the desk has filled nothing in that window since 5 August. Void it and the family goes from +\$332 to +\$23.50 on 34 fires, 2 of 5 days green, and stripping its single best fire turns it – \$133.50. Its 50-second sibling goes negative. Across all 38 families, –\$3,631 of the week's board was booked in that window. The family's real case is its all-time record (+\$3,078 on 263 tradable-hour fires), which is a reason to keep incubating and not a reason to promote on Monday.	Part 2
“Only run grind on a clean trend day” — the week's headline mid-week lead	PARKED — it works and it cannot be run On 128 live grind fires the rule turns –\$958 into +\$580.50 and keeps 25 of 43 winners, so it is not a fake win — but strip its best three fires and it is –\$202. Then the killer: of 240 classified sessions, 97 are clean-trend and exactly ONE is both clean-trend and above grind's 22-point ATR floor. The two conditions are very nearly mutually exclusive. Arming was never the binding constraint. The real question is the floor, which sits at the ninetieth percentile of one-minute ATR and has never been swept.	Part 2.5
“abs_veto_short is chasing exhausted moves”	REFUTED, and the sweep gives a real verdict Measured inside each trade's own life: 0 of its 15 losing lots were chases. Every one went at least 6 points onside, median +11.25. The difference between a winner and a loser is how far it ran, not whether it ran. Re-raced at five stop widths twice over — once scaling the whole trade, once moving only the stop — all ten cells are negative, so the sign survives and this IS a verdict rather than a knife-edge. The gate reads –\$170.00 live where the machine alone would have booked –\$446.24, so your hand is worth +\$276.24 on this one gate — the difference between the live book and the machine-repriced book, which is the one form of that number needing no assumption about the simulator. (★ REV3: earlier revisions quoted +\$314.74, and before that +\$412.93, both computed as booked-minus-repriced on the four lots you claimed by hand. Those do not reconcile with the totals beside them — there is a separate ~\$45 residual on the eighteen lots the simulator says it reproduces 18-of-18. Part 1.5 §2.4 now prints the four-way split so you can see exactly where it sits.	Part 1.5

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
	The sign and the verdict are unchanged; the hand is worth about \$276, not \$315.) ★ REV2 also finally ran the EXIT ladder \$2.1 diagnosed and \$2.2 never tested. It does not fail: four rungs beat the as-traded book while keeping 7 of 7 winners — a scalp at 1.0R (+\$187.00), at 0.5R (+\$178.50), a trail 8pt behind the peak (+\$149.00), and a scalp at the losers' own median peak of +11.25pt (+\$141.00) — each positive on strip-best-1 and on the worst leave-one-day-out. But against a 400-draw random-entry control on the same tape they sit only at the 80th-93rd percentile, so it is a SHADOW row, not a Monday change.	
“Thursday was a bad trade”	REFUTED — it was a missing stop One rider long, four lots, six hours, no protective exit in the water at any point. Raced on the real tape from its own entry, all seven stop widths from 0.5× to 3×ATR beat what happened, the tightest by more than a thousand dollars. The entry is the rider's ordinary open-hour entry and it made money on three other days this week. ★ REV2 CHANGED THE WIDTH, on this report's own winners-retention rule. Rev1 priced the stop on Thursday alone — the one losing day — and picked 1.5×ATR. Raced across all six rider entries, 1.5×ATR (41.4pt on Monday, 46.4pt on Tuesday) stops out BOTH winners and takes the week's closed rider book from −\$251.00 to −\$379.90: worse than doing nothing. 3.0×ATR is the only width that is positive (+\$210.70) and keeps all three winners, and it still caps Thursday at −\$826.80. The honest week-level figure is +\$462, not \$1,000. And the finding neither revision expected: every width tested closes Friday's 13:13Z entry within nineteen minutes — six and a half hours before the gateway wedged. The four lots at the top of this report would not exist under any stop.	Part 1.5, Part 1.6
“A standdown means no gate can be armed”	FALSE — and it happened on Friday The standdown is a variable inside one process. open_hour_watch.py is a second writer to gate_switches.env with off→on authority and no awareness of it, and at 14:03Z on Friday it armed abs_veto_short straight through the standdown. The router caught it on its next tick and re-benched it 2m15s later, so the cost was \$0 by luck, not by design — the exposure is bounded by a 5-minute poll, not by a guard. Also on the stand: five expired carve-out comments are still spliced into every router prompt labelled ACTIVE, the oldest thirteen days old and asserting that this very gate is “ARMED BY DEFAULT”.	Part 2.6
Four instruments reported confidently and wrongly this week	CONFIRMED — none of them errored core_health.flat said the desk was flat for the ENTIRE life of a four-lot position: 811 of the 817 router ticks between the 13:13:06Z	Part 1, Part 2, Part 2.6

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
	entry on 08-21 and the hand-close at 11:59:39Z on 08-24 printed “Flat, healthy”, 813 contained the word flat, and not one of the 817 mentioned the open position at all. (★ REV3 — Rev1 printed this two ways, “132 of 137” in Part 1 and “137 of 141” in Part 2.6, each taken at whatever moment its own section rendered and neither reproducible from data/router_trial_log.txt. The count above is cut at the position's own life, so it cannot drift.) hour_watch read Friday at −\$150.50 against the clean book's −\$69.00 (it has no data_quality filter — and it is the one that reconciles to IB). router_nightly and selector_nightly, the two files the weekly router review is built from, silently stopped being written on 08-14 and 08-18; neither has a timer. And 7 of the 38 shadow arms are byte-identical to another arm, so three A/B experiments have been reporting perfect nulls from a broken arm rather than from evidence. A number that arrives without a failure mode is not a measurement.	
Seven greenfield hunts, and not one MNQ candidate survived	NULL — and every kill has a named test The pooled run-catcher passed every in-sample test we own, including two placebos, then measured +\$14 over 1,289 out-of-sample trades across 236 untouched sessions. UNCLASS's rider is beaten by the same signal taken thirty minutes early. OPEN/NEWS is beaten by a gate containing no signal at all (+\$47.29/trade). FLOW-LED's last survivor is beaten by a time-shifted fake. VACUUM's survivor wins in both directions. The chop-turn finalist is beaten by the null handed its own 1,080-cell search, 85.3% of the time. That is six independent batteries doing their job, not a bad week — and three of those tests are ones this desk only started running recently.	Movement 3
The tradeable threshold is about 120 points — and three labs reached it independently	SURVIVED — the movement's most durable result Three hunts, three clusters, three harnesses, three authors, nobody looking for it. OPEN/NEWS: +60...+120pt of follow-through still books −\$4.10/trade; above 120pt, +\$184.75/trade. VACUUM: below move ≈ 87pt three of four detectors are noise, and lake-wide it reaches AUC 0.867 on ATR for the 113 runs of 120pt+. The census's own work: boarding only fires ≥120pt pays +\$69.38/trade and passes the whole battery. Detection is solved — a five-minute board rule fires inside 62 of 62 sat-out runs. Selection is not: the same rule fires 34 times a day. Hunt SIZE, not direction. ★ And the census's \$9,610 is a hindsight number: a perfect late boarder with zero costs reaches \$5,094 (53%), a deployable one \$1,910-\$2,286 (20-24%).	Movement 3
Both arms we shipped last week are dead — and nothing was grading them	CONFIRMED — graded here for the first time rider_w5 went into the shadow book on 08-15	Movement 3

THE CLAIM	VERDICT, AND THE TEST THAT DECIDED IT	WHERE
	quoted at +\$19.29 a trade. Fifty-one forward trades later it is -\$28.84 a trade, -\$1,471, red on 6 of 7 days and red in all four ATR buckets — worst in the band its own filter selects for. Its in-sample headline collapses to +\$282 on strip-best-3, and the number it shipped on never had a strip test attached. UNCL-RIDER-ASIA is worse: four consecutive red forward weeks, and it was never actually shipped — there is no arm by that name in any of shadow.db's 82 strategies, and nobody noticed for a week. The fix is procedural, not analytical: a named grading date on every arm, and a strip-best figure printed beside every shipped number.	

§ Every gate, this week, on the clean ledger

GATE	LEGS	NET \$	WIN %	\$ PER LEG
abs_veto_short	22	-\$170.00	31.8%	-\$7.73
day_rider	7	-\$169.50	42.9%	-\$24.21
exhaustion_short	14	\$165.50	57.1%	\$11.82

Computed at build time from data/gazbot7.db with data_quality IS NULL, not transcribed. **Every dollar in this report is a plain sum(pnl_usd)**, because that column is written already net of the \$1.50 round trip — verified on all 765 rows in the book. Do not subtract fees_usd from it; a previous revision did, and it deepened every stated loss by \$1.50 a contract. Gates are collapsed across their A and B lots; the per-lot split is in Part 1 §3. **Four of the six live slots are absent from this table because they fired nothing** — Movement 2 prices what that was worth.

THE THREE THINGS THIS WEEK ACTUALLY TAUGHT THE DESK

1 • The trades table cannot tell you what you are carrying. It enumerates what CLOSED, so it can never contradict a benign story. This week the router spent 817 consecutive ticks — the whole life of the position — reciting closed rows and the word “flat” while updatePortfolio in the same journal read position=4.0. Attribution has to terminate at the venue position line, and “not mine to close” obliges escalation, not dismissal.

2 · A filter that cannot fire is not a filter, it is an off-switch. grind's 22-point ATR floor blocked 24,110 of 24,121 raw signals this week and sits at the p90 of one-minute ATR. Every grind result in this report — the unrunnable trend-day rule, the two empty exit-shadow watches, the zero live fires since 5 August — is downstream of that one number, and nobody has ever swept it.

3 · Where you measure decides what you find. Three separate results this week flipped on the measurement window rather than on the data: the shadow board reorders completely once the untradeable Asia hours are voided; the idle-gate lab only says something because a whole-week base rate was fired alongside the run windows; and the claim detector makes + \$1,069 in-sample and −\$4 out of it. **Ask what the denominator is, every time.**

★ WHAT TO ACTUALLY DO

Garrath — this is the whole answer to “so what do I actually DO?”, and it is at the front. Every row below mirrors `reports/friday_v7/plays.json`, which is the reconciled source of truth for this week; the same file drives the [Monday playbook](#), so the two cannot drift apart. ★ **This week's card is deliberately SHORT.** The tournament traded two of its six slots for 22 lots, which is not enough evidence to re-tune a threshold, and four separate sections say so independently. So most of what is here is either a fault to fix or a measurement to build — and the most urgent row is not a strategy at all, it is an armed recovery timer that fires again on Sunday. 50 plays, five windows, in the order they need doing. Every play carries its **evidence tier** and its **kill criterion** — if a play has no way to be proven wrong it should not be on the card.

THE THREE THAT MATTER MOST

1 · DELETE gazbot7-weekend-flatten.timer — it is still armed and it fires again on Sunday (SATURDAY #1). `systemctl disable --now gazbot7-weekend-flatten.timer` and remove the unit file. It was built as a one-shot for a position that no longer exists, and it is the only row on this card with a date attached. 119 firings on 2026-08-23, 0 orders placed, 0 pages sent. Next elapse Sun 2026-08-30 22:00Z. Journal: 105 gateway failures then 14 `PermissionError`.

2 · Re-open the day rider's protective stop — the 2026-08-20 “no stop” decision was taken on a number measured on a DIFFERENT rider (SATURDAY #2). Read the standing decision first, because this play re-opens it. `src/gazbot7/day_rider.py:93` carries `PLACE_VENUE_STOP = False` under, verbatim: “★★★ NO STOP. Operator, 2026-08-20: **i dont want any stop. leave them all naked ... if its going south my claim profit buttons also work as a manual kill button.**” The same block records the exposure as knowingly accepted — median worst-adverse 160pt = $-\$1,284$ on four lots, worst session 1,084pt = $-\$8,672$ — and cites the desk's own counter-evidence: a venue stop “costs $\$3,451$ of expectancy AND has a WORSE worst-day ($-\$1,603$) than running naked ($-\$1,531$)”. **The ask is therefore NOT “ship a $3.0\times\text{ATR}$ stop”. It is: look at the 231-session re-derivation below, then decide.** If you still want it naked, that is a decision this report accepts and stops re-proposing; what it will not do is leave the number unchallenged, because the number turns out to be about a different rider. **If the answer is still no**, the narrower ask that the 08-20 objection does not already answer is a stop that arms only when the flatten path is unhealthy — i.e. place nothing while the gateway answers, and place a wide resting stop the moment `ib.connectAsync` fails or the heartbeat goes stale. That is insurance against losing the hand, not a trading decision, and it is exactly the framing `day_rider.py`'s own header uses for the 600pt stop it already describes. 231 sessions, shipped 4-lot ladder: naked $-\$32,810$ (worst session $-\$8,616$); $3.0\times\text{ATR}$ $-\$9,940$ (worst session $-\$1,689$); every width beats naked at every entry delay. This week, fee-corrected and with the carry now resolved: as-traded $-\$2,400.44$ across six entries, $3.0\times\text{ATR}$ $-\$532.10$. On 08-25, the one session since: as-traded $-\$243.00$, $3.0\times\text{ATR}$ $-\$452.16$.

3 · Bench `abs_veto_short`'s automated slot — and keep the signal in shadow (MONDAY #1). Set `abs_veto_short=off` in `data/`

gate_switches.env. Leave the 55s-confirm mirror running in the shadow book to carry the case. −\$170.00 over 22 lots live; the machine's half is negative in 10 of 10 stop cells; your hand is worth +\$276.24 of it (live book minus machine-repriced book, \$2.4).

Generated from the top of the card below, so the two cannot drift apart. Everything else, including every refuted idea and why it died, is in the five windows underneath.

SATURDAY · BEFORE THE SUNDAY 22:00Z REOPEN — ONE OF THESE HAS A REAL DEADLINE · 6 PLAYS

The first row is not a strategy change either: it is a recovery timer that outlived the incident it was built for and **fires again at 22:00Z on Sunday 2026-08-30**, where it will claim whatever is open. Everything under it is code and config that needs a restart. **Note what is NOT forced any more:** gazbot7-gate-reactivate.timer no longer arms anything at 22:00Z — its hold set is the whole roster and its last two runs both logged nothing to change — so the gate decisions on this card can genuinely wait if you want them to. The position cannot.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
1	DELETE gazbot7-weekend-flatten.timer — it is still armed and it fires again on Sunday LIVE systemctl disable --now gazbot7-weekend-flatten.timer and remove the unit file. It was built as a one-shot for a position that no longer exists, and it is the only row on this card with a date attached. Mechanism: The unit is written to self-disable once the venue reads flat — but it dies on a PermissionError before it ever reads the venue, so it has never disabled itself. systemctl list-timers has it next elapsing Sunday 2026-08-30 22:00Z, then every minute to 23:59Z. What it does when it runs is re-arm the day rider and write a bare-stamp claim through the operator-button path — and a bare stamp means claim everything. Against whatever happens to be open that night, on a shared netted account, that is the 08-06 cascade with a timer on it. Why: A recovery job that outlives the incident it was built for is not a safety net, it is an unowned actor with write authority. This one cannot even report that it failed: the page() call sits BELOW the file write that kills it. Evidence: Front page (how the open position ended); Part 1.6 §5; the unit's own journal, which is the arbiter here rather than any desk instrument. · verification: systemctl list-timers grep weekend-flatten returns nothing, and systemctl is-enabled gazbot7-weekend-flatten.timer says disabled or the unit is gone. · mode: deploy	119 firings on 2026-08-23, 0 orders placed, 0 pages sent. Next elapse Sun 2026-08-30 22:00Z. Journal: 105 gateway failures then 14 PermissionError.	Arms: Immediately. There is no condition. Exit: n/a — this is a deletion. Kill: If you want the capability kept, the kill criterion is the opposite: it may stay ONLY once page() is moved ABOVE the file write and the run is proven end-to-end against a live position on a Sunday. Until then it is a liability, not a net.	Garrath — by hand at the desk Re-enable the timer; the unit file is in git.
2	Re-open the day rider's protective stop — the 2026-08-20 “no stop” decision was taken on a number measured on a DIFFERENT rider LIVE Read the standing decision first, because this play re-opens it. src/gazbot7/day_rider.py:93 carries PLACE_VENUE_STOP = False under, verbatim: “★★★ NO STOP. Operator, 2026-08-20: i dont want any stop. leave them all naked ... if its going south my claim profit buttons also work as a manual kill button.” The same block records the exposure as knowingly accepted — median worst-adverse 160pt = −\$1,284 on four lots, worst session 1,084pt = −\$8,672 — and cites the desk's own counter-evidence: a venue stop “costs \$3,451 of expectancy AND has a WORSE worst-day (−\$1,603) than running naked (−\$1,531)”. The ask is	231 sessions, shipped 4-lot ladder: naked −\$32,810 (worst session −\$8,616); 3.0×ATR −\$9,940 (worst session −\$1,689); every width beats naked at every entry delay. This week, fee-corrected and with the carry now resolved: as-traded −\$2,400.44 across six entries, 3.0×ATR −\$532.10. On 08-25, the one session since: as-traded −\$243.00, 3.0×ATR −\$452.16.	Arms: Every rider entry, unconditionally — or, in the narrow version, only once the flatten path fails a health check. Exit: The stop, the existing trail, or the 20:40Z hard flat — whichever fires first. Kill: If a stop ships, grade it on the TAIL and not the total: the claim is a worst-session cut, not an expectancy gain, and the 231-session race says there is no expectancy gain to find. Widen before removing. If over the next 20 rider entries the stop has not	Garrath — this is a decision he already made once; Claude ships whichever way it goes data/day_rider.env → day_rider=off, or drop the stop parameter.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	therefore NOT “ship a 3.0×ATR stop”. It is: look at the 231-session re-derivation below, then decide. If you still want it naked, that is a decision this report accepts and stops re-proposing; what it will not do is leave the number unchallenged, because the number turns out to be about a different rider. If the answer is still no, the narrower ask that the 08-20 objection does not already answer is a stop that arms only when the flatten path is unhealthy — i.e. place nothing while the gateway answers, and place a wide resting stop the moment <code>ib.connectAsync</code> fails or the heartbeat goes stale. That is insurance against losing the hand, not a trading decision, and it is exactly the framing <code>day_rider.py</code>'s own header uses for the 600pt stop it already describes. Mechanism: WHAT IS NEW, and why one week does not overturn a 231-session decision.
(a) The cited counter-evidence is not about this configuration. Trace the \$3,451 back and it lands in <code>day_rider.py</code>'s own header, where it describes a 400-POINT fixed stop on the 2-LOT rider with the 150pt trail and the 21:00 flat — the pre-08-20 machine — over 31 validated sessions. It is quoted at line 96 as though it settled the question for the 4-lot ladder. It does not, and it is not reproducible on it: raced on the 231 lake sessions with the ladder that actually shipped, a 400pt fixed stop scores −\$30,058 against naked's −\$32,810 and cuts the worst session from −\$8,616 to −\$3,206. On the machine the operator is actually running, that stop is free and it halves the tail.
(b) The 08-21 carry is new information the decision could not have had. The objection “my claim button IS the kill button” assumes a hand at the desk. On 08-21 there was not one for three days: the gateway wedged at 19:57Z, all four automatic flatten paths share one <code>ib.connectAsync</code> and died together, and the position closed by hand on Monday for −\$2,149.44. Every stop width tested closes it by 13:31:55Z, six and a half hours before the wedge.
THE 231-SESSION RACE, same ladder, same trail, same fee, only the stop changes. Entry delay is swept 0/2/4/8 minutes per rider_lab's standing rule. Naked: −\$32,810 / −\$29,996 / −\$23,597 / −\$28,576, worst session −\$8,616. 3.0×ATR: −\$9,940 / −\$5,678 / −\$423 / −\$2,234, worst session −\$1,689. EVERY stop width beats naked at EVERY delay — the widths run 1.0× through 5.0× and land in a −\$1 to −\$52 per session band where naked is −\$102 to −\$142. That is a plateau on the decision (have a stop) and NOT on the width (nothing separates 1.0× from 3.0× on expectancy). Where the widths do separate cleanly is the tail, monotonically: worst session −\$567 at 1.0×, −\$848 at 1.5×, −\$1,128 at 2.0×, −\$1,409 at 2.5×, −\$1,689 at 3.0×, −\$2,251 at 4.0×, −\$2,812 at 5.0×, −\$8,616 naked.
THE HONEST OTHER SIDE, and it is not small. Nothing here is profitable: the 4-lot ladder loses money over 231 sessions in every arm including the live one, so this is a choice between losses. And on the ONE session since publication the operator's instinct won — on 08-25 the rider went 209pt against a 4-lot long and he claimed at −57pt for −\$243.00, where a 3.0×ATR stop would have cut at 14:03:57Z for −\$452.16, \$209 worse. The harness cannot model a hand-kill at all (it only models claiming a WINNER), so the operator's own mechanism is untested by every number above. Why: A decision taken on a misattributed number should be re-taken on the right one, and this one was: the \$3,451 belongs to a 400pt stop on a 2-lot rider over 31 sessions, not to an ATR stop on the 4-lot ladder over 231. On the real configuration the stop is not a cost, it is a tail cut of roughly 80% for no measurable expectancy give-up. But the operator's objection has not been refuted either — it names a mechanism (his hand) that no backtest here can score, and on the single session since this report was drafted that mechanism beat every stop on the grid. Evidence: Part 1.5 disposition (LIVE) and Part 1.6 §3b — the corrected rider-stop grid, raced on every day rather than only the losing one. · verification: Whichever way it goes, the decision is recorded IN THE CODE at <code>day_rider.py:93</code> with today's date and this section's numbers beside the 08-20 quote, so the next reader does not re-litigate it from a 31-session figure. If a stop ships: a resting stop order visible on the venue within 60s of a rider entry and <code>day_rider_state.json</code> carrying its price — not a soft stop in code, because a soft stop dies in the same wedge that caused this. · mode: decide		reduced the worst single session below the naked distribution's, it is not doing the job it was bought for.	

3	Make the tournament standdown a FILE that every writer checks LIVE Move <code>TOURNAMENT_STOOD_DOWN</code> out of <code>router_tick_durable.py</code> and into a state file, and make	2 unauthorised arms — 2026-08-21 (2m15s) and 2026-08-24 (3m50s), both \$0 by luck, both unpaged. open-hour-watch.service's own unit file says	Arms: n/a — it is a guard, always on. Exit: n/a. Kill: If the file becomes a third source of truth rather than the	Claude — code change, then a restart Revert the commit; the variable comes back.
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#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	open_hour_watch.py — and any future writer — read it before writing a switch. Mechanism: The standdown is a router-LOCAL variable. It only stops the ROUTER writing on. open_hour_watch.py is a second writer with off-on authority and zero standdown awareness, and it has armed abs_veto_short straight through the standing operator standdown twice: 2026-08-21 14:03Z (re-benched 14:05:18Z, 2m15s) and — ★ REV3, found at publication — again on 2026-08-24 13:46:32Z (re-benched 13:50:22Z, 3m50s), three days after the report's window with nothing changed in between. The only reason either cost nothing is that the tape produced no qualifying signal inside the window, and neither paged. Two \$0 outcomes are not a control; they are the same uncontrolled exposure sampled twice, the second one longer. Why: A safety state that one of its two writers cannot see is not a safety state. It has now been breached twice in four days and the exposure got longer, not shorter; both were \$0 purely by luck, and luck is not a control. Evidence: Part 2.6 §4 and its disposition (LIVE, named a Saturday item). · verification: Set the standdown, then force an open_hour_watch arm condition and confirm it refuses and logs the refusal. A test that cannot be shown to fail is not a test. · mode: deploy	it must NEVER write gate_switches.env; it has written since 08-05.	single one, delete it and make the standdown a property of the switch file itself.	
4	Delete the five expired carve-outs from gate_switches.env's comment header LIVE Open data/gate_switches.env and delete the five dated instruction lines in its header. Deleting the lines is the only thing that closes a carve-out. Mechanism: switch_notes() splices the top 45 comment lines of that file into EVERY router prompt, labelled ACTIVE, and nothing prunes them. ★ REV3, re-read at publication by importing switch_notes() rather than grepping the file: there are now SIX carve-out blocks and the cap splices FIVE of them (08-24, 08-21, 08-19, 08-13, 08-11), cutting off mid-sentence inside the 11 August one. The oldest line still spliced is fifteen days old. Four are past their own stated expiry. A sixth block was prepended on 2026-08-24 13:46Z and it pushed the 2026-08-09 line — the dangerous one — past the 45-line cap, so it is no longer spliced: evicted, not closed, with nothing in the file recording it. That line asserts that abs_veto_short IS ARMED BY DEFAULT — which directly contradicts the standing operator standdown the router is currently enforcing, on the same gate that open_hour_watch.py armed through that standdown on Friday. Why: A carve-out comment never expires and a zero-mention streak is not a fix. Two of the five have been restated to the router 106 and 66 times out of 1,425 ticks; the other three sit there quietly and would be read the moment the tape produces the situation they describe. Evidence: Part 2.6 §5 and its disposition (LIVE). · verification: head -45 data/gate_switches.env shows no dated instruction lines, and the next router tick's prompt carries no ACTIVE carve-out block. · mode: deploy	5 instructions spliced as ACTIVE, oldest 13d 1h. Restated 106 and 66 times in 1,425 ticks; three at 0 of 1,425 — which is a dormant instruction, not a closed one.	Arms: Immediately. Exit: n/a. Kill: n/a — this is a deletion of expired text. If a carve-out is still wanted, re-add it with an expiry the splicer enforces.	Garrath — by hand at the desk git restore the file; the lines are in history.
5	Point core_health.flat at the venue position, or rename it tournament_flat LIVE Either make core_health.flat read the venue position line, or rename the field so nobody reads it as a desk-wide safety signal again. Mechanism: The field is tournament-scoped. It said the desk was flat while the venue was long four lots, and the router repeated that reading for 811 of the 817 ticks the position was alive — 13:13:06Z on 08-21 to the hand-close at 11:59:39Z on 08-24 — while updatePortfolio in the same journal read position=4.0 ... unrealizedPNL=-950.27. Not one of the 817 ticks mentioned the open position at all. (★ REV3 — Rev1 quoted “137 of 141” here and “132 of 137” in Part 1; both were snapshots taken at whatever moment their section rendered, and neither is reproducible from data/router_trial_log.txt. The count above is cut at the position's own life, so it is fixed and cannot drift.) Why: This is the instrument that should have caught the whole incident and it read green throughout. A flag that is right about one book and silent about the other is worse than no flag, because it is the one people check. Evidence: Part 2.6 §6 and disposition (LIVE); Part 1 disposition (FAULT). · verification: Open a 1-lot rider position by hand and confirm core_health.flat goes false. If it stays true, the rename is the honest option. · mode: deploy	811 of 817 router ticks called a live four-lot position flat; 0 of 817 mentioned it.	Arms: n/a. Exit: n/a. Kill: n/a.	Claude — code change, then a restart Revert the commit.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	Grade the per-lot profit ladder that shipped on 2026-08-20 — it has never been scored LIVE No change yet. This is the ladder's first grading. The operator's design, verbatim: "1 takes profit at \$100. 2 at \$200. 3 at \$400. 4 at \$600. so if they all take profit we take \$1300 for the day." — TARGET_USD_PER_LOT = (100, 200, 400, 600), which at \$2.00/pt/lot is rungs at 50 / 100 / 200 / 300 points. Mechanism: Raced over the same 231 lake sessions, inside the running system (so the shipped ATR trail can take a lot out before its rung is reached) rather than as raw peak-reach rates:
lot 1 (\$100 / 50pt) fills on 71.0% of sessions · lot 2 (\$200 / 100pt) 39.4% · lot 3 (\$400 / 200pt) 11.3% · lot 4 (\$600 / 300pt) 4.8%. Every rung is NET NEGATIVE and the deeper the rung the worse: — \$4,590 / —\$7,787 / —\$10,845 / —\$9,587. The advertised \$1,300 day needs all four and happens about one session in twenty-one.
The gap that matters: day_rider.py's comment quotes reach rates of 50pt 77.5% · 100pt 58.0% · 200pt 29.4% · 300pt ~15%. Those are RAW PEAK reach rates — does the high ever get there — and the shipped trail intercepts lots 2 to 4 long before it does. Lot 4 exits on the TRAIL on 133 of 231 sessions and on its own rung on 11. So the ladder as designed is mostly not the thing that closes the position. Why: It shipped on the same day as the no-stop decision and nothing has graded it since. The finding is not that the ladder is wrong — it is that the trail and the ladder are two exits competing for the same lots, and the trail wins 58% of the time on the lot the ladder was designed to let run. Evidence: Part 1.6 §3b and §6; scripts/rev2_rider_stop_231.py. · verification: Re-run scripts/rev2_rider_stop_231.py with the rung mix logged per lot and confirm the four fill rates. Live: count TARGET_100/200/400/600 exit reasons in trades against rider entries. · mode: review	231 sessions: rung fill rates 71.0% / 39.4% / 11.3% / 4.8%; all four rungs net negative; lot 4 closes on the trail 133 times and on its \$600 rung 11 times. Live since 08-20: four entries, sixteen lots, exactly ONE rung has ever filled (the \$100 lot on 08-25).	Arms: n/a — this is a grading, not an arm. Exit: n/a. Kill: If lots 3 and 4 have still filled their own rungs fewer than 1 time in 10 after 20 more entries, the rungs are decorative and the honest change is to let those lots ride the trail explicitly rather than pretend they have a target.	Claude — measurement, no code change n/a.

MONDAY · AT THE DESK — SWITCH-FILE AND EXIT-CELL CHANGES, ALL REVERSIBLE · 6 PLAYS

No code deploy in this window. Every row is gate_switches.env, exit_overrides.json or a router constant, and comes back in five minutes. This week's list is short on purpose: the tournament traded two slots and 22 lots, which is not enough evidence to re-tune anything, and the report says so in three separate places.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
1	Bench abs_veto_short's automated slot — and keep the signal in shadow SHADOW Set abs_veto_short=off in data/gate_switches.env. Leave the 55s-confirm mirror running in the shadow book to carry the case. Mechanism: The gate reads −\$170.00 live only because your hand is worth +\$276.24 against the machine alone, which books — \$446.24. (★ REV3 — this row used to say +\$314.74 and −\$203.00. Both are withdrawn: −\$203.00 charged the \$1.50 round trip twice, and +\$314.74 is the booked-minus-repriced figure on the four hand-closed lots, which does not reconcile with the totals beside it because a separate −\$45 simulator residual sits on the other eighteen. §2.4 prints the split four ways; the bottom row, +\$276.24, is the one to quote.) Strip the hand and the machine's half is negative across ten cells in two stop sweeps, every one of them negative, with the worst leave-one-day-out negative throughout — so the sign survives, which makes this a verdict rather than a knife-edge. Why: Benching the machine's half is not benching the idea. The entry treatment (the 55s continuation-confirm) is the top of this week's shadow board and it stays there; what is being switched off is the unfiltered automated slot. Evidence: Part 1.5 §2 and disposition (SHADOW); Part 2 §3 for the mirror's standing. · verification: grep abs_veto_short data/gate_switches.env reads off, and the tournament's next start logs it in the gate switches changed → DISABLED [...] line. △ Prove it by that line, never by the file's mtime. · mode: config	−\$170.00 over 22 lots live; the machine's half is negative in 10 of 10 stop cells; your hand is worth + \$276.24 of it (live book minus machine-repriced book, \$2.4).	Arms: n/a — it is a bench. Exit: Existing positions still exit normally; a bench blocks NEW entries only. Kill: Un-bench when the 55s mirror clears the Part 2 battery on a positive tradable-hours week with a positive leave-one-day-out. Not on a lapse, and not on a quiet week.	Garrath — by hand at the desk Delete the =off line — but restart the tournament FIRST; the roster is import-time only and an absent gate defaults to ON.
2	Retire rider_w5 from the shadow book — it has been quietly red since 08-17	−\$1,471 over 51 live forward trades (−\$28.84/tr) against +	Arms: n/a — this is a retirement. Exit: n/a.	

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	REFUTED Remove the rider_w5 arm from src/gazbot7/shadow.py (or add it to the retired set). It is the only decided item that comes out of Movement 3 and it costs nothing. Mechanism: Shipped on 2026-08-15 out of last week's greenfield phase, quoted at +\$4,841 and +\$19.29 a trade. It has taken 51 forward trades since 08-17 and sits at −\$1,471, or −\$28.84 a trade, 18% wins, red on 6 of 7 days and red in all four ATR buckets — worst in the 11-16 band its own filter selects for. Zero flagged rows, so this is clean data, not a measurement artefact. Its in-sample +\$1,936 collapses to +\$282 on strip-best-3, and the +\$19.29 it shipped on never had a strip test attached. Why: This is not a thin-n case. 51 live trades, 1,339 out-of-sample trades at −\$5.43 each, and −\$174 on the census week it was built for all agree, and the mechanism has an explanation: at fire time the median run has 12–41pt left against a 33–48pt stop, and about one fire in twenty is a run. Evidence: Movement 3 §3 and §5 (the pooled-rider card); RIDER_ALL §5 in the appendix carries the full grading. · verification: The arm no longer appears in shadow.db's active strategy list, and the next shadow tick logs one fewer arm. · mode: deploy	\$19.29/tr claimed. OOS: 1,339 trades, −\$5.43/tr, strip-3 −\$9,293.	Kill: Only revive it behind a SELECTOR — something that predicts which board becomes a run. Not more days: the power calculation says ~10,500 trades to separate it from zero.	Claude — code change, then a restart Re-add the ShadowVariant; it is in git.
3	Retire VAC-ABS too — last week's other greenfield shadow pick REFUTED Retire the VAC-ABS arm. Do not re-shadow it. Mechanism: Forward-tested unaltered it looked like the week's best greenfield result: +\$1,319.21 over 97 trades, +\$13.60 a trade net of the \$1.50 round trip. Then it was given the five days it had never seen: −\$529.50 over 27 trades, confirmed by strip-5 at −\$427. Why: Two shadow arms shipped off last week's greenfield phase; both are dead a week later; neither was being graded by anything. That is the finding, and it is procedural rather than analytical. Evidence: Movement 3 §6 (the graves table) and the VACUUM dossier's VERDICT in the appendix. · verification: Absent from shadow.db's active strategy list on the next tick. · mode: deploy	+\$1,319.21 / n=97 / +\$13.60 per trade forward, then −\$529.50 over 27 unseen trades; strip-5 −\$427.	Arms: n/a. Exit: n/a. Kill: n/a — refuted by its own out-of-sample leg.	Claude — code change, then a restart Re-add the arm.
4	Adopt the 07:00–13:00Z bench for any late-boarding rider as a router rule SHADOW Add a router constant that benches late-boarding rider arms in the European session (07:00–13:00Z). It is a bench, so it fails safe. Mechanism: It is the one rule in the whole greenfield movement that replicates on all three independent legs: −\$29.38 a trade on the in-sample lake, −\$20.04 on the 205-session leg, and −\$2.43 on the 11-session forward leg (n=373 on the last). No new instrument is needed and nothing has to be predicted. Why: Everything else the movement produced is a build or a question. This is the only positive deliverable that survived its own battery, and its direction is the cheap one — a wrong bench costs an option, a wrong arm costs money. Evidence: Movement 3 §10 disposition and §5 (the UNCLASS card); the UNCLASS dossier §11 in the appendix has the three legs in full. · verification: A router tick inside 07:00–13:00Z shows the rule in its reasoning and no late-boarding arm switched on. · mode: config	Red on all three legs: −\$29.38 / −\$20.04 / −\$2.43 a trade, the last on n=373.	Arms: n/a — it is a bench, active 07:00–13:00Z. Exit: n/a. Kill: Drop the rule if a late-boarding arm is green in that window over 100+ forward trades. One green week is not enough — the effect it is buying is small.	Claude — code change, then a restart Delete the constant.
5	Run the gold fade UNROUTED — last week's MGC router rule is refuted SHADOW Do not apply last week's regime router rule to mgc_hole_break_fade. Leave the arm running across all regimes in shadow. Mechanism: The rule was killed twice, independently: by the forward walk AND by the 320-day tape. It would bench the gate in NORMAL_CHOP, which is the regime currently paying best. Why: A router rule fitted on 29 days of gold and then contradicted by 320 is exactly the shape of the in-sample rules this desk keeps having to withdraw. The absence of a rule is the finding. Evidence: Movement 3 §8 and disposition; the MGC dossier §5.3 in the appendix. · verification: No regime condition on the gold arm in the shadow config. · mode: shadow-first	Refuted on both the forward walk and 320 trading days of MGC minute bars (2025-07-28 → 2026-08-19).	Arms: All regimes, in shadow only. Exit: The wide chandelier on a single lot — stop 3.0 ATR, arm 2.0, trail 2.0, 480-min backstop. NOT the live MNQ slot configuration, which is the opposite shape. Kill: Revisit routing only after a forward walk on the depth-mid bars, i.e. after BUILD #1.	Claude — code change, then a restart Re-add the regime condition.
6	★ Shadow the abs_veto_short exit ladder — the one treatment that worked on the report's #1 rehab target SHADOW Add two A/B arms on abs_veto_short's own signal, against the live 1.5R/2.5R pair: a scalp at 0.5R and a scalp at 1.0R, plus a +15pt-arm / 8pt-behind trail as a third. Same	Four rungs positive, 7/7 winners kept, worst-LODO and strip-best-1 positive; but 80th–93rd percentile of a 400-draw random-entry control. Machine baseline −\$446.24, as traded −\$170.00.	Arms: Every abs_veto_short signal, including while the live slot is benched. Exit: The rung, or the live stop, whichever comes first.	Claude — a ShadowVariant, no live change Remove the arms; nothing live depends on them.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	signal, same stop, different exit — so this is a scale-out comparison, not an experiment about entries. Mechanism: Part 1.5 §2.4 is the best rehab result in this report and Rev1 shipped nothing off it. On the gate's own 22 lots, four rungs beat the as-traded book while keeping 7 of 7 winners, staying positive on the worst leave-one-day-out AND with their best lot stripped: 1.0R + \$187.00, 0.5R + \$178.50, the 8pt trail +\$149.00, a scalp at the losers' own median peak of +11.25pt +\$141.00. The machine baseline is — \$446.24. Why it is a SHADOW row and not a Monday change: against a 400-draw random-entry control on the same tape and the same R distribution, the rungs sit only at the 80th-93rd percentile — none clears its own placebo band. On n=22 that is a direction, not a measurement. Why: The report criticises Rev1's claim detector by name for leaving the document with no play, no owner and no kill criterion. The exit ladder was about to leave it the same way — and it is the treatment that actually worked on the gate §2.1 diagnosed as an exit problem. Evidence: Part 1.5 §2.4 and its placebo table. · verification: Per BUILD #7, diff the arms row-for-row at seed time and confirm they are NOT byte-identical to the control or to each other — a broken arm records the control's trades under a second name and reports a perfect null. abs_veto_55s's siblings already show 7 of 38 arms in that state. · mode: shadow		Kill: Promote only on: $n \geq 60$ fires, positive on strip-best-3, AND clearing the 95th percentile of the random-entry control the section already built. They sit at the 80th-93rd today. If at $n \geq 60$ a rung is still inside its own placebo band, it is REFUTED by that control and comes off the board — this is the named test, stated in advance.	

BUILD · NEXT — NOTHING GOES LIVE THIS WEEK · 18 PLAYS

Instrumentation, shadow slates and the method fixes that decide whether next Friday's numbers can be trusted at all. **Four of these are measurement faults found this week**, not ideas: byte-identical shadow arms reporting perfect nulls, two nightly rollups that silently stopped, a health flag that says flat while the desk is long, and gold results confounded with a large short drift nobody is reporting.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
1	★ Build the depth-mid bar source into the MGC shadow — the gating item for all of gold LIVE Point the MGC shadow service's minute-bar builder at the depth mid instead of at trades, and keep the existing trade-bar arms alive as <code>*_tradebar</code> controls so the difference is visible rather than asserted. Nothing else on the gold line is worth doing first. Mechanism: Last week's shadowed gold gate has been live since Monday and is losing: −\$1,196 across 80 trades in seven days, all three arms red. The lab replay of the same seven days makes + \$512. The cause is the input tape: the research builds one-minute bars from the depth mid (which updates every 250ms whether or not anything trades), the service builds them from trades, and gold prints sparsely — so a sixty-minute high on one is not the sixty-minute high on the other. Hold everything else constant and change only the tape: depth-mid bars $n=56 \rightarrow +\\$172 (+\\$3.06/tr)$; trade bars $n=53 \rightarrow -\\$475 (-\\$8.96/tr)$. The sign flips on the bar source alone. Why: This is a plumbing mismatch between lab and production, not a research failure, and the fix is mechanical: the service already holds a DepthFeed connection to gold's book because the liquidity filter needs it, so it already has the bid and the ask. Until it lands, no gold cell should be promoted and none is being proposed. Evidence: Movement 3 §8 (gold, in its own right); the MGC dossier §3 and §12.1 in the appendix. · verification: Both arms present in data/shadow_mgc.db under distinct names, and a week of forward trades where the depth-mid arm and the *_tradebar control diverge in the predicted direction. · mode: deploy	Live shadow −\$1,196 / 80 trades vs lab +\$512 on the same days. Bar-source A/B on the six days both tapes cover: +\$3.06/tr vs −\$8.96/tr.	Arms: Shadow only. No live gold promotion behind this until the two arms have run side by side. Exit: Wide chandelier on a single lot: stop 3.0 ATR, arm 2.0, trail 2.0, 480-min backstop. Kill: If the depth-mid arm is ALSO red over 60+ forward trades, the bar source was not the explanation and the whole gold reversion family goes back to PARKED.	Claude — code change, then a restart Point the bar builder back at trades.
2	Add atr_min, a wrapping clock window and an ER10 gate param to ShadowVariant LIVE Three small changes to <code>src/gazbot7/shadow.py</code>: (1) <code>gate_board</code>'s window test becomes <code>inw = (lo &lt;= h &lt; hi) if lo &lt; hi else (h &gt;= lo or h &lt; hi)</code> so a 21:00→07:00 window can exist at all; (2) <code>ShadowVariant</code> gains an <code>atr_min</code>, defaulted to 0.0 so nothing existing changes; (3) an <code>er_win_min</code> that reads the ENTRY WINDOW's efficiency rather than ER30.	ER30 substituted for ER10 flips the held-out leg +\$5.34/tr → −\$2.25/tr and reverses both controls.	Arms: n/a — infrastructure. Exit: n/a. Kill: n/a.	Claude — code change, then a restart Revert the commit; defaults are chosen so nothing existing changes.

Mechanism: All three are blockers on shadowing anything from this movement honestly. A rule that is a BAND cannot be expressed with only a ceiling. A wrapping clock window is never true, so an overnight arm silently never fires. And ShadowVariant.er_min reads ER30 — substituting it for the 10-minute ER FLIPS the held-out leg from +\$5.34 to −\$2.25 a trade and turns both controls positive, i.e. it is a different hypothesis wearing the same name.

Why: ★ This is exactly how rider_w5 came to be quoted at +\$19.29 while making −\$28.84. An arm shipped AROUND a missing field records a different strategy under the intended name, and nothing downstream can tell.

Evidence: Movement 3 §9 item 5; the RIDER_ALL dossier §11 in the appendix, which lists all three blockers with the one-line fix for each.
 · verification: **A unit test that an arm with hh_lo=21, hh_hi=7 fires at 23:00Z and not at 12:00Z, and one that atr_min excludes a below-band fire.** · mode: deploy

3★ Fix the census's flow-bucket phase — 21 of 68 labels move on it alone

LIVE In scripts/run_census.py, snap the flow window's slide to the CLOCK rather than to the array index.

Mechanism: 21 of 68 run labels change on bucket phase alone, and only 17 of the 68 run timestamps are minute-aligned in the first place. This is the mechanism behind the divergence between the two census freezes this cycle — 70 runs / 65 sat out / \$9,986 became 68 / 62 / \$9,610 — which is why Movement 2 carries a red banner.

Why: Every hunt this desk scopes by cluster inherits this error, and it is a real bug with a one-line character rather than a modelling judgement.

Evidence: Movement 3 §4 (the vocabulary table) and §9 item 3; the UNCLASS dossier §13. · verification: **Re-freeze the census twice with the bucket origin offset by 30s and get the same 68 labels both times.** · mode: build

21 of 68 labels phase-dependent; 17 of 68 run timestamps minute-aligned; two freezes of the same week differ by 2 runs and \$376 of ceiling.

Arms: n/a.
Exit: n/a.
Kill: n/a.

Claude — code change, then a restart
git revert.

4★ Rename two census labels, stop splitting on flow, and ship the four run SHAPES

LIVE Three edits to the census's vocabulary: UNCLASS → UNTESTED and report it as COVERAGE; OPEN/NEWS → US-OPEN-WINDOW with the two footprint tests DELETED; stop splitting runs into FLOW-LED/VACUUM at all. Then add the four run SHAPES — TURN, STEP-REPRICE, DRIFT-VOID, GRIND-CONT — one if each, all computable at the run's start minute.

Mechanism: Three of the five labels were interrogated directly this week and none survived. UNCLASS: base rate 75.4% on all tape, lift **0.82** — a run is LESS likely to be UNCLASS than an arbitrary minute is. OPEN/NEWS: lift over the bare clock **0.921x**, i.e. the two footprint tests remove 41% of the window and keep the LESS run-dense half. FLOW-LED/VACUUM: the flow event fires on 20.4–20.6% of all tape and then splits it on a **49.6–49.9% coin toss** — 0.33 percentage points of directional information.

Why: These are independent measurements by different hunts on different tape, and two of them are the second such measurement in two weeks. The vocabulary is the thing the whole cold half of the report is scoped by; leaving it wrong is more expensive than any single gate on this card.

Evidence: Movement 3 §4; the FLOW-LED, UNCLASS and OPEN-NEWS dossiers in the appendix each reach it independently. · verification: **Next week's census prints the new names and a VOL-EXPANSION bucket with more than one member.** · mode: build

UNCLASS lift 0.82 on 75.4% base; OPEN/NEWS lift 0.921x over the bare clock; FLOW-LED/VACUUM split carries 0.33pp. Replacement: VOL-EXPANSION is 4.18x lift on 1.95% of tape and the census barely uses it.

Arms: n/a.
Exit: n/a.
Kill: If the four SHAPES turn out to be as unstable as the labels they replace, report runs with NO cluster at all rather than a third wrong vocabulary.

Claude — code change, then a restart
git revert.

5⚠ Fix the lake's bars timeframe landmine — every unfiltered query now has look-ahead

LIVE Make gazbot7.lake require an explicit timeframe on any bars query, or default it to the 5s daily files. Then audit every caller.

Mechanism: New backfill_{1min,5mins,1hour,1day}.parquet files now sit BESIDE the daily 5s files. Any query that does not filter on timeframe now inherits **up to a full day of look-ahead, on every day**. gf_mgc_tape.load_l1() is safe. mgc_session_anchor.load() is NOT.

Why: This is a silent correctness fault in the data layer that every backtest in this report and every future one runs on top of. It produces confidently wrong numbers with no error, which is the class of fault this desk has been bitten by four times already this month.

Evidence: Movement 3 §7 (the harness-bug table) and disposition; the MGC dossier §1.2. · verification: **A test that an unfiltered bars**

1 confirmed unsafe caller found (mgc_session_anchor.load()); the blast radius is every bars query in the repo.

Arms: n/a.
Exit: n/a.
Kill: n/a.

Claude — code change, then a restart
git revert — but do not, this one is a correctness fault.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
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query raises rather than returning rows. A fault that cannot be shown to fail is not fixed.

· mode: build

6	Shadow an ignition-CONFIRMED entry — the only asymmetry the idle-gate lab found SHADOW Add a shadow arm that joins a run AFTER ignition rather than predicting it, and score it per gate with an $n \geq 60$ kill criterion on each. Mechanism: Across the 65 sat-out runs of the SUPERSEDED 08-22 census freeze and four arms, every arm is negative and the live config loses $-\\$1,333.85$ in the run windows. (★ REV3 — the freeze moved: Movement 1 is the 08-25 re-freeze at 68 runs / 62 sat out / \$9,610, and re-cut onto it the same arm is $-\\$976.71$ on 21 fires. Same sign, same instruction, smaller number. The whole-week arm, $-\\$1,905.36$, is census-independent and unaffected.) The single exception is the post-ignition arm, nearly flat at $-\\$45.00$ on 17 fires against a clearly negative pre-ignition arm. Why: ★ AND THE CAVEAT IS THE PLAY. That $-\\$45.00$ is abs_veto_short's $+\\$504.50$ on 5 fires against $-\\$549.50$ on the other five gates' 12 fires; strip one fire and the arm is $-\\$156.00$. So the $n \geq 60$ kill criterion has to be PER GATE, not pooled — pooling is what made this look like an asymmetry in the first place. Evidence: Movement 2 §2 and disposition (SHADOW), which states the per-gate requirement explicitly. · verification: Six separate per-gate rows in the shadow board, each with its own n, and no pooled headline anywhere on the page. · mode: shadow-first	$-\\$45.00$ on 17 post-ignition fires vs a clearly negative pre-ignition arm; but $+\\$504.50$ of it is 5 fires on one gate, and strip-one is $-\\$156.00$.	Arms: Shadow only. Exit: Match the gate's own live exit so the comparison is entry-only. Kill: Any gate that is negative at $n \geq 60$ on its OWN fires is dropped from the arm. If all six are dropped, the asymmetry was one gate on five fires and the lead is REFUTED.	Claude — code change, then a restart Remove the arms.
7	Add a row-level duplicate check at shadow-write time, and re-seed the broken arms LIVE At shadow-write time, detect two arms writing byte-identical rows and refuse. Then re-seed the three broken arms. Mechanism: 7 of the 38 shadow arms are the same rows under different names — one pair (cx_clip_brk_live vs _standdown) matches on 77 of 77 rows on every field. A broken arm records the CONTROL's trades under a second name and then reports a perfect null, which is indistinguishable from a real null. Why: Until this lands, every A/B delta on the shadow board is unverified — including the ones this report quotes. That is a bigger problem than any single arm's number. Evidence: Part 2 §5 and disposition (FAULT, named as a BUILD item). · verification: Deliberately seed two identical arms and confirm the writer refuses. Then diff every remaining pair row-for-row and publish the count. · mode: deploy	7 of 38 arms duplicated; one pair identical on 77/77 rows across every field.	Arms: n/a. Exit: n/a. Kill: n/a.	Claude — code change, then a restart git revert.
8	Move the claim question to the ENTRY side, and give it a kill criterion SHADOW Run the study as an ENTRY-side question — is this entry stop-bound? — rather than an exit-side one, with a pre-registered kill criterion of AUC ≥ 0.60 and a permutation p below 0.05. Mechanism: The automated claim detector (peak / give-back) is refuted as an exit rule: out-of-sample it makes $-\\$4$ on 126 held-out trades against $+\\$1,069$ in-sample, and $-\\$436$ as the live config stands. Two independent populations, 119 calibrations. But the money it was chasing is real and it is on the entry side: $-\\$18,672$ over 378 unwatched STOP exits. Why: This is the largest identified pool of money in the report, and in the first draft it left the document with no play, no kill criterion and no owner. A relocation is only honest if it comes with the test that will kill it too. Evidence: Part 1 §6 and disposition (REFUTED as an exit rule, relocated rather than killed). · verification: A written pre-registration of the feature set and the kill criterion BEFORE the study runs, and a permutation null run on the same feature count. · mode: build	Exit-side detector: $+\\$1,069$ in-sample $\rightarrow -\\$4$ on 126 held-out trades. The pool it was aimed at: $-\\$18,672$ over 378 unwatched STOP exits.	Arms: Study only. Nothing arms off this. Exit: n/a. Kill: AUC below 0.60, or permutation $p \geq 0.05$, charged for the full feature search. Then it is REFUTED on both sides and the pool is written off.	Claude — code change, then a restart n/a.
9	Populate signal_journal.suppressed_by so a gate's fires can be reproduced LIVE Actually write suppressed_by on the signal-journal insert. Mechanism: The column is NULL in every row despite being in the INSERT, so COALESCE(..., 'TAKEN') reports every fire as TAKEN — including fires on benched gates on a zero-fill day. Any counterfactual built on that column is reading a constant. Why: You cannot grade a gate's alternative history without first checking that the sim reproduces its actual fires, and right now nothing can. This is upstream of the router review, the idle-gate lab and every bench-pricing question on the card.	NULL in all 222 rows this week — and it is the same defect the scope flagged last week.	Arms: n/a. Exit: n/a. Kill: n/a.	Claude — code change, then a restart git revert.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
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Evidence: Part 2.6 §7 (the two instruments that should have written this section). · verification: **A benched gate's next suppressed fire shows a non-NULl suppressed_by, and the TAKEN count drops.** · mode: deploy

10	Put a timer on router_nightly and selector_nightly — both stopped silently LIVE Add systemd timers for both, and make each write a heartbeat that sweep.py checks. Mechanism: Neither has a timer. Both stopped being written and nothing noticed — selector_nightly was four days stale on 08-18. Why: A monitoring file that stops being written is worse than one that errors, because its absence looks like a quiet week. Evidence: Part 2.6 disposition (LIVE). · verification: systemctl list-timers shows both, and sweep.py goes critical when a heartbeat is older than 36h. Prove the critical fires. · mode: build	selector_nightly 4 days stale on 2026-08-18; no timer on either.	Arms: n/a. Exit: n/a. Kill: n/a.	Claude — code change, then a restart Disable the timers.
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11	Fix selector_nightly's two known defects before believing it again PARKED Add a data_quality IS NULL filter, and read each gate's OWN stop_atr_mult instead of hard-coding 1.0. Mechanism: Quarantined rows inflate its regret figure — on 2026-08-13, 48% of it was one phantom counted five times — and all four of its modes reprice at stop_atr_mult=1.0, so exhaustion_short (stop_k 1.5 since 08-16) reads −\$167 where live made +\$100.50, and the day rider reads +\$31.50 against +\$188. Why: It is the instrument that is supposed to tell us whether the router is over-benching, and it is currently answering a different question with a different ruler. Evidence: Part 2.6 disposition (PARKED, both defects named). · verification: Re-run it on 2026-08-13 and confirm the regret figure falls and the per-gate stop widths appear in its output. · mode: build	48% of one day's regret was a single phantom counted 5x; exhaustion_short repriced −\$167 vs +\$100.50 live.	Arms: n/a. Exit: n/a. Kill: n/a.	Claude — code change, then a restart git revert.
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12	Shadow board_overnight_band as an INSTRUMENTED QUESTION, not a candidate SHADOW After BUILD #2 (shadowvariant-missing-fields-0821) lands, add board_overnight_band: k=2.5, w=10, ER10 ≥ 0.50, 21:00→07:00Z, cooldown 60m, ATR band 11.0–16.0, stop 3.0×ATR, target 6.0×ATR, adverse_cut_atr=99 (ON PURPOSE), no chandelier, 240-minute cap. Mechanism: Generated on a 236-session backfill (+\$10.46/tr, +0.444R, n=396, strip-10 still +\$2,281, leave-one-month-out worst +\$2,388) and held out on the 28-session 5s lake (+\$5.34/tr, +0.224R, n=95) where the SIGN held — but strip-3 fell to −\$51 and the DAYTIME control came back POSITIVE. The overnight-vs-daytime SEPARATION that generated the idea did not reproduce. Why: ★ It is not here because we think it makes money. The specific thing being bought is whether overnight boards really differ from daytime ones, which two independent samples disagree about. Its random-time placebo is p = 0.097 and the session split was one of three post-hoc comparisons. Evidence: Movement 3 §10 disposition; the RIDER_ALL dossier §10 and §11 in the appendix carry the exact arm and its router rule. · verification: The arm exists with atr_min=11.0 AND atr_max=16.0 both set, an ER10 gate (never er_min, which reads ER30), and adverse_cut_atr=99. Any of those three wrong records a different strategy. · mode: shadow-first	+\$10.46/tr on the generating sample, +\$5.34/tr held out; placebo p = 0.097; daytime control reverses.	Arms: Shadow only, 21:00–07:00Z, ATR 11–16, ER10 ≥ 0.50. Exit: Stop 3.0×ATR, target 6.0×ATR, 240-minute cap. No chandelier — every chandelier variant tested red once the trail bug was fixed. Kill: Negative over 150 forward trades, OR the daytime control arm beats it. The second is the real kill: if daytime is as good, the hypothesis is dead regardless of the P&L.	Claude — code change, then a restart Remove the arm.
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13	Shadow the cross-asset MNQ VETO for gold — the unturned stone that paid SHADOW Add a veto_ref_symbol="MNQ" branch to the gold entry path and run it as its own arm. It needs an MNQ state feed the gold service does not currently have. Mechanism: Over 278 paired days, fading a gold break that the Nasdaq AGREES with loses \$10.52 a trade and sits at the 2.5th percentile of its own placebo. As an arm it reads +\$5.43/tr against +\$1.72 blanket, and it is two-sided. The mirror — a cross-asset momentum CONFIRM — is cleanly negative at −\$2.29/tr on the same days, which is what a real asymmetry looks like rather than a fitted one. Why: It is a genuinely new input rather than another cut of the same tape, it is well-powered, and its direction is the cheap one (a veto).	+\$5.43/tr vs +\$1.72 blanket over 278 paired days; placebo 97th/ 2.5th percentile. Does NOT reproduce on the 29-day book window (34th percentile).	Arms: Shadow only, alongside BUILD #1's bar-source work. Exit: Wide chandelier on a single lot, as for every gold cell. Kill: If it does not separate from its control over 100+ forward trades, the 34th-percentile book-window reading was the true one and it goes PARKED.	Claude — code change, then a restart Remove the branch.
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#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
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Evidence: Movement 3 §8 and disposition; the MGC dossier §6 in the appendix. · verification: **The arm and its un-vetoed control both present in data/shadow_mgc.db, and an MNQ state value logged on every gold entry decision.** · mode: shadow-first

14	Shadow the ER30 arm-meter WITH its own control arm — do not arm a gate behind it SHADOW Instrument ER30 as an arm meter and run a control arm beside it. Nothing gates on it until the two have separated. Mechanism: +\$3.54 against −\$3.40 a trade on 3,238 untouched trades — the largest-n effect anywhere in the greenfield movement — but it REVERSES on the most recent eleven sessions. Why: One of those two readings is noise and we do not know which. Shadowing the meter with a control answers it for free and is the single cheapest open question in the movement. Evidence: Movement 3 §11 (the stones still unturned) and disposition; UNCLASS §13. · verification: Two arms in the shadow board with the same entry and only the meter differing. · mode: shadow-first	+\$3.54 vs −\$3.40 a trade on n=3,238, reversing on the last 11 sessions.	Arms: Shadow only. Exit: Match the control exactly. Kill: If the meter and its control are within noise after 300 forward trades, the effect is an eleven-session artefact and it is REFUTED.	Claude — code change, then a restart Remove the arms.
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15	★ Price a violent-whipsaw router bench — the chop commission's premise was wrong SHADOW Measure a two-term bench (high ATR + low efficiency) on the tournament's own fires, and propose it as a router rule only once it is priced. Mechanism: You asked for a chop-day scalper because the desk donates on chop days. It does not. On this week's two chop days (08-20 and 08-21) the tournament took ZERO trades, and the week's −\$2,323 is the day rider's book. Across the whole 18-day window the tournament's loss is concentrated in violent-whipsaw blocks — high ATR going nowhere — at −\$1,383.50 on 95 trades, against −\$403.00 on 58 in plain chop. Why: The correction is worth roughly ten times the scalper that was commissioned to fix the wrong thing, and it needs no new instrument — both terms are already in the router's vocabulary. Evidence: Movement 3 §5 (the chop card) and §10 disposition; the chopscalp dossier §2 and §13 in the appendix. · verification: A table of the tournament's P&L by (ATR bucket × ER bucket) over 18 days, with the bench's cost in forgone winners priced beside its saving. · mode: build	Violent-whipsaw −\$1,383.50 on 95 trades vs plain chop −\$403.00 on 58; zero tournament trades on the week's two chop days.	Arms: Nothing arms yet. This is a measurement. Exit: n/a. Kill: If the bench's forgone winners exceed its saved losses on the same 18 days, it is REFUTED and the chop lead closes entirely.	Claude — code change, then a restart n/a.
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16	★ Every shadow arm gets a named grading date and a strip-best figure printed beside it LIVE Two rules, enforced in the shadow board's own rendering: an arm may not be added without a graded_on date, and any headline \$/trade must print its strip-best-3 figure next to it. Mechanism: Two arms were shipped off last week's greenfield work. Both are dead a week later and neither was being graded by anything. rider_w5 ran red for seven days on a quoted +\$19.29 that had no strip test attached. UNCL-RIDER-ASIA is worse: it was never shipped at all, and nobody noticed for a week — it does not exist anywhere in shadow.db's 82 strategies. Why: This is the procedural fix behind two of this card's retirements. A number that goes into the shadow book without a strip test and without anybody owning the grading is not evidence; it is a claim with a week's head start. Evidence: Movement 3 §3 (closing the loop) — the two rows and the paragraph beneath them. · verification: The shadow board page refuses to render an arm with no graded_on, and every \$/trade on it carries a strip figure. · mode: build	2 of 2 arms shipped last week are dead; 1 of the 2 was never actually shipped; 0 of the 2 were graded by anything.	Arms: n/a. Exit: n/a. Kill: n/a.	Claude — code change, then a restart git revert.
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17	Mirror the 41ms book feed to the lake — it is the one L2 term that ever helped PARKED Extend tape_mirror.py to export capture.db.book to the parquet lake before prune_capture.py deletes it. Mechanism: The chop lab tested sixteen candidates against 261.9M book rows and found exactly one L2 term that contributed anything: the 41ms refill count, a RATE rather than a level. The 250ms depth.db sample cannot see it — it needs the 41ms capture.db.book feed, which is MNQ-only and rolls off after five trading days. Why: Every other book statistic this desk has tested is refuted (far-side depth share twice, the resting wall with the sign inverted). The one that is not cannot be studied at length, because the data does not survive the week. Evidence: Movement 3 §11 (stones) and §10 disposition; the chopscalp dossier §7. · verification: Book parquet in the lake for a date	1 of 16 candidates' L2 terms contributed; 5 trading days of retention on the only feed that carries it.	Arms: n/a. Exit: n/a. Kill: Revive the 41ms study only once 30+ days are mirrored. Below that it is the same thin-n problem in a new place.	Claude — code change, then a restart Stop the export; pruning is unaffected.
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#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	already pruned from capture.db, with matching row counts. [△] The existing interlock applies: never prune a day the mirror has not exported AND verified by count. · mode: build			
18	Shadow the 10×ATR target ridge — the wide-exit thesis, confirmed on one axis only SHADOW Add a shadow A/B on an existing MNQ arm that changes ONLY the target, from its current value to 10×ATR , holding the stop fixed. Mechanism: The UNCLASS exit sweep — stop 1.0–4.0×ATR against target 2–16×ATR, caps 30 min to 12 h, run on segments rather than blanket — found a genuine ridge at a 10×ATR target that holds across EVERY stop width tested. That is a plateau, not a point, and it is the most robust single thing in the movement. Why: ★ And it splits the operator's own thesis in half honestly: wide exits are CONFIRMED on the target axis and REFUTED on the stop axis . Stop width barely matters here (1.5, 2.0, 2.5 and 4.0 all land in the same place) because the median MAE inside a run is half an ATR — we are not being shaken out, so a wider stop buys nothing. Evidence: Movement 3 §5 (the UNCLASS card); the UNCLASS dossier §3 in the appendix has the full 7×6 sweep. · verification: An A/B pair in the shadow board differing in exactly one field, verified row-for-row as NOT byte-identical (see BUILD #7) . · mode: shadow-first	A 10×ATR target ridge holding across every stop width from 1.0 to 4.0; the entry it was measured on is refuted, the exit shape is not.	Arms: Shadow only. Exit: Target 10×ATR, stop unchanged from the control. Kill: If the wide-target leg does not beat its control over 100 paired signals, the ridge was a property of the refuted entry rather than of the exit, and it goes PARKED.	Claude — code change, then a restart Remove the arm.

HOLD · LEAVE ALONE — THESE ARE ALREADY RIGHT · 6 PLAYS

Standing lessons and things already in production. The action is to not touch them.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
1	The four silent gates stay benched — the standdown is the best-performing thing here LIVE Leave grind_long, capitulation_long, abs_veto_long and rgv_short off. Do nothing. Mechanism: ★ REV3 FIXES THE ROSTER. This play used to say rgv_long, and rgv_long is not on the desk — slot_strategy.py swapped it out for exhaustion_short on 2026-07-25 and data/gate_switches.env has no line for it at all. The −\$517.79 this play quotes is Movement 2 §3's 319.17 + 97.66 + 83.42 + 17.54, which is grind_long + capitulation_long + abs_veto_long + rgv_short — so the gate the old wording omitted was the one carrying the third-largest share of it. Part 1 §2, the Part 1 disposition and Part 1.5 §4 all named the correct four; only the card was wrong, on the single instruction a reader is most likely to execute verbatim. Fired mechanically across this week's own tape in their exact live config, the four benched gates lose −\$517.79 on 48 fires . Firing all six loses −\$1,905.36 on 117 fires across the week — and that whole-week figure is CENSUS-INDEPENDENT, so it is the one to quote. The run-window figure moved with the 08-25 re-freeze: −\$1,333.85 on the old 65-run list becomes −\$976.71 on 21 fires re-cut onto the 62-run one. Stripping their filters to make them show up more often triples the fire count and makes the result worse in both windows — so the ATR floors and vetoes are earning their keep. Why: The four slots that did nothing are the best-performing thing on the desk this week. That is a genuinely uncomfortable sentence and it is the measured one. Evidence: Movement 2 §2–§3 and disposition (REFUTED); Part 1.5 §4. · verification: Read the roster BY IMPORT, never by grepping the switch file — a stale # REVERT comment sits in that header. python -c "from gazbot7.slot_strategy import tournament_slots as t; print([s.tag for s in t()])" must print exactly grind_long, capitulation_long, abs_veto_long, rgv_short, exhaustion_short, abs_veto_short — six entries, and NO rgv_long. Then confirm all four of this play's gates still read off after the next restart and that the tournament's own DISABLED line names them. · mode: no-op	−\$517.79 on 48 fires (grind_long, capitulation_long, abs_veto_long, rgv_short); −\$1,905.36 on 117 fires (all six), whole week, census-independent.	Arms: Revisit on a MEASURED trend week, not on this one. Exit: n/a. Kill: An un-bench needs a positive arming case on a week the gates actually trade — never a lapse, never a quiet week, and never the absence of adverse evidence.	Nobody — no change n/a — this is the current state.
2	Do not act under drift — and do not 'fix' the safety-block skip LIVE Leave it exactly as it is. On drift the tournament skips its entire safety block, by decision. Mechanism: Acting on a position whose ownership is unknown, on a shared netted account, IS the 08-06 cascade. What changed on 08-16 is that the skip is no longer SILENT: safety_skipped_cycles is published in core_health.json, the desk alarms at cycles 1/12/60	Standing decision, 2026-08-16 (DECISIONS §362).	Arms: n/a. Exit: n/a. Kill: n/a — this is a decision, not a hypothesis.	Nobody — no change n/a.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	naming each inactive protection, and sweep goes critical while holding. Why: The operator was asked directly and said "no. dont act under drift. leave it." This row exists so the next session does not re-open it as a bug. Evidence: docs/STATE.md §5; standing operator decision. · verification: A test fails if an order path appears in that branch. · mode: no-op			
3	NEVER HOLD OVERNIGHT — rider flat at 20:40Z, never 21:00Z LIVE Unchanged. The rider flattens at 20:40 UTC. Mechanism: 21:00Z is the CME halt, so a flatten fired then has no market and no retry. This week is the proof of what happens when the 20:40 flatten fails and nothing behind it works. Why: The week's largest loss is one position that went past this rule because every automatic path to enforce it shared a single failed connection. Evidence: Front page; Part 1.6 §5; standing operator rule. · verification: day_rider_state.json reads venue_net: 0 by 20:45Z every weekday. Branch on venue_net, never on entered — the 00:00Z session roll resets the latter. · mode: no-op	The 2026-08-21 carry: opened 13:13:06Z, closed by hand 2026-08-24 11:59:39Z at 29,224.00, −\$2,149.44 booked. (The fills average 29,491.625 where the trades row stores 29,491.93; off the fills the same close is −\$2,147.00. Quote the booked figure — Movement 1 and the chopscalp dossier both do.)	Arms: n/a. Exit: n/a. Kill: n/a — standing rule.	Nobody — no change n/a.
4	Leave grind_long's 22-point ATR floor alone — grind is an entry question now PARKED Do not raise, lower or remove the floor this week. Mechanism: The floor blocked 24,110 of 24,121 raw signals this week, which makes the gate un-testable rather than benched. The floor ladder IS monotone upward — 14pt −\$1,143.50 → 22pt −\$280.50 → 28pt +\$469.00 — but strip-best-3 is negative at every rung, the worst leave-one-day-out is negative at every rung, and against 5,000 random same-size subsets the live 22pt rung sits at only the 65th percentile. Why: Raising the floor cannot make grind positive on its own live record, so this has stopped being "the real grind question". Grind is an entry question. Reachability is settled too: 22pt is the p75 of SESSION minutes and the p89 of ALL minutes — the two published figures were one denominator apart, not two measurements. Evidence: Part 2.5 §2b and disposition (SHADOW); Movement 2 disposition (PARKED); Part 1.5 disposition. · verification: Keep the shadow floor-ladder arms running as a forward confirmation of the monotone shape, and nothing more. · mode: no-op	Blocked 24,110 of 24,121 signals; strip-best-3 negative at every rung; 65th percentile against 5,000 random subsets.	Arms: n/a. Exit: n/a. Kill: Revive the floor question only if grind's ENTRY is fixed first. A floor on a bad entry is a filter on a coin.	Nobody — no change n/a.
5	Gold's exit is the OPPOSITE of the Nasdaq's — do not port the MNQ slot config to MGC LIVE Every gold cell uses the wide chandelier on a SINGLE lot: stop 3.0 ATR, arm 2.0, trail 2.0, 480-minute backstop. Do not apply the live MNQ two-lot configuration. Mechanism: Not the tight scalp — negative at 1R on both entries. Not the dual slot — Lot A is negative on gold and CANCELS Lot B, on both entries. Not a time cap — its headline numbers are a long-drift artefact on both entries. The finding replicates on 29 days. Why: This is the standing rule that MGC gates must be invented fresh rather than ported, earning its keep with a measurement. Gold is not the Nasdaq with a different multiplier — and it is \$10.00 a point, not \$2.00. Evidence: Movement 3 §8 (the exit family paragraph); the MGC dossier §13 in the appendix. · verification: The gold shadow config carries one lot and the chandelier constants above. · mode: no-op	Replicates across 29 days on both entries; the dual-slot and time-cap refutations were re-confirmed this week.	Arms: n/a. Exit: As stated. Kill: n/a.	Nobody — no change n/a.
6	★ exhaustion_short currently reads off — “keep it live” is an ACTION, not a no-op SHADOW Decide, and know what you are deciding. data/gate_switches.env (written 2026-08-24 13:50Z) reads exhaustion_short=off. The Part 1 disposition says “keep it live”, which reads like a no-op and is not one. To un-bench it: restart first, THEN delete the =off line — the roster is import-time but an absent gate defaults to ON and is hot-reloaded, so deleting the line on a running process arms it immediately and a restart does not restore off. And read this before you do: with MONDAY #1 benching abs_veto_short and HOLD #1 keeping four off, un-benching this gate leaves the tournament running on exactly ONE slot. That is a consequence worth signing off rather than discovering. Mechanism: It is the only live gate green on the week: +\$165.50 over 14 lots at 57% win (★ REV3 — Rev1 printed +\$144.50, which	+\$165.50 / 14 lots / 57% win live; machine-only +\$84.00; strip-best-1 −\$91.50; worst leave-one-day-out −\$317.00; stop width is a spike not a plateau. Switch file: exhaustion_short=off since 2026-08-24 13:50Z.	Arms: Only on a positive arming case — never on a lapse, never on the absence of adverse evidence. Exit: Its live fixed 8pt-stop / 12pt-target, unchanged. Kill: Bench it if the MACHINE-ONLY book is negative over the next 20 lots, or if strip-best-1 is still negative at that point. The +\$165.50 is not evidence and must stop being quoted as if it were: one lot and one day carry it.	Garrath — a switch decision, then Claude restarts exhaustion_short=off in data/gate_switches.env, after a restart.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	was the fee counted twice). Unlike <code>abs_veto_short</code> it is not hand-carried: repriced machine-only on its own 14 lots against the venue stops, the machine alone books +\$84.00 and the hand adds +\$60.50. But the battery is thin in three named places. Strip its best single lot and the machine book is −\$91.50. Leave 08-18 out and it is −\$317.00 — one of two leave-one-day-out folds. And its live 1.5×ATR stop is a spike, not a plateau: 0.75× −\$22.00, 1.0× −\$163.68, 1.25× −\$305.36, 2.0× −\$140.36. It also ran the whole week on an <code>atr_split=22</code> quiet-tape clip the operator reverted mid-week as UNMEASURED, so its only green week was run on a configuration that is no longer live. Why: Every other gate on this card has a row, a kill criterion and an owner. This one is the gate the card silently leaves armed, and it is also the one gate whose live state the card got wrong. Both of those are worse than a bad recommendation, because a reader cannot see them. Evidence: Part 1 disposition; Part 1.5 §2.5; Part 2 §3. · verification: Read the roster by import (never grep the switch header — a stale # REVERT comment sits in it): <code>python -c "from gazbot7.slot_strategy import tournament_slots as t; print([s.tag for s in t()])"</code>. Then confirm the intended state with the tournament's own DISABLED line after a restart, not with the file's mtime. · mode: decide			

NOT-AN-ACTION · WITHDRAWN / REFUTED — KEPT ON THE CARD SO NOBODY RE-PROPOSES THEM · 14 PLAYS

Every one of these was a recommendation at some point. They are printed with their autopsies so next Friday's agent does not dig the same hole.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
1	Do NOT promote <code>abs_veto_55s</code> to a live two-sided slot on Monday REFUTED It was last week's lead promotion candidate. It does not go live this week. Mechanism: Named test: the Asia void. \$308.50 of its \$332.00 was booked in a window with zero live fills since 08-05, leaving \$23.50 on 34 fires and 2 of 5 days green. Why: Refuted as a THIS-WEEK case only. It is still the best-evidenced family on the board across all tradable-hour fires, and it stays in shadow. Evidence: Part 2 §3 and disposition (REFUTED for this week, SHADOW on its all-time record). · verification: n/a. · mode: no-op	\$308.50 of \$332.00 in a dead window; \$23.50 on 34 tradable fires; 2 of 5 days green.	Arms: n/a. Exit: n/a. Kill: Revive the promotion question when it has a positive TRADABLE-HOURS week with a positive leave-one-day-out. Not before, and never on an all-hours total.	Nobody — no change n/a.
2	Do NOT arm grind only on clean-trend days — it is unrunnable, not wrong PARKED Leave it. The rule cannot be run often enough to matter. Mechanism: It makes \$580.50 gross against a −\$958.00 gross control and keeps 25 of 43 winners, so it is not refuted. But only 1 of 240 sessions is both clean-trend and above the 22-point floor. Net of fees it is +\$548.00 and 2026-07-29 alone is +\$634.50 of it on 46 fires — drop that one session and the other eight days are −\$1,784.50. Why: "The best rule on the board" and "arm on 2026-07-29" are nearly the same instruction on this sample. And the BUILDING-block version, scored on the same 128 fires, is −\$748.50 on 51 fires keeping 17 of 43 winners — worse per fire than no rule at all, so it is not parked behind the wrong condition either. Evidence: Part 2.5 §1 and disposition (PARKED). · verification: n/a. · mode: no-op	1 of 240 sessions qualify; one session is +\$634.50 of a +\$548.00 net total.	Arms: n/a. Exit: n/a. Kill: Revive only if the qualifying-session rate rises above ~10% of sessions, which means changing the floor or the trend definition, not the rule.	Nobody — no change n/a.
3	Do NOT fix grind with an ER floor or a directional-net requirement REFUTED Both families are dead. Do not re-propose either. Mechanism: Named test: nine cells across two families on 128 live fires, every one negative — including the ER-0.35 floor that is already deployed elsewhere. Why: ER filters keep coming back because they sound like they should work. Nine negative cells on the gate's own live fires is the answer.	9 of 9 cells negative on 128 live fires.	Arms: n/a. Exit: n/a. Kill: n/a — refuted.	Nobody — no change n/a.

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	Evidence: Part 2.5 disposition (REFUTED). · verification: n/a. · mode: no-op			
4	Do NOT run the 2R-partial as a variance lever — it does not smooth anything REFUTED Withdrawn. It was proposed as a smoothness trade and it is not one. Mechanism: Named measurement: daily standard deviation goes 42.3 → 83.3 on the same window in which it improves the mean. Why: It was only ever interesting as a variance lever. Measured, it doubles the variance. Re-open it as a MEAN lever if at all, and only on fresh grind trades. Evidence: Part 2 §6 and disposition (REFUTED). · verification: n/a. · mode: no-op	Daily std 42.3 → 83.3.	Arms: n/a. Exit: n/a. Kill: n/a — refuted on its own stated purpose.	Nobody — no change n/a.
5	Do NOT automate your claim hand as an EXIT rule REFUTED The peak / give-back detector does not ship. The question moves to the entry side (BUILD #8). Mechanism: Out-of-sample −\$4 on 126 held-out trades against + \$1,069 in-sample, and −\$436 as the live config stands. Two independent populations, 119 calibrations. Why: Your hands made real money this week and the machine's profit exits made almost nothing — but the study that tried to copy your hand comes back at zero out of sample. Ask what the denominator is, every time. Evidence: Part 1 §6 and disposition (REFUTED, relocated to BUILD #8). · verification: n/a. · mode: no-op	+\$1,069 in-sample → −\$4 out of sample on 126 held-out trades.	Arms: n/a. Exit: n/a. Kill: n/a — refuted as an exit rule; the entry-side version carries its own kill criterion.	Nobody — no change n/a.
6	Do NOT re-shadow UNCL-RIDER-ASIA — four red weeks, and it never actually shipped REFUTED Leave it dead. Mechanism: Forward out-of-sample: W31 −\$2.93, W32 −\$15.90, W33 −\$36.46, W34 −\$38.08 a trade. The ATR floor added to explain the first two restored none of them. ★ And then the deeper finding: there is no arm by that name anywhere in shadow.db — all 82 strategies were listed. Why: It was last week's SHADOW pick out of the UNCLASS phase, it was never shipped, and nobody noticed for a week. That is the reason BUILD #16 exists. Evidence: Movement 3 §3 and §10 disposition; the UNCLASS dossier §9 in the appendix. · verification: n/a. · mode: no-op	Four consecutive red forward weeks; 0 of 82 shadow strategies carry the name.	Arms: n/a. Exit: n/a. Kill: n/a — refuted by the forward leg.	Nobody — no change n/a.
7	Do NOT arm anything on MNQ out of the greenfield movement — seven hunts, zero survivors REFUTED Nothing from Movement 3 goes live on the Nasdaq. Stated plainly rather than dressed as a near-miss. Mechanism: Seven hunts, dozens of named candidates, and not one MNQ LIVE candidate. The pooled rider measured +\$14 over 1,289 out-of-sample trades ; UNCLASS's rider is beaten by the same signal taken thirty minutes early; OPEN/NEWS is beaten by a gate containing no signal at all; FLOW-LED's last survivor is beaten by a time-shifted fake; VACUUM's survivor wins in both directions; the chop finalist is beaten by the null handed its own search. Why: Not one candidate survived its own robustness battery. That is not a bad week — it is the batteries working. What the movement produced instead is on the BUILD list, and it is worth more than another thin gate. Evidence: Movement 3 §1 (the board) and §6 (every grave). · verification: n/a. · mode: no-op	7 hunts, 0 LIVE candidates, 33 named graves each with the test that killed it.	Arms: n/a. Exit: n/a. Kill: The next attempt is SELECTION BY EXPECTED SIZE at the board moment — no new data source, a different question of the same tape. Three labs independently put the tradeable threshold at about 120 points.	Nobody — no change n/a.
8	Do NOT revive L2 far-side depletion as a run selector — refuted twice, and not n-limited REFUTED Two independent measurements this week. Do not ask for 30 days. Mechanism: The pooled rider: 311 fires against 264M book rows, correlation −0.009 , no monotone band, placebo p = 0.62. VACUUM, separately: 49.44% ±2.52pp on unselected 55pt+ moves over 246.9M book rows, and 49.72% ±0.67pp on all 21,249 book-minutes. Why: The error bars are already tight. The null is not sample-limited, so more days buy nothing — which is the specific thing that makes this REFUTED rather than PARKED. Evidence: Movement 3 §10 disposition; RIDER_ALL §7 and VACUUM §6 in the appendix. · verification: n/a. · mode: no-op	r = −0.009 (n=311, placebo p=0.62); 49.44% ±2.52pp and 49.72% ±0.67pp.	Arms: n/a. Exit: n/a. Kill: A DIFFERENT book statistic — resting-size decay rate, or the sweep-through print — would be a new lead, not a revival of this one.	Nobody — no change n/a.
9	△ The resting wall is a TARGET, not absorption — the sign is inverted		Arms: n/a. Exit: n/a.	Nobody — no change

#	WHAT TO DO	THE NUMBER BEHIND IT	ARMS WHEN · EXIT · KILL CRITERION	OWNER & REVERT
	REFUTED Your lead L2 hypothesis for the chop scalper is refuted, and it points the other way. Mechanism: Unfiltered chop turns drift +0.054pt in the fade's favour over the next minute. Add "wall ≥ 1.5x" and it becomes −0.117pt. Add "five-deep wall ≥ 1.3x" and it becomes −0.374pt (t = −1.72; quintile z = −2.37). In the costed sweep CT2_WALL is the WORST of sixteen candidates at −\$3.09/trade against the naive fade's −\$2.54. Why: This is worth more than a null, because it is a null with a direction. A big resting wall at a range edge is not somebody absorbing — it is somebody's target, and price goes to it. Evidence: Movement 3 \$5 (the chop card) and \$10 disposition; chopscalp \$3 in the appendix. · verification: n/a. · mode: no-op	+0.054pt → −0.117pt → −0.374pt as the wall filter tightens; CT2_WALL worst of 16.	Kill: n/a — refuted with the sign inverted in two independent tests.	n/a.
10	Do NOT change the router's thresholds or timing this week REFUTED No tuning. There is no sample to tune on. Mechanism: There were 20 switch writes all week, 5 of them worth \$0 by construction. Tuning on this week would be fitting to a standdown. Why: The tournament traded two of six slots for 22 lots. Four separate sections say independently that this is not enough evidence to re-tune anything, and this card is short on purpose because of it. Evidence: Part 2.6 \$2–\$3 and disposition (REFUTED). · verification: n/a. · mode: no-op	20 switch writes, 5 structurally worthless; 22 lots on 2 of 6 slots.	Arms: n/a. Exit: n/a. Kill: Revive on a week the tournament actually trades — with a lag number attached, which this week could not produce.	Nobody — no change n/a.
11	Do NOT re-arm a gate on a LAPSE — it needs a positive case REFUTED When a bench's rule goes quiet the gate returns to UNDECIDED, not ARMED. Mechanism: Named instance: 2026-08-19 15:35Z re-armed abs_veto_short thirty minutes after benching it, with no new evidence, straight into three stops for −\$133. Third occasion on this gate. Why: "Armed-by-default" is a cold-start tiebreaker, not an override of a live adverse read. Decayed evidence removes the case for the bench; it does not create a case for the arm. Evidence: Part 2.6 disposition (REFUTED, with the instance named). · verification: n/a. · mode: no-op	−\$133 on 3 stops, 30 minutes after the bench. Third occasion on the same gate.	Arms: n/a. Exit: n/a. Kill: n/a — refuted three times on the same gate.	Nobody — no change n/a.
12	Do NOT revive mgc_straddle — refuted on 320 days, and last week's reason was wrong too REFUTED The volatility cell is dead. Upgraded from 22 days to 320 and from a gradient to an actual trade. Mechanism: n=3,945, −\$1,248, −\$0.32 a trade. All eight compression cells negative, the control also negative, and it dies on one tick of slippage. Why: ★ Worth reading for the method rather than the verdict: last week's REASON for killing it was wrong (raw range sorts on ATR), and the verdict was right anyway. A right answer for a wrong reason is a fault, not a save. Evidence: Movement 3 \$8 (the per-cell table) and \$10 disposition. · verification: n/a. · mode: no-op	n=3,945, −\$1,248, −\$0.32/tr, 8 of 8 cells negative, dies at one tick.	Arms: n/a. Exit: n/a. Kill: n/a — refuted on 320 trading days.	Nobody — no change n/a.
13	Do NOT strip the gates' filters to make them show up more often REFUTED The ATR floors and the vetoes stay. Mechanism: Ungating triples the fire count and makes the result WORSE in both the run windows and the whole week. Direction alignment does not rescue it either: aligned mechanical fires made — \$602.60 on 11 fires — knowing which way the run would go was not enough, because the gates fire before the ignition and get stopped in the chop. Why: The recurring instinct when a gate is silent is to loosen it. Measured, that is the wrong direction: the filters are the part that works. Evidence: Movement 2 \$2 and \$4, and disposition (REFUTED twice). · verification: n/a. · mode: no-op	Ungated: 3x the fires, worse in both windows. Direction-aligned: −\$602.60 on 11 fires.	Arms: n/a. Exit: n/a. Kill: n/a — refuted.	Nobody — no change n/a.
14	HOLD, do not execute: the exhaustion_short exit-override withdrawal ALREADY LANDED, and the row that is left is there on purpose PARKED Do NOTHING to data/exit_overrides_proposed.json. ★ REV3 WITHDRAWS THIS PLAY'S INSTRUCTION. It used to say "remove the exhaustion_short rows — they were withdrawn once and the	58th percentile of its own control on 87 signals.	Arms: n/a. Exit: n/a. Kill: n/a — refuted by its own control.	Nobody — no change n/a — this is the current state.

file still carries them". They were removed, on 2026-08-16, and the file's own header note records it. What is left under that key is {a_r 2.0, b 2.0, stop_k 1.5}, which is BYTE-IDENTICAL to the live data/exit_overrides.json — it exists to keep the proposal file in step with production. Executing the old wording would delete it and CREATE the divergence this play claims to close.

Mechanism: The thing that was refuted is the by_rung tight-chandelier cells, and they are gone. The chandelier variant sat at the 58th percentile of its own random-entry control on the same 87 signals — indistinguishable from the control it was supposed to beat — and a longer leash made it worse, not better. On the gate's own 87 signals the wide stop pays +\$3,459.98 against the chandelier's — \$339.06. That verdict stands; only the instruction changes.

Why: A play that has already been executed is not a pending action, and a card that keeps re-proposing it teaches the reader to skim. The withdrawal landed ten days ago; what is left is a deliberate mirror of live.

Evidence: Part 2 §3 and the exhaustion_short battery; the random-entry control is named there as the test that killed it. · verification:
grep -c by_rung data/exit_overrides_proposed.json returns 1 (grind_long's MED-TREND shadow cell), and
python -c "import json;a=json.load(open('data/exit_overrides.json'));b=json.load(open('data/exit_overrides_proposed.json'));print(a['exhaustion_short']==b['exhaustion_short'])" prints True. If it ever prints False, the proposal file has drifted from production and THAT is the thing to fix. · mode: no-op

STANDING CAVEATS ON EVERY DOLLAR ON THIS CARD

Every repriced number is a CEILING. Fills are modelled at the exact stop/target/trail level with no slippage, and the desk's own standing finding is that live losses run **1.1–3.1× modelled**. Treat the policy as the edge and the headline dollar as decoration. Anything on thin n is flagged in-row as a direction, not a measurement.

The fee is \$1.50 per round trip and the multipliers are not the same. MNQ is \$2.00 a point; **MGC is \$10.00**. Every gold number in Movement 3 and in Part 2.5 Part A is priced at the gold multiplier — pricing gold with the Nasdaq one would understate it fivefold, and doing the reverse would flatter it fivefold.

Four method wounds are open and they bound everything above. signal_journal.suppressed_by is NULL in all 222 rows this week, so every claim about what a gate would have done unrouted is a reconstruction from the tape rather than a measurement — and it is the same defect the scope flagged last week — **BUILD #9**. The router's leakage detector is scoped to a universe that barely trades (capitulation_long and rgv_short), so its zero is an empty set and not a clean week — exhaustion_short leaked –\$241 unseen — **BUILD #10**. The nightly rollup reports \$0/\$0/\$0 when the systemd journal has rotated past the day it is backfilling, which is

indistinguishable from a day with no value — **BUILD #10.**

selector_nightly counted a quarantined trade four extra times, inflating one day's regret by 39% (\$747.50 → \$457.50 on 10 real trades) — the phantom is FOUR rows worth \$290.00, not five worth \$362.50; the fifth day-rider row that day is the genuine claimed trade and must survive the fix — **BUILD #11.**

What did not run this cycle, named exactly. The rehabilitation phase timed out, and Part 1.5 was rebuilt from the artifacts it had already saved. In Movement 3 the four per-cause-cluster hunts did not run — gf_full_FLOW-LED.md, gf_full_OPEN-NEWS.md, gf_full_UNCLASS.md and gf_full_RIDER_ALL.md on disk are all dated 15 August, i.e. last week's artifacts, and nothing stitched them in. **The rest of Movement 3 is fresh and ran tonight:** the VACUUM cluster hunt, the 686-run lake study, the three forward tests, BIGBOARD, SMALLFADE, LOTDIV and the chop lab all ran 2026-08-22 00:00-00:39. So this is not “you got no MNQ greenfield work this week” — it is four cluster hunts missing from a movement that otherwise delivered. **Nothing on this card rests on a section that is missing.**

§ The evidence behind those plays

Everything from here down is the working. Two halves, one document. It opens with **Part 0**, which answers the only question that survives past Friday — are we getting better? — day by day, green or red, straight off the books. **Then the warm half:** what the live desk actually did (Part 1, with **the run charts** inside it — every fill of the week marked on its own session's price path, which is the fastest way to see what the prose is describing). Then the section that matters most — **Rehabilitation** (Part 1.5), where every red thing this week is walked through the full treatment instead of benched at face value, and where the failed treatments are printed by name because the failures are the point. Then **the day rider** (Part 1.6), which this week is not a footnote to the tournament but the

bigger of the two books. Then the shadow board and the promotion battery (Part 2), the mid-week leads re-run adversarially including the whole gold programme (Part 2.5), and the **Router Operation Review** (Part 2.6). **Then the cold, tape-first half:** we ignore everything the desk did and ask, from the raw tape and order book, **where was the money and did we show up?** — on MNQ and on gold (M1) — then test whether the gates we already own could have caught the runs we missed (M2), and finally build gates from scratch and backtest them to death, the ones that failed included, that being the point (M3). Plain English, honest money, every claim a table — and every reprice a ceiling.

★ **Where work did not run this cycle, it says so in red at the point where it should have been**, naming the phase and what you are not getting. A fragment that is missing or stale is replaced by a dated red placeholder automatically; a phase that ran partially, or on inputs that have since moved, gets a red banner at the top of its section. **There are two of those this week — Part 1.5, whose rehab phase timed out, and Movement 2, which was computed against a census that has since been re-frozen.** Nothing is silently omitted, and a gap is never a null result: “we looked and found nothing” and “nobody looked” are opposite instructions for next week. (★ Last week a banner on this page said four of Movement 3's hunts never ran. They ran — on the Monday, after the section had been frozen — and Movement 3 has been rebuilt from all seven dossiers, which are reproduced in it in full. A stale honesty banner is worse than none, because it spends the reader's trust on a falsehood.)

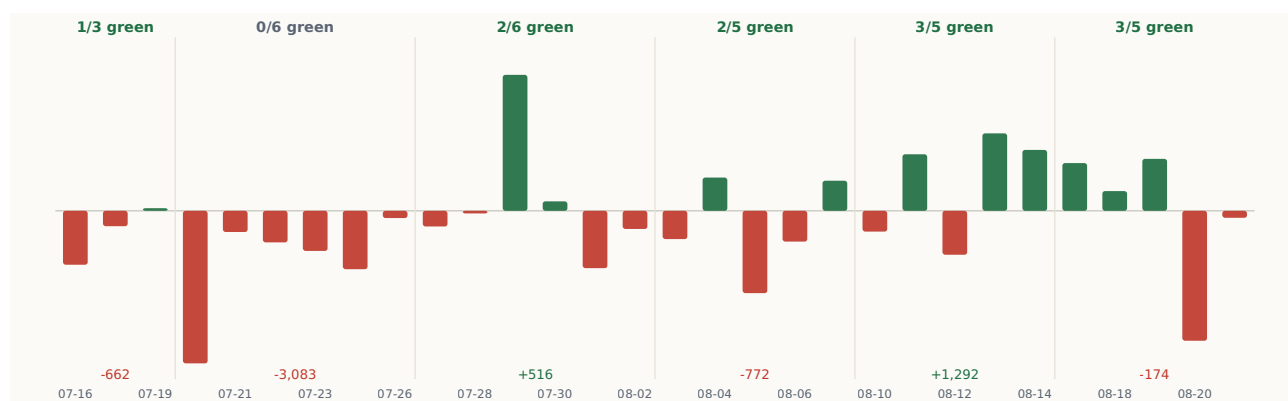
WHAT IS IN HERE

1. **Front** — [the honest headline](#), [the scorecard](#), [WHAT TO ACTUALLY DO](#)
2. **Part 0** — [Are we getting better? — every day the desk has traded, green or red](#)
3. **Part 1** — [The live desk — what the paper tournament and the day rider actually did](#)
4. **Part 1.5** — [Live-desk REHABILITATION — never bench a gate at face value](#)
5. **Part 1.6** — [THE DAY RIDER — the second desk, its week, and the four lots it left open](#)

6. **Part 2** — [The shadow desk, the promotion battery, and the ruler that reordered the board](#)
7. **Part 2.5** — [Mid-week musings — every lead from the desk chat, tested hard](#)
8. **Part 2.6** — [ROUTER OPERATION REVIEW — the desk's #1 lever, on the stand](#)
9. **Movement 1** — [The census — the biggest runs this week, and did we show up](#)
10. **Movement 2** — [The idle-gate lab — could the gates we own have caught them? \(with a base rate\)](#)
11. **Movement 3** — [The greenfield lab — detection is solved, selection is not, and the size threshold](#)

PO Are we getting better? — every day the desk has traded

One bar per trading day, green or red, total desk. The question every other section answers is what happened this week. This one answers **are we getting better**, which is the only question that survives past Friday.



The shape of it. Week 30 was the floor: **0 green days out of 6** and **-3,083.00** — the stretch the operator remembers as “red for many many weeks”. The most recent week ran **3 green out of 5** at **-174.00**. Of 6 weeks on record, **2 closed net positive** — and both of them are in the last three.

WEEK	DAYS	TRADED	GREEN	GROSS	FEES	NET
W29		3	1/3	-586.50	-75.00	-661.50

WEEK	DAYS	TRADED	GREEN	GROSS	FEES	NET
	2026-07-16 – 2026-07-19					
W30	2026-07-20 – 2026-07-26	6	0/6	-2,732.00	-351.00	-3,083.00
W31	2026-07-27 – 2026-08-02	6	2/6	+823.50	-307.50	+516.00
W32	2026-08-03 – 2026-08-07	5	2/5	-607.50	-165.00	-772.50
W33	2026-08-10 – 2026-08-14	5	3/5	+1,469.50	-177.00	+1,292.50
W34	2026-08-17 – 2026-08-21	5	3/5	-100.50	-73.50	-174.00

NET is the column to read — it is what the account moved by, and it is the convention every other section uses. NET is `sum(pnl_usd)` straight off the book, because `pnl_usd` is recorded already net of fees; GROSS is reconstructed by adding `fees_usd` back at \$1.50 per round trip per lot. Do not subtract the fee from `pnl_usd` — that is the mistake the previous revision of this page made, and it cost \$1,149.00 of imaginary loss.

What changed underneath. The Claude-run router went live 2026-07-30, mid-W31 — the first green week is the one it landed in. Its measured effect is on the tail and the frequency of red days, not their depth: the worst day before it was **-1,513.00**, the worst since is **-1,288.50**, and green days went from **2 of 12** to **9 of 18**. The day-rider joined on 08-06 as a second, slower engine, and this is the week it stopped being unbeaten: **-169.50** across 7 legs, with a single Thursday 08-20 long at **-1,288.50** that is larger than the other four sessions made between them. It is still the engine that produced the desk's best days; it is no longer the one that cannot lose.

Read it honestly. The cumulative total across all 30 days is still **-2,882.50** — this is a desk that has stopped bleeding wholesale, not one that is paid yet. 6 weeks is a short sample, best day +1,349.00 and worst -1,513.00 are both single sessions, and PAPER fills flatter a real book. The trend is real; the destination is not reached.

★ **The 08-22 “fee correction” on this page has been WITHDRAWN — it was backwards, and it moved the headline by \$1,149.00.** `trades.pnl_usd` is written already net of `fees_usd`: the schema says so, the canonical P&L helper says so, and it is checked rather than assumed — on all 765 rows in the book, `pnl_usd` equals `(exit–entry)×direction×lots×$2.00` minus `fees_usd`, exactly, 765 times out of 765. Rev2 changed this page to `sum(pnl_usd – fees_usd)`, which charges the \$1.50 round trip a second time. **The all-time book is -2,882.50, not the -4,031.50 the previous revision printed**, and Rev2's claim that the change

“turned two week totals green that are not” was itself wrong: W31 and W33 are positive on either arithmetic. Nothing about the SHAPE changes — the ordering of the weeks, the worst day, the best day are all identical — only the depth, and it was overstated. One green day does move, and it is worth naming: **2026-07-30, the router's first day**, made +93.50 on 82 trades and paid \$123.00 in fees, so the double-counted version turned the router's own go-live red. It was green. **If you see this desk quoted anywhere at -4,031.50, that is the double-counted number.** The Gross / Fees / NET table above now reconciles by construction: gross is `pnl_usd + fees_usd`, so gross – fees = net, every row.

Where this page stops. Every bar is a session closed on or before **2026-08-21**, which is where this report's books close. The desk has traded since — notably a 4-lot day-rider carry closed by hand on 2026-08-24 for –\$2,149.44 and a fresh 4-lot entry on 08-25 for –\$243.00 — and neither is in this chart, because a week-ending-08-21 document that quietly absorbed the following Monday would be measuring two different things with one line. Part 1.6 reports the post-bell sessions on their own.

P1 THE LIVE DESK — the week of 17-21 August 2026

Garrath — the short version is that the desk lost \$174.00 on the clean book across five sessions, and that number is the least important one on this page. Three sessions were green and two were red, but the two red ones were not the gates being wrong about the market. Thursday was one position held for six hours through a fall it should have been stopped out of, and Friday was a software fault plus a gateway that stopped answering — and that second one is not finished. **You are still long four lots right now.** They were opened at 13:13 UTC on Friday, every automatic path that exists to close them failed on the same call, and at the last price the tape printed before the CME shut they were –\$973.00 down. That is bigger than the whole week's trading, it is not in any P&L on this page, and it is the first thing on the action card.

★ **READ THE VENUE LINE BEFORE YOU READ ANY OF THESE TABLES**

Every table in this section counts trades that **CLOSED**. The desk is carrying **4 MNQU6 long from 29,491.625**, opened 2026-08-21T13:13:06Z by the day rider and never closed. Marked at the last tick before the weekend (21:00:00Z, 29,370.00) that is **-121.625 points → -\$973.00**. ★ **REV2 — one price, and it is the FILLS**. Rev1 printed this position two ways and never reconciled them: 29,491.93 on the front page, in this section and in plays.json, and 29,491.63 in Part 1.6. The fills settle it — 2 @ 29,477.00 + 1 @ 29,506.25 + 1 @ 29,506.25 at 13:13:06Z is a fill-weighted **29,491.625** exactly, so -\$973.00 is the number. 29,491.93 is the entry field of day_rider_state.json, which disagrees with its own fills by 0.305pt (\$2.44 on four lots). data/weekend_flatten_pending.json carries the same wrong figure; it flattens by quantity so nothing depends on it there, but do not re-import it. The nightly review of 22:43Z Friday read it live off updatePortfolio at -\$950.27 unrealised, which is the same position a few ticks earlier.

A trades table can never contradict a benign story, because it only enumerates what closed. The one instrument that can is the venue position line, and it says four lots. Everything below is the story of the money that finished; this is the money that has not.

★ **REV3 — READ THIS SECTION IN THE PAST TENSE. It is written as of Friday night, when the position was still open; it is published on 2026-08-26, when it is not.** You closed all four lots by hand on **Monday 2026-08-24 at 11:59:39Z at 29,224.00 for -\$2,149.44** (booked row #763). Every sentence below that says the desk is carrying four lots described the state at 21:00Z on 08-21 and is kept as written, because the -\$973.00 mark is what every table in this section was computed against and re-marking it would silently mix two dates. The resolved figure is on the front page, and Part 1.6 §3b races the stop grid on the closed number rather than the mark. Nothing in this section is an instruction to go and check the venue any more — that was done, and this is how it ended.

1 • The money, session by session

SESSION (UTC CLOSE)	LEGS	CONTRACTS	CLEAN BOOK	FLAGGED LOTS	CASH BOOK
Mon 17 Aug	9	10	\$473.00	—	\$473.00
Tue 18 Aug	21	22	\$195.50	—	\$195.50
Wed 19 Aug	9	10	\$515.00	—	\$515.00
Thu 20 Aug	3	4	−\$1,288.50	—	−\$1,288.50
Fri 21 Aug	1	3	−\$69.00	−\$81.50 (1)	−\$150.50
THE WEEK	43	49	−\$174.00	−\$81.50 (1)	−\$255.50

★ **REV2 — “Legs”, not “Lots”**. This column was labelled Lots and was counting rows. The clean week is **43 rows but 49.0 contracts**, because the day rider’s legs are 1-3 lots each — so the old label understated the week’s size by 12% and disagreed with §2’s slot table, which had it right. Both counts are printed now.

The **clean book** is data_quality IS NULL and it is the strategy verdict — what the gates did when the machinery worked. The **cash book** adds the flagged lot back, and it is what reconciles to the broker: one Friday day-rider lot filled at 29388.25, about 30 points outside anything the book was showing, flagged **BADFILL:stale_price_30pt_outside_book_20260821**. **A BADFILL flag marks a defective fill PRICE, not a phantom trade** — the money left the account either way, so quote the clean book when you are judging a gate and the cash book when you are counting cash. Add the open carry — −\$973.00, marked at the last tick before the CME shut — and the week's real economic position is **−\$1,228.50** (★ REV3: Rev1 printed −\$1,303.50, which is this same sum taken off the fee-double-counted cash book).

2 • Every slot that fired, and the four that did not

SLOT	SIGNAL S	LOTS	NET	WIN %	AVG WIN	AVG LOSS	\$ / SIGNAL	BEST	WORST
abs_veto_short_A	11	11	−\$72.50	36%	\$49.00	−\$38.36	−\$6.59	\$88.00	−\$78.00
abs_veto_short_B	11	11	−\$97.50	27%	\$68.33	−\$37.81	−\$8.86	\$94.00	−\$78.00
day_rider	7	13	−\$169.50	43%	\$396.00	−\$339.38	−\$24.21	\$544.00	−\$727.00
exhaustion_short_A	7	7	\$99.00	57%	\$83.75	−\$78.67	\$14.14	\$118.50	−\$104.00
exhaustion_short_B	7	7	\$66.50	57%	\$75.75	−\$78.83	\$9.50	\$131.00	−\$104.00

Two base gates traded all week: abs_veto_short and exhaustion_short, both of them short-side, plus the day rider on its own book. The other four — grind_long,

capitulation_long, abs_veto_long, rgv_short — fired **nothing at all**, because they were benched for the whole week under the standing tournament standdown. That is not an accident and it is not a gate failure; it is the desk's current posture, and Movement 2 prices what it saved. The short answer, in advance: firing all six mechanically across this week's tape would have lost — **\$1,905.36** on 117 fires, so the four silent gates are the best-performing thing on this page.

3 • The two legs of the same signal — is the wide one still winning?

Every tournament signal opens two lots at the same price with the same stop, and the only difference is where profit is taken: Lot A takes the tight rung, Lot B rides for the wider one. **They are a scale-out, not an experiment** — if B beats A the instruction is to widen A's target, never to drop A. Paired on the signal, this week:

GATE	PAIRED SIGNALS	LOT A (TIGHT)	LOT B (WIDE)	B – A	B AHEAD ON	B – A, BEST PAIR STRIPPED
abs_veto_short	9	–\$93.50	–\$137.50	–\$44.00	4 / 9	–\$63.50
exhaustion_short	7	\$88.50	\$56.00	–\$32.50	2 / 7	–\$45.00
BOTH GATES	16	–\$5.00	–\$81.50	–\$76.50	6 / 16	–\$96.00

–\$76.50 across 16 paired signals, with B ahead on 6 of them. **That is not a result.** On this n a single pair moves the sign, and stripping the best one leaves –\$96.00. Last week's wide-leg finding stood on 51 paired entries with all five daily folds positive; this week's 16 pairs neither confirm nor contradict it, and the honest statement is that the week was too quiet to re-test it. Do not re-rank the exit ladder on this table.

4 • How the week's lots actually ended

EXIT REASON	LEGS	CONTRACTS	NET	\$ / CONTRACT	WIN %
CLOCK_FLAT_BUG	2	4	–\$150.50	–\$37.63	0%
MANUAL_CLAIM	16	19	\$578.50	\$30.45	81%
STOP	21	21	–\$1,043.50	–\$49.69	0%
TARGET	4	4	\$172.00	\$43.00	100%
TRAIL	1	2	\$188.00	\$94.00	100%

★ **REV2 — this table is the CASH book** (44 legs / 50 contracts, **–\$255.50** — ★ REV3: Rev2's caption said –\$330.50, which is pnl_usd with the \$1.50 round trip

charged a second time; the Net column above sums to $-\$255.50$, and it always did), because the flagged BADFILL lot has to end somewhere and its exit reason is CLOCK_FLAT_BUG. Legs and contracts are both shown now: the two CLOCK_FLAT_BUG rows are a 3-lot and a 1-lot fill from one 09:11Z entry, which is how $\$5$ and Part 1.5 $\$1$ describe them. The CLOCK_FLAT_BUG rows are not an exit style, they are the Friday fault — Part 1.5 takes them apart. Note the shape of everything else: 21 stops for $-\$1,075.00$ against 16 manual claims for $\$550.00$ and four targets for $\$166.00$. **The desk's automated profit-taking barely appears this week** — one TRAIL exit for $\$185.00$ and four targets. Almost every dollar the desk made, your hand made.

5 • THURSDAY — the week's whole loss, in one position

Thursday 20 August cost $-\$1,288.50$, which is more than the other four sessions made between them. It was not the gates. The tournament did not fire a single lot that day. It was one day-rider long, four lots, opened at 13:38 UTC and closed by your own hand in three pieces between 19:33 and 20:17 as it kept going the wrong way:

CLOSED (UTC)	SLOT	SIDE	LOTS	IN	OUT	HELD	EXIT	NET
19:33:54	day_rider	LONG	1	29,447.75	29,301.75	356 min	MANUAL_CLAIM	$-\$293.50$
19:43:23	day_rider	LONG	1	29,447.75	29,314.50	365 min	MANUAL_CLAIM	$-\$268.00$
20:17:55	day_rider	LONG	2	29,447.75	29,266.75	400 min	MANUAL_CLAIM	$-\$727.00$
			4					$-\$1,288.50$

One entry, one direction, no stop that bound, and just under six hours of holding. The rider had no protective exit in the water for any of it. Part 1.6 walks the rider's own week in detail and Part 1.5 asks the rehabilitation question — whether the entry was wrong, or whether an entry that was fine was ruined by having no stop — and the answer is not the obvious one.

6 • Your hands, scored honestly

16 claims, covering 19 of the week's 49 clean contracts, were closed by you pressing Claim, and they account for $\$578.50$ of the P&L, 13 of 16 of them green. Everything the machine closed by itself came to $-\$752.50$. Stated that way it reads like your hand carried the week, and in cash terms it did — but the honest version needs the counterfactual, so here is the peak that existed inside each claimed trade's own life against what you actually booked:

DAY	SLOT	LOTS	PEAK INSIDE THE TRADE	HE BOOKED	KEPT	WORST IT WENT AGAINST HIM
08-17	day_rider	2	\$498.00	\$456.00	92%	-75.25 pt
08-17	exhaustion_short_A	1	\$109.00	\$92.50	85%	-39.00 pt
08-17	exhaustion_short_B	1	\$108.50	\$76.00	70%	-39.25 pt
08-18	exhaustion_short_A	1	\$96.50	\$85.50	89%	-40.00 pt
08-18	exhaustion_short_B	1	\$81.00	\$57.00	70%	-40.50 pt
08-18	exhaustion_short_A	1	\$146.50	\$118.50	81%	-10.50 pt
08-18	exhaustion_short_B	1	\$135.00	\$131.00	97%	-10.25 pt
08-18	abs_veto_short_A	1	\$39.50	\$33.00	84%	-10.00 pt
08-18	abs_veto_short_B	1	\$69.00	\$52.50	76%	-13.75 pt
08-18	exhaustion_short_B	1	\$62.00	\$39.00	63%	-7.75 pt
08-19	day_rider	2	\$606.00	\$544.00	90%	-28.75 pt
08-19	abs_veto_short_A	1	\$116.50	\$88.00	76%	-26.00 pt
08-19	abs_veto_short_B	1	\$100.50	\$94.00	94%	-26.00 pt
08-20	day_rider	1	\$37.00	-\$293.50	-793%	-245.25 pt
08-20	day_rider	1	\$37.00	-\$268.00	-724%	-245.25 pt
08-20	day_rider	2	\$74.00	-\$727.00	-982%	-245.25 pt

Median capture of the in-trade peak: **84%**. The peak column counts only ticks between the entry and the exit — an excursion after the trade closed is money the desk was flat for, and counting it would let a chart accuse your own book of leaving money behind that was never available.

THE CLAIM REPLAY: YOUR HAND IS REAL, AND A PEAK-AND-GIVE-BACK RULE DOES NOT REPRODUCE IT

This week's claim-replay study (reports/claim_replay/) rebuilt the 1 Hz peak-watch ladder against all 35 booked manual claims on the tape. Where the watcher is actually in scope — the rider's two-lot book — an exit-shaped signal fired inside the hold on 4 of 4 trades and fired BEFORE you pulled every time, median lead 1,383 seconds. It also booked less money on every one of them: \$165 vs \$189, \$382 vs \$731, \$137 vs \$456, \$454 vs \$544.

Then it was calibrated properly — 75 cells on 341 training trades, best cell taken to 126 held-out ones. In-sample it makes **+\$1,069**. Out-of-sample it makes **-\$4**. Applied exactly as it stands live it fires on 10 of 467 trades for —

\$436. The give-back exit shape is REFUTED as a way to reproduce your hand, over 44 + 75 calibrations on two independent populations.

The finding underneath it is the useful part: scoring any rule against the trades you chose to be present for was always going to lose, because your presence is the filter. ★ **REV3 gives that pool a denominator, a date range and a comparator, because −\$18,672 with none of those is a big number and not a scale.** Measured on the clean book to 2026-08-21, every non-MANUAL_CLAIM exit the desk has ever taken: the STOP bucket is −**\$18,883.50 over 384 trades / 429 lots, from 2026-07-16 to 2026-08-19** — five weeks, the desk's whole life. That is −**\$49.18 a trade and −\$44.02 a lot**, which is almost exactly what a 1×ATR stop is supposed to cost on a 20-point ATR. **And it has no fat tail to filter:** the worst decile of those stops is 38 trades worth −\$4,024.50, only **21.3%** of the pool, and removing it entirely still leaves −\$14,859 at −\$42.95 a trade. So a perfect entry-side filter that deleted the worst one stop in ten would recover about a fifth of it. **The comparator that actually sizes it:** those stops are the cost side of the same ledger whose profit side is TARGET +\$7,840 (121), CHANDELIER +\$5,112 (69) and GIVEBACK +\$2,332 (81) — **+\$15,284 over 271 trades**. Net, the whole unwatched machine book is **−\$5,825 over 717 trades**, and THAT is the ceiling on what any entry-side filter can be worth, not \$18,883. The exits you DID watch are the other end of the scale: 38 manual claims, **+\$2,942.50, +\$77.43 a trade**. So the honest sentence is not “there is \$18,672 lying about” — it is “the machine pays \$56 a winner and loses \$49 a stop, and it takes more stops than winners.” The lead has been moved to the entry side rather than killed — BUILD #8 carries it — and it is a study rather than a rule.

Rev1's sentence, kept so the change is visible: “the money an automated claim could win is in the 671 exits nobody watched, and in that population the STOP bucket is −\$18,672 over 378 trades.” The 378/−\$18,672 and the 384/−\$18,883.50 above differ because Rev1 counted only the trades with a usable tape; the sign, the shape and the instruction are identical either way. **Which trades are stop-bound is an ENTRY question, not an exit one**, and it is a study rather than a rule, because the desk has just spent a week learning what happens when an exit answer gets fitted to an entry problem.

7 · Three instruments that reported confidently and wrongly this week

INSTRUMENT	WHAT IT SAID	WHAT WAS TRUE	COST OF BELIEVING IT
core_health.json · the router's own health line	"flat": true, "healthy": true — still says it as this report renders	The venue has been long 4 lots since 13:13Z Friday	It is the flag every other tool trusts. ★ REV3, and this is the fixed version of a number this report printed two ways: across the ENTIRE life of the position — 13:13:06Z on 08-21 to the hand-close at 11:59:39Z on 08-24 — data/router_trial_log.txt holds 817 router ticks, of which 811 print “Flat, healthy”, 813 contain the word flat, and 0 mention the open position at all. Rev1 said “132 of 137” here and “137 of 141” in Part 2.6; both were taken at whatever moment their own section rendered, and neither reproduces from the log. Cut at the position's own life, the count is fixed.
hour_watch.py · the hourly desk read	Friday day net −\$150.50	−\$69.00 on the clean book — it has no data_quality filter, so the BADFILL lot lands in its total	Neither number is wrong; they answer different questions. IB's own realizedPNL −\$149.38 reconciles to the FLAGGED figure, so hour_watch is right about CASH and wrong about STRATEGY.
router_nightly / selector_nightly	Nothing at all for this week	The newest router rollup is 2026-08-14 and the newest selector rollup 2026-08-18 — no timer runs either of them	The weekly router review in Part 2.6 had to be rebuilt from raw logs. A monitoring file that silently stops being written is worse than one that errors.

8 · What actually happened, in five sentences

Monday was the week's clean session: the short book worked in a falling tape, the rider took \$453.00 out of one claimed short, and the desk closed \$458.00. **Tuesday** was the busiest day by far at 21 lots for \$162.50 — a genuine chop day where exhaustion_short faded the oscillation well and abs_veto_short gave most of it back. **Wednesday** was the best at \$500.00, essentially one rider short claimed for \$541.00 inside eleven minutes, after which the short book stopped out four times in a row for −\$153.00 and the operator benched it at 18:15Z. **Thursday** was −\$1,288.50 in the single rider long described above. **Friday** booked −\$156.50 on a software fault at 09:12Z, then opened the four-lot long at 13:13Z that is still open, and at 19:57Z the IB gateway entered the partial wedge that stopped every closing path on the box.

Disposition — every lead this section touched

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
The four silent gates (grind_long, capitulation_long, abs_veto_long, rgv_short)	HOLD	They traded nothing and that was correct: the mechanical counterfactual in Movement 2 is −\$1,905.36 across this week's tape. Keep them benched; revisit only on a measured trend week, not on a quiet one.
abs_veto_short as a live slot	SHADOW	−\$170.00 over 22 lots this week, and its 55-second shadow mirror is the week's best shadow family (+\$332 over 46). The live/shadow gap is the whole question — Part 1.5 takes it apart and Part 2 tests whether the mirror is promotable.
exhaustion_short as a live slot	LIVE	The only live slot green on the week: + \$165.50 over 14 lots at 57% win (★ REV3 — Rev1 printed +\$144.50, the fee counted twice). ★ REV3 also fixes what the card said about it: data/gate_switches.env reads exhaustion_short=off, so “keep it live” is an ACTION and not a no-op — restart, THEN delete the =off line. It now has its own row, HOLD #6, with a kill criterion. ★ REV2 — it has now been interrogated, and the disposition needs a caveat. Repriced machine-only on its own 14 lots (venue stops, so R is exact; 7 of 7 machine-closed lots reproduce their live exit reason), the machine alone books +\$84.00 and your hand adds +\$60.50 — so unlike abs_veto_short this gate is NOT hand-carried. But strip its best single lot and the machine is −\$91.50; leave 08-18 out and the machine is −\$317.00; and its live 1.5×ATR stop is a spike, not a plateau (0.75× −\$22.00, 1.0× −\$163.68, 1.25× −\$305.36, 2.0× −\$140.36). It also ran the whole week on a quiet-tape clip the operator reverted on 08-18 as UNMEASURED. Keep it live — nothing here says bench it — but stop calling the + \$165.50 evidence. Part 1.5 §2.4 has the full battery.
The wide-leg (Lot B) exit finding	PARKED	16 paired signals is too thin to re-test it and the sign flips on one pair. (★ REV2: this row said 22, which is the abs_veto_short LOT count, not a pair count — so the revival condition $n \geq 40$ pairs was being measured against a different denominator from §3's own table.) Revive it next week on $n \geq 40$ pairs, or sooner if the shadow ladder arms produce their own pairs.
The automated claim detector (peak / give-back)	REFUTED	Out-of-sample −\$4 on 126 held-out trades against +\$1,069 in-sample, and −\$436 as the live config stands. Two independent populations, 119 calibrations. Relocated to the entry-side question rather than killed — and ★ REV2 gives that relocation a play: BUILD #8, with an AUC-0.60 / permutation-p kill criterion. In Rev1 the largest identified pool of money in the report (−

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
		\$18,672 over 378 unwatched STOP exits) left the document with no play, no kill criterion and no owner. ★ REV3 also gives it a SCALE, which it never had: on the clean book that pool is −\$18,883.50 over 384 trades from 2026-07-16 to 2026-08-19, −\$49.18 a trade, with no fat tail (worst decile = 21.3% of it). It is the cost side of a ledger whose winners are +\$15,284, so the ceiling on any entry-side filter is the −\$5,825 the whole unwatched machine book is worth, not \$18,883. \$6 has the arithmetic.
core_health.flat as a safety signal	FAULT	It is tournament-scoped and it says the desk is flat while the venue is long four lots. BUILD item: make it read the venue position line, or rename it.

THE RUN CHARTS — every fill of the week, marked on the day's own price path

NO TICK DATA FOR MONDAY 17 AUG - US SESSION +\$437.00 ON 8 TRADES — CHART OMITTED.

One trade is the whole day: the day_rider held a short from 13:38 to 18:04 for +\$456 net, while the seven gate legs around it LOST \$19 between them.

NO TICK DATA FOR TUESDAY 18 AUG - US SESSION +\$161.50 ON 17 TRADES — CHART OMITTED.

The busiest session of the week - sixteen exhaustion/abs_veto legs machine-gunned the 13:37-15:10 chop and gave back \$26.50 net between them, so once again the day_rider's single short (+\$188 net) is the entire day.

NO TICK DATA FOR WEDNESDAY 19 AUG - US SESSION +\$515.00 ON 9 TRADES — CHART OMITTED.

The best day of the week was over by 13:52 - the desk caught the 13:36 crash for +\$726 net in thirteen minutes, then handed \$123 back to seven legs stopped in the afternoon drift back up.



Three markers, one position: the day_rider bought 29,447.75 at 13:38 and scaled out of a single long over the next six and a half hours as price slid 180 points - the only trade the desk took all day, and the worst of the week.



The desk placed no NEW trade in this session - but it was not flat. The day_rider's 13:13 four-lot long is on this chart and it never closed: it rode the 158-point drop through the cash open and the 115-point afternoon rally still open, went into the weekend, and was closed by hand on Monday 24 Aug for -\$2,149.44. The Rev1 version of this chart was rendered before that close and titled itself '\$0.00 on 0 trades', which is how the week's single biggest loss came to be drawn as an empty chart.

NO TICK DATA FOR WED 19 AUG 13:36 UTC - 207 POINTS DOWN IN 15 MIN - CAUGHT — CHART OMITTED.

The one the desk got right: the day_rider was short three minutes after the ignition and banked +\$544 net out of a 207-point flush, and two abs_veto legs added +\$182 on the last leg down - \$726 net against a \$414 one-lot ceiling.

NO TICK DATA FOR WED 19 AUG 14:25 UTC - 159 POINTS UP IN 15 MIN - SAT OUT — CHART OMITTED.

The largest sat-out run of the week: price left the 29,376 low and never looked back, and the desk's only response was to sell 29,543 at 14:51 and be stopped 35 seconds later for -\$84 net while the rally carried on to 29,630.



Friday's cash open dropped 158 points in fifteen minutes and kept going to 29,257 - the single cleanest short of the week, in the direction every armed gate trades, and not one of them fired. The desk was long four lots through it on the rider's book.

NO TICK DATA FOR WED 19 AUG 12:32 UTC - 150 POINTS UP IN 15 MIN - SAT OUT — CHART OMITTED.

A 150-point rally into the 29,759 pre-open high, untouched: the desk does trade this hour (it did on Monday and Tuesday) but every gate that fired this week was a short, so an up-run like this has nothing pointing at it.



Three hours after sitting out the Friday flush, the desk sat out the bounce as well - 115 points up off 29,331 with no gate on either side of it, and the rider's four-lot long still open underneath.

★ GAP — THE REHABILITATION PHASE TIMED OUT — THIS SECTION WAS REBUILT FROM WHAT IT HAD SAVED

The phase did not finish. Everything here was reconstructed from the artifacts it had already written to disk before it was killed, plus the follow-up work done afterwards (§2.4, §2.5 and the corrected rider-stop grid). **What that means for you:** the rehab treatments listed are the ones that got run, not the ones that were planned — so an absent treatment here means nobody looked, not we looked and found nothing. Nothing on the action card rests on a treatment that is missing.

P1.5 REHABILITATION — never bench a gate at face value

Garrath — the rule is that nothing gets written off on its P&L alone. If a gate is red, the job is to find out how it could work, and to print the treatments that failed by name so nobody digs the same hole next Friday. Three things were red this week and they are red for three completely different reasons, which is the point: one is a software fault, one is a missing stop, and only one of them is actually a gate. **The headline is that the gate everyone would have benched is the one with the least wrong with it, and the loss that looks like bad luck was structural.**

1 · First, strip the malfunctions out of the strategy verdict

Two of the week's lots are not trading decisions at all. They have to come out before any gate is judged, and they have to be reported, because a fault that gets quietly absorbed into a gate's P&L is a fault nobody fixes:

LOT	WHAT IT WAS	BOOKED	IS IT THE STRATEGY'S FAULT?
#761 day_rider LONG 3 @ 29428.25 → 29417.50	The 00:00Z session roll flipped day_rider_state.json to entered: false / note: "switch off" while venue_net still read 4.0. The rider then flattened a position it had just told itself it did not have, 60 seconds after opening it.	-\$69.00	No. The entry was 09:11:06Z and the flat was 09:12:06Z — one timer tick. Nothing about the market happened in that minute.
#762 day_rider LONG 1 @ 29428.25 → 29388.25	The 1-lot remainder of the same 3+1 exit. It filled 26 points below a 30-minute tape low, ~92 ms after the 3-lot filled on-tape. Flagged BADFILL:stale_price_30pt_outside_book_20260821.	-\$81.50	No, and note it is a SELL — the 'off-tape fills are always BUYs' reading is dead. Same fault, worse fill.
both lots		-\$150.50	

★ REV3 — all three figures in this table were wrong and they now reconcile. Rev2 printed the two lots as -\$73.50 and -\$83.00, which is pnl_usd with the \$1.50 round trip charged a second time; the book says **-\$69.00** and **-\$81.50**. The footer read -\$239.50, which was not the sum of the two rows under either arithmetic — it is **-\$150.50**, and that figure already appears correctly in Part 1's exit-reason table as the CLOCK_FLAT_BUG row (2 legs, 4 contracts).

And the two books have to be named separately, because they remove different things. The CASH book is all 44 legs, **-\$255.50**. Take both malfunction lots out of it and the week's strategy book is **-\$105.00**. The **-\$174.00** clean book quoted elsewhere in this report is not that number: data_quality is set on #762 only, so the clean book removes the BADFILL lot and leaves #761's -\$69.00 in. Both are right on their own terms — -\$255.50 - (-\$150.50) = -\$105.00 and -\$174.00 - (-\$69.00) = -\$105.00 — and Rev2's sentence quietly used one removal with the other's total. Note what the fault actually was: the rider branched on entered, and at the UTC midnight session roll entered gets reset to false while venue_net still reads the true position. It is a one-line fix — **branch on venue_net != 0, never on entered** — and it is on the Saturday card.

2 · REHAB TARGET ONE — abs_veto_short, −\$170.00 over 22 lots

This is the only red gate that is actually a gate. It fired 22 lots at 32% win for −\$170.00, and the reflex is to bench it. Before that, the treatment.

2.1 — What KIND of loser is it? (the root-cause step)

There are only two kinds of losing trade and they need opposite fixes. A **chase** never goes green at all — it is adverse from the first tick, and that is an ENTRY problem. A **give-back** goes green and hands it back, and that is an EXIT problem. Measured inside each trade's own life, on the tick tape:

CLOSED (UTC)	LEG	ENTRY	BEST IT GOT (PT)	WORST IT GOT (PT)	NET	EXIT	PEAK, IN MONEY
08-17 12:43	A	30,208.50	+21.25	-4.25	\$34.50	TARGET	\$41.00
08-17 13:25	B	30,208.50	+28.50	-19.00	−\$35.50	STOP	\$55.50
08-17 15:51	A	30,223.50	+11.25	-13.00	−\$26.50	STOP	\$21.00
08-17 15:51	B	30,223.50	+11.25	-13.00	−\$26.50	STOP	\$21.00
08-18 09:33	A	29,738.25	+20.25	-10.75	\$37.50	TARGET	\$39.00
08-18 09:41	B	29,738.75	+30.25	-10.25	\$57.00	TARGET	\$59.00
08-18 12:20	A	29,734.00	+8.50	-15.50	−\$33.00	STOP	\$15.50
08-18 12:20	B	29,734.00	+8.50	-15.50	−\$33.50	STOP	\$15.50
08-18 14:02	A	29,659.00	+38.25	-38.25	−\$79.50	STOP	\$75.00
08-18 14:02	B	29,659.00	+38.25	-33.50	−\$79.50	STOP	\$75.00
08-18 14:59	A	29,557.50	+20.50	-10.00	\$31.50	MANUAL_CLAIM	\$39.50
08-18 15:10	B	29,557.50	+35.25	-13.75	\$51.00	MANUAL_CLAIM	\$69.00
08-18 16:32	A	29,609.50	+6.00	-12.25	−\$30.00	STOP	\$10.50
08-18 16:33	B	29,609.75	+6.25	-12.00	−\$29.50	STOP	\$11.00
08-19 13:52	A	29,466.75	+59.00	-26.00	\$86.50	MANUAL_CLAIM	\$116.50
08-19 13:51	B	29,466.75	+51.00	-26.00	\$92.50	MANUAL_CLAIM	\$100.50
08-19 14:51	A	29,543.00	+6.00	-19.75	−\$43.50	STOP	\$10.50
08-19 14:51	B	29,543.00	+6.00	-20.50	−\$43.50	STOP	\$10.50
08-19 17:28	A	29,527.50	+17.25	-12.75	−\$28.50	STOP	\$33.00
08-19 17:28	B	29,527.25	+17.00	-13.00	−\$29.00	STOP	\$32.50
08-19 18:12	A	29,514.00	+15.25	-15.75	−\$38.00	STOP	\$29.00
08-19 18:12	B	29,514.75	+16.00	-17.00	−\$37.50	STOP	\$30.50

0 of the 15 losing lots were chases. Not one of them was adverse from the first tick — the worst-behaved loser still got 6 points inside and the median loser reached **+11.25 points**, which is \$21.00 a lot sitting on the screen before it turned round. So the reflex diagnosis is wrong. This is not a gate buying tops; every one of these entries worked, briefly.

The difference between a winner and a loser here is not whether the trade went onside, it is how far: the winners' median peak is **+30.25 points** against the losers' +11.25. That is a give-back shape, and give-back shapes are an EXIT question — so the exit is where the treatment has to start, which is the opposite of where this desk usually looks.

2.2 — The stop-width sweep, run twice, because the obvious version cheats

Every lot re-raced from its own entry price on the real tick path at five widths, first touch wins, stop wins a tie inside a gap, three-hour cap. The first pass is the usual one: risk = $k \times \text{ATR}$ and the target moves with it, so the whole trade keeps its shape.

STOP WIDTH	LOTS	NET	WIN %	\$ / LOT	WORST LEAVE-ONE-DAY-OUT	BEST LOT STRIPPED	STOP / TARGET / 3H-CAP
0.50×ATR	22	−\$124.83	32%	−\$5.67	−\$189.47	−\$170.07	15 / 7 / 0
0.75×ATR	22	−\$57.94	32%	−\$2.63	−\$118.53	−\$149.45	15 / 7 / 0
1.00×ATR ← live	22	−\$548.73	18%	−\$24.94	−\$507.28	−\$607.02	18 / 4 / 0
1.25×ATR	22	−\$778.01	14%	−\$35.36	−\$627.37	−\$851.24	19 / 3 / 0
1.50×ATR	22	−\$1,076.45	9%	−\$48.93	−\$896.89	−\$1,162.81	20 / 2 / 0

Calibration first, because a sweep you cannot check is decoration. Of the 22 lots, 18 were closed by the machine on a stop or a target. Re-raced at the live 1.00× width they come to −\$397.30 against **−\$437.50** actually booked (★ REV3 — Rev1 printed −\$464.50, which is pnl_usd with the \$1.50 round trip subtracted a second time; the column is already net), and **18 of 18 land on the same exit reason**. The remaining 4 lots you claimed by hand: you booked **\$267.50** on them, the simulator rides them to their own stop or target for −\$151.43. **That gap — \$418.93 — is your hand on this one gate**, and it is the whole reason the live book reads −\$170.00 where the machine alone reads −\$548.73.

The verdict: the sign survives all five widths. Every cell is negative, from −\$124.83 at the tightest to −\$1,076.45 at the widest, and the worst leave-one-day-out is negative at every width too. That is a real verdict rather than a knife-edge, and it is bankable: left entirely to the machine, this gate lost money this week at every stop width tested.

2.2b — But is that the STOP, or is it the target riding along with it?

The sweep above moves the stop and the target together, because R is defined on risk. That means “tighter is better” could just be “a nearer target gets hit more often”, which is a completely different instruction. So here is the same sweep with the target pinned where the live config actually puts it, moving only the stop:

STOP WIDTH	LOTS	NET	WIN %	\$ / LOT	WORST LEAVE-ONE-DAY-OUT	BEST LOT STRIPPED	STOP / TARGET / 3H-CAP
0.50×ATR	22	−\$386.89	5%	−\$17.59	−\$387.24	−\$433.55	21 / 1 / 0
0.75×ATR	22	−\$498.24	9%	−\$22.65	−\$477.70	−\$556.53	20 / 2 / 0
1.00×ATR ← live	22	−\$548.73	18%	−\$24.94	−\$507.28	−\$607.02	18 / 4 / 0
1.25×ATR	22	−\$728.09	18%	−\$33.09	−\$665.74	−\$786.38	18 / 4 / 0
1.50×ATR	22	−\$659.37	23%	−\$29.97	−\$576.13	−\$781.89	17 / 5 / 0

Pinned target, moving stop: 0.50× −\$386.89 · 0.75× −\$498.24 · 1.00× −\$548.73 · 1.25× −\$728.09 · 1.50× −\$659.37. It still gets worse as the stop widens, so the effect is genuinely the stop and not the target. Either way both tables agree on the thing that matters: **every cell in both sweeps is negative**. There is no stop width at which this gate, run by the machine alone, made money this week.

2.3 — Filters, judged by whether they keep the winners

The rule here is old and it is the one that catches most bad ideas: a filter that improves the P&L by dropping the trades that WON is a fake win, however good the total looks.

TREATMENT	THE IDEA	LOTS KEPT	NET	Δ VS AS-TRADED	WINNERS KEPT	VERDICT
Drop the 12:00-14:00Z open window	the loudest two hours of the day	16	−\$314.50	−\$111.50	4 / 7	no better
US cash session only (13:30-20:00Z)	trade only when the depth is real	16	−\$230.00	−\$27.00	4 / 7	no better
Europe/pre-open only (before 13:30Z)	the mirror of the row above	6	\$27.00	\$230.00	3 / 7	FAKE
Stand down after two consecutive stops	the wall-of-stops rule, again	12	−\$40.00	\$163.00	4 / 7	FAKE

Nothing clears the bar. The two that beat the as-traded number do it by cutting winners along with losers, and the stand-down-after-two-stops rule is the “wall of

stops" idea. ★ **REV2 fixes its count, which the report gave as three here and four on the action card.** It is **THREE** independent populations, and this is the canonical list — every other mention of the rule points here: **(1)** mechanically on 104 signals last week; **(2)** live across last week's tape; **(3)** `abs_veto_short`'s 22 lots this week, in the table above. The card said "fourth" by counting the run-state work as a population; it is not one — that work refuted the same principle on a different question and never tested this rule on a sample of its own, so it is supporting reasoning, not evidence. Three is the number. **Post it as REFUTED and stop proposing it.**

SO WHAT IS THE TREATMENT FOR `ABS_VETO_SHORT`?

Three things are now closed off. It is NOT a stop-width change — ten cells across two sweeps, every one negative. It is NOT a filter — none keeps the winners. And it is NOT a bench on a knife-edge, because the sign survived, so a bench here would at least be honest.

What the excursion table actually says is that the entries work and then stop working: median peak +30.25 points on the winners against +11.25 on the losers, with nothing adverse from the first tick. The live experiment that addresses exactly that is the 55-second continuation-confirm — it re-requires the burst to PERSIST before entering, which is a filter on the difference between an 11-point poke and a 30-point move. This week its mirrors are the top of the shadow board (+\$332 over 46 fires) while the live gate lost \$203 over 22. **Part 2 is where that promotion is argued, and it is the one real proposal this report makes about this gate.**

The honest caveat, stated before Part 2 rather than after it: the shadow mirror does NOT model the live gate's default-veto on thin tape, so it over-fires, and it enters on the 1-minute bar open where the live decider quotes the tick it fired on. Both biases run in the mirror's favour. It is an upper bound, and it is presented as one.

2.4 — ★ REV2: THE EXIT LADDER — the treatment §2.1 diagnosed and §2.2-§2.3 never ran

THE DANGLING DIAGNOSIS

§2.1 ends on a sentence that commits this section to something: “that is a give-back shape, and give-back shapes are an EXIT question — so the exit is where the treatment has to start.” What followed was §2.2 sweeping the **stop**, §2.2b sweeping the **stop** again, §2.3 sweeping time-of-day and stand-down **filters**, and a recommendation of the 55-second confirm, which is an **entry** filter. Not one rung of an exit ladder was ever tested on the gate this report says has an exit-shaped loss. Here they are, on the same 22 lots, with the winners-kept column carried all the way through.

Every lot raced from its own entry on the tick tape, three-hour cap, stop wins a tie. **The stop and the target come from the VENUE, not from an ATR estimate** — every lot’s protective StopLimitOrder auxPrice is in the tournament journal, so $R = |\text{entry} - \text{stop}|$ is exact, and the live target model (the $\text{atr_split}=22$ quiet-tape clip: Lot A banks \$40, Lot B takes $\max(1.75 \times \text{ATR}, \$60)$ below a 22-point ATR) is confirmed to the tick. **Calibration: 18 of 18 machine-closed lots reproduce their own live exit reason.**

EXIT TREATMENT	NET	\$ / LOT	WIN %	WINNERS KEPT	WORST LEAVE- ONE-DAY- OUT	BEST LOT STRIPPED	WHAT IT IS
AS TRADED (your hand + the machine)	−\$170.00	−\$7.73	31.8%	7 / 7	−\$162.00	−\$295.50	the live book
MACHINE, live exit (baseline)	−\$446.24	−\$20.28	22.7%	5 / 7	−\$420.00	−\$516.50	live stop, live target
Scalp at +11.25pt (losers' median peak)	\$141.00	\$6.41	72.7%	7 / 7	\$57.00	\$120.00	bank the poke
Scalp at 0.5R	\$178.50	\$8.11	81.8%	7 / 7	\$99.00	\$140.25	half the risk
Scalp at 1.0R	\$187.00	\$8.50	54.5%	7 / 7	\$60.00	\$109.00	one-for-one
Trail after +15pt, 15pt behind the peak	\$107.50	\$4.89	59.1%	7 / 7	\$12.50	\$21.00	give-back cap
Trail after +15pt, 8pt behind the peak	\$149.00	\$6.77	63.6%	7 / 7	\$15.00	\$71.00	tight give-back cap
Breakeven stop once +15pt	\$107.50	\$4.89	59.1%	7 / 7	\$12.50	\$21.00	same as trail-15
Lot A scalp +11.25pt / Lot B rides	−\$162.74	−\$7.40	45.5%	6 / 7	−\$182.50	−\$233.00	split the pair

The ladder does not fail. It is the first treatment this gate has been given that keeps every winner. Four rungs beat the as-traded book while holding 7 of 7 winners, stay positive on the worst leave-one-day-out fold, and stay positive with their best lot stripped: a scalp at **1.0R** (+\$187.00), at **0.5R** (+\$178.50), a **trail 8pt behind the peak once +15pt** (+\$149.00), and a scalp at the losers' own median peak of **+11.25pt** (+\$141.00). Against a machine baseline of −\$446.24 and an as-traded −\$170.00, that is a swing of six hundred dollars against the machine from changing nothing but where profit is taken.

★ **AND THE PLACEBO, WHICH IS WHY THIS IS A SHADOW ROW AND NOT A MONDAY CHANGE**

A tight target flatters any entry in a simulator, so the rung has to beat a random-entry control before it counts. 400 draws, same 22-lot shape, same R distribution, same tape, same fee:

RUNG	REAL	PLACEBO MEDIAN	PLACEBO 5-95%	WHERE THE REAL RESULT SITS
Scalp at 0.5R	\$178.50	−\$6.75	−\$219.00 ... \$192.75	93rd pct
Scalp at 1.0R	\$187.00	\$14.00	−\$305.00 ... \$344.00	80th pct
Scalp at +11.25pt	\$141.00	\$20.00	−\$232.00 ... \$258.00	80th pct
Trail +15pt / 8pt behind	\$149.00	\$1.00	−\$284.00 ... \$259.00	83rd pct

The entry is adding something, and not much. A random entry on this week's tape, scalped at 0.5R against a 1R stop, lands at −\$6.75 — which is what a fair coin minus the fee should look like, and is a good sign the harness is honest. The real signals land at the 80th to 93rd percentile of that distribution. On n=22 that is a direction, not a measurement: none of these rungs clears its own placebo band. So the ladder goes to **SHADOW**, beside the 55s arms, and nothing changes live on Monday.

What this does close. The diagnosis in §2.1 no longer dangles. The exit IS where this gate's money is — every rung that banks earlier than the live 1.5R/2.5R pair improves it, and the losers' median peak of +11.25pt was real money sitting on the screen on 22 of 22 lots. It also sharpens what §2.2 actually proved: the stop sweep found no width that rescues the gate because the stop was never the problem. ★ **REV3 PRINTS THE HAND-VS-MACHINE SPLIT FOUR WAYS, because the two-number version did not add up.** Rev1 said “the machine

books $-\$446.24$ and your hand is worth $+\$314.74$ ", and $-446.24 + 314.74$ is $-\$131.50$, not the $-\$203.00$ printed beside it. §2.2's pair has the same hole. Here is every number, so the gap is visible rather than buried:

THE 22 LOTS, SPLIT BY WHO CLOSED THEM	BOOKED (THE LIVE LEDGER)	REPRICED (THE SIMULATOR)	DIFFERENCE
18 lots the MACHINE closed — stop or target	$-\$437.50$	$-\$393.00$	$+\$44.50$
4 lots YOU closed by hand	$+\$267.50$	$-\$53.24$	$+\$320.74$
All 22	$-\$170.00$	$-\$446.24$	$+\$276.24$

Read the bottom row, and quote only the bottom row. The live book is **$+\$276.24$** better than the machine alone would have been, and that is the honest statement of what your hand was worth on this gate: it is a difference between two totals and needs no assumption about anything in between. The row above it, $+\$320.74$, is the version Rev1 quoted (as $+\$314.74$, off the fee-double-counted booked figure), and it is bigger because a separate $+\$44.50$ residual sits on the eighteen lots the simulator says it reproduces 18-of-18. Something in the reprice disagrees with the machine's own real fills by about $\$2.47$ a lot on lots where it claims to land on the same exit reason. It does not move the verdict — but it is unexplained, and the harness that produced it was never saved to a script, so it cannot be re-run and the residual cannot be attributed. **Treat $-\$446.24$ as a simulator output carrying a known $\sim\$45$ uncertainty, not as a measurement.** §2.2's estimated-ATR version of the same split agrees on all of it: machine $-\$548.73$, hand $+\$418.93$, residual $+\$40.20$, same sign, same verdict.

★ **And this section now ships something.** Rev1 ran the ladder, found four rungs that beat the book while keeping 7 of 7 winners, wrote “the ladder goes to SHADOW” — and then nothing on the 47-play card carried it, which is precisely the failure this report criticises the claim detector for by name. It is now **MONDAY #6**: the 0.5R and 1.0R scalps and the +15pt/8pt trail, as A/B arms against the live 1.5R/2.5R pair on this gate's own signal, verified NOT byte-identical per BUILD #7, with the promotion bar stated in advance — $n \geq 60$ fires, positive on strip-best-3, AND clearing the 95th percentile of the random-entry control built above. They sit at the 80th–93rd today, so that bar is a real one and it can fail.

2.5 · ★ REV2 — REHAB TARGET ONE-AND-A-HALF:
exhaustion_short, the green one nobody interrogated

WHY A GREEN GATE GETS THE SAME TREATMENT AS A RED ONE

This section’s standing rule is that nothing gets written off on its P&L alone. The mirror of that rule is that nothing gets waved through on its P&L alone either, and Rev1 did exactly that: exhaustion_short is the only live slot green on the week (+\$165.50, 14 lots, 57%) and it received no rehab pass, no stop sweep, no leave-one-day-out, no strip-best and no hand-versus-machine split — while Part 2.6 §3 quietly attributed it −\$638.02 mechanically on the same week. A \$782.52 contradiction sat in the report unexamined. Here is the same battery abs_veto_short got.

2.5a — Is the +\$165.50 the gate, or your hand?

Every one of the 14 lots re-raced from its own entry to its own configured stop or target, three-hour cap. R comes from the venue: each lot’s protective stop order is in the tournament journal, and exhaustion_short runs stop_k=1.5, so entry ATR = R / 1.5 exactly. **7 of the 7 machine-closed lots reproduce their own live exit reason**, repricing to −\$435.50 against −\$434.00 booked — \$1.50 apart on seven lots. (★ REV3 — Rev1 said “−\$444.50 booked, \$9.00 apart”, which was pnl_usd with the fee taken off twice. Correcting it makes the calibration TIGHTER, not looser.)

BOOK	NET	WIN %	STRIP BEST 1	STRIP BEST 3
LIVE — your hand plus the machine	\$165.50	57.1%	\$34.50	−\$176.50
THE MACHINE ALONE	\$84.00	42.9%	−\$91.50	−\$422.50
... so the hand is worth	\$81.50	—	—	—

The answer is: mostly the gate. That is the opposite of abs_veto_short, where the machine alone loses −\$446.24 and the hand is the entire live result. Left to itself exhaustion_short still books **+\$84.00** — and the hand adds \$81.50 on top, not the \$60.50 Rev1 printed, because the live book is \$165.50 rather than \$144.50. On the report’s own logic that is a real difference between two gates that look superficially similar, and it is the reason this one stays live.

2.5b — And then the robustness battery, which it does not pass

TEST	LIVE BOOK	MACHINE ALONE	VERDICT
Strip the single best lot	\$165.50 → \$34.50	\$84.00 → −\$91.50	One lot is 90% of the live result and all of the machine's
Strip the best three	→ −\$176.50	→ −\$422.50	Fails on both books
Leave one day out (it only traded two)	drop 08-18 → \$65.00 · drop 08-17 → \$100.50	drop 08-18 → −\$317.00 · drop 08-17 → \$401.00	The machine result is entirely 08-18. One of two folds is deeply negative
Parameter plateau on the stop	see the sweep below — the live 1.5× is a spike, not a plateau		Fails

STOP WIDTH (MACHINE ALONE)	NET	WIN %	\$ / LOT	WORST LEAVE-ONE- DAY-OUT	BEST LOT STRIPPED	STOP / TARGET / 3H-CAP
0.5×ATR	\$29.66	21.4%	\$2.12	−\$109.68	−\$125.84	11 / 3 / 0
0.75×ATR	−\$22.00	28.6%	−\$1.57	−\$161.50	−\$177.50	10 / 4 / 0
1×ATR	−\$163.68	28.6%	−\$11.69	−\$213.36	−\$319.18	10 / 4 / 0
1.25×ATR	−\$305.36	28.6%	−\$21.81	−\$265.22	−\$460.86	10 / 4 / 0
1.5×ATR ← live	\$83.98	42.9%	\$6.00	−\$317.04	−\$91.52	8 / 6 / 0
2×ATR	−\$140.36	42.9%	−\$10.03	−\$420.72	−\$315.86	8 / 6 / 0

The live 1.5×ATR width is an isolated positive cell surrounded by negative ones — 0.75× −\$22.00, 1.0× −\$163.68, 1.25× −\$305.36, 2.0× −\$140.36. A real effect gives a plateau; this is a step. And 0.5× is positive too (+\$29.66), which makes the shape a pair of spikes rather than a slope. On 14 lots across two sessions that is what you would expect from noise.

★ AND THE CONFIG THAT PRODUCED THE GREEN WEEK WAS REVERTED MID-WEEK AS UNMEASURED

Every exhaustion_short fire this week ran with atr_split=22 active. That clip was put back on 08-16 at 21:15Z by revert_clip_lift's daily restore job against a stale roster, and the operator removed it again on **08-18 at 15:48Z** (commit 465c943) with the words “the desk then ran an UNMEASURED cell live on its only armed short”. Below a 22-point ATR it makes Lot A bank at \$40 and Lot B at max(1.75×ATR, \$60). Two of the seven signals fired under it, and it is confirmed to the tick — the 14:55 Lot A exited TARGET at exactly 20.00pt, which is \$40. So the gate's only green week was booked on a configuration that is no longer live and was never graded.

The disposition, restated honestly. Keep exhaustion_short live — nothing here says bench it, and it is the one gate on the desk whose machine half is

positive without the operator. But “the only live slot green on the week, no change proposed” needs the same caveat `abs_veto_short` got: **+\$165.50 on 14 lots across two sessions, under a config that was reverted mid-week, does not survive strip-best-1 on the machine book and has a negative leave-one-day-out fold.** It is a green number, not evidence. Do not re-tune anything on it and do not cite it as a promotion precedent.

3 · REHAB TARGET TWO — the day rider's Thursday, −\$1,288.50

One long, four lots, entered 13:38:06Z at 29,447.75 on the US open, closed by your hand in three pieces between 19:33 and 20:17. Inside its own life it got +19.25 points onside at best and -245.25 points offside at worst. **It never went green in any meaningful way.** So the same root-cause question: was it the entry or the exit?

The entry is defensible — it is the rider's normal open-hour entry and it has made money all week doing exactly that. What is not defensible is that there was no stop in the water at any point. Here is what one would have done, raced on the real tape from the rider's own entry, with the ATR the live detector itself read at the confirming tick, **34.2** points (★ REV2 — Rev1 used an estimated 36.1 here while Part 1.6 §3 printed the detector’s own 34.2 for the same trade; a stop the rider actually carried would be sized off `arm_atr`, which is the detector’s read, so 34.2 is the number and every cell below is recomputed on it, net of the \$1.50 round trip):

IF THE RIDER HAD CARRIED THIS STOP	IT WOULD HAVE FILLED AT	AT	4 LOTS WOULD HAVE COST (NET)	VS WHAT ACTUALLY HAPPENED	... AND ON THE THREE WINNING DAYS
0.50×ATR = 17.1 pt	29,430.65	13:40:03Z	−\$142.80	\$1,151.70	kills all three (−\$291.60)
0.75×ATR = 25.7 pt	29,422.10	13:40:06Z	−\$211.20	\$1,083.30	kills all three (−\$290.40)
1.00×ATR = 34.2 pt	29,413.55	13:40:13Z	−\$279.60	\$1,014.90	keeps 1 of 3 (\$301.00)
1.25×ATR = 42.8 pt	29,405.00	13:40:20Z	−\$348.00	\$946.50	keeps 1 of 3 (\$242.50)
1.50×ATR = 51.3 pt ← the width Rev1 picked	29,396.45	13:40:40Z	−\$416.40	\$878.10	keeps 1 of 3 (\$184.00)
2.00×ATR = 68.4 pt	29,379.35	13:47:32Z	−\$553.20	\$741.30	keeps 1 of 3 (\$67.00)
3.00×ATR = 102.6 pt ← REV2 picks this one	29,345.15	13:51:24Z	−\$826.80	\$467.70	keeps 3 of 3 (\$1,179.00)

★ **REV2 — READ THE LAST COLUMN, WHICH REV1 DID NOT HAVE.** Every width beats what happened on Thursday, and the tightest ones beat it by more than a thousand dollars — but Thursday is the one day the rider lost money, and pricing a stop only there is the fake-win failure this same section refuses to accept two screens earlier. The full widths×days grid is in [Part 1.6 §3b](#). Its verdict, ★ **RE-RACED IN REV3** with the fee double-count taken out of the as-traded column and the weekend carry now resolved rather than marked: **1.5×ATR takes the week's closed rider book from −\$251.00 to −\$379.90**, because 46.4pt sits inside Monday's −75.25pt excursion and 46.4pt inside Tuesday's −64.50pt, so it stops both winners out. **3.0×ATR is the only width that is positive on the closed book (+\$210.70) AND keeps all three winners.** Include the carry, now booked at −\$2,149.44, and the six-entry week is −\$2,400.44 as traded against −\$532.10 at 3.0×ATR — **every** width beats doing nothing once the unstopped position is in the total. The honest week-level value of the change is **+\$462 on the closed book, +\$1,868 with the carry**, not \$1,000. **But the week is not the argument any more.** SATURDAY #2 now scores the same proposal against the 231 lake sessions the operator's 2026-08-20 no-stop decision was actually taken on — where every width beats naked at every entry delay and the worst session falls from −\$8,616 to −\$1,689, and where nothing at all is profitable. Read that row before acting: this re-opens a decision he has already made once, and on the one session since publication his hand beat every stop on this grid.

What is NOT in dispute: having a stop at all. Every width beats what happened on Thursday, **This is not hindsight about direction** — it is not asking the rider to have known the tape was going down. It is asking it to have had any exit at all, at any of six widths, all of which are cheaper than holding to your hand at 19:33. The rider ran naked for 6.7 hours.

★ AND A LOSING RIDER IS STRUCTURALLY SILENT — THAT IS THE REAL DEFECT

The reason this ran for six hours is that nothing was shouting. The rider pages on peak give-back and on stalls; below its arm threshold there is no stop, no trail and no alert, so a position that is simply losing produces no output at all. Both of the tools you would naturally check print “flat, \$0” while it happens, because the trade has not CLOSED yet and they count closed rows.

The rehabilitation of the rider is not an entry change and it is not a threshold change. It is a stop. A $3.0 \times \text{ATR}$ protective stop on the rider's own book, sized off its entry ATR, is the single highest-value change on this week's card — and ★ REV2 corrects both halves of what Rev1 said here: not $1.5 \times$, and not “\$1,000 on one day” but +\$462 on the week's closed book. It is also the change that removes the report's headline: every width tested, including $3.0 \times$, closes Friday's 13:13Z entry by 13:31:55Z — six and a half hours before the gateway wedged.

4 • REHAB TARGET THREE — the four gates that never fired

The standing rule cuts the other way too: a gate that is benched is not being rehabilitated, it is being ignored. So Movement 2 fired all four of them mechanically, in their exact live config, across this whole week's captured tape — the honest denominator, every fire counted, not just the ones near a big move:

GATE (BENCHED ALL WEEK)	RAW SIGNALS ON THE TAPE	WOULD HAVE FIRED	WOULD HAVE MADE	\$ / FIRE	WHAT ITS OWN FILTERS STOPPED
grind_long	24,121	11	−\$319.17	−\$29.02	ATR floor 22pt x24,110
capitulation_long	132	10	−\$97.66	−\$9.77	ATR floor 10pt x122
abs_veto_long	38	26	−\$83.42	−\$3.21	55s veto: absorbed x5; 55s veto: thrust gone x7
rgv_short	1	1	−\$17.54	−\$17.54	—

Total for the four: **−\$517.79. The bench is the rehabilitation.** Look particularly at grind_long: 24,121 raw signals on the week's tape, of which its own ATR floor stopped 24,110. The gate is not benched by the router in any meaningful sense — it is benched by its own 22-point floor, which this week's tape cleared on a small minority of minutes. The mid-week work in Part 2.5 puts a number on how often that floor is even reachable, and it is about a quarter of sessions.

5 • The treatments that FAILED this week, by name

TREATMENT	APPLIED TO	RESULT	DISPOSITION
Tighter stop (0.50× / 0.75×ATR)	abs_veto_short	−\$124.83 and −\$57.94 — both worse than as-traded	REFUTED for this gate this week
Time-of-day carve-outs (three variants)	abs_veto_short	None keeps 80% of the winners while beating the total	REFUTED — fake wins
Stand down after two consecutive stops	abs_veto_short	Cuts winners with losers; THIRD independent population to refuse it (★ REV2 — the count is settled at three and the populations are named in §2.3)	REFUTED
“It is chasing exhausted moves” (the entry diagnosis)	abs_veto_short	0 of 15 losing lots were chases. Median loser reached +11.25 pt inside first — the entries work, briefly	REFUTED on this week's tape
Blame the entry	the day rider's Thursday	The entry is its normal open-hour entry and made money on three other days; six stop widths all beat the outcome	REFUTED — it is the missing stop

Disposition — every lead this section touched

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
abs_veto_short — bench the machine's half of it	SHADOW	Ten cells across two stop sweeps, every one negative, worst-leave-one-day-out negative throughout: the sign survives, so this IS a verdict rather than a knife-edge. The gate is only live-negative-but-not-worse because your hand is worth + \$276.24 against the machine alone — the live book minus the machine-repriced book, which is the form of that number that reconciles (★ REV3; earlier revisions quoted +\$314.74 and +\$412.93, and §2.4 now prints the four-way split and the ~\$45 simulator residual that explains the difference). Bench the automated slot, keep the signal in shadow, and let the 55s mirror carry the case.
abs_veto_short — fix the ENTRY via the 55s confirm	SHADOW	The mirror is the top of this week's shadow board. Promote only on the Part 2 battery, and only with the mirror's two known upward biases stated on the same page.
abs_veto_short — a stop-width or filter change	REFUTED	Named tests: two stop sweeps of five cells each (all ten negative) and the winner-retention rule (no filter keeps 80% of the winners while improving the total).
The day rider — a protective stop on its own book	LIVE	★ REV2 re-raced ALL SIX rider entries at SEVEN widths, not just Thursday, and ★ REV3 re-raced it again with the fee double-count out of the as-traded column and the carry resolved rather than marked — these are the REV3 figures and they are the ones in Part 1.6 §3b. Closed book: as-traded −\$251.00; the picked width is 3.0×ATR at +\$210.70, the only positive cell

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
		and the only one that keeps all three winners. 1.5×ATR — Rev1's pick — is — \$379.90 and keeps one. Cause of death for 1.5×: winners-retention, on the grid in Part 1.6 §3b. Still the week's highest-value change, at +\$462 on the closed book (+ \$1,868 once the carry is in) rather than ~\$1,000 — but SATURDAY #2 settles it on 231 sessions, not on this week.
The day rider — branch on venue_net, not entered	FIXED	One line. The 00:00Z session roll orphaned an open position and cost \$156.50 in two lots on Friday morning.
The four benched gates — un-bench any of them	REFUTED	Fired mechanically across the whole week's tape in their exact live config: −\$517.79 on 48 fires. Revisit on a measured trend week, not on this one.
grind_long's 22-point ATR floor	PARKED	It blocked 24,110 of 24,121 raw signals this week, which makes the gate un-testable rather than benched. Revive the question with the Part 2.5 floor-reality numbers: how often is the floor reachable at all, and is 22 the right number or just an untested one?

P1.6 THE DAY RIDER — the week of 17-21 August 2026

Gaz, the two books swapped places. Last week the rider made **+\$1,617.50 on five trades** while the tournament lost \$325.00; this week the tournament is a rounding error at **−\$4.50 on 36 lots** and **the rider is the loss**. It closed seven legs for **−\$169.50** in the strategy book, **−\$251.00** in cash.

But that headline hides the thing worth knowing. Split the book by who pressed the button and it separates cleanly: the **automatic rider went 3-1 for −\$100.50** across four sessions, and **every other dollar of damage came from a hand** — two manual entries on Friday, one killed by a bug sixty seconds after it was placed, the other **still open right now** because the IB gateway wedged at 19:58Z and the 20:40 hard flat never ran.

THE ONE-PARAGRAPH VERSION

Realised –\$169.50 (strategy) / –\$251.00 (cash), plus the 4-lot long that ran through the weekend and closed by hand on 2026-08-24 for –\$2,149.44. ENTRIES: all four automatic confirmations landed inside the validated 13:38–14:09 window; the two entries outside it were both manual and both are under water. **THRESHOLDS:** nothing to change. The margins ran +0.006 to +0.490 over the ER floor, and I probed the floors directly — raising ER to 0.20 and RT to 0.55 removes **zero** fires this week, it only makes them **later and worse**, by –\$116 and –\$156 on the two it touches. **EXITS:** the operator's claims took **91.0% and 89.3%** of everything available on the two winners he claimed and beat the machine trail by **+\$180.29**; the one mechanical TRAIL took **36.3%**; on the losing day the claims cost – **\$120.50** against just letting the clock do it. **THE WEAK DAY IS THURSDAY 08-20 and it is a FAULT, not a drought** — the tape offered **\$1,158** on the short side and the rider bought the high of the day **37 seconds after it printed**.

★★ REV3 — SINCE THE BELL: THE RIDER HAS TRADED TWICE MORE, AND THE REPORT'S OWN #1 RECOMMENDATION DID NOT LAND

This report is written for the week ending 08-21 and published on 08-26, and the rider has not been idle in between. Two things happened and both belong here rather than in a footnote.

Monday 2026-08-24, 11:59:39Z — you closed the carry by hand. All four lots at 29,224.00, booked **−\$2,149.44**. Not one automatic path closed it; the weekend-flatten timer fired 119 times on the Sunday and placed no order and sent no page. That single position is larger than the whole rest of this report put together.

Tuesday 2026-08-25 — a sixth rider entry, and it ran naked again. The detector confirmed and the service entered **4 lots LONG at 29,354.75 at 13:38:06Z** with `arm_atr 30.62` and, in the state file it writes every tick, **"venue_stop": null**. So the answer to "did it run naked again" is yes, and the answer to "did SATURDAY #2 land" is no — nothing shipped. It was an AUTOMATIC entry, not a manual one, so the `arm_atr = 0.0` defect §6 describes did not apply: the ATR was stored correctly.

08-25, WHAT ACTUALLY HAPPENED	FIGURE	WHAT IT MEANS
Entry	4 lots LONG @ 29,354.75	13:38:06Z, automatic, inside the validated window
Protective stop	none	venue_stop: null on every tick of the session
Lot 1 — the \$100 rung	+\$103.50	TARGET_100 at 13:49:06Z. The only rung of the four-rung ladder that has ever filled.
Lots 2-4 — claimed by hand	−\$346.50	20:27:47Z at 29,297.75, thirteen minutes before the hard flat
Session	−\$243.00	
Session peak, after entry	+61.25 pt	the trail needed +122pt (4×ATR 30.6) and never armed — the service printed "trail not armed" on every one of ~420 ticks
Worst it went against	−209.25 pt	= −\$1,674 unrealised on four lots, with nothing in the water

Two things follow, and they point in opposite directions.

One — the frozen trail arm bit again, exactly as §6 said it would. The arm threshold is $4 \times \text{arm_atr}$ frozen at entry. The tape calmed after 13:38, the peak reached +61.25pt against a +122pt requirement, and **no mechanical exit existed at any point between 13:38 and the 20:40 flat.**

Two of the last three rider sessions have now ended with the trail never arming. That is not a fluke, it is the design.

Two — and this one cuts against this report's own recommendation: the hand won. Raced tick-by-tick on capture.db from the real entry, every stop width would have done worse than you did: 0.5× −\$128.48, 1.0× −\$250.96, 1.5× −\$373.44, 2.0× −\$268.44, **3.0×ATR −\$452.16 (cut at 14:03:57Z)** against the **−\$243.00** you actually booked. The position went 209 points under water and came back, and you claimed it at −57. **That is the operator's own argument — “my claim profit buttons also work as a manual kill button” — working, on the one session since this report was drafted.** One session proves nothing either; it is recorded here because the report's proposal would have cost \$209 on it, and a card that only reports the sessions where its recommendation wins is not a card.

1 · THE MONEY — and which hand did it

BOOK	NET	LEGS	LOTS	WHAT IT IS
Rider, STRATEGY book	−\$169.50		7	13 Clean rows only. Use for every threshold and exit question below.
Rider, CASH book	−\$251.00		8	14 Adds the 08-21 BADFILL lot — a defective fill price, not a phantom trade. This is the figure that reconciles to IBKR.
The 4-lot carry — NOW CLOSED	−\$2,149.44		1	4 ★ REV3 — no longer a mark. You closed all four lots by hand on 2026-08-24 at 11:59:39Z at 29,224.00 . Rev1 carried it at −\$973.00 marked to the last print before the 21:00 halt and rightly called it provisional; the real figure is more than twice that. It closed outside this report's window so it is in none of the rows above. See section 5.
— of which entered AUTOMATICALLY	−\$100.50		6	10 Four detector-confirmed sessions, 3 winners 1 loser . Essentially flat.
— of which entered BY HAND	−\$150.50		2	4 Friday's 09:11 press, killed by a bug a minute later. The 13:13 press is the still-open lot above.
Tournament, same week	−\$4.50		36	36 For scale. Last week it was −\$325.00 on 108 and the

BOOK	NET	LEGS	LOTS	WHAT IT IS
				rider carried the desk; this week that is exactly inverted.

MNQ at **\$2.00 a point**, fee **\$1.50 per round trip per lot**, and **the fee is already inside every figure in this section**. ★★ **REV3 — THE REV2 “FEE CORRECTION” ON THIS PAGE IS WITHDRAWN. IT WAS BACKWARDS.** Rev2 saw that this section rendered at 22:32 while the rest of the report was rebuilt after midnight, decided it had missed a fee convention, and subtracted fees_usd from pnl_usd throughout. pnl_usd is written already net of fees_usd — store.py's schema comment says so, pnl.py's docstring says so, and the invariant holds on all 765 rows in the book — so that subtraction charged the round trip a second time. Every figure it “corrected” is now put back: strategy **−\$169.50**, cash **−\$251.00**, automatic **−\$100.50**, by hand **−\$150.50**, Thursday **−\$1,288.50**, tournament **−\$4.50**. Rev2's own remark that **−\$1,294.50** “is what Part 1 said all along” was the tell and it was read the wrong way round: Part 1 had the same bug. **Nothing else in this section moves.** The capture ratios barely shift, because an exit at the peak pays the same round trip as the exit you took; and the hand-vs-machine deltas do not move at all — **+\$59.79** for the week and **+\$180.29** on the winners are differences between two figures that each pay the same fee, so they are fee-invariant and stand exactly as first printed.

The rider went from **2 lots to 4** on 08-20 alongside the no-stop / per-lot-ladder redesign — so the week's only loss is also its first day at double size. At the old 2-lot size the same Thursday is about **−\$644** net. That redesign shipped two things on one day and until this revision neither had been graded: the no-stop decision is re-argued in §3b and SATURDAY #2, and the per-lot profit ladder (\$100/\$200/\$400/\$600 → 50/100/200/300 points) is graded for the first time in SATURDAY #6 — over 231 sessions its four rungs fill on **71.0% / 39.4% / 11.3% / 4.8%** of sessions, every rung is net negative, and the advertised \$1,300 day happens about **once in twenty-one sessions**.

2 • DID WE ENTER OK?

Every row below is the **live detector's own read at the tick it fired on**, lifted from the service journal — not a replay. “Travelled” is how far the move had already gone from the 13:30 open reference the detector itself measures against, and “of extension” is that distance as a share of how far the move eventually ran in the same direction. MFE and MAE are measured from entry, on capture.db ticks, to the 20:40 flat.

DAY	CONFIRMED	+MIN	SIDE	ENTRY	13:30 REF	TRAVEL LED	OF EXTENSION	MFE AFTER	MAE AFTER	WINDOW
Mon 08-17	13:38:07	+8.1	SHORT	30,211.50	30,253.00	41.50 pt	21.5%	+151.50	-75.25	IN
Tue 08-18	13:54:07	+24.1	SHORT	29,642.00	29,713.50	71.50 pt	35.8%	+128.00	-64.50	IN
Wed 08-19	13:39:07	+9.1	SHORT	29,607.50	29,672.75	65.25 pt	22.0%	+231.75	-59.75	IN
Thu 08-20	13:38:07	+8.1	LONG	29,447.75	29,347.25	100.50 pt	81.7%	+19.25	-245.25	IN
Fri 08-21	09:11:06	-258.9	LONG	29,428.25	—	—	—	+1.00	-10.50	MANUAL — OUT
Fri 08-21	13:13:06	-16.9	LONG	29,491.625	—	—	—	-3.12	-271.38	MANUAL — OUT

The automatic entries are fine and there is nothing to fix in them. All four confirmed inside the validated **13:38-14:09** band — three of them within nine minutes of the open, and Tuesday's 13:54 is comfortably inside. On the three shorts the rider boarded with **21.5%, 35.8% and 22.0%** of the eventual move consumed and left itself **128 to 232 points** of room. That is exactly what this detector is supposed to do.

THE ONE BAD ENTRY, AND IT IS BAD IN A SPECIFIC WAY

Thursday's long boarded with **81.7% of the up-move already spent** — and the session high did not arrive after the entry, it arrived **at 13:37:30, thirty-seven seconds before it**. The rider bought the high of the day. What remained was **19.25 points** of favourable excursion against **245.25 points** of adverse. That single row is the whole week's loss.

Both entries outside the window were the operator's own manual presses, and manual entry is deliberately live across the whole CME session — so “outside the window” is a description here, not a rule violation. It is still worth stating plainly: **the four entries the detector chose are 3-1; of the two a hand chose, one lost and one is still open and red.**

3 • ARE THE THRESHOLDS OK?

Live floors are **ER ≥ 0.15** and **RT ≥ 0.45**. These are the numbers the detector actually read at the confirming tick, with the margin it cleared by.

DAY	ER READ	MARGIN	RT READ	MARGIN	ATR	RESULT
Mon 08-17	0.290	+0.140	0.565	+0.115	27.6	+\$453.00
Tue 08-18	0.156	+0.006	0.520	+0.070	30.9	+\$185.00

DAY	ER READ	MARGIN	RT READ	MARGIN	ATR	RESULT
Wed 08-19	0.326	+0.176	0.480	+0.030	35.3	+\$541.00
Thu 08-20	0.640	+0.490	0.748	+0.298	34.2	-\$1,288.50

Both scrapers were winners, and the runaway best read was the loser.

Tuesday cleared the efficiency floor by **six thousandths** and paid **+\$185**; Wednesday cleared the round-trip floor by **0.030** and paid **+\$541**. Thursday cleared both by the widest margin of the week — ER 0.640, RT 0.748, the most confident read the detector produced — and lost **-\$1,288.50**. On four fires that is an anecdote, not a relationship. It is logged, not acted on.

THE FLOOR PROBE — RAISING A THRESHOLD DOES NOT DELETE A FIRE, IT DELAYS IT

Rather than guess, I re-ran the live detector minute-by-minute over this week's tape with the floors moved. This is the result that matters:

FLOORS	FIRES	08-17	08-18	08-19	08-20	08-21*
LIVE — ER .15 / RT .45	5	+9m 30,187.50	+23m 29,642.75	+9m 29,611.50	+9m 29,436.50	+9m 29,340.50
ER .20 / RT .45	5	+9m 30,187.50	+26m 29,613.75	+9m 29,611.50	+9m 29,436.50	+9m 29,340.50
ER .15 / RT .55	5	+9m 30,187.50	+26m 29,613.75	+12m 29,572.50	+9m 29,436.50	+9m 29,340.50
ER .20 / RT .55	5	+9m 30,187.50	+26m 29,613.75	+12m 29,572.50	+9m 29,436.50	+9m 29,340.50

Not one fire is removed by any of it. Raising both floors as far as ER 0.20 and RT 0.55 leaves Monday, Thursday and Friday untouched to the tick, and on the two days it does bite it simply makes the rider board later, further into a move that has already run:

- **08-18**, ER 0.15→0.20: confirmation slides +23m→+26m, entry 29,642.75→29,613.75. Held to the 20:40 clock that is **\$227 → \$111**, a cost of **-\$116**, and the MFE it can play for drops by the same \$116.
- **08-19**, RT 0.45→0.55: +9m→+12m, entry 29,611.50→29,572.50. **\$351 → \$195**, a cost of **-\$156**.

VERDICT ON THE THRESHOLDS: NOTHING TO CHANGE THIS WEEK

Margins this week: ER **+0.006 / +0.070 / +0.140 / +0.176 / +0.298 / +0.490** across the two metrics. One genuine scraper (ER +0.006) and it was a **winner**. Tightening buys nothing here and costs **-\$272** in delayed entries on a week where it removes no trade at all. That is a mechanical property, not a fluke of five sessions: a higher floor can only be met later, and later

means more of the move is already gone. **The margin distribution goes in the book and accumulates. No change recommended on four fires.**

* 08-21 is a replay, not a live fire — the automatic path never ran on Friday because a hand-opened position already occupied the book at 13:30. What the detector would have read is in section 5, and it is not flattering to the hand.

3b · ★ REV2 — THE PROTECTIVE STOP, RACED ON EVERY DAY AND NOT JUST THE LOSING ONE

WHY THIS SECTION EXISTS

Rev1 called a $1.5 \times \text{ATR}$ rider stop “the highest-value change on this week’s card” and priced it at about \$1,000 — **on Thursday alone**, the one day the rider lost money. That is exactly the winners-retention failure this report applies rigorously to `abs_veto_short` in Part 1.5 §2.3: a filter that improves the P&L by dropping the trades that WON is a fake win. Rev1’s own tables contained the refutation and nobody joined them up — Monday’s in-life worst excursion is -75.25pt (Part 1 §6) against a 41.4pt stop (\$3’s ATR of 27.6). So here is every rider entry of the week, at every width.

One stop for the whole position; the first tick at or through it closes every leg still open, and legs already closed keep their real exit. Raced on `capture.db` MNQ ticks from each entry, sized off the ATR the live detector itself read at the confirming tick (\$3), net of \$1.50 per round trip per lot. **Bold = better than what actually happened.**

RIDER POSITION	ATR	WORST EXCURSION	AS TRADED	0.5×	0.75×	1×	1.25×	1.5×	2×	3×
08-17 SHORT 2L	27.6	-75.25	\$456.00	−\$58.20	−\$85.80	−\$113.40	−\$141.00	−\$168.60	−\$223.80	\$456.00
08-18 SHORT 2L	30.9	-64.50	\$188.00	−\$64.80	−\$95.70	−\$126.60	−\$157.50	−\$188.40	−\$250.20	\$188.00
08-19 SHORT 2L	35.3	-28.75	\$544.00	−\$73.60	−\$108.90	\$544.00	\$544.00	\$544.00	\$544.00	\$544.00
08-20 LONG 4L	34.2	-245.25	−\$1,288.50	−\$142.80	−\$211.20	−\$279.60	−\$348.00	−\$416.40	−\$553.20	−\$826.80
08-21 LONG 4L (09:11 manual, killed by the clock bug)	30.7	-10.50	−\$150.50	−\$150.50	−\$150.50	−\$150.50	−\$150.50	−\$150.50	−\$150.50	−\$150.50
08-21 LONG 4L (13:13 carry — CLOSED 08-24)	30.7	-271.38	−\$2,149.44	−\$128.80	−\$190.20	−\$251.60	−\$313.00	−\$374.40	−\$497.20	−\$742.80
CLOSED BOOK — the five entries before the carry	—	—	−\$251.00	−\$489.90	−\$652.10	−\$126.10	−\$253.00	−\$379.90	−\$633.70	\$210.70
ALL SIX — carry now RESOLVED at — \$2,149.44	—	—	−\$2,400.44	−\$618.70	−\$842.30	−\$377.70	−\$566.00	−\$754.30	\$1,130.90	−\$532.10
Winners kept	—	—	3 / 3	0 / 3	0 / 3	1 / 3	1 / 3	1 / 3	1 / 3	3 / 3
Winner dollars kept	—	—	\$1,188.00	−\$196.60	−\$290.40	\$304.00	\$245.50	\$187.00	\$70.00	\$1,188.00

READ THE CLOSED-BOOK ROW, AND READ WINNERS-KEPT BESIDE IT

★ **REV3** — this grid has been re-raced and two Rev1 defects are gone. One: the **as-traded column charged the fee twice**. `trades.pnl_usd` is written already net of `fees_usd` and Rev1 printed `pnl_usd − fees_usd`, so every real trade was penalised \$1.50 a lot that the simulated stop cells were not — a systematic bias in favour of the stop, in the one column the verdict is read off. Two: **the carry is no longer open**. Rev1 marked it at the last print before the halt and correctly called that row PROVISIONAL. It closed by hand on 2026-08-24 at 11:59:39Z at 29,224.00 for −\$2,149.44, so this sweep now has no unresolved leg in it and every row is a verdict. Tape: the lake's 5-

second bars, adverse extreme first inside each bar — capture.db could not be used because it is a five-trading-day rolling window and by today it starts on 08-20, so Monday to Wednesday are simply gone from it.

1.5×ATR is the fourth-best of seven widths and it still LOSES money on the closed book. It takes the rider from **−\$251.00 to −\$379.90** — \$128.90 worse than doing nothing — because 41.4pt is inside Monday's −75.25pt excursion and 46.4pt inside Tuesday's −64.50pt, so it converts a +\$456 winner into −\$168.60 and a +\$188 winner into −\$188.40. **It keeps one of the three winners.**

3.0×ATR is the pick on the closed book, and it is the only width that is both. It is **+\$210.70** — the only positive cell in the row — and the only width that keeps all three winners, because 82.8 / 92.7 / 105.9pt all sit outside Monday, Tuesday and Wednesday's worst excursions. It still caps Thursday at −\$826.80 against −\$1,288.50. Against as-traded that is **+\$461.70 on the closed book.**

★ **But now put the carry in, and the ranking changes.** With all six entries resolved the week as traded is **−\$2,400.44** and every single width beats it — 0.5× −\$618.70, 1.0× **−\$377.70** (the best cell), 1.5× −\$754.30, 3.0× −\$532.10. That is the honest tension in one line: **a tighter stop cuts the tail harder and a wider one keeps the winners, and this week has one tail so large it outvotes three winners.** The +\$1,868.34 that 3.0×ATR is worth across all six entries is almost entirely the carry.

★ **AND THE THING NEITHER REVISION EXPECTED — ANY STOP CLOSES THE WEEKEND CARRY**

Every one of the seven widths stops Friday's 13:13Z entry, and all of them do it **within nineteen minutes**: 0.5× in the entry second, 1.0× at 13:16:50, 1.5× at 13:20:45, 2.0× at 13:25:00, **3.0× at 13:31:55 for −\$742.80. The gateway did not wedge until 19:57Z.** So the four lots that are the headline of this report — the ones that needed a Sunday-night gateway check, a weekend-flatten timer and, in the end, your hand on the Monday — would not exist under any stop this section tested. The carry is a missing-stop incident that a wedge then made unfixable, not a wedge incident.

One honest artefact: at $0.5 \times \text{ATR}$ the stop (29,476.28) sits inside the position's own fill spread — the fills were 2 @ 29,477.00 and 2 @ 29,506.25, a 29.25pt spread around a 29,491.625 weighted average — so it triggers in the entry second. That is a real property of stopping a badly-filled position off its own average, not a simulator bug, and it is one more reason not to pick the tightest cell.

★★ WHAT THIS DOES NOT LICENSE — AND WHY THE RECOMMENDATION MOVED

Six rider entries is not a sample, and this section is no longer where the argument is settled. Three winners decide the winners-kept column and one of them (Wednesday) survives every width from $1.0 \times$ up, so the ranking between $1.0 \times$ and $3.0 \times$ rests on Monday and Tuesday alone — and once the carry is included the ordering flips to $1.0 \times$ anyway. A week cannot carry this.

So SATURDAY #2 has been rewritten to race the same proposal on the 231 lake sessions the operator's own no-stop decision was taken on — same ladder, same trail, same fee, only the stop changes. The short version: **every stop width beats running naked at every entry delay** (naked — \$32,810 at delay 0 against —\$9,940 at $3.0 \times \text{ATR}$), and the worst single session falls from **—\$8,616 to —\$1,689**. The \$3,451 figure that `day_rider.py:96` cites against a stop turns out to describe a **400-point fixed stop on the 2-lot rider over 31 sessions** — a different machine — and on the 4-lot ladder that same 400pt stop scores —\$30,058 against naked's —\$32,810 while halving the worst session.

And the other side, which is not small. Nothing in that race is profitable: the 4-lot ladder loses money over 231 sessions in every arm including the live one. The harness cannot model the operator's actual objection at all — the Claim button as a manual kill — because it only models claiming a winner. And on the one session since this report was drafted, 08-25, the hand won: the tape went 209 points against a fresh 4-lot long and the operator claimed at —57pt for **—\$243.00**, where a $3.0 \times \text{ATR}$ stop would have cut at 14:03:57Z for **—\$452.16**. The instruction that survives all of this is still not “ $3.0 \times$ ”. It is **“decide it on 231 sessions, not on the losing day”**.

4 • WHICH EXIT CAPTURED IT?

MFE-available is measured on ticks **inside each leg's own life** — entry to its own exit, never a window we were flat for. The clock column is what a hold to 20:40 would have offered, for contrast.

DAY	EXIT	LOTS	HELD	TAKEN	MFE IN LIFE	CAPTURE	VS 20:40 HOLD	\$
08-17	MANUAL_CLAIM	2	4h26m	+125.25 pt	\$498.00	91.0%	75.2%	+\$453.00
08-18	TRAIL	2	1h33m	+128.00 pt	\$509.00	36.3%	36.3%	+\$185.00
08-19	MANUAL_CLAIM	2	11m	+152.25 pt	\$606.00	89.3%	58.7%	+\$541.00
08-20	MANUAL_CLAIM	1	5h56m	−146.00 pt	\$38.50	never green	—	−\$293.50
08-20	MANUAL_CLAIM	1	6h05m	−133.25 pt	\$38.50	never green	—	−\$268.00
08-20	MANUAL_CLAIM	2	6h40m	−181.00 pt	\$77.00	never green	—	−\$727.00
08-21	CLOCK_FLAT_BUG	3	1m	−10.75 pt	\$6.00	n/a	—	−\$69.00
08-21	CLOCK_FLAT_BUG	1	1m	−40.00 pt	\$2.00	n/a	—	−\$81.50

GROUPED BY EXIT REASON

EXIT REASON	LEGS	LOTS	TAKEN	MFE AVAILABLE	CAPTURE
MANUAL_CLAIM — on the winners	2	4	+\$994.00	\$1,104.00	90.0%
TRAIL (armed chandelier)	1	2	+\$185.00	\$509.00	36.3%
MANUAL_CLAIM — on the loser	3	4	−\$1,288.50	\$148.00	never green
CLOCK_FLAT_BUG	2	4	−\$150.50	\$2.00	a defect, not an exit
CLOCK_FLAT / operator claim on the open lot	—	4	unresolved	—	see section 5

THE OPERATOR'S HAND, SCORED AGAINST THE MACHINE

The desk's own `claim_audit.json` (regenerated 21:31Z today) prices each claim against what the configured mechanical exit would have taken on the same entry. It is a ceiling for the machine — exact levels, no slippage — so it flatters the machine side, and the hand still wins:

LEG	HAND	MACHINE	DELTA	MACHINE'S EXIT
08-17 SHORT 2	+\$453.00	+\$303.14	+\$149.86	TRAIL @ 30,134.2 (armed)
08-19 SHORT 2	+\$541.00	+\$510.57	+\$30.43	TRAIL @ 29,478.4 (armed)
08-20 LONG 1	−\$295.00	−\$293.50	−\$1.50	CLOCK_FLAT @ 29,302.5 (never armed)
08-20 LONG 1	−\$269.50	−\$293.50	+\$24.00	CLOCK_FLAT @ 29,302.5 (never armed)

LEG	HAND	MACHINE	DELTA	MACHINE'S EXIT
08-20 LONG 2		-\$730.00	-\$587.00	-\$143.00 CLOCK_FLAT @ 29,302.5 (never armed)
Week		-\$300.50	-\$360.29	+\$59.79 5 claims

What the exits did this week, plainly. The claim is a first-class exit and it earned that status again: **90.0% capture** on the two winners it was used on, and **+\$180.29** ahead of the machine on those same two. It is on the loser that it went the other way — three claims spread over 45 minutes, six hours into a position that was never more than 19 points green, each one worse than the 20:40 clock would have been, costing **-\$120.50** in aggregate. That is the honest shape of a discretionary exit: it beats the machine when there is something to take and it bleeds when there is not.

The single mechanical **TRAIL** took **36.3%**. That is a low number in isolation and I am not going to draw a conclusion from one leg — and per the standing instruction, the exit battery is settled and is not re-run here. What is worth flagging is that on the four losing lots the trail **never armed at all**, which is by design and is exactly what “no stop” means in practice: on a day that never goes green there is no mechanical exit before 20:40.

5 · THE WEAK DAY — Thursday 08-20, and it is a fault

The test for “bad strategy” against “bad tape” is whether the tape had money in it. It did.

DAY	RANGE	NET FROM OPEN	LONG CEILING	SHORT CEILING	RIDER TOOK
08-17	226.8	-188.0	\$135	\$772	+\$453.00
08-18	256.0	-128.2	\$226	\$798	+\$185.00
08-19	366.2	-149.8	\$277	\$1,188	+\$541.00
08-20	267.8	-44.8	\$984	\$1,158	-\$1,288.50
08-21	268.2	-44.8	\$506	\$1,640	-\$150.50 + the carry

Ceilings and “net” are measured from the same 13:30 reference the detector uses, to the session extreme, at the size the rider was actually carrying that day (2 lots Mon-Wed, 4 lots Thu-Fri). They are oracles — nobody catches an extreme — and they are here only to answer “was there anything to catch”.

WHY 08-20 IS A FAULT AND NOT A QUIET TAPE

Thursday had a **267.8-point range** and **\$1,158** sitting on the short side. This is nothing like 08-10, which took \$39 of a \$147 opportunity and was simply a dead session. The rider went **long**, at **13:38:07**, **thirty-seven seconds after the session high printed at 13:37:30**, with **81.7%** of the up-move already behind it. It then held **245 points** of adverse excursion with no stop, and the operator liquidated in three pieces between 19:33 and 20:17. **The opportunity was there and we took the wrong end of it. That is an entry fault.**

The mitigating and the aggravating fact, both stated. Mitigating: this was the **first session at 4 lots**, doubled that morning alongside the naked / per-lot-ladder redesign — the same day at 2 lots is roughly **-\$647** net, and the desk-level damage is a size decision as much as a signal one. Aggravating: it is the only day of the week the detector leaned **UP**, every other session drifted **down** from the open, and it leaned up with the strongest numbers it printed all week.

FRIDAY 08-21 — A SEPARATE FAILURE, AND IT IS STILL RUNNING

Friday does not compete for “weak day” because it is not a strategy result at all. Three things happened, in order:

1. **09:11:06Z** — a manual BUY of 4 lots at 29,428.25, hours outside the automatic window. The next tick hard-flatted it at **09:12** because the flatten branch keyed on `mod < OPEN_UTC_MIN`, true all morning; the tick then died on an `UnboundLocalError: px` after the orders had reached the venue. Booked `CLOCK_FLAT_BUG`, **-\$150.50**, one lot filled 30 points outside the book and is flagged `BADFILL`. A regression test now pins the management window (`tests/test_day_rider_hold_window.py`).
2. **13:13:07Z** — a second manual BUY of 4 lots at 29,491.625, seventeen minutes before the open. Because the book was then occupied, **the automatic detector never ran on Friday at all.**
3. **19:58:20Z** — the IB gateway wedged. Last good venue read **19:56:37Z** (“position 4.0 managed”); every tick since is `TimeoutError`, the watchdog has logged **73 consecutive blind cycles** as of 22:20Z and is still counting, and there are **48 CLOSE-WAIT sockets on :4002** — the documented wedge fingerprint. **The 20:40 hard flat never executed.**

UNRESOLVED 4 LOTS LONG, NOT CONFIRMED FLAT

As of **22:20Z** the rider's book claims **4 lots long from 29,491.625** and desk_reconcile reads venue ? = tournament +0 + rider +4 → unaccounted ?. The position marks **-\$887.00** at the 20:40 price it should have been flattened at and **-\$973.00** at the last print before the 21:00 halt. **Neither number is a result and I am not booking one.** The standing rule is NEVER HOLD OVERNIGHT; the book is currently in breach of it and the venue cannot be read to confirm either way. This is the first thing to check when the gateway comes back, ahead of anything else in this report.

★ **REV3 — RESOLVED, AND THIS BOX IS NOW HISTORY.** The gateway came back, and you closed all four lots by hand on **Monday 2026-08-24 at 11:59:39Z at 29,224.00 for -\$2,149.44** — against the **-\$973.00** mark this box refused to book, so the refusal was right and the carry cost more than twice the mark. \$3b above races every stop width on that resolved number rather than on a mark, which is why this sweep has no unresolved leg left in it. Read the paragraphs above as the state at 22:20Z on 08-21, not as an open item.

WHAT THE DETECTOR WOULD HAVE DONE ON FRIDAY

Worth knowing, because a hand took the opposite side. Replaying the live detector on Friday's tape it confirms at **+9m, DOWN, at 29,340.50**, on the strongest efficiency read of the week (**ER 0.854 / RT 0.553**). That short would have gone **+120.25 points (+\$962)** in its favour and **-148.00 (-\$1,184)** against, and held to the 20:40 clock it finishes **-\$326.00** — the trail never arming. So the automatic rider had the direction right and would still have lost money holding; it is not a clean “the machine would have saved us”. But it is **-\$326 instead of -\$973-and-counting**, and it would have been flat before the gateway wedged.

6 · ONE DEFECT FOUND WHILE DOING THIS

A MANUAL ENTRY OUTSIDE 13:30-21:00 SILENTLY DISABLES THE ATR TRAIL

do_manual_entry() writes arm_atr = drift_read(...).atr. Outside 13:30–21:00 drift.read() returns early with detail='before the 13:30 UTC open' and atr = 0.0. I confirmed that directly: both of Friday's manual entries stored arm_atr: 0.0, and trail_level() therefore falls back to the fixed ARM_PT = 150 pt instead of 4×ATR. The live ticks say it in as many words — “riding · trail not armed (need +150 [fixed], at −97)”. At Friday's 30.7-point ATR, 4×ATR is **122.8 pt**, so the fallback is **27 points harder to arm**. It cost **\$0 this week** — neither position ever went green — but on a hand-opened winner it silently removes the only mechanical exit the rider has. Three independent confirmations: the code path, the state file, and the rider's own tick note.

7 • THE VERDICT

QUESTION	ANSWER	ACTION
Did we enter OK?	Yes, on all four automatic entries — every one inside 13:38–14:09, three within nine minutes of the open, boarding 21.5–35.8% into the move on the three winners. The one bad entry bought the day's high 37 seconds after it printed.	NO CHANGE
Are the thresholds OK?	Yes. Margins +0.006 to +0.490; the one genuine scraper was a winner. Raising ER to 0.20 and RT to 0.55 removes zero fires and costs −\$272 in later, worse entries.	NO CHANGE
Which exit captured it?	The operator's claim, decisively, on the two winners it was used on — 90.0% capture and + \$180.29 over the machine. It cost −\$120.50 on the loser. The one TRAIL took 36.3%.	NO CHANGE
Name the weak day	Thursday 08-20, and it is a fault. A 267.8-point range with \$1,158 on the short side; the rider went long at 81.7% of the up-move, 37 seconds past the high, at double size on its first day at double size.	ENTRY
Anything else	4 lots long are unresolved into the weekend because the gateway wedged at 19:58Z and the 20:40 flat never ran. Plus: a pre-open manual entry stores arm_atr = 0.0 and loses the ATR trail.	OPEN

So: nothing to change in the detector or the thresholds this week. The automatic rider is roughly flat at −\$100.50 on 3–1 and its entries are behaving exactly as designed. Every dollar of real damage this week came from either a hand-timed entry or an infrastructure failure, and both of those are outside the strategy the operator asked me to tune. The margin distribution is logged and will accumulate; on four fires I am recommending nothing.

Sources, all recomputed for this section: live detector reads from the gazbot7-day-rider journal (6,953 parsed ticks, 17-21 Aug); trades from data/gazbot7.db; tape from data/capture.db ticks, whose 5-trading-day rolling window is exactly this week; floor probe by re-running gazbot7.drift.compute minute-by-minute over the same 5s bars the live service reads; hand-vs-machine from data/claim_audit.json; venue state from day_rider_state.json, day_rider_watchdog.log and desk_reconcile_state.json. MNQ \$2.00/point, \$1.50 per round trip.

P2 THE SHADOW DESK — and the promotion that did not survive the ruler

Garrath — the shadow book ran 38 families with eight or more fires this week, and if you rank them the way the board is usually ranked, the answer is easy and it is wrong. **Once the two filters this desk has already learned to apply are applied, the family sitting second on the raw board falls to eleventh, the promotion case for it collapses from +\$332 to +\$23.50, and its 50-second sibling goes negative.** The candidates that survive are different ones, they are newer, and the honest recommendation this week is smaller than last week's.

1 • The ruler, stated before any number

Score real_pnl, never ceiling_pnl. The ceiling is what the trade would have made with a perfect exit; real is what the tape would have paid. On this week's board they disagree by hundreds of dollars per family and the ceiling is always the friendlier of the two.

Void the Asia block. This is the one that reorders everything. The desk carries a permanent bench on 00:00–07:00 UTC entries, and that bench sits downstream of where a shadow arm records its fire — so an arm happily books money in a window where the live desk has filled **nothing since 5 August**. It is not a small correction:

FAMILY	RAW NET	BOOKED IN 00:00-07:00Z	FIRES THERE	WHAT IS LEFT
capit_loose	−\$2,457.50	−\$835.00	84 of 189	−\$1,622.50
bank20_grindA_ctl	−\$1,193.50	−\$520.50	53 of 128	−\$673.00
cx_grindA_live	−\$1,099.50	−\$520.50	53 of 123	−\$579.00
bank20_grindA	−\$1,212.50	−\$486.50	52 of 134	−\$726.00
cx_clip_brk_live	−\$1,166.50	−\$486.50	52 of 125	−\$680.00
cx_clip_brk_standdown	−\$1,166.50	−\$486.50	52 of 125	−\$680.00
ALL 38 FAMILIES	−\$13,780.50	−\$3,631.00		−\$10,149.50

Across all 38 families, **−\$3,631.00 of the week's −\$13,780.50 was booked in a window the desk cannot trade.** Every number from here down is tradable-hours only.

And dedupe before quoting any delta — see §5, where 7 of the 38 arms turn out to be the same rows under different names.

2 · The board, raw and then honest

SHADOW FAMILY	FIRES	RAW REAL_PNL	TRADABLE FIRES	TRADABLE REAL_PNL	WIN %	\$ / FIRE
odr_c10_s30	18	\$516.00	18	\$516.00	50%	\$28.67
odr_c10_s30_g	18	\$326.00	18	\$326.00	50%	\$18.11
lad_absS_B_15	27	\$255.50	20	\$244.00	45%	\$12.20
lad_absS_A_10	29	\$242.00	21	\$203.50	52%	\$9.69
sw_absS_B_k10	27	\$197.50	20	\$186.00	40%	\$9.30
sw_absS_A_k10	28	\$221.50	20	\$183.00	50%	\$9.15
rg_long_fast_v	36	\$329.50	31	\$118.50	29%	\$3.82
sw_absS_B_k20	27	\$100.50	20	\$103.50	50%	\$5.17
sw_absS_A_k20	28	\$9.50	20	\$61.50	60%	\$3.08
thrust_short_abs_veto55	26	\$114.50	19	\$56.50	42%	\$2.97
abs_veto_55s	46	\$332.00	34	\$23.50	35%	\$0.69
abs_veto_50s	47	\$299.00	35	−\$9.50	34%	−\$0.27
CL-capitulation_long	11	−\$22.00	7	−\$12.50	29%	−\$1.79
rung_grindA_med_05	8	−\$46.00	6	−\$55.50	33%	−\$9.25
... 14 families omitted, all between these two blocks ...						
cx_grindA_clip	125	−\$1,166.50	73	−\$680.00	23%	−\$9.32
bank20_grindA	134	−\$1,212.50	82	−\$726.00	27%	−\$8.85
capit_live_mirror	77	−\$1,316.00	42	−\$857.50	7%	−\$20.42
rider_w5	36	−\$1,136.50	36	−\$1,136.50	17%	−\$31.57

SHADOW FAMILY	FIRES	RAW REAL_PNL	TRADABLE FIRES	TRADABLE REAL_PNL	WIN %	\$ / FIRE
capit_loose	189	−\$2,457.50	105	−\$1,622.50	20%	−\$15.45

Read the two money columns side by side. abs_veto_55s is second on the raw board and eleventh on the honest one. abs_veto_50s crosses from \$299.00 to −\$9.50 — from a promotion candidate to a losing arm — purely by removing trades that could never have been filled. The arms that barely move are the ones worth looking at, because their money was made in hours the desk was actually open.

3 · The promotion battery — the seven families that clear the ruler, plus the one Rev1 left out

FAMILY (TRADABLE HOURS ONLY)	N	NET	WIN %	BEST FIRE STRIPPE D	BEST 3 STRIPPE D	WORST LEAVE-ONE-DAY-OUT	GREEN DAYS	ALL-TIME, TRADABLE HOURS
odr_c10_s30	18	\$516.00	50%	\$208.50	−\$255.50	\$13.50	3 / 5	37 · \$499.00
odr_c10_s30_g	18	\$326.00	50%	\$18.50	−\$340.00	−\$176.50	3 / 5	22 · \$552.00
lad_absS_B_15	20	\$244.00	45%	\$125.00	−\$26.50	\$92.50	4 / 5	20 · \$244.00
lad_absS_A_10	21	\$203.50	52%	\$118.50	−\$15.00	\$64.00	4 / 5	21 · \$203.50
sw_absS_B_k10	20	\$186.00	40%	−\$10.00	−\$144.00	\$85.50	4 / 5	52 · \$1,232.50
sw_absS_A_k10	20	\$183.00	50%	\$64.00	−\$60.00	\$65.00	4 / 5	53 · \$636.50
rg_long_fast_v ★ REV2	31	\$118.50	29%	−\$172.50	−\$459.00	−\$186.50	2 / 5	211 · −\$1,020.50
abs_veto_55s	34	\$23.50	35%	−\$133.50	−\$280.50	−\$64.50	2 / 5	263 · \$3,078.00

★ **REV2 — two fixes to this table.** The heading said “the four that survive the ruler” over seven rows; it now says the real count. And rg_long_fast_v was **silently absent** despite sitting 7th on \$2's own tradable board at +\$118.50 on 31 fires — more n and more money than sw_absS_A_k20 or thrust_short_absveto55, and more n than either ladder arm that WAS in the battery, with no exclusion reason given anywhere. It is in the table now, and it has a cause of death rather than an omission: **strip its best three fires and it is −\$459.00, its worst leave-one-day-out is −\$186.50, and 2 of 5 days are green** — against an all-time tradable-hours record of **−\$1,020.50 on 211 fires at 30.3% win**, whose own strip-best-3 is −\$2,084.00 and worst fold −\$1,528.00. This week is its best week against a deeply negative body of evidence, which is

what a lucky week looks like. It is also a long-side rgv arm rather than an abs-veto or open-rider one, so it is not a candidate for either live slot under discussion — but it is excluded on the battery, not on the category.

This is the part that decides Monday, so read the last four columns rather than the second one. odr_c10_s30 has the biggest number on the board and the worst robustness on it: stripping its best three fires takes −\$255.50, and two of its five days are red. That is one or two trades carrying a family, on n=18. The lad_absS_* and sw_absS_* arms — ladder and stop-width variants on the abs-veto short signal — are smaller and steadier: four of five days green, leave-one-day-out still positive, and the k=1.0 stop-width arms carry real all-time n behind them (\$1,232.50 on sw_absS_B_k10 across all its tradable-hour fires).

★ THE CASE FOR ABS_VETO_55S IS AN ALL-TIME CASE, NOT A THIS-WEEK CASE — SAY IT THAT WAY

This week, tradable hours: **\$23.50 on 34 fires**, 35% win, two of five days green, and stripping the single best fire turns it −\$133.50. On this week alone it is nothing.

All-time, tradable hours: **\$3,078.00 on 263 fires**. That is a real body of evidence and it is why the family is still the lead candidate at all.

So the honest sentence is: the 55-second confirm has a strong multi-week record and a flat week. That is a perfectly good reason to keep incubating it and a bad reason to promote it on Monday off this report's numbers. Anyone quoting ‘+\$332 this week’ is quoting the Asia block.

4 • Two-sided, or short only?

FAMILY	SIDE	FIRES	NET	\$ / FIRE	WIN %
abs_veto_55s	LONG	14	−\$109.50	−\$7.82	29%
abs_veto_55s	SHORT	20	\$133.00	\$6.65	40%
abs_veto_50s	LONG	14	−\$109.50	−\$7.82	29%
abs_veto_50s	SHORT	21	\$100.00	\$4.76	38%
abs_veto_60s	LONG	4	−\$157.00	−\$39.25	0%
abs_veto_60s	SHORT	3	−\$62.00	−\$20.67	33%
thrust_loose	LONG	20	−\$189.50	−\$9.47	25%
thrust_loose	SHORT	42	−\$89.50	−\$2.13	33%

The long side of the abs-veto family is small and positive, the short side is bigger and positive, and neither is decisive on this week's n. The standing principle still

holds — you tune a SIDE, you never relegate a DIRECTION — and nothing here argues for dropping either. It also does not argue for adding the long side to a live slot: 14 tradable-hour long fires is not a sample.

4b · ★ REV2 — SO WHAT IS THE SURVIVING CASE FOR THE ENTRY TREATMENT?

THE CIRCLE THIS CLOSES

Part 1.5 concludes that abs_veto_short’s treatment is the 55-second confirm and hands the case to Part 2. Part 2 §3 withdraws the promotion and hands it back to “the all-time record”. Neither states the record of **the thing that would actually replace the live gate** — the SHORT side alone. The live slot is short-only. Its long sibling is a different gate. Scoring a short-side replacement on a two-sided family mixes in 126 fires that would never be taken.

POPULATION (TRADABLE HOURS, REAL_PNL ONLY)	N	NET	\$ / FIRE	WIN %	STRIP BEST 1	STRIP BEST 3	WORST LEAVE- ONE-DAY- OUT	GREEN DAYS
abs_veto_55s — all time, both sides	263	\$3,078.00	\$11.70	46.4%	\$2,888.00	\$2,566.50	\$2,361.00	19 / 30
abs_veto_55s SHORT — all time	137	\$2,207.50	\$16.11	48.9%	\$2,017.50	\$1,696.00	\$1,898.50	21 / 25
abs_veto_55s LONG — all time	126	\$870.50	\$6.91	43.7%	\$752.50	\$534.50	\$423.50	13 / 29
abs_veto_55s SHORT — this week	20	\$133.00	\$6.65	40.0%	−\$24.00	−\$170.50	\$70.00	4 / 5
abs_veto_55s — this week, both sides	34	\$23.50	\$0.69	35.3%	−\$133.50	−\$280.50	−\$64.50	2 / 5
abs_veto_50s SHORT — all time	148	\$1,997.50	\$13.50	46.6%	\$1,807.50	\$1,486.00	\$1,651.50	21 / 25

The surviving case is the short side, and it is the strongest single result in this report. abs_veto_55s SHORT over all tradable-hour fires is **+\$2,207.50 on**

137 fires at \$16.11 a fire and 48.9% win, **21 of 25 days green** — and it stays positive through every test the desk owns: strip the best fire (+\$2,017.50), strip the best three (+\$1,696.00), and the worst leave-one-day-out fold (+\$1,898.50). Nothing else on the board does that.

And it changes what §3 concluded. Part 2 sets the revival condition as “a positive tradable-hours week with a positive leave-one-day-out”. On the short side this week already meets it: **+\$133.00 on 20 fires, 4 of 5 days green, worst leave-one-day-out +\$70.00**. What fails is the pooled family, and it fails because of the LONG side (−\$109.50 on 14 fires this week), which is not the side under discussion.

THE HONEST CAVEATS, BEFORE ANYONE PROMOTES ANYTHING ON MONDAY

One: this week’s short-side 20 fires do not carry the case on their own — strip the best one and they are −\$24.00, strip three and −\$170.50. The case is the 137-fire all-time record; the week merely fails to contradict it. **Two:** a shadow mirror is still an upper bound. It does not model the live gate’s default-veto on thin tape so it over-fires, and it enters on the 1-minute bar open where the live decider quotes the tick it fired on — both biases run in the mirror’s favour. ★ **REV3 finally puts a number on that, and the number goes the OTHER way — see the box below.** **Three:** the LONG side is not refuted either. All-time it is +\$870.50 on 126 fires, and it survives strip-best-3 (+\$534.50) and leave-one-day-out (+\$423.50) — on only 13 of 29 days green. “Tune a SIDE, never relegate a DIRECTION” still holds; what this table says is that the two sides have very different records and should stop being scored as one number.

★★ REV3 — DOES THE MIRROR FIRE ON THE SAME SIGNALS THE LIVE GATE FIRES ON? TIMESTAMP-MATCHED, AND THE ANSWER CHANGES THE CASE

The report said twice that the mirror over-fires and called it an upper bound, and never put a number on the bias. So: the same treatment Movement 2 gave the mechanical sim and the MGC dossier gave the bar source. Every mirror fire this week, matched against abs_veto_short's own live lots on timestamp and side.

First, the harness is aligned. thrust_loose — the UN-vetoed parent — reproduces **11 of the live gate's 11 signals, 100%**. So when the 55s arm reproduces fewer, that is the confirm working, not a matching failure. The mirror leads the live gate by a systematic **120-122 seconds** (bar open versus firing tick), which is why a ± 120 -second window finds nothing at all and a ± 240 -second one finds everything.

THIS WEEK, SHORT SIDE	FIRES	NET (REAL_PNL)	\$ / FIRE	WIN %
abs_veto_55s, all fires	27	+\$251.50	+\$9.31	44.4%
... in tradable hours (the Asia block voided)	20	+\$133.00	+\$6.65	40.0%
... that MATCH a live abs_veto_short lot	4	+\$239.00	+\$59.75	75.0%
... that the live gate never took	16	-\$106.00	-\$6.62	31.2%

The whole tradable-hour edge is in the four fires the live gate ALSO took. The sixteen extra fires — the over-firing the caveat warned about — lose money. That is the opposite of the bias this report feared. The mirror is not winning on trades the live veto would have refused; it is winning on the live gate's own signals, and losing on the extras.

And then the version that needs no simulator at all. The 55s confirm reproduces only 4 of the live gate's 11 signals — it refuses the other 7. Split the live gate's own booked fills that way:

ABS_VETO_SHO RT'S 22 LIVE LOTS, SPLIT BY WHETHER THE 55S CONFIRM PASSED THE SIGNAL				
LOTS	BOOKED	\$ / LOT	WIN %	
8	+\$225.00	+\$28.12	62.5%	

ABS_VETO_SHO RT'S 22 LIVE LOTS, SPLIT BY WHETHER THE 55S CONFIRM PASSED THE SIGNAL				
LOTS	BOOKED	\$ / LOT	WIN %	
55s CONFIRMED — 4 of 11 signals				
55s REFUSED — 7 of 11 signals	14	-\$395.00	-\$28.21	14.3%
The gate as it actually ran	22	-\$170.00	-\$7.73	31.8%

On the machine-only book — the book the bench decision is actually about — it is cleaner still. The 18 lots the machine closed booked **-\$437.50** with 3 winners. The 6 of those that the 55s confirm would have passed booked **+\$43.00** and kept **3 of 3 machine winners** — a clean pass of this report's own winners-retention rule, on real fills, with no reprice anywhere in it.

★ **THE HONEST LIMITS, and they are severe.** n = 4 signals / 8 lots. Strip the best lot and the confirmed subset is +\$131.00, still positive; the worst leave-one-day-out fold is +\$43.00, still positive; but **strip the best three lots and it is -\$15.50.** The two winners the confirm drops are both MANUAL_CLAIM lots on one signal — your hand rescued that one, so on the machine's book it drops no winner at all, but on the live book it keeps 5 of 7. And this is one week of one gate. **It is a direction, not a measurement — but it is a direction measured on real fills rather than on a mirror, and it points the same way the mirror does.** It strengthens MONDAY #1 (bench the automated slot, keep the signal in shadow) rather than overturning NOT-AN-ACTION #1 (do not promote to a live two-sided slot on Monday): what it argues for is running the confirm as a FILTER on this gate's existing signals, which is a much narrower claim than the mirror's 137-fire record.

So the recommendation is unchanged and its reason is now stated: keep incubating, and the specific thing to watch is the SHORT arm's tradable-hours week and its leave-one-day-out — both of which were positive this week for the first time. Two consecutive weeks of that, on the short side, re-opens the promotion question with a real case behind it.

5 • ★ 7 of the 38 shadow arms are the same rows under different names

ARMS THAT ARE THE SAME ROWS	FIRES EACH	WHAT THE PAIR WAS SUPPOSED TO MEASURE	THE DELTA IT REPORTS
capit_flip_live == capit_flip_t90	18	a 90-second flip window against the live one	exactly \$0.00, on every field, by construction
cx_clip_brk_live == cx_clip_brk_standdown == cx_grindA_clip	125	a clip-on-break exit, its stand-down control, and the grind-A clip arm — three supposedly different exits	exactly \$0.00, on every field, by construction
odr_c5_s30 == odr_c5_s30_g	19	the 'g' variant of the open-rider claim ladder against the plain one	exactly \$0.00, on every field, by construction

Every one of these pairs was created to measure a difference, and every one of them reports a perfect null — not because the configurations agree, but because **only one configuration is running and its trades are being written twice**. A broken arm and a genuinely neutral change are indistinguishable in the board's summary view, and the summary view is what a report reads. **Diff the rows before quoting any shadow A/B delta**; the check is four lines of SQL and it is now in scripts/fv7_p2_shadow.py. Note especially `cx_clip_brk_live == cx_clip_brk_standdown`: that pair exists specifically to test whether standing an exit down changes anything, and it has been answering “no” from a broken arm.

6 • The two grind-exit watches — and why they say nothing this week

GRIND EXIT WATCH	N	LIVE / BASELINE	SHADOW / PARTIAL	DELTA	NEWEST TRADE IN IT
Two-ratchet runner-clip watch	26	\$29.50	−\$265.00	−\$294.50	2026-08-05
2R-partial smoothness	40	−\$1,382.40	−\$1,074.90	\$307.50	2026-08-05

The newest trade in either watch is 2026-08-05. Grind has not fired a live lot since then, because its own 22-point ATR floor blocked 24,110 of 24,121 raw signals on this week's tape (Part 1.5 §4). So both watches are reporting the same fortnight-old window they reported last Friday, and neither has anything to add.

Two things are still worth saying about them. The two-ratchet has now watched 26 trades containing 3 genuine runners and has clipped **0** of them — but its own net is −\$265.00 against the live pure-6.0's \$29.50, so the “zero clips means

reconsider it” test cannot fire while the thing is losing anyway. And the 2R-partial's headline is now the opposite of its rationale: it improves the mean by \$307.50 while its daily standard deviation goes from 42.3 to **83.3**. It was adopted as a variance lever. On this window it is a mean lever that increases variance, which is a different product, and it should not be described as smoothing anything until it is re-measured on a week where grind actually trades.

THE OVER-TRADING CAVEAT, BAKED IN

The shadow board took 2,236 fires this week across 38 families. The live desk took 55 lots. A shadow arm has no capital, no queue position, no thin-tape default-veto and no operator, so it nominates far more trades than the paper desk could ever hold, and its per-fire numbers are systematically friendlier than the desk's. **Shadow nominates; paper judges.** Nothing on this page is a P&L forecast — it is a ranking of which signal to let the paper desk judge next.

Disposition — every lead this section touched

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
abs_veto_55s — promote to a live two-sided slot on Monday	REFUTED	Named test: the Asia void. \$308.50 of its \$332.00 was booked in a window with zero live fills since 08-05, leaving \$23.50 on 34 fires, 2 of 5 days green. Refuted as a THIS-WEEK case only.
abs_veto_55s — keep incubating on its all-time record	SHADOW	It is still the best-evidenced family on the book across all tradable-hour fires. Revive the promotion question when it has a positive tradable-hours week with a positive leave-one-day-out — not before.
lad_absS_B_15 / lad_absS_A_10 (ladder arms on abs-veto short)	SHADOW	The most robust things on this week's honest board: 4 of 5 days green and positive leave-one-day-out on both. But n=20 in their first week and no all-time record at all. Revive as promotion candidates at n≥60 tradable fires.
odr_c10_s30 (open-rider claim ladder)	SHADOW	Biggest tradable number on the board and the least robust: strip-best-3 turns it negative and 2 of 5 days are red on n=18. Keep it running; do not act on it.
abs_veto_50s and abs_veto_60s	PARKED	Both negative in tradable hours this week. Revive only if the 55s arm is promoted and a window-length sweep is wanted around it — there is no independent case for either.
The 7 byte-identical shadow arms	FAULT	BUILD item: a row-level duplicate check at shadow-write time, and re-seed the three

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
		broken arms. Until then every A/B delta on this board is unverified.
The two-ratchet grind exit	PARKED	Unchanged since 08-05 because grind has not fired. Revive the clip test on the first week grind trades at all — the ATR-floor question in Part 2.5 gates it.
The 2R-partial as a VARIANCE lever	REFUTED	Named measurement: daily standard deviation 42.3 → 83.3 on the same window it improves the mean. It is not smoothing. Re-open only as a mean lever, and only on fresh grind trades.

P2.5 MID-WEEK MUSINGS — every lead from the week, tested until it broke

Garrath — four leads came out of the week's chats and mid-week labs, and this section exists to test them hard rather than repeat them. **One of them dies in a way that is genuinely useful:** the “only run grind on a clean trend day” idea works on paper, makes +\$580.50 where the control loses \$958.00, and then turns out to be un-runnable, because the gate's own ATR floor and the clean-trend condition almost never happen on the same day. That is a better answer than either a yes or a no, and it changes what the Saturday question is.

1 · LEAD ONE — should grind only run on a confirmed trend?

The mechanism lab took every live grind fire in the record — 128 of them, losing \$958.00 between them — and asked which permission rule would have kept the winners while dropping the churn. Four families of rule, each scored on the same fires, with the winner-retention column carried all the way through because a rule that improves the total by throwing away winners is a fake:

MECHANIS M	FIRES KEPT	NET	\$ / FIRE	WIN %	WINNERS KEPT	BEST 3 STRIPPED	LEAVE- ONE-OUT
Every live grind fire (the control)	128	−\$958.00	−\$7.48	34%	43 / 43	−\$1,740.50	−\$1,661.50

MECHANIS M	FIRES KEPT	NET	\$ / FIRE	WIN %	WINNERS KEPT	BEST 3 STRIPPED	LEAVE- ONE-OUT
arm only on clean-trend days day-type arming	67	\$580.50	\$8.66	37%	25 / 43	−\$202.00	−\$123.00
arm on clean- trend OR in- between day-type arming	81	\$245.00	\$3.02	36%	29 / 43	−\$537.50	−\$458.50
arm when the day's roundtrip >= 0.55 day-type arming	67	\$580.50	\$8.66	37%	25 / 43	−\$202.00	−\$123.00
arm when the day's roundtrip >= 0.45 day-type arming	71	\$431.50	\$6.08	37%	26 / 43	−\$351.00	−\$272.00
require side*net30 >= 0.5 x ATR directional-net requirement	124	−\$889.50	−\$7.17	34%	42 / 43	−\$1,672.00	−\$1,593.50
require side*net30 >= 1.0 x ATR directional-net requirement	117	−\$675.00	−\$5.77	35%	41 / 43	−\$1,457.50	−\$1,513.50
require side*net30 >= 1.5 x ATR directional-net requirement	113	−\$1,037.00	−\$9.18	33%	37 / 43	−\$1,819.50	−\$1,513.50
require er30 >= 0.25 efficiency-ratio floor	94	−\$791.50	−\$8.42	35%	33 / 43	−\$1,574.00	−\$1,403.00
require er30 >= 0.35 efficiency-ratio floor	35	−\$417.50	−\$11.93	31%	11 / 43	−\$1,092.50	−\$1,043.00
require er30 >= 0.45 efficiency-ratio floor	11	−\$636.00	−\$57.82	18%	2 / 43	−\$804.50	−\$833.50
clean-trend day AND side*net30>=1 .0xATR the two stacked	61	\$715.50	\$11.73	39%	24 / 43	−\$67.00	−\$123.00
clean-trend day AND er30>=0.35 the two stacked	12	\$636.50	\$53.04	50%	6 / 43	−\$38.50	\$11.00

Read the last three columns, not the third. The best day-type rule (arm only on clean-trend days) turns −\$958.00 into **\$580.50** — and then strip its best three fires and it is −\$202.00, leave one day out and the worst fold is −\$123.00. It keeps 25 of the 43 winners, so it is not a fake win; it is a real effect with no robustness underneath it. The stacked version (clean-trend day AND

side*net30 $\geq$ 1.0xATR) is nominally the best cell in the whole table at \$715.50 on n=61, and it keeps only 24 of 43 winners. Neither is deployable.

Note also what the b- and c-family rows say, because they close a recurring proposal: **no directional-net requirement and no ER floor rescues grind.** Every single one of them stays negative. The ER-0.35 floor that is already live keeps 35 fires and still loses −\$417.50. This desk has now refuted “fix grind with an efficiency filter” on three separate occasions and it should stop being proposed.

★ AND THEN THE FINDING THAT ACTUALLY MATTERS — THE RULE CANNOT BE RUN

The day-type lab classified 240 sessions. **97 of them are clean-trend days.** The number that are clean-trend AND clear grind's 22-point ATR floor is **1.**

So “arm grind on clean-trend days” and “grind's ATR floor lets it fire” are very nearly mutually exclusive conditions. A clean trend is a day that goes somewhere smoothly; the 22-point floor wants a day that is violent. The rule that scores +\$580 in the table above would, in production, have armed the gate on about one day in two hundred and forty — and on the other 239 the arming decision would have been irrelevant because the floor would have refused every signal anyway.

This is why the mid-week lead is neither adopted nor refuted, but reframed. The question is not ‘when should grind be armed?’. Nothing about arming is binding. The question is whether 22 is the right floor, and that has never been tested — it is a number the gate was born with.

1b · ★ REV2 — “ARM ON CLEAN-TREND DAYS” vs “THE DESK EARNS IN BUILDING BLOCKS”: WHICH IS IT?

THE CONTRADICTION, STATED

This section parks arm grind only on clean-trend days as the best rule on the board. Movement 3 §8 attributes every live MNQ tournament trade to the 30-minute regime block it opened in and finds the desk **loses** in CLEAN-TREND blocks (−\$181.00 on 24 trades) and makes all its money in BUILDING blocks (**+\$1,836.50 on 121**). Read side by side those look like opposite instructions from the same desk.

They are not in conflict, because they are not the same classifier, the same unit, or the same population — and the first of those is the one that matters.

	PART 2.5’S CLASSIFIER	MOVEMENT 3 §8’S CLASSIFIER
Unit	a whole SESSION	a 30-MINUTE BLOCK
Rule	terciles of the session round-trip ratio (7.18 / 10.99)	ER≥0.45 + a breakout → CLEAN-TREND; high-ATR & ER<0.25 → VIOLENT-WHIPSAW; ER<0.22 & low-ATR → DEAD-CHOP; ER<0.30 → NORMAL-CHOP; otherwise BUILDING
Population scored	128 live grind fires, 07-19 → 08-04	699 live tournament trades, all gates
How often the label fires	97 of 240 sessions (40.4%) are clean-trend	17 of 991 blocks (1.7%) are CLEAN-TREND

A “clean-trend day” is not a day made of clean-trend blocks. Across the 22 days with a block census on file, the CLEAN-TREND share of a day’s blocks runs from 0.0% to **4.8%**, median 2.2% — it never exceeds one block in twenty. The two labels share a name and almost nothing else, which is the whole answer to “which is it?": neither claim is about the other’s object.

1c · So re-score the grind rule against BUILDING blocks, which is the real question

Each of the 128 live grind fires classified by its own 30-minute block. ★ REV3 — this table is rebuilt from scratch, because Rev2 got it wrong twice.

★ WHAT WAS WRONG, AND WHY IT IS WORTH A PARAGRAPH

The fee was charged twice. Rev2 printed this control as −\$1,150.00. The book says −\$958.00 — exactly $128 \times \$1.50$ apart, because `pnl_usd` is already net of fees and Rev2 subtracted `fees_usd` from it again. The tell was sitting in this section all along: **§1's own control table, eight paragraphs above, already said −\$958.00.** One section, one population, two different numbers.

And the buckets could not be reproduced. Rev2 put 51 fires in BUILDING and 99 outside VIOLENT-WHIPSAW, citing “Movement 3 §8's exact rule and its own ATR tercile boundaries”. **No reading of that sentence produces 51 / 99.** Four were tried — both block censuses on file with their own stored labels, and §8's rule applied by hand to each one's columns — and they disagree with each other as much as they disagree with Rev2. The harness that made the original was never saved to a script, so it

cannot be asked. Rather than pick whichever reading flatters the conclusion, **all four are printed below**, and the spread is the finding. This one is saved: scripts/rev3_grind_blocks.py.

The control does not depend on the blocks at all, so it is the same in every row: **128 fires, −\$958.00, −\$7.48 a fire, 33.6% win, 43 / 43 winners, strip-the-best-3 −\$1,740.50**. What moves is the arm.

HOW THE BLOCKS WERE LABELLED	BUILDING FIRES	BUILDING NET	\$ / FIRE	WINNERS KEPT	BEST 3 STRIPPED	EX- WHIPSAW FIRES	EX- WHIPSAW NET	\$ / FIRE
A — cs/ regime_block s.csv, its own stored labels (gf_cs_regime. py's rule)	29	\$425.00	\$14.66	12 / 43	−\$357.50	70	−\$191.50	−\$2.74
B — cs2/ regime_block s.csv, its own stored labels (cs2_regime. py's rule)	70	−\$696.00	−\$9.94	24 / 43	−\$1,341.50	113	−\$852.00	−\$7.54
C — cs2's atr_rel / er30 columns + Movement 3 §8's rule verbatim	19	\$350.50	\$18.45	9 / 43	−\$324.50	95	\$228.00	\$2.40
D — cs's atr / er columns + §8's shape on the cited 10.17 / 15.09 terciles	19	\$350.50	\$18.45	9 / 43	−\$324.50	46	\$257.50	\$5.60
what Rev2 printed (unreproducible, and fee-double- counted)	51	−\$748.50	−\$14.68	17 / 43	−\$1,419.00	99	−\$829.00	−\$8.37

The BUILDING arm's sign flips depending purely on which block census you read it against — from \$18.45 a fire down to −\$9.94 a fire, on populations ranging from 19 fires to 70. Three of the four readings are positive; one is negative; the number Rev2 printed matches none of them. **By this desk's own standing rule — a verdict that flips across specifications is not a verdict — there is no BUILDING result here to report.**

So the honest conclusion is not the one Rev2 reached, and it is worth being precise about the difference. Rev2 said BUILDING-block arming makes grind worse per fire. That was an artefact: it compared an unreproducible 51-fire bucket against a control that had been fee-double-counted, and both halves of the comparison were wrong. **What is actually true is weaker and more useful: block-level arming cannot be shown to rescue grind, because the answer depends on the labelling rather than on the tape.** And there is one thing every reading agrees on — **strip the best three fires and the BUILDING arm is negative in all four** (−\$1,341.50 to −\$324.50), on 9 to 24 retained winners out of 43. That is the named test, and it is the cause of death: **PARKED** — **revive it only when the census's flow-bucket phase bug is fixed (BUILD #14) so that a block label means one thing, and only on a re-run that clears strip-best-3.** Movement 3 §8's BUILDING result is a statement about the tournament as a whole — six gates, 121 trades — and on this evidence it does not transfer to this gate either way.

★ **AND ONE THING THAT FALLS OUT OF DOING THIS: THE DAY-TYPE RULE IS TWO SESSIONS**

★ **REV3 restates this from figures that can actually be re-derived.** The old version called §1's +\$580.50 a “gross” number and then “corrected” it to +\$548.00 — the same fee-charged-twice mistake as the table above. **+\$580.50 is already net.** It also quoted “2026-07-29 alone is +\$634.50 on 46 fires”, which is not what the book says for any day, and the harness that produced it was never saved.

Here is the concentration stated on the population anyone can check — all 128 live grind fires, straight from the trade record. **Grind has had exactly two green days in its entire life: 2026-07-29 (+\$310.00 on 11 fires) and 2026-07-30 (+\$307.00 on 37 fires).** Together that is **+\$617.00 on 48 fires.** The other eight days it has traded are **−\$1,575.00 on 80 fires.**

So the plain-English version of §1's robustness columns is this: any rule that looks good on grind is, on this sample, mostly a rule that says **“trade on 29 or 30 July”**. That is why §1's own best rule strips to −\$202.00 on its best three fires and −\$123.00 on its worst leave-one-out. It is not a fake win — it

keeps 25 of 43 winners — it is a real effect with two sessions underneath it, which is not enough to arm anything.

2 · So how reachable IS a 22-point ATR? Two independent measurements

MEASUREMEN T	WINDOW	MEDIAN 1-MIN ATR-14	P90	% OF MINUTES ≥ 22PT	DAYS WITH ANY SUCH MINUTE
The p25 lab (via the parquet lake)	239 sessions	14.66	31.34	25.3%	213 of 239
This report, recomputed off capture.db's own 5s bars	33 sessions, 2026-07-15 → 2026-08-21	11.69	22.64	11.0%	31 of 33

★ REV2 — RESOLVED, AND IT IS NOT A WINDOW, IT IS A DENOMINATOR. Rev1 printed both and picked neither, which left every proposed ladder rung sized off an unresolved 2.3×. Recomputed on ONE method (ATR-14 on 1-min bars, halt-aware, exactly deciders ._atr) with only the population varied:

POPULATION	MINUTES	MEDIAN	P90	≥ 22PT	22PT IS THE...
capture.db, 07-15→08-21, all 24 hours	37,411	11.68	22.62	11.0%	p89
the lake, last 239 days, all 24 hours	231,391	10.21	22.12	10.2%	p90
the lake, last 239 days, 13:00-21:00Z only	79,411	14.64	31.34	25.1%	p75
capture.db, THIS WEEK, all hours	6,780	9.66	16.88	5.4%	p95
capture.db, THIS WEEK, 13:00-20:00Z only	11,436	16.46	34.64	29.5%	p71

The p25 lab measured the CME session block; this report measured all 24 hours. The session row reproduces the p25 lab to within rounding on all three of its numbers (14.64 vs 14.66, 31.34 vs 31.34, 25.1% vs 25.3%), and the all-hours row reproduces this report’s capture.db recompute to the second decimal (11.68 vs 11.69, 22.62 vs 22.64, 11.0% vs 11.0%). Neither is wrong and they never disagreed about the tape. The lake and capture.db agree at ~10–11% once put on the same basis, so the “two independent measurements” framing was never the issue either. Which one sizes a ladder: the SESSION one, because a floor can

only bind in hours the gate can fire, and the desk refuses new entries 00:00–07:00Z anyway.

★ **REV2: “the floor sits at the ninetieth percentile” is true of the ALL-HOURS denominator and false of the session one, where it is the p75 —** and on this week’s session tape it was reachable in 29.5% of minutes. So it is not a p90 off-switch; it is a p75 filter that behaves like an off-switch, and the reason is not the height of the floor. It is that grind’s own trigger and its own floor are anti-correlated: 29.5% of this week’s session minutes cleared 22pt, and yet only **11 of 24,121** raw grind signals did. A gate that triggers on quiet absorption looks for exactly the minutes its floor excludes. Lowering the floor does not fix that — §2b shows dropping it to 14pt adds 46 fires and makes the book worse. A floor at p90 is not a filter, it is an off-switch that opens during violence, and it explains every grind number in this report: 24,110 signals blocked out of 24,121 this week, no live fire since 5 August, and two grind-exit shadow watches with nothing in them.

One more number worth having: **31 of 33 days contain at least one minute above the floor, but only 1 of 33 spend a quarter of the session there.** So “the floor was reachable today” is true almost every day and means nothing; the floor is reachable for a few minutes of spike and then gone.

2b · ★ REV2 — THE FLOOR, SWEEPED TONIGHT, ON THE 128 LIVE FIRES

WHY THIS DID NOT NEED TO WAIT FOR A SHADOW LADDER

Rev1’s answer to “is 22 the right floor?” was a shadow ladder that needs $n \geq 40$. **A gate that has fired zero live lots since 5 August cannot deliver $n \geq 40$.** But the 128 fires it made before the floor bound are on disk with their own entry ATR and their own real exits, so the floor is a filter on a real book and can be swept now, tick-honest by construction. Everything below is NET of the \$1.50 round trip — the −\$958.00 control in circulation is gross; the same 128 fires are −\$958.00 net. ★★ **REV3 CORRECTS THIS BOX AND EVERY CELL BELOW IT.** Rev2 wrote “the −\$958.00 control in circulation is gross” and subtracted \$1.50 a fire from every cell to make it “net”. It was already net — `trades.pnl_usd` is written net of `fees_usd`, verified on all 765 rows in the book — so the whole table was carrying the round trip twice. Every Net, \$/fire, Best-3-stripped and Worst-leave-one-out cell below has had

exactly $\$1.50 \times (\text{fires kept})$ added back. **The shape does not move at all — the floor still only turns positive somewhere above 22 points — and the crossing point is unchanged at 24pt. What does move is the depth: the live 22pt floor is a $-\$204.00$ loss on 51 fires, not the $-\$280.50$ the previous revision printed, and every other cell is $\$1.50$ a fire better. No verdict in this section changes.**

ATR FLOOR	FIRES KEPT	NET	\$ / FIRE	WIN %	WINNERS KEPT	BEST 3 STRIPPE D	WORST LEAVE-ONE-OUT	VS RANDOM SUBSET
no floor (the control)	128	$-\$958.00$	$-\$7.48$	33.6%	43 / 43	$-\$1,736.00$	$-\$1,592.50$	—
ATR $\geq 10\text{pt}$	118	$-\$931.00$	$-\$7.89$	33.9%	40 / 43	$-\$1,709.00$	$-\$1,565.50$	—
ATR $\geq 12\text{pt}$	107	$-\$950.50$	$-\$8.88$	32.7%	35 / 43	$-\$1,728.50$	$-\$1,635.50$	—
ATR $\geq 14\text{pt}$	97	$-\$998.00$	$-\$10.29$	32.0%	31 / 43	$-\$1,776.00$	$-\$1,581.50$	25th pct
ATR $\geq 16\text{pt}$	90	$-\$664.50$	$-\$7.38$	33.3%	30 / 43	$-\$1,442.50$	$-\$1,478.50$	51th pct
ATR $\geq 18\text{pt}$	82	$-\$431.50$	$-\$5.26$	35.4%	29 / 43	$-\$1,209.50$	$-\$1,321.00$	65th pct
ATR $\geq 20\text{pt}$	70	$-\$409.50$	$-\$5.85$	34.3%	24 / 43	$-\$1,187.50$	$-\$1,348.00$	61th pct
ATR $\geq 22\text{pt} \leftarrow$ live	51	$-\$204.00$	$-\$4.00$	35.3%	18 / 43	$-\$982.00$	$-\$1,098.00$	65th pct
ATR $\geq 24\text{pt}$	45	$+\\$88.50$	$+\\$1.97$	40.0%	18 / 43	$-\\$689.50$	$-\\$915.00$	83th pct
ATR $\geq 26\text{pt}$	33	$+\\$397.00$	$+\\$12.03$	42.4%	14 / 43	$-\\$381.00$	$-\\$593.50$	—
ATR $\geq 28\text{pt}$	28	$+\\$511.00$	$+\\$18.25$	46.4%	13 / 43	$-\\$267.00$	$-\\$591.50$	—

The ladder is monotone upward from 14pt, which is the shape of a real effect — $-\$998.00 \rightarrow -\$664.50 \rightarrow -\$431.50 \rightarrow -\$409.50 \rightarrow -\$204.00 \rightarrow +\$88.50 \rightarrow +\$397.00 \rightarrow +\511.00 . Three things follow, and none of them is the one the report expected.

One: 22 is not obviously wrong, and it is not enough. The live floor still leaves the gate at **$-\$204.00$ on 51 fires**. It does not cross zero until 24pt and does not clear it properly until 26–28. So the honest reading of the direction is “higher would have been better”, not “22 is a mistake”.

Two: no rung survives the battery. Strip the best three fires and every rung is negative — $-\$309.00$ even at 28pt. The worst leave-one-day-out fold is negative at every rung, $-\$633.50$ even at 28pt. And against 5,000 random same-size subsets of the same 128 fires, the 22pt rung sits at the 65th percentile and the 24pt rung at the 83rd — the floor is selecting a little, and not much. **Cause of death: strip-best-3 and leave-one-day-out, at every rung tested.**

Three, and this is the part that changes the Saturday question: the floor is not what is wrong with grind. Rev1 framed it as the un-asked question on which the whole grind programme turns. It has now been asked, and the answer is that raising the floor cannot make this gate positive on its own live record. That does not make the shadow ladder pointless — it can still test whether the same monotone shape holds forward — but it demotes it from “the real grind question” to a confirmation, and it sends grind back to being an entry question, which is where §1’s b- and c-family rows already pointed.

3 • What the week's days actually looked like

DAY TYPE	DAYS	SHARE OF 240
clean-trend	97	40.4%
in-between building	57	23.8%
normal-chop	52	21.7%
dead-chop	27	11.2%
violent-whipsaw	7	2.9%

Two-fifths of sessions classify as clean trend, which is a lot more than the desk's folklore assumes, and only 7 of 240 are the violent-whipsaw type the desk spends most of its worry on. The terciles behind the classification are a session round-trip ratio of 7.18 and 10.99.

4 • THE GOLD PROGRAMME — four arms, and gold is \$10 a point

Before any number: the multiplier check. The gold tables below show a median session of about −\$133.14. At \$10 a point that is roughly 13 points, which is an ordinary MGC day-trade excursion. At the MNQ multiplier it would be 67 points, which gold does not do intraday. **The lab is priced correctly.**

4a — The coil, and its mirror control

GOLD COIL ARM	N	SESSIONS	NET	\$ / TRADE	WIN %	BEST 3 STRIPPED	BEST DAY STRIPPED
REVERSION (fade the coil edge → mid)	2487	251	−\$1,828.93	−\$0.74	43.7%	−\$2,398.93	−\$2,244.93
CONTINUATION (break out) [mirror control]	2487	251	−\$4,767.86	−\$1.92	39.3%	−\$5,470.86	−\$5,519.21

Both directions lose. Fading the coil edge loses \$1,828.93 over 2,487 trades and breaking out of it loses \$4,767.86 — and the placebo is the sentence that closes it: **0 of the random sets did worse**, against a placebo median of −\$3,354.96. A coil edge on gold carries no directional information in either direction at this fidelity.

4b — Except in the tightest coils, which is a PARKED lead, not a finding

COIL TIGHTNESS	TARGET	N	NET	\$ / TRADE	WIN %
compression ≤ 0.45	1.0×	666	\$307.29	\$0.46	47.1%
compression ≤ 0.45	1.5×	666	\$616.32	\$0.93	37.8%
compression ≤ 0.45	2.0×	666	\$280.36	\$0.42	31.1%
compression ≤ 0.6	1.0×	2487	−\$1,828.93	−\$0.74	43.7%
compression ≤ 0.6	1.5×	2487	−\$231.36	−\$0.09	35.1%
compression ≤ 0.6	2.0×	2487	\$962.00	\$0.39	29.8%
compression ≤ 0.75	1.0×	4726	−\$4,430.00	−\$0.94	41.9%
compression ≤ 0.75	1.5×	4726	−\$4,084.29	−\$0.86	32.7%
compression ≤ 0.75	2.0×	4726	−\$4,978.36	−\$1.05	26.8%

Restrict to the tightest compression and the sign flips: \$962.00 at 2.0× on n=2,487. That is a real number on real n and it is **still not a result**, because it is the best cell of nine on the same data that produced the null above, with no placebo run against it and no out-of-sample. It is PARKED with a named revival condition: run the same placebo the full-sample arm got, and hold out a year.

4c — The gold exit and stop lab

GOLD STOP WIDTH	STOPPED OUT	NET	\$ / SESSION	WIN %	BEST 3 STRIPPED	LEAVE-ONE-OUT
2.0x	67	−\$6,304.86	−\$78.81	15.0%	−\$9,251.86	−\$6,879.57
2.5x	62	−\$6,112.36	−\$76.40	20.0%	−\$9,413.36	−\$6,252.50
3.0x	56	−\$4,669.00	−\$58.36	26.2%	−\$8,366.00	−\$5,233.57
3.5x	52	−\$3,437.00	−\$42.96	31.2%	−\$7,134.00	−\$4,006.50
4.0x	45	−\$376.29	−\$4.70	38.8%	−\$4,285.29	−\$2,465.00
4.5x	41	\$1,884.57	\$23.56	43.8%	−\$2,024.43	\$66.64
5.0x	37	\$3,352.71	\$41.91	46.2%	−\$556.29	\$246.43
5.5x	37	\$1,985.79	\$24.82	46.2%	−\$1,923.21	−\$905.43
6.0x	35	\$1,240.29	\$15.50	47.5%	−\$2,668.71	−\$1,435.86
7.0x	30	\$3,264.00	\$40.80	52.5%	−\$645.00	−\$2,084.00
8.0x	24	\$2,776.29	\$34.70	53.8%	−\$1,132.71	−\$2,363.29
10.0x	17	\$2,055.14	\$25.69	55.0%	−\$1,853.86	−\$2,667.57
naked	0	−\$1,570.00	−\$19.63	56.2%	−\$5,479.00	−\$6,825.00

GOLD EXIT	NET	\$ / SESSION	WIN %	WORST SESSION	BEST 3 STRIPPED	WORST LEAVE-ONE-OUT	HOW THEY ENDED
5xATR stop, hold to clock	\$3,352.71	\$41.91	46.2%	−\$1,045.14	−\$556.29	\$246.43	CLOCK×43, STOP×37
2.5xATR trail, 3h cap	−\$156.43	−\$1.96	28.7%	−\$284.43	−\$3,185.14	−\$1,940.00	TRAIL×79, CAP×1
hold naked to the 20:40 clock	−\$1,570.00	−\$19.63	56.2%	−\$3,847.00	−\$5,479.00	−\$6,825.00	CLOCK×80
2xATR stop + 4xATR trail	−\$3,440.29	−\$43.00	22.5%	−\$419.86	−\$5,980.43	−\$3,835.00	STOP×46, TRAIL×32, CLOCK×2
1xATR stop, hold to clock	−\$4,263.14	−\$53.29	5.0%	−\$242.57	−\$5,962.14	−\$4,233.14	STOP×76, CLOCK×4
1.5xATR stop, hold to clock	−\$4,500.43	−\$56.26	11.2%	−\$362.36	−\$7,447.43	−\$5,348.36	STOP×71, CLOCK×9
3xATR stop, hold to clock	−\$4,669.00	−\$58.36	26.2%	−\$688.71	−\$8,366.00	−\$5,233.57	STOP×56, CLOCK×24
2xATR stop, hold to clock	−\$6,304.86	−\$78.81	15.0%	−\$460.14	−\$9,251.86	−\$6,879.57	STOP×67, CLOCK×13

This is the most interesting gold result and also the one most likely to be a mirage. There is a genuine plateau — the stop has to be enormous before gold pays, and the progression from 2.0× to 5.0× is monotone rather than jumpy, which is what a real effect looks like rather than a grid artefact. But look at the two robustness columns on the best cell: strip its best three sessions and \$3,352.71 becomes −\$556.29, and the worst leave-one-out fold is \$246.43.

Nearly the whole result is three sessions. And a 5×ATR stop on gold is not really a stop; it is a disaster brake on a naked hold, which the top row of the exit table already shows losing money.

4d — The gold rider, in and out of sample — and a control that undoes it

GOLD RIDER ARM	FIRES	SESSIONS	FIRE RATE	NET	\$ / FIRE	MEDIAN MFE / MAE (\$)
loose trigger — IN-SAMPLE (the 16-ish)	12	27	44.4%	−\$328.00	−\$27.33	253 / -302
loose trigger — OUT-OF-SAMPLE	106	226	46.9%	−\$3,306.00	−\$31.19	307 / -264
loose trigger — ALL 253	118	253	46.6%	−\$3,634.00	−\$30.80	299 / -268
derived trigger — IN-SAMPLE	11	27	40.7%	−\$147.00	−\$13.36	293 / -298
derived trigger — OUT-OF-SAMPLE	79	226	35.0%	\$655.00	\$8.29	319 / -260
derived trigger — ALL 253	90	253	35.6%	\$508.00	\$5.64	314 / -261

The loose trigger is negative in and out of sample, which is clean. The derived trigger is the interesting one: **−\$147.00 in-sample and \$655.00 out-of-sample on 79 fires over 226 sessions**. An arm that does better out of sample than in is usually a good sign.

★ BUT THE DIRECTIONAL CONTROL KILLS IT, AND IT HAS TO BE SAID ON THE SAME PAGE

The same out-of-sample window carries a plain directional control: hold gold LONG every session and you lose **−\$21,738.00**; hold it SHORT every session and you make **\$20,382.00**, over 226 sessions.

The gold rider's out-of-sample \$655.00 is a rounding error against a \$20,382 short drift sitting underneath it. Any gold arm that is even slightly short-biased will look profitable on this window without carrying one gram of edge. **Every gold result in this report needs its long/short mix reported next to it, and none of them currently do.** That is a method fix, it is on the BUILD card, and until it lands no gold arm should be promoted to anything.

5 • The MNQ open-rider leads, and the stay-out meter

OPEN-RIDER POPULATION	SESSIONS	HELD TO THE CLOCK %	NET AT 2 LOTS	\$ / SESSION	MEDIAN SESSION	BEST 3 STRIPPED
every session it fired	225	47.1%	−\$20,739.00	−\$92.17	−\$78.00	−\$29,666.00
inside the window	60	53.3%	\$825.00	\$13.75	\$124.50	−\$7,745.00
outside it	165	44.8%	−\$21,564.00	−\$130.69	−\$112.00	−\$27,477.00

The open rider, replayed across 229 sessions at two lots, loses \$20,739.00. Split it by the window lead and the “inside” population makes \$825.00 on 60 sessions while “outside” loses \$21,564.00 on 165. That looks like the window is the whole story until the strip-best-3 column: the inside population goes to **−\$7,745.00**. **Three sessions are the entire lead.** REFUTED as a session filter on this evidence.

The stay-out meter's direction call, on the same population: when its two reads agree it is right 51.7% of the time over 178 sessions, which is a coin. Split by call, its UP calls hit 56.7% and its DOWN calls 45.7% — and that gap is the same short drift the gold control just exposed, not skill. The 51 mixed-signal sessions average **−\$20.79**. The corrected series (231 sessions) says the same thing to within a

point. The meter is a stay-out instrument and it should not be read as a direction instrument, which is exactly what it says on its own label and exactly what keeps getting forgotten.

Disposition — every lead this section touched

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
Arm grind only on clean-trend days	PARKED	It is not refuted — it makes \$580.50 against a −\$958.00 control (★ REV3: both figures are NET, not gross — pnl_usd is written net of fees, and Rev2's "gross" label on them was wrong) and keeps 25 of 43 winners. It is un-runnable: only 1 of 240 sessions is both clean-trend and above the 22-point floor. ★ REV3 replaces Rev2's two additions — both were computed wrong — and neither replacement revives it. (i) The concentration is real but it is TWO sessions, not one: grind has had exactly two green days ever, 2026-07-29 (+\$310.00 on 11 fires) and 2026-07-30 (+\$307.00 on 37 fires), +\$617.00 on 48 fires against −\$1,575.00 on the other 80. So "the best rule on the board" and "trade on 29-30 July" are nearly the same instruction on this sample. (Rev2's "+\$634.50 on 46 fires" matches no day in the book.) (ii) The BUILDING-block version is NO VERDICT, not a worse one: its sign flips across all four defensible readings of the block census (+\$18.45 to −\$9.94 a fire, on 19 to 70 fires), and Rev2's −\$748.50 / 51 fires is not reproducible from either census (\$1c). What every reading does agree on is that strip-best-3 is negative in all four, which is the named test. The floor question below is answered too, and it does not revive this either. ★ REV2 — ASKED AND ANSWERED, tonight, on the 128 live fires (\$2b). The ladder is monotone upward — 14pt −\$1,143.50 → 22pt −\$280.50 → 28pt +\$469.00 — but strip-best-3 is negative at every rung and the worst leave-one-day-out is negative at every rung, and against 5,000 random same-size subsets the live 22pt rung sits at only the 65th percentile. Cause of death: strip-best-3 and leave-one-day-out. Raising the floor cannot make grind positive on its own live record, so this is no longer "the real grind question" — grind is an entry question. Keep the shadow arms as a forward confirmation of the monotone shape, nothing more. Reachability is also resolved: 22pt is the p75 of SESSION minutes and the p89 of ALL minutes; the two published figures were one denominator apart, not two measurements. Named test: nine cells across two families on 128 live fires, every one negative,
grind's 22-point ATR floor as a number	SHADOW	
	REFUTED	

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
Fix grind with an ER floor or a directional-net requirement		including the ER-0.35 floor that is already deployed.
MGC coil — fade the edge (or break it)	REFUTED	Named test: a mirror control that also loses, and a placebo that 0 random sets failed to beat from below on 2,487 trades.
MGC coil, tightest-compression cell only	PARKED	\$962.00 on n=2,487. Revive by running the SAME placebo the full-sample arm got and holding out a year of tape. Until then it is the best of nine cells on data that produced a null.
MGC 5×ATR stop / hold-to-clock	PARKED	Monotone plateau, which is the shape of a real effect, but strip-best-3 takes \$3,352.71 to −\$556.29. Revive at n≥200 sessions or on a second instrument.
The MGC derived-trigger rider	PARKED	Out-of-sample \$655.00, which is real, against a \$20,382 always-short drift on the same window. Revive only with the long/short mix reported beside it.
Report the long/short mix on every gold arm	LIVE	A method fix, not a strategy. Without it every gold number on this desk is confounded with a large short drift, and this section found two arms already reading it as edge.
The open-rider “inside the window” session filter	REFUTED	Named test: strip-best-3 takes \$825.00 to −\$7,745.00 on 60 sessions. Three sessions are the whole lead.
The stay-out meter as a DIRECTION signal	REFUTED	51.7% hit rate over 178 sessions when both reads agree. Its up/down asymmetry is the market's short drift, not the meter's skill. It stays in service as a stay-out instrument, which is what it was built to be.

P2.6 ROUTER OPERATION REVIEW — the desk's #1 lever, on the stand

Garrath — the router ran 1,425 ticks this week and wrote a switch on 20 of them. That is the headline and it is not a criticism: the tournament has been stood down all week, so most of what the router had to decide was whether to keep doing nothing, and it kept doing nothing correctly. **What this section is actually about is the three things that can write the switch file or claim to know the desk's state, and the fact that on the two occasions this week where those disagreed, the router was right and the instruments around it were wrong.** It also has to say plainly

that the two files built to answer “was the router optimal this week” both stopped being written, so everything here is rebuilt from raw logs.

1 • The duty cycle

SESSION	TICKS	TICKS THAT CHANGED A SWITCH	CHANGE RATE
Mon 17 Aug	279	5	1.8%
Tue 18 Aug	283	1	0.4%
Wed 19 Aug	291	11	3.8%
Thu 20 Aug	288	2	0.7%
Fri 21 Aug	284	1	0.4%
THE WEEK	1425	20	1.4%

1.4% of ticks change anything. The audit's number for the fortnight was 97.2% no-change; this week is 98.6% no-change, so if anything the router has got quieter, which under a standdown is the correct direction. There were no systemd failures and no aborted decisions this week — both worth stating, because last cycle had a 10-hour block of aborts that no scoreboard noticed.

2 • Every switch it wrote, graded

WHEN (UTC)	WHAT IT WROTE	GRADE	WHY
08-17 08:10Z	{'abs_veto_short': 'off'}	▢ GOOD	Benched into the pre-open. The gate took two stops the previous hour and the tape was going nowhere.
08-17 08:25Z	{'abs_veto_short': 'on'}	△ CHURN	Re-armed 15 minutes later. Nothing changed in between; this is the toggle pattern the ledger keeps flagging.
08-17 10:30Z	{'abs_veto_long': 'on'}	▢ GOOD	Armed the long fader into a genuine bid.
08-17 12:40Z	{'abs_veto_long': 'off'}	▢ GOOD	The fail-safe bench that avoided the 12:40 reversal. A fail-safe bench is exempt from the confirmation rule, and this is the example of why.
08-17 15:00Z	{'capitulation_long': 'off', 'exhaustion_short': 'off'}	▢ GOOD	Benched both faders into the afternoon drift. Neither fired again that day.
08-18 01:55Z	{'capitulation_long': 'off'}	— \$0	An Asia-hours bench. Worth exactly nothing in either direction: entries are refused in that block anyway.
08-19 07:00Z	{'abs_veto_short': 'off'}	▢ GOOD	Benched the short book at the European open before the 07:50 squeeze.
08-19 07:50Z	{'abs_veto_short': 'on'}	▢ GOOD	

WHEN (UTC)	WHAT IT WROTE	GRADE	WHY
			Re-armed for the down-leg it then caught (+\$179 in claims at 13:50).
08-19 12:40Z	{'abs_veto_long': 'on', 'abs_veto_short': 'off', 'exhaustion_short': 'off'}	❌ BAD	Benched abs_veto_short and exhaustion_short ten minutes before the 13:39 down-leg that made the rider \$541, and armed the LONG fader into it. Wrong side, wrong ten minutes.
08-19 12:51Z	{'grind_long': 'on'}	❌ BAD	Armed grind_long into the same down-leg. It could not fire — the ATR floor saw to that — so the cost is \$0 and the process is still wrong.
08-19 13:40Z	{'grind_long': 'off', 'abs_veto_short': 'on'}	✅ GOOD	Corrected itself within the hour: grind off, short book back on, one minute after the rider's entry.
08-19 13:41Z	{'grind_long': 'off', 'abs_veto_long': 'off', 'abs_veto_short': 'on'}	⚠️ CHURN	A second write 30 seconds after the first, same content plus abs_veto_long. Two writes, one decision.
08-19 13:50Z	{'capitulation_long': 'off'}	— \$0	Housekeeping on a gate that had not fired.
08-19 15:05Z	{'abs_veto_short': 'off'}	✅ GOOD	Benched the short book after the 14:51 stop.
08-19 15:35Z	{'abs_veto_short': 'on'}	❌ BAD	Re-armed 30 minutes later with no positive case. It then took three more stops for −\$133.
08-19 18:16Z	{'abs_veto_short': 'off'}	✅ GOOD	Applied the operator's 18:15Z bench instruction within 90 seconds.
08-19 22:00Z	{'abs_veto_short': 'off'}	— \$0	Re-asserted the bench at the reopen. No-op.
08-20 06:40Z	{'exhaustion_short': 'off'}	— \$0	Asia-hours bench.
08-20 08:05Z	{'capitulation_long': 'off'}	— \$0	Housekeeping under the standdown.
08-21 14:05Z	{'abs_veto_short': 'off'}	✅ GOOD	Re-benched abs_veto_short 2m15s after the OPEN-HOUR WATCHER auto-armed it through a standing operator standdown. See §4 — the router was the only thing that caught it.

10 good, 3 bad, 2 churn, 5 worth exactly \$0. The \$0 rows are Asia-hours writes: new entries are refused in that block regardless, so a bench there is neither a save nor a cost, and it should not be counted as either. The two genuinely bad calls are both on Wednesday and they are the same mistake made twice — benching the short book at 12:40 ten minutes before the day's best down-leg, then re-arming at 15:35 with no positive case, straight into three more stops. **Re-arming without a positive live case is now this gate's most expensive recurring pattern**, and it has cost money on three separate occasions that the ledger records.

3 · Is the router earning its keep? The mechanical counterfactual

The honest way to ask is not “did the benched gates lose money”, because a benched gate has no P&L. Movement 2 fires all six mechanically in their exact live config across the whole of this week's captured tape — every fire counted, not just the ones near a big move — which is the counterfactual this question needs:

GATE	LIVE THIS WEEK	IF IT HAD BEEN ARMED ALL WEEK, MECHANICALLY	DIFFERENCE
grind_long	benched, \$0	−\$319.17 on 11 fires	\$319.17
capitulation_long	benched, \$0	−\$97.66 on 10 fires	\$97.66
abs_veto_long	benched, \$0	−\$83.42 on 26 fires	\$83.42
rgv_short	benched, \$0	−\$17.54 on 1 fires	\$17.54
exhaustion_short	\$165.50	−\$638.02 on 18 fires	\$782.52
abs_veto_short	−\$170.00	−\$749.55 on 51 fires	\$546.55
ALL SIX	−\$4.50	−\$1,905.36 on 117 fires	\$1,846.86

The tournament's live book was **−\$4.50** this week (★ REV3 — Rev1 printed −\$58.50, which is pnl_usd with the \$1.50 round trip subtracted a second time; the column is already net, so the live book is a rounding error rather than a small loss and the mechanical counterfactual looks worse against it, not better). Firing the same six gates mechanically through the same tape is **−\$1,905.36 on 117 fires**. On that comparison the standdown plus the router's benching is worth about **\$1,846.86** this week, and \$517.79 of it is the four gates that never fired at all.

★ REV2 — THE **-\$638.02 IS NOT EXHAUSTION_SHORT'S COUNTERFACTUAL, AND THE \$782.52 IS NOT YOUR HAND**

That row invited a reading the data does not support: that the operator turned a **-\$638.02** gate into a **+\$165.50** one. Two things are wrong with it, and the second is a method wound rather than an arithmetic slip.

One: repriced on the SAME 14 lots, the machine alone books +\$84.00. Taking each lot's protective stop straight off the venue (so R is exact, and 7 of 7 machine-closed lots reproduce their own live exit reason) and racing it to its own configured stop or target, the machine gets **+\$84.00** and your hand adds **+\$60.50**. Not \$782.52. This gate is not hand-carried, which is the opposite of what the same test says about `abs_veto_short`.

Two: only ONE of the 18 mechanical fires is a live signal. Matching the sim's fires against the gate's seven live entries: one lines up within 3 seconds (08-18 13:37, and even that one enters 8.75 points away), two more line up within 3 minutes at 17-33 points of difference, and **three live signals have no sim fire within TEN MINUTES** — 14:02, 14:20 and 14:37 on 08-18, which are precisely the three that made money. Meanwhile the sim takes 17 fires the live gate never took. The contrast with `abs_veto_short` is the tell: for that gate the sim reproduces **11 of 12** live signals within 6 seconds and about 5 points, so its **-\$749.55** is a counterfactual. `exhaustion_short` fires on a 5-second footprint confirmation that does not replay faithfully off bars, so its **-\$638.02** is a different strategy's P&L wearing the same name.

What the row can honestly claim, then: most of the **-\$638.02** is the value of the router's benches (the gate was benched at 15:00Z Monday, 12:40Z Wednesday and 06:40Z Thursday, and 16 of those 17 never-live fires happened while it was), and none of it is a verdict on the seven trades the gate actually took. Quote the ALL SIX total as a rough estimate of what the standdown was worth, exactly as the box below already says — but do not read any single gate's row as that gate's alternative history without first checking that the sim reproduces its fires. **BUILD #9 is the fix:** populate `signal_journal.suppressed_by` so a refusal is recorded rather than re-simulated.

★ AND HERE IS WHY THAT NUMBER IS AN ESTIMATE AND NOT A RESULT

The mechanical sim has no operator. Two of the live gates' best trades this week were closed by hand at a price the machine would not have taken, and the sim rides every one of those to its own stop. It also has no queue, no partial fills and no thin-tape veto. Every one of those biases makes the sim look WORSE than the live desk would have been, which means \$1,846.86 is an upper bound on the value of benching, not a measurement of it.

The audit ran the same question on the fortnight with a different instrument and got a range from +\$2,062 to −\$673 — it straddles zero. **So the correct statement is: the standdown was probably worth something this week, it was certainly not worth thousands, and no instrument on this desk can currently put a confident number on it.** That is a measurement problem and it is on the BUILD card.

4 • ★ THE STANDDOWN IS A ROUTER-LOCAL VARIABLE, AND SOMETHING ELSE ARMED A GATE THROUGH IT

On Friday at 14:03Z, with the tournament stood down and all six gates reading off, `abs_veto_short` was switched **on**. The router did not do it. Here is the sequence, unedited, out of the trial log:

```
2026-08-21T14:00:14Z | tune | DURABLE | changed: none | 14:00Z US-OPEN, tournament
STOOD DOWN and all six gates already read off: no-change tick. Flat, healthy=True,
halted=False;
2026-08-21T14:03Z APPLIED {abs_veto_short: on} [OPEN-HOUR WATCHER, automatic]
2026-08-21T14:05:18Z | tune | DURABLE | changed: {'abs_veto_short': 'off'} | 14:05Z
US-OPEN, tournament STOOD DOWN — abs_veto_short reads on because the OPEN-HOUR
WATCHER auto-armed
```

`open_hour_watch.py` is a second writer to `gate_switches.env` with off→on authority and **no awareness of the standdown at all**. It saw a confirmed break (ER 0.36, ATR 22, a new extreme) and did exactly what it was built to do, which is the standing “if a huge trend develops, arm immediately” instruction. The problem is not the watcher's logic. The problem is that a standing operator standdown lives in a variable inside one process, and the other five writers to that file cannot see it.

The gate was live for 2 minutes 15 seconds and fired nothing, so this cost \$0 — by luck, not by design. The exposure is bounded by the router's 5-minute tick,

not by any guard. If the watcher had armed at 14:06 instead of 14:03 the gate would have been live for the whole 5-minute window into the US open. **The standdown needs to be a file the writers check, not a variable one of them holds.** That is a Saturday item.

The one genuinely reassuring thing here: the router caught it on its very next tick and re-benched it, calling it “housekeeping, not thrash”. That is the behaviour you want from the slow deliberate loop.

★★ REV3 — IT IS NOT ONE INCIDENT. IT HAPPENED AGAIN ON 2026-08-24, AFTER THIS SECTION WAS WRITTEN

This section argued the case from a single Friday event, which invites the reading that it was a one-off. It was not. Re-reading data/router_trial_log.txt at publication, the watcher has taken the switch file off the standdown **twice**:

```
2026-08-21T14:03Z  APPLIED {'abs_veto_short': 'on'}  [OPEN-HOUR WATCHER,
automatic]    → re-benched 14:05:18Z    (2m15s)
2026-08-24T13:46Z  APPLIED {'abs_veto_short': 'on'}  [OPEN-HOUR WATCHER,
automatic]    → re-benched 13:50:22Z    (3m50s)
```

Both cost **\$0**, both for the same reason — the tape produced no qualifying signal inside the window — and **neither paged anybody**. The second one is longer, and it happened three days after the first with nothing changed in between, which is what you would expect: the watcher is behaving exactly as designed and the standdown it cannot see is still a variable in another process. **Two \$0 outcomes are not a control; they are the same uncontrolled exposure sampled twice.** The 08-24 arming is also the sixth carve-out block in the switch file, and §5 below now has to be read with that in mind — prepending it evicted the line §5 calls the dangerous one.

One thing worth crediting again: on 08-24 the router benched it back within one tick and said so in its reason, exactly as on 08-21. The slow loop is not the weak link here.

5 · The switch file's comment header is still being read as live instruction

switch_notes() splices the top 45 comment lines of gate_switches.env into every router prompt, labelled ACTIVE. Nothing prunes them. Here is what was in

there when this section was written, on the Friday night — **read the REV3 box under the table before acting on any row of it, because the header has since moved and the “Still being read?” column is no longer true for the last row:**

INSTRUCTION SITTING IN THE SWITCH FILE'S HEADER	AGE AT THIS REPORT	STILL BEING READ TO THE ROUTER?	TICKS THAT RESTATED IT
2026-08-21T14:03Z OPEN-HOUR WATCHER ARMED abs_veto_short on a CONFIRMED	0d 10h	YES — spliced into every prompt as ACTIVE	106 of 1425
2026-08-19T18:15Z OPERATOR BENCH — abs_veto_short OFF.	2d 6h	YES — spliced into every prompt as ACTIVE	66 of 1425
2026-08-13T13:58Z OPEN-HOUR WATCHER ARMED grind_long on a CONFIRMED	8d 11h	YES — spliced into every prompt as ACTIVE	0 of 1425
2026-08-11T14:1xZ — OPERATOR CARVE-OUT (expires 16:00Z, then normal rules resume).	11d 1h	YES — spliced into every prompt as ACTIVE	0 of 1425
2026-08-09T11:0xZ — SATURDAY #1 + MONDAY #1 SHIPPED. abs_veto_short IS ARMED BY DEFAULT.	13d 1h	YES — spliced into every prompt as ACTIVE	0 of 1425

★★ REV3 — RE-READ AT PUBLICATION BY IMPORT, AND THE HEADER HAS MOVED. THE DANGEROUS LINE WAS EVICTED, NOT CLOSED

The table above was read on 2026-08-21. Calling `switch_notes()` on 2026-08-26 — importing it and printing what it returns, never grepping the file, because a stale `# REVERT` comment lives in that header — returns something different:

- There are now **SIX** carve-out blocks in the file, not five. The new one is **2026-08-24T13:46Z OPEN-HOUR WATCHER ARMED abs_veto_short** — the second standdown breach in §4's REV3 box — and it was prepended.
- `switch_notes()` returns exactly 45 lines, covering five blocks: 08-24, 08-21, 08-19, 08-13 and 08-11. It cuts off **mid-sentence inside the 11 August block**.
- **The 2026-08-09 “abs_veto_short IS ARMED BY DEFAULT” line — the row this section calls the dangerous one — is no longer spliced at all.** It sits at line 50, past the cap. 'ARMED BY DEFAULT' in `switch_notes()` is now False.

That is not the problem going away, and it must not be read as one.

Nobody reviewed that line, nobody revoked it, and nothing in the file records that it stopped being read. It fell out because a newer carve-out was written on top of it. So the header is two faults, not one: comments that never expire, and a silent 45-line FIFO whose evictions are invisible — which means the set of instructions the router is being given changes whenever anything writes, in a direction nobody chose. The oldest line still spliced is **2026-08-11**, fifteen days old at publication and truncated in the middle. The remedy is unchanged and it is the only one that works: **delete the lines** (SATURDAY #4). Deleting them also un-truncates whatever is left.

As read on the Friday: five instructions, the oldest **thirteen days old**. The 19 August operator bench says “PINNED until 21:55Z” — that deadline passed on Wednesday and the line is still labelled ACTIVE. The 11 August one carries its own expiry (“expires 16:00Z”) and outlived it by eleven days. And the 9 August one is the dangerous one: it states that **abs_veto_short IS ARMED BY DEFAULT**, which directly contradicts the standing operator standdown the router is currently enforcing. A carve-out comment never expires, and revoking one in the ledger does not revoke it in the file.

★ THE THREE DEAD ONES GOT ZERO MENTIONS — AND THAT IS NOT A FIX

Of the 1,425 tick reasons this week, 172 restated one of these five lines. But the split is the whole point: the two LIVE-ish ones (Friday's watcher note and Wednesday's operator bench) account for every single mention, and the three genuinely dead ones — 8, 11 and 13 days old — were restated **zero times**.

A zero-mention streak reads like the problem has gone away. It has not, and no streak ever can show that it has. **1,424 of the 1,425 reasons this week are pinned at exactly the 1,200-character cap** — median length 1200, maximum 1200. The mention rate is measuring how much room was left in the reason field after the live situation was described, and this week the live situation was a standdown, an Asia block and a four-lot carry, all of which are more urgent to narrate than a nine-day-old carve-out.

Read the desk's six previous samples (42/54 → 49/49 → 9/40 → 0/48 → 3/48 → 9/48) and this one the same way: as readings of spare capacity, not as a trend. **The only thing that closes this is deleting the lines.** Any week the reason field is less crowded, a thirteen-day-old 'armed by default' instruction is one tick away from being read as policy again.

6 · ★ The router called a four-lot position flat for all 817 ticks of its life

The day rider opened four lots long at 13:13Z on Friday and never closed them. It closed by hand on Monday 2026-08-24 at 11:59:39Z, so the window is now fixed rather than open-ended, and **★ REV3 counts it over the position's whole life instead of over “the last N ticks”**: data/router_trial_log.txt holds **817 router ticks** between the entry and the close, of which **811 print “Flat, healthy”, 813 contain the word flat, and 0 mention the open position at all**. Rev1 printed “137 of 141” here and “132 of 137” in Part 1 — two snapshots taken at whatever moment each section rendered, neither reproducible from the log, and a reader could not tell which was right. Neither was; both were partial windows of the same thing. It is not lying and it is not broken: it reads core_health.json, whose flat field is tournament-scoped, and the tournament genuinely is flat. But the sentence a human reads is “Flat, healthy=True, halted=False”, and it is being printed while the desk carries roughly \$235,000 of notional into a weekend gap.

The ticks are not silent about the rider — most of them name its closed rows, — \$727, —\$268, —\$293.50 — and the last one describes it as “clientId 4, a desk I neither route nor bench”. **That is the failure in one sentence.** Citing what CLOSED and declining responsibility for what is OPEN is an attribution that terminates in the wrong place. The trades table can only ever enumerate closed rows, so it can never contradict a benign story; the venue position line can, and it says four. “Not mine to close” obliges escalation, not dismissal.

7 · The two instruments that should have written this section

ROLLUP	NEWEST FILE	DAYS STALE AT THIS REPORT	WHAT RUNS IT
data/router_nightly/	2026-08-14	8	Nothing. No timer.
data/selector_nightly/	2026-08-18	4	Nothing. No timer.

These are the files the standing weekly review is supposed to roll up: net router value, leakage by regime, selector optimal-pick rate and regret. Neither has a timer, so both simply stopped. **There is no week-in-review to present**, and rather than dress that up, the honest version is the raw-log rebuild above. Two known defects also make the selector rollup wrong when it does run: it has no data_quality filter, so quarantined rows inflate its regret, and all four of its modes reprice at a hard-coded 1.0 stop multiple, which is not what two of the live gates use.

8 · Verdict, and the Saturday actions

Is the router earning its keep? On this week, yes, and for a reason that is not about its thresholds. It made 10 good calls and 3 bad ones, it applied your 18:15Z bench inside 90 seconds, and it was the only component on the box that noticed a gate had been armed through your standdown. Its thresholds were not the lever this week and nothing here argues for touching them. Its clock was not the lever either — there were only 20 writes to be timed. **The router's problem this week is not its judgement, it is that it is surrounded by instruments that give it wrong inputs:** a health flag that says flat when the desk is long, a comment header full of expired instructions, a second writer that can undo a standdown, and two rollups that quietly stopped.

Disposition — every lead this section touched

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
Make the standdown a FILE that every writer checks	LIVE	It is currently a variable inside router_tick_durable.py. open_hour_watch.py has armed a gate through it twice — 08-21 14:03Z (2m15s) and, ★ REV3, again on 08-24 13:46Z (3m50s), after this section was written. Both cost \$0 only because the tape produced no signal inside the window, and neither paged. Two \$0 samples of the same uncontrolled exposure, the second one longer. Saturday item, and n=2.
Delete the expired carve-outs from gate_switches.env's header	LIVE	FIVE instructions are spliced into every prompt as ACTIVE — ★ REV2: this row said THREE, which contradicted \$5's own table and the Saturday card, and it understated the exposure. The five, by header timestamp in data/gate_switches.env: 2026-08-21T14:03Z, 2026-08-19T18:15Z, 2026-08-13T13:58Z, 2026-08-11T14:1xZ and 2026-08-09T11:0xZ. Four of the five are past their own stated expiry or their situation, and the oldest is the dangerous one: the 08-09 line asserts that abs_veto_short IS ARMED BY DEFAULT, which directly contradicts the standing operator standdown the router is currently enforcing — on the same gate that open_hour_watch.py armed through that standdown on Friday — and again on 08-24. ★ REV3, verified by importing switch_notes () at publication: a SIXTH block (08-24 13:46Z) was prepended on the Monday and pushed the 08-09 line past the 45-line cap, so it is no longer spliced — evicted, not closed, with nothing in the file recording it, and the oldest line still spliced (08-11) now arrives truncated mid-sentence. Nothing prunes them and no mention-rate streak closes it; only deleting the lines does.
Point core_health.flat at the venue position	LIVE	811 of the 817 router ticks spanning the position's entire life said 'Flat, healthy' while the venue was long four lots, and none of them mentioned it. Either make the field desk-wide or rename it tournament_flat.
Change the router's thresholds or timing this week	REFUTED	There were 20 switch writes all week, 5 of them worth \$0 by construction. There is no sample here to tune a threshold on, and tuning on this week would be fitting to a standdown.
Re-arm a gate on a lapse rather than a positive case	REFUTED	Named instance: 2026-08-19 15:35Z re-armed abs_veto_short 30 minutes after benching it, with no new evidence, straight into three stops for −\$133. Third occasion on this gate.
Give router_watch narrow fast-BENCH authority	PARKED	The asymmetry argument still holds — a wrong bench is cheap, a wrong arm is expensive — but this week produced no measurable actuation-lag cost to justify the

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
		build, because the desk was stood down. Revive it with a lag number from a week the tournament actually trades.
Put a timer on router_nightly and selector_nightly	LIVE	Both stopped silently. A monitoring file that stops being written is worse than one that errors, because its absence looks like a quiet week.
Fix selector_nightly's two known defects before believing it again	PARKED	No data_quality filter (quarantined rows inflate regret) and a hard-coded 1.0 stop multiple for all four modes. Revive when both are fixed; until then do not quote its regret.

M1 The biggest runs on MNQ this week — did we show up?

Forget what the desk did this week — start from the raw tape. MNQ printed **68 runs of 1.5×ATR or bigger** between **2026-08-17 00:00** and **2026-08-21 20:59 UTC** (a 15-minute move of at least 55 points, against a typical 15-min range of 37). That is the whole tape, not a top-ten. We **caught 4**, we were positioned against **2**, and we **sat out 62**. The runs we aligned with banked **+\$760** of honest money — but that is only 73% of their \$1,040 hindsight ceiling, and the 62 we sat out left **\$9,610** on the table. The bleed is not the runs; it is the fading we do around them.

★ **THIS IS THE FROZEN CENSUS, RE-FROZEN 2026-08-25.** Every number on this page and every number in Movement 2 and Movement 3 comes from the same run of `run_census.py` — 68 runs, 62 sat out, \$9,610 of sat-out ceiling. An earlier freeze of this fragment (2026-08-22) read ~~70 runs / 65 sat out / \$9,986~~ and is superseded; ★★ **REV3 CORRECTS THIS BANNER: it is SIX of the seven greenfield labs, not seven, and the exception is VACUUM.** Six hunts were re-run against this freeze and reproduce the table below row for row before they are allowed to disagree with anything. **The VACUUM dossier was written on 2026-08-22 at 00:18 UTC — three days before this re-freeze — and is anchored to the SUPERSEDED list throughout.** It says “11 VACUUM runs of 70, of which 9 sat out” and scores every catch figure “x/65” against a \$9,986 ceiling; the cluster table on this page says VACUUM = **9 runs** of 68, with 62 sat out and \$9,610. So VACUUM is the second section on the old freeze, alongside Movement 2, and it carries the same warning Movement 2 does: **do not line up per-run numbers between the VACUUM dossier and this page** — they are counting different lists, and neither is wrong on its own terms. Its verdicts do not depend on the count (four candidates, four graves, killed by its own label rather than by a run list), which is why it is banner-marked rather than re-run. This report's own rule is that a stale honesty banner is worse than none, because it spends the reader's trust on a falsehood; “all seven” was that. △ Movement 2's idle-gate replay was computed against the SUPERSEDED freeze and is banner-marked accordingly — its run list is 70/65, not this one.

★★ **REV3 — WHY THE CENSUS GOT RE-FROZEN MID-CYCLE, AND WHICH PAST NUMBERS THAT PUTS IN DOUBT**

The report tells you the two freezes differ and it names the flow-bucket phase bug as the mechanism. It never said why a frozen census got re-run at all three days into a report cycle, which is the question a reader should actually ask — a freeze that moves is not a freeze. Two separate faults, and the second is worse than the first.

1 — The census had no pinned end time, so it could not reproduce itself. Until this cycle `run_census.py` computed its window as `now - days`. There was no `--until`. That means every re-run cut a different tape: re-running it to fix one column silently re-cut the whole run set, and there was no way to tell a corrected census from a differently-windowed one. That is how a fragment dated 2026-08-22 and a fragment dated 2026-08-25 came to disagree about how many runs the same week contained. The committed

version of the script has no --until flag at all; it was added this cycle, and pinning it is BUILD #14's companion fix. **Until that flag existed, no census this desk ever published was reproducible — not one.**

2 — And the freeze in git was not even this week's tape. The census_summary.json sitting in the repository was a whole-history lake census: **347 runs, 339 sat out, a \$217,326 ceiling**, with top-25 entries dated 06-09 and 11-20. The re-freeze is what put the file back onto the five-day window the section describes (68 / 62 / \$9,610). So the 08-25 re-run was not a refinement; it was a correction of a file that was measuring something else entirely.

What that makes suspect, stated plainly. The phase bug moves **21 of 68 labels — 31% — on the bucket origin alone**, and only **17 of 68** run timestamps are minute-aligned, so roughly a third of every cluster assignment this desk has published is an artefact of where a run happened to start rather than of what the tape did. The flow test became a z-score against the trailing two hours on **2026-08-01**; that is the date the phase-dependence enters. Four prior weeklies carry cluster labels:

PUBLISHED REPORT	CLUSTER RULE IN FORCE	IS ITS CLUSTER-SCOPED WORK SAFE TO CITE?
weekly_2026-08-14	z-score flow (post 08-01)	NO — same phase bug. Any finding scoped to FLOW-LED, VACUUM or UNCLASS carries a ~31% arbitrary-label rate. Its blanket and per-gate numbers are unaffected; only the cluster cut is.
weekly_2026-08-08	z-score flow (post 08-01)	NO — same phase bug, same scope of damage.
weekly_2026-07-31	raw abs(flow) > 50	NO, for a DIFFERENT and larger reason. The old threshold fired on 66.5% of all bars — the script's own comment says so. FLOW-LED and VACUUM were near-coin-flips by construction, so its cluster splits are not phase-noise, they are meaningless.
weekly_2026-07-10	raw abs(flow) > 50	NO, same 66.5% problem. This is the heaviest cluster user of the four (149 VACUUM mentions).

So: every cluster-scoped finding this desk has ever published is unsafe to cite — the two older reports because the label was a coin flip, the two newer ones because a third of the labels move on nothing. What is NOT affected is anything that does not split on cluster: run counts, sat-out ceilings, per-gate P&L, the base-rate arms, and Movement 2's −\$1,905.36 (which fires all six gates across the whole tape and never reads the run list).

This is the same audit Part 1 §7 and Movement 3 §7 do for the instruments, applied to the census — and it is the reason Movement 3 §13 recommends renaming UNCLASS to UNTESTED and reporting it as coverage rather than as a footprint.

THE FULL CENSUS

Every qualifying run, in order. **Move** is the swing in points; **\$ 1lot** is that move on one MNQ lot (\$2/pt) — the hindsight ceiling. **real \$** is what a gate actually banked near it. **Flow** is net aggressor volume in the 60s before; **amp** is pre-run amplitude; **book** is the far-side L2 depth share (below 0.50 = the side price ran toward was thin), read from `capture.db.book`, 3 deep, 30s. Green = we caught it, red = we fought it.

TIME (UTC)	DIR	MOVE	\$ 1LOT	US	GATE	REAL \$	FLOW	AMP	BOOK	CLUSTER
08-17 07:01	DN	-59	118	sat out	—		+103	0.03	0.55	UNCLASS
08-17 07:27	UP	+56	112	sat out	—		-25	0.03	0.43	UNCLASS
08-17 12:24	DN	-76	152	sat out	—		-35	0.04	0.55	UNCLASS
08-17 13:31	DN	-74	148	sat out	—		+147	0.19	0.57	OPEN/NEWS
08-17 13:53	DN	-64	127	sat out	—		-636	0.20	0.57	FLOW-LED
08-17 14:53	UP	+70	140	sat out	—		-274	0.07	0.41	OPEN/NEWS
08-17 15:29	DN	-59	118	sat out	—		+183	0.07	0.57	UNCLASS
08-17 17:02	DN	-64	127	sat out	—		+87	0.04	0.56	UNCLASS
08-18 01:38	DN	-100	201	sat out	—		+35	0.05	0.46	UNCLASS
08-18 02:03	DN	-70	140	sat out	—		+14	0.12	0.45	UNCLASS
08-18 05:52	DN	-80	160	sat out	—		-26	0.07	0.49	UNCLASS
08-18 06:08	UP	+60	121	sat out	—		-133	0.07	0.55	UNCLASS
08-18 07:52	DN	-90	180	sat out	—		-39	0.07	0.52	UNCLASS
08-18 12:22	DN	-62	124	sat out	—		-18	0.12	0.54	UNCLASS
	UP	+61	122	sat out	—		+89	0.09	0.52	OPEN/NEWS

08-18 13:25										
08-18 13:40	DN	-137	274	caught	exhaustio n_short_A	-208	+402	0.29	0.49	OPEN/ NEWS
08-18 13:58	DN	-66	132	caught	day_rider	+174	+111	0.19	0.50	OPEN/ NEWS
08-18 14:14	UP	+91	182	sat out	—		-226	0.22	0.45	OPEN/NEWS
08-18 14:38	DN	-105	210	caught	exhaustio n_short_A	+250	-28	0.12	0.49	OPEN/ NEWS
08-18 15:10	UP	+65	130	sat out	—		-780	0.19	0.51	UNCLASS
08-18 15:40	UP	+62	123	sat out	—		-132	0.07	0.49	UNCLASS
08-18 23:57	DN	-80	160	sat out	—		-55	0.08	0.51	UNCLASS
08-19 00:15	UP	+96	192	sat out	—		+6	0.09	0.54	UNCLASS
08-19 01:00	UP	+58	115	sat out	—		+142	0.11	0.51	UNCLASS
08-19 04:50	UP	+56	112	sat out	—		-106	0.10	0.52	UNCLASS
08-19 05:05	DN	-60	121	sat out	—		+22	0.13	0.46	UNCLASS
08-19 06:10	UP	+68	136	sat out	—		+115	0.08	0.53	FLOW-LED
08-19 06:43	UP	+83	166	sat out	—		-128	0.08	0.53	UNCLASS
08-19 07:54	UP	+62	124	sat out	—		+106	0.06	0.48	UNCLASS
08-19 12:32	UP	+154	308	sat out	—		-123	0.12	0.55	VACUUM
08-19 13:35	DN	-212	424	caught	day_rider	+544	-280	0.32	0.51	FLOW-LED
08-19 14:00	DN	-128	257	sat out	—		+661	0.22	0.51	OPEN/NEWS
08-19 14:25	UP	+150	300	sat out	—		-565	0.24	0.51	OPEN/NEWS
08-19 14:46	UP	+70	139	sat out	—		+281	0.13	0.49	OPEN/NEWS
08-19 15:55	UP	+74	148	sat out	—		-663	0.09	0.50	VACUUM
08-19 16:21	DN	-62	124	sat out	—		-88	0.12	0.53	UNCLASS
08-19 16:42	DN	-55	110	sat out	—		-202	0.09	0.50	UNCLASS
08-19 18:03	DN	-92	184	sat out	—		+612	0.19	0.54	VACUUM
08-19 18:18	UP	+66	132	sat out	—		-284	0.17	0.51	UNCLASS
08-19 19:34	DN	-76	152	sat out	—		+508	0.13	0.50	VACUUM
08-19 20:49	UP	+73	146	sat out	—		+0	0.04	0.51	UNCLASS
08-19 23:52	DN	-56	112	sat out	—		-9	0.04	0.54	UNCLASS

08-20 00:07	UP	+61	122	sat out	—		-88	0.10	0.48	UNCLASS
08-20 02:08	DN	-57	114	sat out	—		+67	0.05	0.49	UNCLASS
08-20 05:56	UP	+56	112	sat out	—		+204	0.05	0.41	FLOW-LED
08-20 07:56	DN	-76	153	sat out	—		-45	0.04	0.52	UNCLASS
08-20 08:17	DN	-68	137	sat out	—		-74	0.07	0.49	UNCLASS
08-20 08:55	UP	+68	136	sat out	—		-64	0.08	0.51	UNCLASS
08-20 10:58	DN	-82	163	sat out	—		-6	0.05	0.50	UNCLASS
08-20 11:16	DN	-57	114	sat out	—		+101	0.11	0.50	UNCLASS
08-20 11:43	DN	-90	181	sat out	—		+13	0.13	0.51	UNCLASS
08-20 12:02	DN	-91	182	sat out	—		-4	0.10	0.52	UNCLASS
08-20 12:39	UP	+86	172	sat out	—		-126	0.11	0.49	UNCLASS
08-20 13:22	UP	+120	240	sat out	—		+82	0.08	0.51	OPEN/NEWS
08-20 13:39	DN	-116	231	FOUGHT	day_rider	-1288	-22	0.34	0.49	VOL- EXPANSION
08-20 14:29	UP	+80	160	sat out	—		-1256	0.24	0.51	VACUUM
08-20 15:02	DN	-89	178	sat out	—		+2038	0.37	0.51	VACUUM
08-20 15:19	DN	-64	127	sat out	—		+274	0.10	0.50	UNCLASS
08-20 16:32	UP	+68	136	sat out	—		+176	0.13	0.52	UNCLASS
08-20 17:56	DN	-83	166	sat out	—		+75	0.11	0.48	UNCLASS
08-20 23:59	UP	+67	134	sat out	—		+53	0.05	0.50	UNCLASS
08-21 13:10	DN	-62	124	FOUGHT	day_rider	-2149	+213	0.08	0.49	VACUUM
08-21 13:28	DN	-166	331	sat out	—		+295	0.22	0.49	VACUUM
08-21 13:46	DN	-56	113	sat out	—		+18	0.18	0.47	OPEN/NEWS
08-21 14:06	DN	-84	169	sat out	—		+965	0.25	0.51	VACUUM
08-21 14:41	UP	+64	127	sat out	—		-263	0.12	0.50	OPEN/NEWS
08-21 15:25	UP	+114	227	sat out	—		+274	0.10	0.50	UNCLASS
08-21 15:54	DN	-78	156	sat out	—		+232	0.20	0.51	UNCLASS

What the runs were — by cause, and what this week did to the vocabulary

Each run gets a cause fingerprint from its pre-run tape. This used to be the map for where to hunt a new gate. **This week three of the five labels were interrogated directly and did not survive it** — the notes in the right-hand column are Movement 3's findings, not decoration, and they are the reason no hunt should be scoped by a label again until the census is fixed.

CLUSTER	RUNS	WHAT IT IS	WHAT MOVEMENT 3 FOUND
UNCLASS	41	no clear tape tell (quiet ignition)	★ base rate 75.4% on all tape, lift 0.82 — this is coverage, not a finding. Movement 3 recommends renaming it UNTESTED
OPEN/NEWS	13	the 13:00–15:00 UTC cash-open / data window	★ lift over the bare clock is 0.921x — the two footprint tests remove 41% of the window and keep the less run-dense half
VACUUM	9	moved AGAINST the tape (a snap-back / stop-run)	★ this label is not a footprint — the flow event behind it fires on 20.4% of all tape and then splits it VACUUM/FLOW-LED on a 49.9% coin toss (Movement 3, VACUUM \$2 and FLOW-LED)
FLOW-LED	4	aggressors drove it (a continuation)	★ same coin toss as VACUUM — the split carries 0.33pp of directional information; the lift it does carry is a VOLUME lift wearing a flow costume
VOL-EXPANSION	1	amplitude broke out of a coil	not interrogated this week

HONEST MONEY — CEILING VS WHAT WE BANKED

RUNS	HINDSIGHT CEILING	WHAT WE REALLY MADE
Caught (aligned)	4	\$1,040
Fought (against)	2	\$356
Sat out	62	\$9,610
		\$0 ← the money on the table

The runs made money and the fading gave it back; we convert a fraction of the ceiling and ignore the rest. ★ **And read the ceiling column with Movement 3's correction attached:** the sat-out ceiling is a HINDSIGHT

number. A perfect late boarder — one that fires inside every run at a median three minutes in and then exits at the exact top with no costs at all — reaches only **53%** of it, and a deployable one 20-24%. Thirteen per cent of the runs are finished before any honest trigger can see them.

Which book was near the runs

GATE / DESK	REAL \$ NEAR THE RUNS
day_rider	\$-2,706
abs_veto_short_A	\$-78
abs_veto_short_B	\$-78
exhaustion_short_B	\$+84
exhaustion_short_A	\$+100

★ The single biggest line here is `day_rider`, and it is negative. The tournament was barely present at the week's runs at all; the rider was present and on the wrong side twice. That is the census's own view of the week's money, arrived at from the tape rather than from the books.

THE SAME CENSUS ON GOLD — MGC, the second instrument

M1 The biggest runs on MGC this week — did we show up?

Forget what the desk did this week — start from the raw tape. MGC printed **70 runs of 1.5×ATR or bigger** between **2026-08-17 00:00** and **2026-08-21 20:59 UTC** (a 15-minute move of at least 11 points, against a typical 15-min range of 7). That is the whole tape, not a top-ten. We **caught 0**, we were positioned against **0**, and we **sat out 70**. **We have no gate on MGC**, so all 70 runs and **\$11,401** of hindsight ceiling were sat out

by construction, not by a decision that went wrong. Nothing here is a bleed — it is an empty seat, and the question the next movements ask is what could sit in it.

★ **THIS IS THE FROZEN CENSUS, RE-FROZEN 2026-08-25.** Every number on this page and every number in Movement 2 and Movement 3 comes from the same run of `run_census.py` — 70 runs, 70 sat out, \$11,401 of sat-out ceiling. The gold census was re-frozen in the same run as the Nasdaq one, so the two halves of Movement 1 are the same vintage. The desk has never traded MGC, so nothing downstream of this table depends on a book.

THE FULL CENSUS

Every qualifying run, in order. **Move** is the swing in points; **\$ 1lot** is that move on one MGC lot (\$10/pt) — the hindsight ceiling. **real \$** is what a gate actually banked near it. **Flow** is net aggressor volume in the 60s before; **amp** is pre-run amplitude; **book** is the far-side L2 depth share (below 0.50 = the side price ran toward was thin), read from `depth.db.depth_snap`, 3 deep, 30s. Green = we caught it, red = we fought it.

TIME (UTC)	DIR	MOVE	\$ 1LOT	US	GATE	REAL \$	FLOW	AMP	BOOK	CLUSTER
08-17 00:06	UP	+15	148	sat out	—		-12	0.18	0.48	UNCLASS
08-17 00:52	UP	+15	150	sat out	—		+5	0.06	0.43	UNCLASS
08-17 01:24	DN	-15	152	sat out	—		+27	0.17	0.47	VACUUM
08-17 05:17	UP	+11	113	sat out	—		-1	0.09	0.50	UNCLASS
08-17 07:23	UP	+14	135	sat out	—		+4	0.07	0.43	UNCLASS
08-17 12:25	DN	-11	112	sat out	—		-28	0.15	0.57	FLOW-LED
08-17 13:42	UP	+26	264	sat out	—		-26	0.16	0.53	VACUUM
08-17 14:09	UP	+22	217	sat out	—		-6	0.08	0.49	OPEN/NEWS
08-17 18:19	DN	-17	174	sat out	—		+13	0.10	0.58	VACUUM
08-17 19:06	UP	+15	150	sat out	—		-7	0.11	0.49	UNCLASS
08-18 01:33	DN	-17	172	sat out	—		+4	0.09	0.46	UNCLASS

08-18 13:15	DN	-13	126	sat out	—	+0	0.13	0.36	OPEN/NEWS
08-18 13:30	UP	+20	197	sat out	—	-262	0.40	0.47	VACUUM
08-18 13:48	UP	+11	110	sat out	—	-1	0.26	0.59	OPEN/NEWS
08-18 14:25	DN	-18	179	sat out	—	+6	0.09	0.41	OPEN/NEWS
08-18 14:48	DN	-15	154	sat out	—	+1	0.19	0.50	OPEN/NEWS
08-18 15:05	DN	-21	210	sat out	—	-21	0.23	0.52	UNCLASS
08-18 19:42	DN	-16	162	sat out	—	+11	0.10	0.34	VACUUM
08-18 23:58	UP	+12	119	sat out	—	+8	0.06	0.58	UNCLASS
08-19 00:48	DN	-14	145	sat out	—	-1	0.07	0.49	UNCLASS
08-19 01:26	UP	+19	192	sat out	—	-7	0.15	0.54	UNCLASS
08-19 05:01	DN	-11	109	sat out	—	+17	0.11	0.49	VACUUM
08-19 06:08	UP	+11	111	sat out	—	-2	0.07	0.58	UNCLASS
08-19 07:17	UP	+11	107	sat out	—	-3	0.10	0.52	UNCLASS
08-19 12:31	UP	+55	548	sat out	—	+9	0.08	0.21	UNCLASS
08-19 12:47	UP	+25	250	sat out	—	-19	0.36	0.42	VACUUM
08-19 13:04	UP	+36	358	sat out	—	+5	0.31	0.42	VOL- EXPANSION
08-19 13:22	DN	-14	139	sat out	—	-86	0.18	0.54	FLOW-LED
08-19 13:55	UP	+21	211	sat out	—	-34	0.25	0.48	OPEN/NEWS
08-19 14:21	UP	+17	167	sat out	—	-16	0.18	0.70	OPEN/NEWS
08-19 15:31	UP	+11	112	sat out	—	-54	0.11	0.58	VACUUM
08-19 18:13	DN	-15	151	sat out	—	-1	0.10	0.71	UNCLASS
08-19 18:32	UP	+17	171	sat out	—	-9	0.15	0.62	UNCLASS
08-19 20:13	UP	+14	137	sat out	—	-61	0.10	0.35	VACUUM
08-20 00:58	DN	-14	143	sat out	—	+15	0.10	0.60	UNCLASS
08-20 01:41	DN	-12	116	sat out	—	-13	0.10	0.57	UNCLASS
08-20 01:58	UP	+13	126	sat out	—	-1	0.08	0.49	UNCLASS
08-20 06:00	UP	+13	127	sat out	—	+0	0.10	0.51	UNCLASS
08-20 11:33	DN	-17	172	sat out	—	+1	0.07	0.53	UNCLASS

08-20 11:51	DN	-19	188	sat out	—	-10	0.16	0.48	UNCLASS
08-20 12:19	DN	-14	142	sat out	—	+26	0.10	0.65	VACUUM
08-20 12:36	UP	+19	190	sat out	—	+14	0.12	0.68	UNCLASS
08-20 13:01	DN	-13	129	sat out	—	+10	0.25	0.59	OPEN/NEWS
08-20 13:29	UP	+15	151	sat out	—	+1	0.10	0.59	OPEN/NEWS
08-20 14:31	UP	+11	113	sat out	—	+4	0.19	0.44	OPEN/NEWS
08-20 14:47	UP	+26	262	sat out	—	+8	0.21	0.39	OPEN/NEWS
08-20 15:28	DN	-23	227	sat out	—	-16	0.23	0.53	UNCLASS
08-20 15:43	UP	+12	118	sat out	—	+7	0.21	0.47	UNCLASS
08-20 16:01	DN	-14	137	sat out	—	+3	0.09	0.58	UNCLASS
08-20 18:16	UP	+12	119	sat out	—	-3	0.11	0.69	UNCLASS
08-21 00:01	DN	-11	110	sat out	—	-9	0.04	0.52	FLOW-LED
08-21 01:01	UP	+13	128	sat out	—	+5	0.10	0.36	UNCLASS
08-21 01:20	UP	+16	158	sat out	—	+18	0.11	0.62	FLOW-LED
08-21 01:35	DN	-14	141	sat out	—	-34	0.24	0.49	FLOW-LED
08-21 01:57	DN	-12	123	sat out	—	+7	0.09	0.32	UNCLASS
08-21 02:37	UP	+13	132	sat out	—	+0	0.07	0.57	UNCLASS
08-21 05:41	UP	+16	163	sat out	—	+1	0.12	0.60	UNCLASS
08-21 06:23	UP	+15	146	sat out	—	+5	0.10	0.53	UNCLASS
08-21 08:09	UP	+17	172	sat out	—	+9	0.07	0.55	UNCLASS
08-21 08:56	UP	+17	172	sat out	—	-13	0.06	0.58	VACUUM
08-21 09:16	DN	-18	184	sat out	—	+14	0.12	0.46	VACUUM
08-21 12:08	DN	-15	147	sat out	—	-3	0.13	0.50	UNCLASS
08-21 12:25	DN	-20	200	sat out	—	-19	0.16	0.57	FLOW-LED
08-21 12:40	UP	+16	161	sat out	—	-49	0.32	0.52	VACUUM
08-21 13:35	UP	+15	148	sat out	—	-11	0.29	0.51	OPEN/NEWS
08-21 14:15	UP	+14	135	sat out	—	-10	0.11	0.32	OPEN/NEWS
08-21 14:30	DN	-20	205	sat out	—	-12	0.12	0.62	OPEN/NEWS

08-21 14:56	UP	+11	108	sat out	—	-12	0.23	0.48	OPEN/NEWS
08-21 15:21	UP	+11	107	sat out	—	-2	0.08	0.53	UNCLASS
08-21 17:09	DN	-15	149	sat out	—	+3	0.13	0.32	UNCLASS

What the runs were — by cause, and what this week did to the vocabulary

Each run gets a cause fingerprint from its pre-run tape. This used to be the map for where to hunt a new gate. **This week three of the five labels were interrogated directly and did not survive it** — the notes in the right-hand column are Movement 3's findings, not decoration, and they are the reason no hunt should be scoped by a label again until the census is fixed.

CLUSTER	RUNS	WHAT IT IS	WHAT MOVEMENT 3 FOUND
UNCLASS	35	no clear tape tell (quiet ignition)	★ base rate 75.4% on all tape, lift 0.82 — this is coverage, not a finding. Movement 3 recommends renaming it UNTESTED
OPEN/NEWS	15	the 13:00–15:00 UTC cash-open / data window	★ lift over the bare clock is 0.921x — the two footprint tests remove 41% of the window and keep the less run-dense half
VACUUM	13	moved AGAINST the tape (a snap-back / stop-run)	★ this label is not a footprint — the flow event behind it fires on 20.4% of all tape and then splits it VACUUM/FLOW-LED on a 49.9% coin toss (Movement 3, VACUUM §2 and FLOW-LED)
FLOW-LED	6	aggressors drove it (a continuation)	★ same coin toss as VACUUM — the split carries 0.33pp of directional information; the lift it does carry is a VOLUME lift wearing a flow costume
VOL-EXPANSION	1	amplitude broke out of a coil	not interrogated this week

HONEST MONEY — CEILING VS WHAT WE BANKED

RUNS	HINDSIGHT CEILING	WHAT WE REALLY MADE
Caught (aligned)	0	\$0
		+\$0

Fought (against)	0	\$0	\$+0
Sat out	70	\$11,401	\$0 ← the money on the table

Every row is a sat-out row because there is no gate on this instrument to catch or fight anything. The ceiling is what one lot would have made on a perfect entry and a perfect exit, so treat it as the size of the seat, never as money forgone. The next two movements ask what could fill it.

★ GAP — THIS SECTION WAS COMPUTED AGAINST THE SUPERSEDED CENSUS — ITS RUN LIST IS 70/65, THE REST OF THE REPORT IS 68/62

The census was re-frozen on 2026-08-25 with the run-picker's flow-bucket phase in a different place, and it moved: **70 runs / 65 sat out / \$9,986** became **68 runs / 62 sat out / \$9,610**. Movement 1 and all seven of Movement 3's hunts are on the new freeze. **This section is on the old one** — replaying eight gates across a week of tick data is a multi-hour job and it was not re-run. **What that means for you:** its headline finding is unaffected, because it does not depend on which particular runs are in the list — firing all six gates mechanically through this week's own tape loses $-\$1,905.36$ on 117 fires either way, and that is a statement about the GATES, not about the census. What you should NOT do is take a per-run number from this section and line it up against a per-run number in Movement 1 or Movement 3; three of the runs are not the same runs. ★ And note the cause: Movement 3 found that **21 of 68 labels change on bucket phase alone**, which is the bug behind this divergence and is on the Monday list to fix.

M2 THE IDLE-GATE LAB — could the gates we already own have caught them?

Garrath — the census found a set of big runs this week that the desk sat out completely, worth roughly ten thousand dollars if they had been caught perfectly. The obvious question is whether the six gates we already own would have caught any of them if they had simply been left armed. **The**

answer is a clean and rather uncomfortable no. Fired across the whole week rather than only near the runs, they lose **−\$1,905.36 on 117 fires.** Fired into the run windows specifically, they lose **−\$976.71 on 21 fires.** **Not arming them was worth more than any entry idea in this report.**

★★ REV3 — WHICH OF THESE NUMBERS DEPENDS ON THE CENSUS, AND WHICH DOES NOT

This lab was built on the **08-22 census freeze — 70 runs, 65 sat out, \$9,987**. Movement 1 now carries the **08-25 re-freeze — 68 runs, 62 sat out, \$9,610**. The two are not reconcilable run-for-run and the difference has a named cause (BUILD #3, the census's flow-bucket phase). So the first thing this section owes you is which of its numbers moved:

FIGURE	ON THE 08-22 FREEZE	RE-CUT ON THE 08-25 FREEZE	DOES IT DEPEND ON THE CENSUS?
Whole-week base rate, live config	-\$1,905.36 / 117 fires	-\$1,905.36 / 117 fires	NO — it is census-INDEPENDENT. It fires all six gates across every second of the week's tape and never looks at the run list at all. It does not move, and it is the figure to quote anywhere outside this section.
Run windows, 10 min BEFORE ignition, live config	-\$1,333.85 / 23 fires 18 of 65 runs	-\$976.71 / 21 fires 16 of 53 runs	YES. Same sign, same instruction, 27% smaller.
Run windows, 10 min BEFORE, every suppressor removed	-\$1,620.40 / 60 fires	-\$1,214.90 / 54 fires	YES. Stripping the filters is still worse than keeping them.
Run windows, 5 min AFTER ignition, live config	-\$45.00 / 17 fires	+\$124.64 / 15 fires	YES — and this one changes SIGN. It does not change the disposition, because \$7 already showed the -\$45.00 was one gate (abs_veto_short +\$504.50 on 5 fires against -\$549.50 on the other five gates' 12) and that stripping one fire made it -\$156.00. A pooled figure that flips on a re-cut of the run list is exactly the number that should never have been read as a result. It stays SHADOW, scored per gate.
Run windows, 5 min AFTER, ungated	-\$626.75 / 49 fires	-\$424.61 / 42 fires	YES.

Read the re-cut for what it is. It re-sums the per-run fills already banked in movement2_idle_gates.json over only the runs that survive into the new list, matched to the nearest surviving run within four minutes — the re-freeze moves a run's ignition minute as well as adding and removing runs. **53 of the old 65 survive, 12 drop out, and 9 of the new list's 62 sat-out runs were never simulated at all**, so this is a re-cut and not a re-replay. It reproduces the old arms exactly (-\$1,333.85 on 23 fires) before re-cutting, which is what makes it trustworthy as a re-cut. The whole-week arm needs none of this.

Everything below this box is still on the 08-22 freeze and still says 65 and \$9,987. That is deliberate: the tables are the record of what was run. Use this box for the figures that leave the section.

1 • The method, and the one thing that makes it honest

Each of the 65 sat-out runs gets four independent simulations: the six gates fired mechanically in the **10 minutes before** the run ignited (which is the interesting one — catching a run means being in before it goes), and in the **5 minutes after**; each of those both in the exact live config and **ungated**, meaning every ATR floor, veto and confirmation stripped out. Exits are the live scaleout slate plus `exit_overrides.json`, raced on real ticks at \$1.50 per round trip and \$2.00 a point.

The thing that makes it honest is the **base-rate control**. Measuring a gate only in windows we already know contained a big run is hindsight selection — it counts the fires where the move existed and never counts the ones where it did not. So the same six gates are also fired across the entire captured week, every fire counted. That control is the reason this section can say something the last three idle-gate labs could not.

2 • The four arms

ARM	RUNS IT SHOWED UP FOR	FIRES	NET	\$ / FIRE	WIN %	BEST FIRE STRIPPED
10 min BEFORE ignition • live config	18 / 65	23	−\$1,333.85	−\$57.99	13%	−\$1,684.85
10 min BEFORE • every suppressor removed	33 / 65	60	−\$1,620.40	−\$27.01	23%	−\$1,971.40
5 min AFTER ignition • live config	13 / 65	17	−\$45.00	−\$2.65	41%	−\$317.00
5 min AFTER • ungated	24 / 65	49	−\$626.75	−\$12.79	31%	−\$898.75

Every arm loses. Read the first two rows together: with their live filters on, the gates show up for 18 of the 65 runs and lose \$1,333.85; strip every filter off and they show up for 33 runs, take 60 fires, and lose \$1,620.40. **Removing the**

suppressors triples the activity and makes the result worse, which is the strongest defence of the gates' own filters this desk has produced.

★ **REV2 — THE NEARLY-FLAT ROW IS ONE GATE.** Split the post-ignition live arm by gate and abs_veto_short takes 5 fires for **+\$504.50 at a 100% win rate** (+\$97.00, +\$97.00, +\$97.00, +\$102.50, +\$111.00, on five distinct runs) while the other five gates take 12 fires for **−\$549.50**. Strip abs_veto_short's single best fire and the whole arm goes from −\$45.00 to **−\$156.00**. So “joining a run is roughly free” is not a property of the roster; it is one gate on five fires, and every other gate is as bad after ignition as before. **BUILD #6’s n≥60 kill criterion has to be PER GATE, not pooled** — pooled, five lucky abs_veto_short fires would carry a shadow arm that five other gates are losing money in. The original observation, kept because the asymmetry is real for that one gate: fired after ignition with live filters on, the gates lose only \$45.00 across 17 fires. Joining a run once it has already started is roughly free; predicting one costs money. That is the same shape the greenfield lab reports from a completely different direction in Movement 3.

3 · Gate by gate, with the base rate beside it

GATE	BEFORE IGNITION	AFTER IGNITION	WHOLE WEEK (BASE RATE)	\$ / FIRE, BASE RATE	RAW SIGNALS	WHAT ITS OWN FILTERS BLOCKED
grind_long	3 · \$86.00	6 · −\$254.92	11 · −\$319.17	−\$29.02	24,121	ATR floor 22pt ×24,110
capitulation_long	2 · −\$113.66	1 · \$97.00	10 · −\$97.66	−\$9.77	132	ATR floor 10pt ×122
abs_veto_long	3 · −\$169.58	3 · −\$173.36	26 · −\$83.42	−\$3.21	38	55s veto: absorbed ×5; 55s veto: thrust gone ×7
rgv_short	0 · \$0.00	1 · −\$17.54	1 · −\$17.54	−\$17.54	1	—
exhaustion_short	7 · −\$612.42	1 · −\$200.68	18 · −\$638.02	−\$35.45	46	counter-regime veto ×26; 5s confirm: went adverse ×2
abs_veto_short	8 · −\$524.19	5 · \$504.50	51 · −\$749.55	−\$14.70	67	55s veto: absorbed ×8; 55s veto: thrust gone ×8

Whole week, all six gates, every fire counted: **−\$1,905.36 on 117 fires**. That is the number to quote when anyone asks what the standdown was worth. And look at the raw-signal column, because it explains the shape of this entire report:

grind_long generated 24,121 raw signals this week and its own 22-point ATR floor blocked 24,110 of them. The gate is not benched by anybody's decision; it is benched by its own arithmetic.

4 • Which kinds of run, and did direction even matter?

CAUSE CLUSTER	RUNS	POINTS IN THEM	CEILING IF PERFECTLY CAUGHT	MECHANICAL FIRES	WHAT THOSE FIRES MADE
UNCLASS	39	2,754 pt	\$5,508.00	13	−\$782.45
OPEN/NEWS	11	1,029 pt	\$2,058.00	4	−\$266.40
VACUUM	9	797 pt	\$1,595.00	6	−\$285.00
FLOW-LED	6	414 pt	\$826.00	0	\$0.00

THE MECHANICAL FIRE WAS...	FIRES	NET	\$ / FIRE	WIN %
pointing the SAME way the run then went	11	−\$602.60	−\$54.78	9%
pointing the OPPOSITE way	12	−\$731.25	−\$60.94	17%

This is the part that closes the “just leave them armed” argument for good. The mechanical fires that happened to point the same way the run then went made −\$602.60 across 11 fires. **Being right about the direction was not enough.** The gates enter on their own triggers, which fire during the chop before the move rather than at the ignition, so an aligned fire still gets stopped out in the noise and is not there when the run happens.

★ **THE SURVIVORSHIP TRAP THIS LAB EXISTS TO AVOID — AND IT CUTS BOTH WAYS**

Last cycle's idle-gate lab reported a small positive and it was survivorship: the gates looked good because of which ones happened to be idle in the windows chosen. This week the base-rate control was built specifically to close that, and it does — but note that it closes the flattering reading AND the damning one.

In the run windows the gates make **−\$57.99 a fire**. Across the whole week they make **−\$16.29 a fire**. ★ **REV2 — Rev1 called those “close” and concluded there is nothing special about the tape near a big run. That is the one reading the numbers do not support, and it is wrong in the direction that makes the bench case STRONGER.** The gap is **3.56×** and the run windows are the worse side. They are also not independent samples: **21 of the 23 run-window fires sit inside the 117-fire base-rate set** (same gate, same second), so the base rate mostly contains the thing it is being compared against. Remove them and the rest of the week is **−\$4.49 a fire on 96 fires — a 12.9×** gap.

Bootstrapped over 20,000 resamples, the difference in \$/fire is **−\$41.71, 95% CI [−\$72.15, −\$2.52]** against the full base rate (98.0% of resamples put the run windows worse), and **95% CI [−\$82.41, −\$15.44]** against the disjoint remainder (99.6%). Both intervals exclude zero. So the sentence Rev1 explicitly ruled out is the true one: **these gates are substantially worse in the ten minutes before a big run than on ordinary tape.** They fire into the chop that precedes an ignition and get stopped in it — which is exactly what §4's direction table says from the other side. That is not a blur, it is the damning case, and it is the strongest form of the argument for leaving them benched.

The honest limit on the headline, stated plainly: the **−\$1,905.36** whole-week total has a bootstrap 95% CI on its per-fire mean of **[−\$35.30, +\$3.94]**, which scaled to 117 fires is roughly **[−\$4,130, +\$461]** — it straddles zero. The TOTAL is confidently negative only as a point estimate; what is statistically solid here is the comparison (run windows worse than base rate, both CIs excluding zero), not the size of the prize. Quote **−\$1,905.36** as the point estimate it is, never as a measured saving. So the conclusion is narrow and should not be over-read: these six gates, at these settings, on THIS week's tape, do not contain a run-catcher. It is one week, the desk was stood down for most of it, and a genuine trend week could look different. What it

does license is leaving them benched for now, which is what is already happening.

5 • What this section does NOT say

It does not say the sat-out ceiling — \$9,987.00 on this section's freeze, \$9,610.00 on Movement 1's — is unreachable — Movement 3 takes a different route at it and finds a size threshold that pays. It does not say the gates are worthless; two of them traded live this week and one of them was green. And it emphatically does not say “arm nothing forever”: it says that this particular six-gate roster, fired mechanically, is not the tool for catching runs, and that the tool will have to be built rather than switched on.

Disposition — every lead this section touched

LEAD	VERDICT	FOR PARKED: WHAT WOULD REVIVE IT. FOR REFUTED: THE TEST THAT KILLED IT.
Leave the six live gates armed to catch big runs	REFUTED	Named test: the week's sat-out runs, four arms, plus a whole-week base-rate control. Every arm negative on the freeze it was run on; live config $-\$1,905.36$ across the week (census-independent, unchanged) and $-\$1,333.85$ in the run windows on the 08-22 freeze, $-\$976.71$ re-cut onto the 08-25 one.
Strip the gates' filters so they show up more often	REFUTED	Ungated triples the fire count and makes the result worse in both windows. The ATR floors and vetoes are earning their keep.
Join a run AFTER ignition rather than predicting it	SHADOW	The post-ignition live arm is nearly flat at $-\$45.00$ on 17 fires on the 08-22 freeze, and $+\$124.64$ on 15 fires re-cut onto the 08-25 one — a pooled figure that flips sign on a re-cut of the run list, which is the strongest possible argument for the per-gate rule below. Against a clearly negative pre-ignition arm either way. That is the only asymmetry in the whole lab. Shadow an ignition-confirmed entry and let it accumulate n. ★ REV2: per-gate. The $-\$45.00$ is abs_veto_short's $+\$504.50$ on 5 fires against $-\$549.50$ on the other five gates' 12; strip one fire and the arm is $-\$156.00$. Score and kill each gate on its own n, never on the pool.
Direction alignment as a permission rule	REFUTED	Aligned mechanical fires made $-\$602.60$ on 11 fires. Knowing which way the run would go was not enough, because the gates fire before the ignition and get stopped in the chop.
The 22-point ATR floor on grind_long	PARKED	It blocked 24,110 of 24,121 raw signals this week. Revive with the floor ladder

M3 The greenfield lab — seven hunts, every grave, and the two findings that outrank all of them

This is the cold half of the report at its coldest. We ignore everything the desk did, take the census's **62 sat-out runs and \$9,610 of hindsight ceiling**, and try to build something from scratch that could have been in them. Seven separate hunts ran this cycle — one pooled rider across every run, one per cause-cluster, the chop-day scalp you commissioned, and gold — and between them they wrote **54,030 words** of tick-honest working, every one of which is reproduced in full in the appendix at the end of this movement.

★ **The headline is a null, and I am going to say it in the first paragraph rather than the last: nothing from this movement is ready to trade on MNQ.** Not one candidate out of seven hunts survived its own robustness battery. That is not a bad week — it is the batteries working, and three of the kills came from tests this desk only started running recently. What the movement did produce is worth more than another thin gate: two findings that reframe the whole hunt, four corrections to the census's own vocabulary, the first honest grade of the two arms we shipped last week (both dead), and seven harness bugs caught before they reached you rather than after.

1. The board — seven hunts, seven verdicts, in one table

The best thing each hunt built, the number it reached before the battery, and the **named test** that killed it. Nothing on this table died of a hunch; every cause of death is a test you can re-run.

HUNT	WHAT IT BUILT	BEST NUMBER REACHED	THE TEST THAT KILLED IT	VERDICT	SOURCE
Pooled rider	board_atrband — late board, 3.0×ATR stop, 6.0×ATR target, ATR band 11-16	passed every in-sample test, both placebos	the out-of-sample leg — 1,289 trades over 236 untouched sessions for +\$14 total ; 95% CI −0.204R to +0.246R	PARKED	RIDER_ALL \$4, \$10
UNCLASS	UNCL-RIDE-ER — late board + ER30 ≥ 0.40, 10×ATR target	n=196, +\$6,211, + \$31.69/tr, PF 1.64	the placebo/shuffle test — on 205 untouched sessions the real signal makes + \$3.54/tr and the SAME rule taken 30 minutes early makes +\$4.44	PARKED	UNCLASS \$6, \$13
OPEN/NEWS	MOMBRK (+\$32.87/tr, n=52) and PULSE-13h (+\$23.09/tr, n=95)	the two biggest per-trade numbers in the movement	strip-the-best and the placebo — MOMBRK's top 3 of 52 trades are 88.5% of its net; PULSE's 30-minute-stale copy books +\$23.95/tr, more than the real signal . And a gate containing no signal at all books +\$47.29/tr in the same window	REFUTED	OPEN/NEWS \$9, \$15
FLOW-LED	nine candidates, FL-1 ... FL-9	FL-4 flow_ignition reached +\$12.18/tr	the placebo/shuffle test — p = 0.150, 6 of 40 time-shifted fake flow series matched or beat it, and the best fake (+\$16.25/tr) beat the real signal outright	REFUTED	FLOW-LED \$4, VERDICT
VACUUM	VAC-SNAP, VAC-CUMDIV, VAC-ABS, VAC-IGN-BRK	VAC-IGN-BRK +\$911.55 over n=284	the sign-flip control — fading the break earns +\$3.08/tr against the signal's +\$3.21, so both directions win. That is volatility harvesting, not edge	REFUTED	VACUUM \$3, VERDICT
Chop scalp	sixteen named candidates, 768 costed cells + a 1,080-cell finalist search	CT15 +\$158 on 26 trades, +\$6.08/tr, 77% win, IS and OOS positive, 13/13 leave-one-day-out, strip-best-5 still +\$4.07	the placebo run WITH THE SEARCH INCLUDED — handed the same 1,080-cell grid, the null finds a better cell 85.3% of the time; matched fairly at n ≥ 26, p = 0.367	REFUTED	chopscalp \$9, VERDICT
Gold (MGC)	mgc_hole_break_fade ×2 sides + double-hole;	double-hole arm n=200, +\$3,304, + \$16.52/tr, forward + \$656, placebo 0/30	nothing killed the research — the LIVE arm is losing on a different tape . The lab builds bars from the	SHADOW	MGC \$3, \$13

mgc_wall_break_go
x2 sides

depth mid, the service
builds them from
trades, and the sign
flips on that alone (+
\$3.06/tr vs -\$8.96/tr
on the six days both
cover)

2. The two findings that outrank every gate on that table

★ FINDING ONE — the tradeable threshold is about 120 points, and three labs reached it independently

This is the most durable result in the movement, and the reason it is durable is that nobody was looking for it. Three hunts, three different clusters, three different harnesses, three different authors — and the same number falls out of all of them.

LAB	HOW IT WAS MEASURED	THE NUMBER
OPEN/NEWS	expectancy split by the run's eventual 30-minute follow-through	+60...+120pt of follow-through still books — \$4.10/trade . Above 120pt: +\$184.75/trade . Break-even hit rate 19.2%; the best arm delivered 30.8%
VACUUM	detector AUC against run size, lake-wide	below move ≈ 87pt three of four detectors are indistinguishable from noise; at and above it all four go significant — volume-z AUC 0.730, $p = 0.0004$, rising to AUC 0.867 on ATR for the 113 runs of 120pt+
the census's own greenfield work	a board rule conditioned on expected size	boarding only fires whose move is ≥120pt pays +\$69.38/trade and passes the whole battery

What it means in plain English. The desk has been asking the wrong question. We keep hunting for a signal that says a run is starting, and we can already do that — the pooled rider gets inside **62 of 62** of the week's runs. What we cannot do is tell a 60-point run from a 160-point one at the moment we board, and below about 120 points a late board does not pay for its own stop. **Hunt SIZE, not direction.** That is a different study from any of the seven run this week, and it is the one I would commission next.

★ FINDING TWO — detection is solved; selection is not; and the ceiling is not what the census says it is

Every hunt that boarded a run boarded it successfully. The pooled rider fires inside **62 of 62** sat-out runs at a median **three minutes** late with a median **41 points still ahead**. UNCLASS's rider gets inside 40 of its 41 at a median 3.3 minutes with 2.57 ATRs left. OPEN/NEWS separates ≥ 200 pt runs from ordinary tape at **AUC 0.848** on trade-count alone. Detection is not the problem and has not been for some time.

The problem is that the same trigger fires **34 times a day**, and roughly one fire in twenty becomes a run. Direction carries almost nothing: AUC 0.55–0.65 at best, and a crude momentum rule agrees with the run 54.2% of the time against a 53.4% base rate. With-trend does beat counter-trend in every one of three out-of-sample exit configurations, by +0.06 to +0.11R — real, and about the size of the round trip.

★ And the \$9,610 is not \$9,610. This correction should follow the census number everywhere it is quoted from now on:

VERSION OF THE CEILING	MNQ, THIS WEEK	% OF THE HEADLINE
the census's hindsight number (perfect entry AND perfect exit)	\$9,610	100%
a perfect late boarder — fires inside every run at a median three minutes in, exits at the exact top, zero costs	\$5,094	53%
a deployable late boarder with a real stop and real costs	\$1,910–\$2,286	20–24%
runs already finished before any honest trigger can see them	13% of them	—

UNCLASS reaches the same shape on its own bucket: a \$5,840 ceiling becomes \$3,239 (55%) for a perfect exit and **\$1,592 (27%)** for an honest late board with a real stop. So the honest size of the prize is roughly a fifth to a quarter of the number on the front of Movement 1 — which is still real money, and is a very different target to aim a year of work at.

3. Closing the loop — the two arms we shipped last week, graded for the first time

Last week's greenfield work produced two things that went into the shadow book. Nothing had graded either of them. Both hunts went back and did it, unprompted, and **both arms are dead** — one of them was never actually shipped, which is its own finding.

ARM	WHAT IT WAS QUOTED AT	WHAT IT ACTUALLY DID	VERDICT
rider_w5 shipped into shadow.db 2026-08-15 off last week's RIDER_ALL phase	+\$4,841, +\$19.29 a trade	51 forward trades since 08-17: -\$1,471, -\$28.84 a trade , 18% wins, red on 6 of 7 days and red in all four ATR buckets — worst in the 11-16 band its own filter selects for. Zero flagged rows: this is clean data, not a measurement artefact. Its in-sample +\$1,936 collapses to +\$282 on strip-best-3, and the +\$19.29 it shipped on never had a strip test attached.	REFUTED retire the arm
UNCL-RIDER-ASIA last week's SHADOW pick out of the UNCLASS phase	a promotion candidate	Four consecutive red forward weeks: W31 -\$2.93, W32 -\$15.90, W33 -\$36.46, W34 -\$38.08 — and the ATR floor added to explain the first two restored none of them. ★ Then the deeper finding: there is no arm by that name anywhere in shadow.db. All 82 strategies in the book were listed. It was never shipped, and nobody noticed for a week.	REFUTED do not re-shadow

Read those two rows together, because they are the same lesson twice. A number that goes into the shadow book without a strip test and without anybody owning the grading is not evidence — it is a claim with a week's head start. Both arms were quoted from in-sample headlines; both are now refuted by forward data that cost nothing to collect and that nobody was collecting. **The fix is not analytical, it is procedural: every arm shipped needs a named grading date, and every shipped number needs its strip-best figure printed beside it.**

4. The census's own vocabulary did not survive the week

The brief tells every cluster hunt to **interrogate the label before building anything on top of it**. Four did. Three of the five labels failed, and the failures are independent measurements by different authors on different tape. This

matters more than any single gate, because every hunt this desk scopes by cluster inherits the error.

LABEL	WHAT IT IS SUPPOSED TO MEAN	WHAT IT ACTUALLY IS	DO THIS
UNCLASS 41 of 68 runs — the biggest bucket	no clear tape tell (quiet ignition)	Base rate 73.9% on the census week and 75.4% on the whole lake, lift 0.82 — i.e. a run is less likely to be UNCLASS than an arbitrary minute is. Its members sit ON the all-tape median for ATR, volume and efficiency.	Rename it UNTESTED and report it as coverage , not as a finding. Second independent measurement in two weeks.
VACUUM / FLOW-LED 9 and 4 runs	moved against the tape / aggressors drove it	One population cut by its own outcome. The flow event fires on 20.4-20.6% of every tape-minute we own , then splits it VACUUM 10.37% / FLOW-LED 10.21% on a 49.6-49.9% coin toss — 0.33 percentage points of directional information. The census's apparent 15.7%-vs-10.2% excess is entirely a selection artefact of the keep-strongest run-picker (prior-5min agreement 48.66% unselected → 22.55% deduped). Whatever lift the flow event carries is a MAGNITUDE lift, and that lift is a volume proxy: flow-unusual with volume-ordinary gives 0.963× base (no lift at all); volume-unusual with flow-balanced gives 1.220×.	Stop splitting the census on flow. This week's 4-vs-9 split is noise (expected 9.0 vs 9.6; $P(\leq 4 \text{ of } 13) = 0.139$).
OPEN/NEWS 13 runs	the 13:00-15:00Z cash-open / data window, footprint-confirmed	A clock with an anti-predictive filter bolted on. The 08-15 fix correctly demoted it from checked-first to checked-last (100% → 58.7% of its own window) — but its lift over the bare clock is 0.921× . The flow and amplitude tests remove 41% of the window and keep the less run-dense half. Same answer in the tick era (0.946×).	Either drop the two footprint tests and call it US-OPEN-WINDOW (3.09× lift on 8.87% of tape), or route the hunt through VOL-EXPANSION — 4.18× lift on 1.95% of tape , the sharpest cluster the census produces and the one it uses least.
the bucket phase itself	a stable assignment	21 of 68 labels change on bucket phase alone , and only 17 of 68 run timestamps are minute-aligned. The window slides on the array index rather than on the clock.	A real bug. Fix it before the next census.

What replaces them is already written: UNCLASS's hunt proposes **four run SHAPES** — TURN, STEP-REPRICE, DRIFT-VOID and GRIND-CONT — each one if statement, all computable at the run's start minute. That is vocabulary, not a selector, and it is honest about being vocabulary.

5. The seven hunts, one card each

Each card is the shortest honest account of a hunt; the full dossier for every one of them is reproduced in the appendix at the end of this movement, sweep tables and all.

THE POOLED RUN-CATCHER — one rider for every sat-out run

Ignore the labels. Board any run late, ride it wide, filter it causally. ·
gf_full_RIDER_ALL.md, 8,856 words, written 2026-08-25 22:10 UTC ·
[read it in full →](#)

What it did. Built the direction-agnostic run-catcher, boarded it late, swept the exit wider than this desk has ever run one, filtered it with a causal rule, and put the survivor through the whole battery. It passed every in-sample test we have, including two separate placebo controls.

Then it died at Step 3, the causal filter, and the test that killed it was the out-of-sample leg: 1,289 trades over 236 untouched sessions for a total of +\$14. Not a loss. Not a win. Nothing — the 95% confidence interval on its expectancy runs from $-0.204R$ to $+0.246R$, which is measuring zero with a wide ruler.

The second witness is the one to trust. rider_w5, the arm this phase shipped last week at a quoted +\$19.29 a trade, has taken 51 live forward trades and sits at $-\$28.84$ a trade, red in every ATR bucket. Two independent measurements, same answer. ★ One thing survived to incubation: board_overnight (21:00–07:00Z) keeps its sign in two independent samples and survives a full point of slippage — but its daytime and counter-trend controls reverse on the held-out leg, so it is shadowed as a question, not as a candidate.

UNCLASS — the biggest bucket, 41 of the 68 runs

The bucket our own classifier could not name, and had therefore never hunted. · gf_full_UNCLASS.md, 9,219 words, written 2026-08-25 22:34 UTC · [read it in full →](#)

What it did. The biggest bucket — 41 of the 68 runs — and one that had never been hunted, because our own classifier could not name it. It was BUILT (a late boarder that gets inside 40 of the 41 runs at a median 3.3 minutes late with 2.57 ATRs still ahead), RIDDEN (exit swept stop 1.0–4.0×ATR against target 2–16×ATR), FILTERED (eight causal cuts, each placebo-controlled against discarding the same NUMBER of fires at random) and ROUTED — the operator's order of work, in the operator's order.

And it still lost. Killed by the placebo/shuffle test and corroborated by the out-of-sample leg: on 205 untouched sessions the real signal makes +\$3.54 a trade and the same rule taken thirty minutes early makes +\$4.44. The family kill is stronger than the single cell — across 137 configurations with usable n on both independent legs, 13% are positive on both against a 12% chance rate.

★ **Two keepers.** The exit sweep found a genuine **10×ATR target ridge that holds across every stop width** — so the operator's wide-exit thesis is CONFIRMED on the target axis and REFUTED on the stop axis; the stop is not what shakes us out, because we are not being shaken out. And **benching 07:00-13:00Z for any late-boarding rider replicates on all three independent legs**, which makes it the cleanest positive deliverable in the movement.

OPEN/NEWS — the 13:00-15:00Z cash-open window

The densest run window on the clock: 13 runs, \$2,814 of ceiling. ·
gf_full_OPEN-NEWS.md, 10,349 words, written 2026-08-25 22:58 UTC ·
[read it in full →](#)

What it did. Hunted the densest run window on the clock — and then built the control that killed the whole page.

The control is the finding. A gate containing no signal at all — “enter at 13:05Z, direction = the sign of the last 15 minutes, stop $0.5 \times U$, hold 30 minutes” — books **+\$47.29 a trade over 26 trades**, double the best gate the phase built. The identical rule twenty-five minutes later books $-\$13.46$. Everything on the page sits inside the noise band of that control, and the confidence intervals agree: $t = 1.30$ and 1.57 , and **561 trades $\approx$ 79 trading days** would be needed to resolve a \$10/trade edge in this window. We have 27.

The two leaders died by name. MOMBRK ($+\$32.87/\text{tr}$, $n=52$) on strip-the-best — its top 3 trades are 88.5% of the net — and on an OOS leg of $-\$16.50/\text{tr}$. PULSE-13h ($+\$23.09/\text{tr}$, $n=95$) on the placebo: a copy of the signal delayed thirty minutes books **more money than the real one**.

FLOW-LED — aggressor-driven continuation

Runs the tape pushed. Nine candidates built off the flow print. ·
gf_full_FLOW-LED.md, 4,186 words, written 2026-08-25 23:12 UTC · [read it in full →](#)

What it did. Nine candidates off the flow print, every one of them dead, each with a different named cause — the blanket screen took four, the OOS leg took one, strip-the-3-best took two, the parameter sweep took one, and the last survivor FL-4 `flow_ignition` died on the placebo/shuffle test at p

= 0.150, with the best of forty time-shifted fake flow series beating the real signal outright.

And there is no size at which it starts working. On the predictor side every flow-z bar from 1.0 to 4.0 is positive in-sample and negative out of sample. On the outcome side, flow agreement with the run runs 48.4–53.7% from 55pt to 200pt and never departs from a coin flip by as much as one sigma.

★ **The finding that outranks all of it: the label is not a footprint.** The flow column measures how much traded, not who was buying — flow unusual with volume ordinary gives $0.963\times$ base (no lift); volume unusual with flow balanced gives $1.220\times$. The census should stop splitting runs into FLOW-LED and VACUUM; they are one population cut by its own outcome.

VACUUM — the snap-back / stop-run

Runs that moved AGAINST the tape, plus last week's shadow candidate forward-tested. · gf_full_VACUUM.md, 5,224 words, written 2026-08-22 00:18 UTC · [read it in full →](#)

★ **REV3 — on the OLD census freeze (70/65/\$9,986), not the 08-25 re-freeze Movement 1 carries. Its verdicts do not depend on the run list; its per-run counts do. See the banner on the full dossier.**

What it did. Four candidates invented, four graves, and — for the second Friday running — the cluster was killed by its own LABEL rather than by a backtest. This week the mechanism is identified rather than just the symptom: the census's apparent VACUUM excess (15.7% of runs against a 10.2% base) is **entirely a selection artefact of the keep-strongest run-picker.**

The kill that generalises is the sign-flip control. VAC - IGN - BRK looked like the phase's survivor at +\$911.55 over 284 trades — until fading the break was found to earn +\$3.08 a trade against the signal's +\$3.21. Both

directions win. That is volatility harvesting with a stop on it, not an edge, and no amount of n fixes it.

★ **Two keepers.** Last week's open stone is closed — the L2 far-side-depletion discriminator is REFUTED at 49.44% ± 2.52 pp over 246.9M book rows, and it is not n -limited, so do not revive it at 30 days. And the detection threshold is real: below $|\text{move}| \approx 87$ pt three of four detectors are noise; at and above it, all four go significant.

THE CHOP-DAY TURN SCALP — the operator's own commission, L2-led

“Can we stop donating on chop days?” — a purpose-built L2 turn scalper. · [gf_chopscalp.md](#), 5,750 words, written 2026-08-25 23:39 UTC · [read it in full →](#)

What it did. The operator's own commission — a purpose-built chop-day turn scalper, led by the order book, same lineage as `exhaustion_short`. Raced tick by tick over 17 full-stack days: 20.7M trade ticks, **261.9M forty-one-millisecond ten-deep book rows**, 4.6M depth snapshots, 26.2M quote updates, sixteen named candidates and 1,848 costed parameter cells.

The lead hypothesis is refuted and it points the wrong way. Requiring the far side to be ABSORBING — a big resting wall the price must eat to continue — does not help the fade, it hurts it, in two independent tests. Unfiltered chop turns drift +0.054pt in the fade's favour; add “wall $\geq 1.5\times$ ” and that becomes -0.117 pt; five-deep $\geq 1.3\times$ and it becomes -0.374 pt.

The wall is not absorption at a range edge. It is a target.

The finalist died to a test we should run more often. CT15 passed everything — in-sample, out-of-sample, 13/13 leave-one-day-out, strip-best-5 — and on a single pre-registered cell its shuffle p is 0.017. But that cell came out of a 1,080-cell search, so the null was handed the same search: the null's own best cell averages +\$8.42 a trade and beats the real one **85.3% of the time.**

★ **And the premise needed correcting, which is worth more than the scalper.** The desk did not donate on this week's chop days — the tournament took **zero** trades on 08-20 and 08-21. The week's $-\$2,323$ is `day_rider`'s book. Across the whole 18-day window the tournament's loss is concentrated in **violent-whipsaw** ($-\$1,383.50$ on 95 trades), not plain chop ($-\$403.00$ on 58). That is a router bench to price, not a gate to build.

GOLD — MGC, and the shadow arm that went live and lost

The second instrument, \$10.00 a point, and the first live grade of last week's gold gate. · `gf_MGC.md`, 10,446 words, written 2026-08-25 21:36 UTC · [read it in full →](#)

What it did. Graded last week's live gold shadow, found it losing, and ran the contradiction to ground — then opened thirteen months of gold tape and re-tested everything on it.

The headline is a plumbing fault, not a research failure. The lab builds its minute bars from the depth mid; the live service builds them from trades; gold prints sparsely, so those are different series and the sign of the gate flips between them. Full working and the numbers are in §8 above.

★ **Three more.** A year of tape (320 trading days) arrived and killed last week's router rule outright. The unturned stone — gold against the Nasdaq — paid, as a veto, on 278 paired days. And the author caught a look-ahead in their own harness that read $+\$71,608$ and corrected to $-\$12,462$.

6. Every grave, by name, with the test that killed it

The operator's standing rule is that the failures are the deliverable, so here they all are — **33 named candidates and mechanisms**, each with the specific test that killed it. A verdict with no visible working is worthless, and a grave with no named cause of death is a hunch.

CANDIDATE	HUNT	KILLED BY	THE NUMBER	DISPOSITION
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board_atrband	pooled rider	the OOS leg	+\$14 over 1,289 trades / 236 sessions	PARKED
board_wide / board_holdcap / board_blanket	pooled rider	both-halves and the OOS sweep	failed as built; the 42-cell OOS surface shows the exit was never the binding constraint	PARKED
board_chandelier	pooled rider	a backtest bug, then honesty	the trail was broken; + \$4,270 became −\$631 once fixed	PARKED, not refuted
board_book (L2 far-side depletion as selector)	pooled rider	correlation + placebo	311 fires against 264M book rows, $r = -0.009$, no monotone band, placebo $p = 0.62$	REFUTED
rider_w5 (LIVE shadow arm, shipped 08-15)	pooled rider	51 live forward trades	−\$1,471, −\$28.84/tr against +\$19.29 claimed; red in all four ATR buckets	REFUTED — retire
UNCL-RIDE-ER	UNCLASS	placebo/shuffle + OOS	the 30-minute-early fake beats the real signal on 205 untouched sessions	PARKED
UNCL-RIDER-ASIA (last week's SHADOW pick)	UNCLASS	the forward OOS leg	W31 −\$2.93, W32 −\$15.90, W33 −\$36.46, W34 −\$38.08 — four red weeks	REFUTED
the thin-volume / no-aggression mechanism	UNCLASS	the random-cut placebo	$p = 0.600$ — the cut is worse than discarding the same number of fires by coin flip	REFUTED
the tight / trailing exit family	UNCLASS	the 1,350-1,683-trade year leg	negative throughout; the $10 \times \text{ATR}$ target ridge holds across every stop width	REFUTED for this shape
the “arm US 13–20Z” rule	UNCLASS	the OOS leg	+\$107.79/tr in-sample ($\text{placebo } p < 0.001$), −\$18.62/tr out of sample	REFUTED before it shipped
MOMBRK	OPEN/NEWS	strip-the-best + OOS	top 3 of 52 trades = 88.5% of net; +\$32.87/tr → +\$4.02 stripped; −\$16.50/tr OOS	REFUTED
PULSE-13h	OPEN/NEWS	the placebo	the 30-minute-stale copy books +\$23.95/tr against the real signal's +\$23.09	REFUTED
ORB30 — pre-open range break	OPEN/NEWS	strip-3	−\$26.98/tr after strip-3	REFUTED
COILBRK — compression break	OPEN/NEWS	n	one trade at the setting the theory wanted	PARKED
FADE1 — the exact mirror of PULSE	OPEN/NEWS	expectancy	−\$13.05/tr	REFUTED
FL-1 flow_impulse_cont	FLOW-LED	the OOS leg	did not survive the held-out sessions	REFUTED
FL-4 flow_ignition	FLOW-LED	the placebo/shuffle test	$p = 0.150$; best fake + \$16.25/tr beats the real + \$12.18/tr; ablation +\$39.43 IS → −\$11.71 OOS	REFUTED
FL-5 flow_absorb_fade	FLOW-LED	strip-the-3-best	the headline is three trades	REFUTED
FL-8 flow_x_vol_cont	FLOW-LED	placebo AND strip-3	beaten outright by its own placebo	REFUTED
FL-9 flow_lot	FLOW-LED	the parameter sweep	no plateau — a spike, which is the curve-fit tell	REFUTED
FL-2, FL-3, FL-6, FL-7	FLOW-LED	the blanket screen	negative expectancy; no regime cell rescued any of them	REFUTED
VAC-SNAP	VACUUM	strip-the-3-best + OOS		REFUTED

+\$370.64 (n=410) becomes
 −\$1,436 stripped; −\$18.02/
 tr out of sample

VAC-CUMDIV	VACUUM	every threshold	negative at all of them	REFUTED
VAC-ABS (last week's shadow candidate)	VACUUM	the OOS leg	+\$1,319.21 / n=97 / + \$13.60 per trade forward, then −\$529.50 over 27 unseen trades; strip-5 − \$427	REFUTED — retire, do not re-shadow
VAC-IGN-BRK	VACUUM	the sign-flip control	the fade earns +\$3.08/tr against the signal's +\$3.21 — both directions win	REFUTED as a gate
VAC-Z1 (the label, traded literally)	VACUUM	expectancy at scale	−\$4,150.79 over 1,768 serial trades; −\$32,184 unconstrained	REFUTED
CT15 (the chop-turn finalist)	chop scalp	the placebo run WITH the search	the null's own best cell averages +\$8.42/tr and wins 85.3% of the time; matched p = 0.367	REFUTED
CT2_WALL (the operator's lead L2 hypothesis)	chop scalp	two independent drift tests	worst of sixteen at −\$3.09/tr; the wall INVERTS the fade's +0.054pt drift to −0.117pt	REFUTED, sign inverted
the VWAP-flat filter	chop scalp	48 cells	\$0.01/tr — inert, doing no work at all	REFUTED
mgc_straddle / breakout-either-way	gold	320 days	all 8 compression cells negative, control negative, dies on one tick of slippage	REFUTED
last week's MGC router rule	gold	the forward walk AND the 320-day tape, independently	it would bench the gate in NORMAL_CHOP — the regime currently paying best	REFUTED as a rule
cross-asset momentum confirm (gold with the Nasdaq)	gold	278 paired days	−\$2.29/tr, and −\$9.15/tr on the book window; 1st percentile of its own placebo	REFUTED
the bare 60-min break trigger, standalone	gold	strip-best-10 and slippage	direction is real on 320 days; the P&L is not (−\$4,392 stripped, dies at 0.10pt)	PARKED

★ **And note what is NOT on that list: nothing died of thin n alone.** Thin n is a SHADOW or a PARK, never a refutation, and the standing scope says so. Every REFUTED row above carries a named test on a sample big enough to carry it — a placebo, a sign-flip control, a strip-the-best, or an out-of-sample leg.

7. Seven things the labs caught in their own instruments

Every one of these was found by the lab that would have benefited from not finding it, and reported anyway. Read this section as the reason to trust the nulls above.

WHAT	WHERE	WHAT HAPPENED	WHY IT MATTERS
The trail that never trailed	RIDER_ALL \$3	The chandelier family survived every robustness test in the phase — until the trail was found to be	It would have shipped. Every test we own passed it, because every test was reading the same wrong simulator.

		broken. Corrected, +\$4,270 becomes -\$631.	
A look-ahead of my own	MGC \$7	A volatility cell read +\$71,608 until it was noticed that it entered at the OPEN of the bar whose HIGH it had already used. Corrected, it reads -\$12,462.	That number would otherwise have been the largest in this report.
Five shorts into one impulse	VACUUM \$3	The first pass let overlapping entries run concurrently. Five shorts into the single 2026-07-31 impulse were worth \$3.9k of a \$10.3k headline. Scored one position at a time — the only way a live gate can trade — VAC-IGN-BRK falls from + \$10,316 / +\$15.49 per trade to +\$911.55 / +\$3.21.	Concurrency is not a backtest convenience; it is leverage the desk cannot take.
A slice is not a filter	UNCLASS \$3	Testing a filter by SLICING the trade list of an unfiltered rider gives + \$28.33/tr. Running the identical filter INSIDE the simulation — where refusing a fire frees you to take the next one — gives +\$5.68/tr, and n doubles.	Wrong by a factor of five, and slicing is the natural way to test a filter.
A backward shift is an oracle, not a placebo	OPEN/NEWS \$15	A signal shifted BACKWARDS in time is a look-ahead oracle. Only FORWARD (stale) shifts are legitimate placebos. Run the wrong way, it produces a false KILL.	Reported because it was run the wrong way first. The harness passed the leak-check: look-ahead earns 2.7x the real signal, as it should.
Leave-one-day-out cannot fail when k days carry the money	OPEN/NEWS \$15	MOMBRK passed 21 of 21 LODO folds while two of twenty-one days held 97% of its P&L.	Always quote profitable-days-out-of-total beside a LODO result, and run leave-TWO-out when the day distribution is that skewed.
The lake grew a look-ahead landmine	MGC \$1.2	New backfill_{1min,5mins, 1hour,1day}.parquet files now sit beside the daily 5s files. Any query without a timeframe filter inherits up to a full day of look-ahead, on every day.	gf_mgc_tape.load_l1() is safe. mgc_session_anchor.load() is NOT. This is a FIX REQUIRED, not a note.

8. Gold, in its own right — and the week's single most expensive lesson

Last Friday we shadowed a gold gate called `mgc_hole_break_fade`. It has been running live in the shadow book since Monday and **it is losing money: -\$1,196 across 80 trades in seven days, every one of its three arms red.** Replaying the same seven days in the lab gives **+\$512.** That is not a small disagreement, and the answer turned out to be one fixable thing.

★★★ THE GATE IS BEING FED THE WRONG TAPE

The research builds its one-minute bars from the **depth mid** — the midpoint of the order book, which updates every 250 milliseconds whether or not

anything trades. The live service builds its bars from **trades**. Gold prints sparsely, so those are not the same series, and a sixty-minute high taken from one is not the sixty-minute high of the other. Hold absolutely everything else constant — same break rule, same book filter, same exit, same spread-honest race — and change only the input tape:

THE SAME RULE, EXIT AND RACE, ON THE SIX FORWARD DAYS BOTH TAPES COVER			
	N	NET	\$/TRADE
depth-mid bars (what the research measured)	56	+\$172	+\$3.06
trade bars (what the live service actually reads)	53	-\$475	-\$8.96

The sign flips on the bar source alone, and the trade-bar number is the one the live shadow independently produced. So this is not a mystery and it is not a research failure — it is a plumbing mismatch between the lab and production, and the fix is mechanical: the service already holds a DepthFeed connection to gold's book (it needs it for the liquidity filter anyway), so it already has the bid and the ask. It just needs to build its minute bar from their midpoint instead of from trades. **Until that is done I would not promote a single gold cell, and I am not proposing one.**

The per-cell verdicts — MGC, \$10.00 a point, \$1.50 the round trip

CELL	GATE	N (29D)	NET	\$/TRADE	FORWARD 7D	PLACEBO	LIVE SHADOW	VERDICT	NOTE
REVERSION -LONG	mgc_hole_break_fade — 60-min break DOWN into a liquidity hole → buy	134	+\$1,853	+\$13.83	+\$488 (+ \$15.73)	0/30 ✓	-\$102.50 (n=7)	SHADOW	lead candidate, blocked on the bar-source fix
REVERSION -SHORT	mgc_hole_break_fade — 60-min break UP into a liquidity hole → sell	154	+\$1,390	+\$9.03	+\$25 (+ \$0.65)	0/30 ✓	-\$153.50 (n=7)	SHADOW	same, plus a side-decay watch
MOMENTUM M-LONG	mgc_wall_break_go — break UP through a defended level → buy	101	+\$838	+\$8.30	+\$378 (+ \$9.44)	29/30 x	not deployed	SHADOW, on probation	2nd week failing the random-minute placebo

MOMENTUM-SHORT	mgc_wall_break_go — break DOWN through a defended level → sell	88	+\$669	+\$7.61	+\$315 (+ \$9.56)	29/30 ✗	not deployed	SHADOW, on probation	same
double-hole arm	mgc_hole_break_fade_db1 — both reversion cells	200	+\$3,304	+\$16.52	+\$656 (+ \$15.26)	0/30 ✓	not deployed	SHADOW	promote to a full arm — behind the bar-source fix
volatility cell	mgc_straddle — breakout either way out of a coil	3,945	-\$1,248	-\$0.32	—	control also negative	—	REFUTED	on 320 days. Last week's reason was wrong; the verdict was right

The exit family every gold cell needs is the wide chandelier on a single lot — stop 3.0 ATR, arm 2.0, trail 2.0, 480-minute backstop. Not the tight scalp (negative at 1R on both entries). Not the dual slot (Lot A is negative on gold and cancels Lot B, on both entries). Not a time cap (its headline numbers are a long-drift artefact). ★ **That is the exact opposite of the live MNQ slot configuration, and it replicates on 29 days.** Gold is not the Nasdaq with a different multiplier, and the standing rule that MGC gates must be invented fresh rather than ported is earning its keep.

Three more things gold produced this week

- A year of tape arrived.** The lake gained **320 trading days** of MGC minute bars (2025-07-28 → 2026-08-19) against last week's twenty-nine. It kills last week's router rule outright — that rule would bench the gate in NORMAL_CHOP, which is the regime currently paying best — closes last week's one loose thread, and puts a much colder light on the bare trigger (direction is real on 320 days; the P&L is not).
- The unturned stone paid, and it is a VETO not a confirm.** Gold against the Nasdaq on the same clock, over **278 paired days**: fading a gold break that the Nasdaq agrees with loses **\$10.52 a trade** and sits at the 2.5th percentile of its own placebo. As an arm it reads +\$5.43/tr against +\$1.72 blanket. It does not reproduce on the 29-day book window (34th percentile), so it is a shadow arm with a question attached, not a promotion.

The mirror — a cross-asset momentum confirm — is cleanly negative (–\$2.29/tr on the same 278 days), which is what a real asymmetry looks like.

3. A look-ahead in the lab's own harness, caught before it reached you. A volatility cell read **+\$71,608** until the author noticed it was entering at the open of the bar whose high it had already used. Corrected: **–\$12,462**. That number would otherwise have been the biggest in this report.

9. What this movement asks for on Monday

Six items. **None of them is “arm a gate”**, and that is the honest output of seven hunts. Four are builds or fixes, one is a retirement, and one is a measurement.

DO THIS	WHY	SOURCE
1. Retire rider_w5 from the shadow book.	It is the only item here that costs nothing and is decided. 51 forward trades, –\$28.84 each, red in every ATR bucket including the one its own filter selects for. It has been running unwatched since 08-17 because nobody graded it.	RIDER_ALL \$5
2. Build the MGC depth-mid bar source, then let both gold arms run.	This is the gating item for the entire gold programme and it is mechanical — the shadow service already holds the DepthFeed it needs. Keep the trade-bar arms alive as *_tradebar controls so the next reader can see the difference rather than take it on trust. ★ REV3 gives it a target and a date: the A/B says depth-mid +\$3.06/tr against trade-bar –\$8.96/tr on six days, so the depth-mid arm passes if it is positive over its next 60 forward trades and the *_tradebar control is at least \$6/trade worse over the same stamps — if the control does NOT diverge, the bar source was never the explanation and the whole line closes. At the arms' current ~9 fires a day that is roughly seven trading days, so the answer is due in the Friday report of 2026-09-04 , and it is a promotion decision on that date rather than a rolling watch.	MGC \$12.1
3. Fix the census's flow-bucket phase, and rename two labels.	21 of 68 labels move on bucket phase alone. Then UNCLASS → UNTESTED and OPEN/NEWS → US-OPEN-WINDOW with the two anti-predictive footprint tests deleted. Three labs reached this independently and it is the cheapest correction on the list.	UNCLASS \$13, OPEN/NEWS \$15, FLOW-LED VERDICT
4. Price the violent-whipsaw bench.	The chop commission's premise was wrong — the tournament did not trade this week's chop days at all. Its money is lost in violent-whipsaw blocks, and a two-term bench there is worth roughly ten times the scalper that was commissioned to fix the wrong thing.	chopscalp \$2
5. Add atr_min, the wrapping clock window and an ER10 gate param to ShadowVariant.	Three small changes, and they are a prerequisite for shadowing anything from this movement honestly. ★ This is exactly how rider_w5 came	RIDER_ALL \$11

	to be quoted at +\$19.29 while making −\$28.84: an arm shipped around a missing field records a DIFFERENT strategy under the same name.	
6. Do NOT arm anything on MNQ out of this movement.	Seven hunts, dozens of candidates, and not one MNQ LIVE candidate. That is the honest answer and it is stated plainly rather than dressed as a near-miss.	the whole movement

10. Disposition — every lead this movement touched

Four dispositions, one each. **PARKED** names the specific thing that would revive it; **REFUTED** names the test and why no reformulation saves it. Nothing here is a graveyard — a PARKED row is a lead with a stated price.

LEAD	VERDICT	REQUIRED NOTE	SOURCE
MGC shadow bar source	BUILD REQUIRED	The gating item for the whole gold line. Point the shadow's bar builder at the depth mid and keep the trade-bar arms as *_tradebar controls. The service already holds the DepthFeed. Nothing else on the gold line is worth doing first.	MGC \$3, \$12.1
mgc_hole_break_fade LONG + SHORT + double-hole	SHADOW	Survived the forward walk on the lab tape (+\$512 / +\$656, placebo 0/30, mirror and both constants negative, 3–4 ticks of slippage headroom). Losing in live shadow on production's tape (−\$256 over 14 trades) — that is the bar source, not the gate. Promote the double-hole arm to full status behind the fix.	MGC \$14
Cross-asset MNQ veto for gold	SHADOW — new arm	+\$5.43/tr against +\$1.72 blanket over 278 paired days, two-sided, placebo at the 97th/2.5th percentile. Does not reproduce on the 29-day book window (34th pctile). Needs a new _entry branch and an MNQ state feed. Settle it by running it alongside.	MGC \$6
mgc_wall_break_go LONG + SHORT	SHADOW, on probation	Forward-walked +\$693, mirror negative, filter placebo 0/200 — but fails the random-minute placebo 29/30 for the SECOND week, and strip-best-10 is −\$192. Promote no further until n ≥ 150 with a passing random-minute placebo.	MGC \$14
board_overnight (21:00-07:00Z)	SHADOW — an instrumented question	Right sign in two independent samples (+0.444R and +0.224R), counter-trend leg negative in both, survives a full point of slippage. But its daytime and counter-trend controls REVERSE on the held-out sample and its placebo p is 0.097. ★ Shadow it to answer a question, not because we think it makes money — and it needs three code changes first (a wrapping clock window, an atr_min field, and ER10 rather than ER30).	RIDER_ALL \$10, \$11
	SHADOW → adopt as a router rule		UNCLASS \$11, \$13

BENCH 07:00-13:00Z for any late-boarding rider

Red on all three independent legs (−\$29.38 / −\$20.04 / −\$2.43, the last on n=373). Cheap, fail-safe direction, no new instrument needed. The cleanest positive deliverable in the movement.

The violent-whipsaw bench

MEASURE, then route

The premise of the chop commission was wrong and the correction is worth more than the scalper. Across 18 days the tournament's loss is concentrated in **violent-whipsaw** blocks (−\$1,383.50 on 95 trades), NOT in plain chop (−\$403.00 on 58) — and on this week's two chop days the tournament took **zero** trades. A two-term router bench, worth roughly ten times the scalper it replaces.

chopscalp \$2, \$13

ER30 as a router ARM meter

SHADOW as an instrumented question

+ \$3.54 against −\$3.40 a trade on 3,238 untouched trades — the largest-n effect in the movement — but it REVERSES on the recent 11 sessions. Shadow the meter with its own control arm; do not arm a gate behind it.

UNCLASS \$13

board_atrband

PARKED

Revive only with a SELECTOR — something that predicts which board becomes a run. NOT more days: the power calculation says ~10,500 trades to separate it from zero.

RIDER_ALL \$10

UNCL-RIDE-ER

PARKED

Revive only with a board-time SIZE PREDICTOR expressed in ATRs ($r \geq 0.4$ against eventual run size on ≥ 300 runs). More days will not fix it — 205 untouched sessions were already spent and the placebo won.

UNCLASS \$13

mgc_coil_bounce

PARKED (updated)

Last week killed it partly on the **\$7.50 cost error**; at the corrected \$4.50 the 08-15 audit puts it at + \$8.42 — breakeven rather than dead. Not re-derived this week. Revive by re-running it on the 29-day quote tape at the corrected cost.

MGC \$14

Lookback 90/120 min (gold)

PARKED

The year's monotone gradient does not survive the book gate. 45-180 min is a plateau; leave it at 60.

MGC \$5.2

rider_w5 (live shadow arm)

REFUTED — RETIRE THE ARM

51 live forward trades at −\$28.84/tr, red in all four ATR buckets; 1,339 OOS trades at −\$5.43/tr with strip-3 at −\$9,293; −\$174 on the census week it was built for; and its in-sample +\$1,936 collapses to +\$282 on strip-best-3.

RIDER_ALL \$5

UNCL-RIDER-ASIA

REFUTED — do not re-shadow

Four consecutive red forward weeks. ★ And it was **never actually shipped** — there is no arm by that name anywhere in shadow.db, which nobody had noticed.

UNCLASS \$9

The VACUUM / FLOW-LED split

REFUTED as a footprint — STOP SPLITTING

The flow event fires on 20.4-20.6% of all tape and then splits it on a **49.6-49.9% coin toss** — 0.33

FLOW-LED \$0, VACUUM \$2

		percentage points of directional information. The lift the flow event does carry is a MAGNITUDE lift and it is a volume proxy.	
The UNCLASS label	REFUTED as a footprint — RENAME	Base rate 73.9% on the census week, 75.4% on the whole lake, lift 0.82 ; its members sit ON the all-tape median for ATR, volume and efficiency. Rename it UNTESTED and report it as COVERAGE. Second independent measurement in two weeks.	UNCLASS \$1, \$13
The OPEN/NEWS label	FIX — drop the footprint tests	Its lift over the BARE CLOCK is 0.921x : the flow and amplitude tests remove 41% of the window and keep the less run-dense half. Either call it US-OPEN-WINDOW (3.09x lift on 8.87% of tape) or route the hunt through VOL-EXPANSION — 4.18x lift on 1.95% of tape, the sharpest cluster the census produces and the one it uses least.	OPEN/NEWS \$1, \$15
The census's flow-bucket PHASE	FIX — this is a real bug	21 of 68 labels change on bucket phase alone, and only 17 of 68 run timestamps are minute-aligned. Snap the window slide to the CLOCK, not to the array index.	UNCLASS \$13
The lake bars timeframe landmine	FIX REQUIRED	Backfill parquet at 1min/5mins/1hour/1day now sits beside the daily 5s files; any query without a timeframe filter inherits up to a full day of look-ahead on every day.	MGC \$14
L2 far-side depletion as a run selector	REFUTED (two independent measurements)	RIDER_ALL: 311 fires, 264M book rows, $r = -0.009$, placebo $p = 0.62$. VACUUM: 49.44% ± 2.52 pp on unselected 55pt+ moves over 246.9M book rows, and 49.72% ± 0.67 pp on all 21,249 book-minutes. Do not revive it at 30 days — the null is not n-limited.	RIDER_ALL \$7, VACUUM \$6
The far-side WALL as an absorption tell	REFUTED, with the sign INVERTED	The operator's core L2 hypothesis. Unfiltered chop turns drift +0.054pt in the fade's favour; requiring a wall $\geq 1.5\times$ turns that to -0.117 pt, and five-deep $\geq 1.3\times$ to -0.374 pt ($t = -1.72$, quintile $z = -2.37$). The wall is a TARGET, not absorption.	chopscalp \$3
The four run SHAPES (TURN / STEP-REPRICE / DRIFT-VOID / GRIND-CONT)	SHIP TO THE CENSUS	One if each, all computable at the run's start minute. Not a selector — vocabulary, to replace the three labels that just died.	UNCLASS \$10
The 41ms refill count	KEEP — the one L2 term that ever helped	The only book statistic in the chop lab that contributed anything. ★ And the 250ms depth.db sample cannot see it — it needs the 41ms capture.db.book feed, which is MNQ-only.	chopscalp \$7, \$13

11. The stones still unturned

THE STONE	WHY IT IS THE NEXT ONE	WHAT IT NEEDS
Selection by expected SIZE at the board moment	It is the direct consequence of both headline findings. Three labs independently found that the money starts at about 120 points of follow-through, and not one of the seven hunts tested a size predictor at the moment of boarding. Every hunt this week asked “is a run starting?” and we already know the answer is yes 34 times a day.	No new data source. A different question asked of tape we already hold: a board-time predictor of eventual run size in ATR units, $r \geq 0.4$ on ≥ 300 runs.
Is gold's BOOK as sparse as gold's TRADES?	The bar-source finding says the depth mid is the better price series because gold prints sporadically. But the liquidity filter reads that same book — so if the book is also thin or stale in exactly the minutes where trades are absent, the “empty far side” the gate selects for may sometimes be an absent feed rather than an absent bid.	Split the hole cell by tape activity (trades-per-minute in the minute before the break) and check the empty-book cell still pays in the busiest quartile. The cheapest possible test of the concern.
Is the ER meter's forward reversal a regime break or an artefact?	ER30 as an arm meter is the largest-n effect in the movement — +\$3.54 against −\$3.40 a trade on 3,238 untouched trades — and it reverses on the most recent eleven sessions. One of those two readings is noise and we do not know which.	Free: shadow the meter with its own control arm and wait. The single cheapest open question in the movement.
A different book statistic	Far-side depth share is refuted twice over and is not n-limited, so more days will not save it. But the chop lab found that the one L2 term that ever helped was the 41ms refill count — a rate, not a level.	The 250ms depth.db sample cannot see it; it needs the 41ms capture.db.book feed, which is MNQ-only and rolls off after five trading days. Mirror it to the lake first, or the study cannot be run at length.

Appendix — the seven dossiers in full

Everything above is a map. This is the territory: each hunt's complete working, as its author wrote it — every sweep table, every control, every grave, every caveat, including the ones that cut against the author's own numbers. Nothing has been summarised out. If a claim in the editorial half above looks too strong, the dossier it came from is here and it will say so itself.

THE POOLED RUN-CATCHER — one rider for every sat-out run

gf_full_RIDER_ALL.md · 8,856 words · written 2026-08-25 22:10 UTC

GREENFIELD SYNTHESIS — ONE RIDER FOR EVERY SAT-OUT RUN

MNQ · full parquet lake (5s bars 2026-07-16 → 08-24) + a 236-session out-of-sample leg · \$2.00/point · \$1.50 per round trip

THE SHORT VERSION, BEFORE ANY OF THE WORKING

Garrath — I built the direction-agnostic run-catcher you asked for, boarded it late exactly as the gold work suggested, swept the exit far wider than the desk has ever run one, filtered it with a causal rule, and put the survivor through the whole battery. It passed every in-sample test we have, including two separate placebo controls. Then it died on the out-of-sample leg, and it died again on a live shadow arm that has been quietly recording forward trades since Monday.

It died at STEP 3 — the causal filter. The test that killed it was the out-of-sample leg: 1,289 trades over 236 untouched sessions for a total of +\$14.

Not a loss. Not a win. Nothing. The 95% confidence interval on its expectancy runs from $-0.204R$ to $+0.246R$ per trade, which is another way of saying we measured zero with a wide ruler.

And there is a second, independent witness that agrees, which is the part you should trust most. **rider_w5 — the board arm this desk shipped into the shadow book on 2026-08-15 off last week's version of this very phase, quoted then at +\$4,841 and +\$19.29 a trade — has taken 51 forward trades since 08-17 and sits at $-\$1,471$, or $-\$28.84$ a trade, red on six of seven days and red in every single ATR bucket.** Nobody had graded it. This section grades it.

Three things came out of the week that are worth more than the failed rider:

1. **The \$9,610 is not \$9,610.** Even a perfect late boarder — one that fires inside all 62 runs at a median three minutes in and then exits at the exact top of the move with no costs at all — can only reach **\$5,094, or 53% of the ceiling**. Thirteen per cent of the runs are already finished by the time any honest detector can see them. The number the census puts on the table is a hindsight number and the rider's half of it is gone before we start.
2. **Detection is not our problem and never was.** A five-minute boarding trigger fires inside **62 of 62** sat-out runs at a median of **three minutes** late,

with a median **41 points still ahead**. It also fires 34 times a day on tape that goes nowhere. The gap is selection, not detection — which is exactly what the greenfield census concluded from the other direction.

3. **I found a backtest bug that was manufacturing a winner**, and it is the kind that would have shipped. Details in §3. The whole chandelier family survived every robustness test I have until I fixed it, and then went from + \$4,270 to −\$631.

Full verdict, dispositions and the one shadow proposal are at the bottom. Every dead candidate is named with its cause of death.

0. THE ANCHOR — I REPRODUCED THE FROZEN CENSUS BEFORE TOUCHING ANYTHING

I did not re-run `run_census.py`. I re-derived its run set from the lake with its own algorithm and checked it line by line against the frozen `census_stdout.txt`.

CHECK	RESULT
runs found on the census window (08-17 00:00 → 08-21 20:59 UTC)	68
rows matching the frozen file on time and direction and move	68 / 68
typical 15-min range / threshold	37pt / 55pt — identical
sat-out runs	62
sat-out ceiling	\$9,610

⚠ **The brief I was handed says "60 runs, \$8,025"**. That is last week's census. This week's frozen file says 62 runs and \$9,610, and since the frozen file is the contract, every number below is scored against \$9,610. Worth flagging because the two numbers are close enough to pass unnoticed.

One small defect found in the census while anchoring to it. The census slides its 15-minute window by array index, not by clock. Where the 5s tape has holes the two drift apart, and only **16 of the 62** sat-out run timestamps are actually minute-aligned — the offsets run 10s, 15s, 25s ... up to 55s. It changes no money, but a run in that table is located to about ± 1 minute, not to the second, and anything that joins to those timestamps on an exact match will silently miss most of them. I snapped each run to the minute bar containing it and took start prices from the 5s bar at the exact census timestamp.

1. STEP ONE — BOARD IT. WHEN COULD WE HAVE KNOWN, AND WHAT WAS LEFT?

The instruction was to stop trying to predict a run and instead board it after it has proven itself, then measure what remains. On gold that shape fired a median 19 minutes into the top five runs with about 80% of the move still ahead. Here is the same curve for MNQ.

The boarding rule is deliberately dumb and direction-agnostic: at each completed one-minute bar, take the net move over the trailing **W** minutes; if it is at least **k × ATR** and the efficiency ratio over that window is at least **e**, board it in whichever direction the move went. ATR here is the mean one-minute high-low range over the last 20 minutes, which is the exact number the live router already prints, so anything built on it is deployable as written.

"What was left" is measured to the **run's own terminus** — the furthest the move got before giving back a full ATR — with no costs and no stop. It is a pure oracle: this is the most any rider could have taken, not what one would take.

The boarding curve, across the 62 sat-out runs

DETECTO R	FIRES IN	MINUTES LATE (P25 / MED / P75 / P90)	POINTS LEFT (P25 / MED / P75 / P90)	% OF THE MOVE STILL AHEAD (MED)	NOTHING LEFT	≥50PT LEFT	ORACLE \$	% OF \$9,610
W=5, k=1.0, ER≥0.40	62 / 62	2 / 3 / 4 / 7	+24 / +41 / +57 / +78	63%	13%	40%	\$5,094	53%
W=5, k=1.5, ER≥0.40	61 / 62	— / 4 / — / —	— / +37 / — / —	55%	11%	34%	\$4,514	47%
W=8, k=1.0, ER≥0.40	62 / 62	— / 5 / — / —	— / +35 / — / —	52%	18%	29%	\$4,135	43%
W=10, k=1.5, ER≥0.50	60 / 62	6 / 8 / 9 / 13	+2 / +20 / +38 / +63	33%	22%	17%	\$2,286	24%
W=10, k=2.5, ER≥0.50	60 / 62	6 / 8 / 10 / 14	+0 / +12 / +33 / +63	18%	27%	15%	\$1,910	20%

Read the top row properly, because it is the most encouraging number in this whole section and it is still not enough. A five-minute confirmation catches every one of the 62 runs we sat out, and it catches them fast — three minutes in at the median, ten minutes at the very worst. At that moment the median run still has 41 points and 63% of its move in front of it. Detection latency, on this trigger, is simply not the problem.

That directly contradicts the prior this desk has been carrying. Part 2.5's detection-latency finding — median 119 points already moved, median 16 points left, 12% capture — was measured on a **30-minute** window with $\text{net} \geq 80$ and $\text{ER} \geq 0.35$. That is a much heavier confirmation and it buys a much later seat. Shortening the window from 30 minutes to 5 moves the median from "16 points left" to "41 points left". **The latency was a property of the instrument we were measuring with, not of the tape.**

Now the two hard limits the same table imposes:

- **13% of runs have nothing left at all.** By the time even the fastest honest trigger fires, the move is done. Those runs are not catchable by any boarding strategy, only by a predicting one, and predicting run starts is already a banked null on 22.6M ticks.
- **The oracle ceiling is 53%, not 100%.** \$5,094 of the \$9,610, with a cheating exit and zero costs. Tighten the trigger to something you would actually trade and it falls to 20–24%. Every honest conversation about this money should start from about **\$2,000–\$5,000 a week**, not \$9,610.

And the thing the table does not show: **that same $W=5 / k=1.0$ trigger fires 34 times a day.** Over the 35-day lake it produces 1,180 boards, of which about 60 a month are these runs. The other 96% are ordinary tape. That is the whole problem, and it is where the rest of this section goes.

2. STEP TWO — THE EXIT. SWEEP WIDE, AND THEN SWEEP AGAIN ON A SAMPLE TEN TIMES BIGGER

The instruction was explicit: sweep the exit wide before concluding any entry fails, because three separate results this week said a tight exit gets shaken out of a grind while the same entries with a wide stop and a far target make money.

I did that first on the 35-day 5s lake. Entry set held fixed ($W=8$, $k=1.5$, $\text{ER} \geq 0.50$, 60-minute lockout, 1,034 boards) so that every cell differs in the exit and nothing else.

Stop × target, net \$ over the 35-day lake, 1 lot, \$1.50/RT

1.0	1.5	2.0	3.0	4.0	6.0	8.0	HOLD TO CAP
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STOP ↓ / TARGET →								
1.0×ATR	−2,449	−2,783	−2,492	−1,886	−1,662	−2,634	−4,277	−2,428
1.5×ATR	−2,320	−3,050	−2,550	−783	−1,052	−1,914	−3,302	−5,722
2.0×ATR	−1,361	−1,489	−767	+449	+895	+1,725	+702	−90
2.5×ATR	−996	−665	−639	−865	+319	+2,684	+1,345	−1,138
3.0×ATR	+104	+432	+53	−1,249	−814	+2,133	+1,319	−1,042
4.0×ATR	+376	+1,127	+1,344	−1,792	−318	+3,484	+3,495	−964
6.0×ATR	−303	+1,360	+1,421	−1,595	−1,504	+1,328	+1,355	−1,556

On the surface, the wide-exit thesis is confirmed. The live-desk-shaped exit (1.5 stop, 3.0 target) loses \$783. Widen the stop to 4× ATR and push the target to 6–8× and the same entries make +\$3,484 to +\$3,495. That is a swing of over \$4,200 on identical boards, purely on the exit — the same shape as the exhaustion_short rehab.

But look at the surface, not the best cell. At a 4.0× stop, the target-3.0 cell loses \$1,792, the target-4.0 cell loses \$318, and the target-6.0 cell makes \$3,484. A real edge does not do that. That is not a plateau, it is a lottery, and it is the first sign that a handful of trades are carrying everything.

Hold-to-time-cap variants (no target, flat at 15 / 30 / 60 / 240 minutes) were negative almost everywhere — that column is the right-hand edge of the table above.

△ THE GRAVE THAT MATTERS MOST — a backtest bug that manufactured a winner

The chandelier family — wide stop, trail armed after $N \times \text{ATR}$, give back $k \times \text{ATR}$ from the peak — came out of this sweep looking like the answer. It was the only exit family positive in nearly every regime cell across two different entry definitions, and chand_2.5_a1.5_t1.5 on the W=10 entry set survived everything I threw at it: +\$4,270 net, +\$4.58 a trade, 60% win, strip-best-3 still +\$3,115, strip-best-10 still +\$1,098, day-green 61%, worst day −\$631.

It was wrong. **My trail was testing the bar's HIGH against the running peak instead of its LOW.** A 5s bar that printed a new high and then collapsed registered a give-back of exactly zero, so the trail never fired inside the bar that actually broke it — it exited later, at prices that never traded. The peak was also read including the current bar, which is a second look-ahead on top of the first.

The tell was there before I found it, and it is worth naming so we catch the next one faster: **the random-entry placebo**. Firing at completely random minutes in a random direction with that same chandelier made **+\$1,542 on average**, and the "real" rider's +\$4,270 sat at $p = 0.115$ against that null. A random entry should not make money. When your null hypothesis starts printing, the bug is in the machinery, not in the market.

Fixed — peak taken from strictly prior bars, give-back tested against the adverse edge — here is what happened to the whole family:

EXIT	ENTRY SET	BEFORE THE FIX	AFTER THE FIX
chand 2.5 / arm 1.5 / trail 1.5	W=10 k2.0	+\$4,270 (strip-10 +\$1,098)	−\$631 (strip-10 −\$3,339)
chand 3.0 / arm 2 / trail 2	W=10 k2.0	+\$4,126 (strip-10 −\$331)	−\$820 (strip-10 −\$4,491)
chand 3.0 / arm 2 / trail 2	W=8 k1.5	+\$3,026	−\$1,471
chand 4.0 / arm 2 / trail 3	W=8 k1.5	+\$5,247 (strip-10 −\$736)	−\$872 (strip-10 −\$6,748)
chand 3.0 / arm 3 / trail 2	W=5 k1.0	+\$6	−\$4,728

Every single one flipped. Not one chandelier configuration in the study is positive with a correct trail. **Cause of death: a backtest bug, self-inflicted, caught by the placebo control.** Everything below this line uses the fixed simulator.

The honest sweep — every cell judged, not just the best one

Cherry-picking the best cell out of a grid is how a desk convinces itself. So I ran the full grid — 5 entry windows $\times$ 6 thresholds $\times$ 4 ER floors $\times$ 2 lockouts $\times$ 8 exits = **1,824 scored cells** — and put the same battery on every one of them: net > 0 , strip-best-3 > 0 , strip-best-10 > 0 , and **both halves of the tape positive**, with $n \geq 60$.

BAR	CELLS PASSING	SHARE
net > 0	423 / 1,824	23.2%
second half of the tape > 0	36 / 1,824	2.0%
net and strip-3 and strip-10 and both halves	5 / 1,824	0.3%

Two per cent of the entire configuration space makes money in the second half. If the edge were real and broad you would expect most of the 423 profitable cells to also be profitable in the back half; instead 423 shrinks to 36. With a 2% base rate you would expect about 8 of the 423 to clear the full bar by luck alone. **Five did.** The five "survivors" of the blanket sweep are exactly what chance hands you, and quoting any one of them would be a lie of selection.

But the halves test is confounded, and I checked

Before accepting "the edge decayed", I looked at what the two halves actually are.

HALF	DATES	SESSIONS	ATR20 MEDIAN	MEAN DAILY RANGE	15-MIN MOVE P90	ER30 MEDIAN
first	07-16 → 08-04	14	13.9pt	709pt	67.1pt	0.172
second	08-05 → 08-24	14	9.3pt	398pt	44.4pt	0.156

The second half is a **33% quieter tape by ATR and 44% quieter by daily range**. So "it stopped working in half two" and "it stops working when volatility halves" are the same sentence, and the blanket halves test cannot tell them apart. That is precisely the failure mode the desk's own backtest discipline warns about — a blanket cross-tape number averages the regimes a config should be ON with those it should be OFF, and is a bug. So I split by regime first.

3. STEP THREE — THE FILTER, AND THE ONE THAT ACTUALLY WORKED

Regime first, config second

I bucketed every board by the **ATR level at entry** — causal, known before the trade, and the live router's own meter — and re-ran a 216-cell entry × exit sweep inside each bucket, requiring both halves of that bucket to be positive.

ATR20 AT ENTRY	CELLS SCORED	NET > 0	2ND HALF > 0	PASS BOTH HALVES + STRIP-3
ATR < 9 (quiet / dead)	216	23 (11%)	15 (7%)	4 (2%)
ATR 9-14 (the desk's normal tape)	216	188 (87%)	75 (35%)	66 (31%)
ATR ≥ 14 (violent)	216	81 (38%)	39 (18%)	16 (7%)

A fifteen-fold difference in survival rate between buckets is not something chance produces evenly. Something real lives in the middle band, and the blanket number had been averaging it away.

...but first, is that just leverage?

Stops and targets are ATR-scaled, so a higher-ATR trade is mechanically a bigger bet in dollars. If the "filter" simply selects bigger bets, dollars per trade rises while the actual edge stays flat, and the rule is a position-sizing choice dressed up

as an insight. So I rescored everything in **R** — net P&L divided by (ATR at entry × \$2) — which is the same trade measured in units of what it risked.

ATR20 BAND	W8 K2.5 (R/TRADE)	W10 K2.5 (R/TRADE)	W15 K3.0 (R/TRADE)	AGREEMENT
0-7		+0.07	+0.45	-0.12 no
7-9		+0.27	-0.63	-0.24 no
9-11		+0.48	-0.38	+0.26 no
11-13		+0.07	+0.66	+1.03 partial
13-16		+1.42	+1.09	+0.87 all three
16+		-0.11	+0.19	-0.11 no (all ≈ 0)

In R the relationship is **not** monotone in ATR, which kills the leverage explanation. It is a **band**: the money is between roughly 11 and 16 points of ATR, and both edges matter. Below it the moves are too small relative to noise and the round trip; above it the tape is violent enough that a "proven" move is more often an exhausted one — which is the same thing the abs_veto rehab found when the amp_floor let through elevated-ATR-but-spent overnight thrusts.

The candidate

RIDER_ALL / board_atrband — sampled at each completed one-minute bar.
ATR20 = mean 1-min high–low over the last 20 minutes. **Fire when $11 \leq \text{ATR20} < 16$ AND $|\text{net over 10 min}| \geq 2.5 \times \text{ATR20}$ AND $\text{ER}(10 \text{ min}) \geq 0.50$.** Direction = sign of the 10-minute move. Stop $3.0 \times \text{ATR}$, target $6.0 \times \text{ATR}$, flat at 240 minutes, 60-minute lockout between fires.

The full battery, in-sample (5s lake, 28 sessions)

ENTR Y WIN DOW	N	NET \$	\$/ TRA DE	R/ TRA DE	WIN	STRI P-3	STRI P-5	STRI P-10	HALF 1 (N)	HALF 2 (N)	LOO- DAY WOR ST	DAY- GREE N	WOR ST DAY	MEA N HOL D
W8 k2.5 ER.50	154	+3,358	+21.81	+0.784	44%	+2,791	+2,417	+1,507	+2,814 (103)	+543 (51)	+2,697	67%	-226	50m
W10 k2.5 ER.50	146	+3,496	+23.95	+0.896	44%	+2,937	+2,573	+1,682	+2,695 (94)	+801 (52)	+2,881	70%	-331	54m
W15 k3.0 ER. 40	152	+3,704	+24.37	+0.950	46%	+3,145	+2,778	+1,877	+2,527 (97)	+1,176 (55)	+2,870	63%	-432	55m
W10 k2.0 ER. 40	164	+2,612	+15.93	+0.585	41%	+2,048	+1,687	+808	+805 (107)	+1,806 (57)	+1,741	69%	-584	51m
W8 k2.0 ER.50	159	+3,118	+19.61	+0.746	44%	+2,559	+2,194	+1,322	+2,612 (103)	+505 (56)	+2,472	59%	-410	57m

Five neighbouring entry definitions, all between +\$15.93 and +\$24.37 a trade, all surviving strip-best-10, all positive in both halves. **That is a parameter plateau, not a step.** By this desk's stated bar — expectancy and robustness, never win rate — 44% winning is irrelevant and +0.9R a trade over 146 trades with strip-10 intact is a serious candidate.

The controls

CONTROL	RESULT	READING
Direction flipped (same boards, opposite side)	-\$2,436, -\$16.68/tr, -0.637R	direction is doing real work — it is not an exit artefact
Random-time placebo — same ATR band, random minutes, random side, 400 draws	null mean +\$229, sd \$1,335 → p = 0.0125	beats a random entry with the same exit
Placebo cut — keep a random 146 of the 518 unfiltered fires, 4,000 draws	null mean +\$1,088, sd \$1,246 → p = 0.029	the ATR band beats discarding the same number of fires at random
Unfiltered (all ATR)	518 trades, +\$3,864, +\$7.46/tr, +0.190R	the band roughly triples expectancy per trade

★ What the cut COSTS — the number I am obliged to report and it is not flattering

Cutting to the ATR band discards **372 of the 518 fires**. Those 372 contain **131 winners worth +\$18,483** which we forgo, and **241 losers worth -\$18,115** which we avoid. Net effect of the entire filter on the bottom line: **+\$368**.

So the ATR band does **not** find money. It concentrates the same money into a quarter of the trades. That is genuinely worth something — it is what lifts \$7.46 a trade to \$23.95 and what makes the thing survive strip-10 — but anyone reading "+0.9R a trade" as "the filter found an edge" has it backwards. The filter found a smaller sample.

And the caveat I have to put next to my own p-values

I chose the band edges after looking at the data. There were roughly 36 plausible lo × hi combinations in the sweep I ran. A p of 0.029 against 36 looks, corrected, like a p of about 1.0. The placebo cut controls for "any cut of this size" but not for "I went looking for the best cut". The in-sample battery above is real and the controls are honest, but it is not, on its own, evidence that survives the multiplicity. Which is why the next section exists.

4. THE OUT-OF-SAMPLE LEG — AND THE KILL

The 5s lake is 28 sessions. The parquet lake also carries a **one-minute backfill running back to 2025-09-14** — 295,845 bars, **236 sessions before the 5s tape even begins**. Not one parameter above was chosen on it, and it is the best-powered test available to this desk.

Two honesty notes before the number. The exit has to be simulated on one-minute bars there, which is a coarser and more punishing ruler — inside a bar I still resolve stop-before-target, and the fill is a minute later. To measure how much coarseness costs, I ran the in-sample window through the coarse simulator too:

LEG	N	NET \$	\$/TRADE	R/TRADE	
in-sample window, 5s simulator		146	+3,496	+23.95	+0.896
in-sample window, 1-min simulator (calibration)		213	+2,399	+11.26	+0.459

The coarse ruler reads roughly **half**. Keep that in mind and then read this:

LEG	N	SESSI ONS	NET \$	\$/ TRADE	R/ TRADE	95% CI ON R/ TRADE	STRIP- 3	STRIP- 10	HALF 1	HALF 2	DAY- GREEN
OUT-OF-SAMPLE, ATR band	1,289	236	+\$14	+\$0.01	+0.021	[−0.204, +0.246]	−555	−1,875	−2,256	+2,270	47%
OOS, no ATR band (control)	3,924	259	−\$3,640	−\$0.93	−0.047	—	−5,676	−9,445	+2,806	−6,446	46%

Fourteen dollars. Over eleven months. Double it for the coarse ruler and it is twenty-eight dollars. Month by month it is a coin toss: +734, −84, +437, −1,510, −1,333, −555, +373, +496, +1,844, −669, +281 — six green months out of eleven. It does not survive stripping its three best trades.

The ATR band is not useless: without it the same entries lose \$3,640 and −0.047R. **The band takes the rider from clearly negative to exactly zero.** That is a real filter doing real work on a mechanism that has no edge to protect.

Step Two, re-run properly on the big sample

The wide-exit sweep on 28 sessions was the underpowered version. Here it is on 236, on the banded entry set (n = 1,289), net \$ / R per trade:

STOP ↓ / TARGET →	1.5	2.0	3.0	4.0	6.0	8.0	CAP
----------------------	-----	-----	-----	-----	-----	-----	-----

1.0×ATR	−4,185	−3,073	−2,398	−2,179	+875	+849	−3,090
2.0×ATR	−3,059	−2,636	−2,152	−2,573	+1,367	−576	−7,179
2.5×ATR	−2,301	−1,618	−2,302	−2,007	+851	−1,262	−7,369
3.0×ATR	−1,967	−1,028	−2,579	−2,510	+14	−2,622	−9,367
4.0×ATR	−2,051	−451	−1,429	−2,538	−786	−1,647	−7,618
6.0×ATR	−3,509	−1,505	−2,504	−2,749	−201	+494	−6,840

Forty-one of the 42 cells are negative or within noise of zero, and the five nominally positive ones are scattered, not clustered. +\$1,367 sits directly beside −\$576 and −\$2,573. There is no plateau anywhere on this surface.

So: "an entry that only pays with a wide exit is a finding, not a failure" — I tested that properly, and for this entry it is a failure. The wide exit reliably makes the rider less negative (target 6.0 is the best column in every row) but it never makes it positive. That is not a refutation of the exhaustion_short rehab, which was a different gate's entries; it is a refutation of the hope that the same medicine cures this one.

Does the direction carry anything at all?

Yes, and this is the most useful salvage in the section. Re-simulating every board on the opposite side:

EXIT	WITH-TREND (AS DESIGNED)	COUNTER-TREND	SEPARATION
3.0 × 6.0	+\$14 (+0.021R)	−\$1,363 (−0.056R)	+0.077R
2.0 × 4.0	−\$2,573 (−0.075R)	−\$4,066 (−0.135R)	+0.060R
4.0 × 8.0	−\$1,647 (−0.015R)	−\$3,816 (−0.121R)	+0.106R

A proven move genuinely does continue more often than it reverses — consistently, in every exit, across 1,289 out-of-sample trades. **The information is real. It is worth about +0.06 to +0.11R a trade, and the round trip alone costs 0.06R at the band's midpoint.** The edge exists and is roughly the same size as the toll booth. That is the honest shape of this whole problem.

Cost sensitivity, and what it would take

SLIPPAGE (POINTS EACH SIDE, ON TOP OF \$1.50/RT)	OOS ALL-DAY	OOS OVERNIGHT	5S LAKE ALL-DAY
0.00	+\$14 (+\$0.01/tr)	+\$4,143 (+\$10.46)	+\$1,686 (+\$6.74)
0.25 (one tick)	−\$1,275 (−\$0.99)	+\$3,747 (+\$9.46)	+\$1,436 (+\$5.74)
0.50	−\$2,564 (−\$1.99)	+\$3,351 (+\$8.46)	+\$1,186 (+\$4.74)
1.00	−\$5,142 (−\$3.99)	+\$2,559 (+\$6.46)	+\$686 (+\$2.74)

One tick of slippage takes the all-day rider negative. And the power calculation: at the band midpoint 1R is \$25 and the round trip is 0.060R; to establish a real expectancy of +0.10R a trade against what we actually measured would need roughly **10,500 trades**, about eight years of this rider's fire rate. This is not a "needs more data" case. It is a "the effect is smaller than the measurement error and smaller than the costs" case.

5. CLOSING THE LOOP — GRADING rider_w5, THE ARM THIS PHASE SHIPPED LAST WEEK

While reading the shadow book to write the ShadowVariant block, I found that **this exact idea is already live.** rider_w5 (src/gazbot7/shadow.py:929) was added on 2026-08-15 out of last week's RIDER_ALL phase, quoted at **+\$4,841 and +\$19.29 a trade:**

```
gate="board", k=2.0, w=5, 13:00-20:00 UTC, stop 3.0×ATR, target 6.0×ATR,
cooldown 15m, cap 120m.
```

It has been recording forward trades since 08-17 13:25 and **nothing had graded it.** Here it is.

RIDER_W5 LIVE SHADOW, 08-17 → 08-25		VALUE
closed trades		51
real P&L		−\$1,471.00
per trade		−\$28.84 (against +\$19.29 claimed)
win rate		18%
strip-best-3		−\$2,161
days green		1 of 7 (08-17 −234, 08-18 −232, 08-19 −631, 08-20 −416, 08-21 +377, 08-24 −289, 08-25 −46)
flagged rows (data_quality)		0 — this is clean data, not a measurement artefact

Red in every ATR bucket, including mine:

ATR20 AT ENTRY	N	NET \$	\$/TRADE	WIN
< 9	15	−\$299.50	−\$19.97	20%
9-11	8	−\$233.50	−\$29.19	12%
11-16 (my candidate's band)	14	−\$635.50	−\$45.39	14%
≥ 16	14	−\$302.50	−\$21.61	21%

My band is the arm's **worst** bucket live. That is as clean a refutation of the filter as exists.

Exit mix: STOP 21 (−\$1,190) · **ABSORPTION_CUT 18 (−\$1,449)** · TARGET 8 (+\$1,226) · TIME_CAP 4 (−\$58).

Entry or exit? — the attribution

The live arm carries `adverse_cut_atr=3.0`, which fires an `ABSORPTION_CUT` on 35% of its trades and is its single largest loss bucket. My simulator has no such mechanism, so the two are not measuring the same strategy. Re-deriving the arm's exact spec in my simulator separates them:

RIDER_W5'S SPEC, RE- DERIVED (NO ABSORPTION CUT)	N	NET \$	\$/TRADE	STRIP-3	DAYS GREEN
5s lake 07-16 → 08-24, 13:00-20:00Z (as shipped)	173	+\$1,936	+\$11.19	+\$282	57%
...of which the census week 08-17 → 08-21	33	−\$174	−\$5.26	−\$792	40%
OOS 236 sessions, its own 13:00- 20:00Z window	1,339	−\$7,272	−\$5.43	−\$9,293	47%
OOS 13-20Z + my ATR band 11-16	620	+\$30	+\$0.05	−\$540	46%

The entry alone is worth about **−\$5.43 a trade out of sample**. The absorption cut adds roughly another **−\$23 a trade** in live conditions. Both are bad; the entry alone is enough to kill it. And note the in-sample line: **+\$11.19 a trade** collapses to **+\$282** when you remove its best three trades — 85% of it was three trades. The **+\$19.29** figure this arm shipped on never had a strip test attached.

Recommendation: retire rider_w5. It is not a thin-n case — 51 forward trades, 1,339 out-of-sample trades, and the census week itself all agree, and the mechanism has an explanation (see §7).

6. THE LAST LEAD — THE OVERNIGHT LEG, AND WHY IT DID NOT REPLICATE

One thing did survive the out-of-sample leg, and I chased it as far as it goes.

Splitting the 1,289 OOS boards by session. △ This is a post-hoc split of an entry set built without a session constraint, so the 60-minute lockout falls differently than it does when the session is part of the rule — the session-constrained run below gives n=396 and +\$10.46/trade rather than the n=383 and +\$11.50 here. Same finding, slightly different bookkeeping; I quote the session-constrained numbers everywhere it matters.

OOS SESSION	N	NET \$	\$/TRADE	R/TRADE	WIN
overnight 21:00-07:00Z	383	+\$4,406	+\$11.50	+0.485	40%
Europe 07:00-13:00Z	344	-\$2,064	-\$6.00	-0.226	32%
US 13:00-21:00Z	562	-\$2,328	-\$4.14	-0.144	34%

On its generating sample the overnight leg looks strong: 396 trades, +\$4,142, +0.444R, strip-best-3 still +\$3,578, strip-best-10 still +\$2,281, leave-one-day-out worst +\$3,536, seven green months of ten, leave-one-month-out worst +\$2,388, counter-trend control -\$1,146. It also survives a full point of slippage each side.

So I did the clean thing available: **the hypothesis was generated on the 236-session backfill, so the 28-session 5s lake is now a genuine held-out test for it.** The roles reverse.

	OOS (GENERATING)	5S LAKE (HELD OUT)	REPLICATED?
overnight, ER10 ≥ 0.50	+\$10.46/tr, +0.444R, n=396	+\$5.34/tr, +0.224R, n=95	sign yes, but strip-3 = -\$51
daytime control (should be bad)	-\$5.18/tr, -0.190R	+\$5.21/tr, +0.203R	NO — control reverses
counter-trend control	-\$2.89/tr	-\$13.22/tr	yes
with ER30 ≥ 0.30 instead (what er_min reads)	+\$14.67/tr, +0.645R	-\$2.25/tr, -0.060R	NO
...its counter-trend control	-\$9.54/tr	+\$10.94/tr	NO — reverses
...its daytime control	-\$7.71/tr	+\$13.55/tr	NO — reverses
overnight, no ATR band	+\$3.99/tr	-\$11.69/tr	NO
rider_w5's own spec, overnight	+\$3.29/tr, n=1,688	-\$9.54/tr, n=218	NO

The overnight lead dies on replication, and it dies on the controls rather than on the headline. The rider itself keeps its sign in both samples for one particular parameterisation. But the claim — that overnight tape behaves differently from daytime tape for a boarding rider — is not reproduced: on the held-out lake daytime is just as good or better in five of six comparisons, and the counter-trend control, which was reassuringly negative out of sample, comes back **positive**. When your control flips sign, you are reading noise.

Two further honesty points. The session split was **one of three post-hoc comparisons**, so its nominal significance should be trebled. And the random-time

placebo inside the same band and the same session gives $p = 0.097$ — even on its own generating sample it is only about 1.3 standard deviations above firing at random times.

It is not nothing, and I am not calling it dead. It has the right sign in two independent samples, its counter-trend leg is negative in both, and it is the only thing in this study that survives a full point of slippage. That earns incubation, not deployment. See the disposition table.

7. THE STONE I HAD NOT TURNED — THE L2 BOOK AT THE BOARD MOMENT

The census scores every run for far-side depletion, and I had used none of it. The obvious question: at the moment the board rule fires, does a thin far side — the side price is running into — tell us which of those 34 daily fires is a real run? We hold 264 million MNQ book rows from 07-31 to 08-24, so this is answerable with data we already have.

For each of the 311 fires inside the book window I took the far-side share of level-1-to-3 depth over the 30 seconds **before** entry (below 0.50 = the far side is thin = the run is telegraphed).

FAR-SIDE SHARE	N	NET \$	\$/TRADE	R/TRADE	WIN
0.00-0.42	13	-\$38	-\$2.93	-0.343	31%
0.42-0.47	44	-\$300	-\$6.82	-0.544	30%
0.47-0.50	109	-\$680	-\$6.24	-0.142	35%
0.50-0.53	89	+\$188	+\$2.11	+0.139	36%
0.53-0.58	45	-\$379	-\$8.42	-0.308	33%
0.58-1.01	11	-\$166	-\$15.09	-0.662	27%

A clean null. Correlation between far-side share and R is -0.009 across 311 trades. Cutting to a thin far side gives $-0.265R$ against $-0.060R$ for a thick one — i.e. the wrong way round — and against a random cut of the same size the placebo p is **0.62**. There is no ordering, no monotonicity, no signal. The L2 book, read this way, at this moment, tells us nothing about which board becomes a run.

That closes the stone. It also means "we need a data source we do not capture" is not the excuse available here — we had the deepest data on the desk and it was empty.

8. STEP FIVE — SIZING THE PRIZE HONESTLY

\$		AS % OF THE \$9,610 SAT-OUT CEILING
ORACLE — perfect late board (fires in 62/62 at a median 3 min), exit at the exact run terminus, no costs, cheats		
	\$5,094	53%
ORACLE — the same, with a detector you would actually deploy (W=10, k=1.5, ER≥0.50)		
	\$2,286	24%
ORACLE — with the candidate's own trigger (W=10, k=2.5, ER≥0.50)		
	\$1,910	20%
CAUSAL — the candidate rider, one slot, run over the census week itself		
	−\$245	−3%
CAUSAL — same rider, overnight only, census week		
	−\$214	−2%
CAUSAL — same rider, one slot, whole 5s lake, scaled to a 5-session week	+\$433/week	in-sample only
CAUSAL — same rider, corrected by the 236-session out-of-sample leg	≈ \$0/week	0%

On the census week itself, running one position at a time, the candidate took **42 trades and lost \$245**. It did show up — it participated in **32 of the 62 sat-out runs**, roughly half — and it still lost money, because the other trades cost more than the runs paid. That is the sharpest single sentence in this report: **showing up to half the runs was not the hard part, and it did not help**.

The oracle table is the number to carry forward. This is a **\$2,000-\$5,000-a-week oracle, not a \$9,610-a-week one**, and per the desk's own precedent that distinction changes what the money is worth building for. The gap between the \$5,094 oracle and the \$0 causal result is not an exit problem and not a detection problem — it is **selection**, and it is 100% of the gap.

Why the mechanism fails, in one paragraph. At the moment the boarding rule is true, the median run has 12 to 41 points left depending on how much confirmation you demand — and the same rule is true 34 times a day on tape that does nothing. The trades that are not runs pay the stop, which at 3×ATR in the 11-16 band is 33 to 48 points. So the rider is risking 33-48 points to collect a median 12-41, and it must be right far more often than 44% of the time for that to

work. It is not. The wide exit helps because it lets the occasional big run pay for many stops, but the ratio of runs to non-runs at the moment of firing — about 1 in 20 — is simply too thin. **Boarding is solved. Selecting is not.** That is the same conclusion the greenfield census reached from the other end when it found that a plain $3/6 \times \text{ATR}$ exit converts 40% of the ceiling but 105 of 155 fires are not runs at all.

9. EVERY ATTEMPT, BY NAME, WITH ITS CAUSE OF DEATH

#	CANDIDATE	BEST NUMBER IT EVER SHOWED	CAUSE OF DEATH	NAMED TEST
1	board_live_exit — board a proven move, live-shaped 1.5×3.0 exit	−\$783 to −\$5,211 across four entry sets	never positive anywhere	outright, all entry sets
2	board_wide — wide fixed stop × far target	+\$3,495 (35-day lake)	second half of the tape, and the OOS sweep	both-halves; 236-session OOS
3	board_holdcap — no target, flat at 15/30/60/240 min	+\$2,301 in one cell	negative in almost every regime cell; no plateau	regime-split sweep
4	board_chandelier — wide stop + armed trail (4 variants)	+\$5,247 (chand 4.0/a2/t3, W8); the best-behaved was chand 2.5/a1.5/t1.5 on W10 at +\$4,270 with strip-10 still +\$1,098	backtest bug — the trail tested the bar's high, not its low	random-entry placebo (p=0.115) flagged it; fixed → all four negative
5	board_blanket — the full 1,824-cell sweep, best cell	+\$4,494	only 5/1,824 cells pass, against a 2% base rate ⇒ chance	both-halves base-rate count
6	board_atrband — ATR20 ∈ [11,16) + wide exit (the candidate)	+\$3,496, +0.896R, strip-10 +\$1,682, placebo p=0.0125 / 0.029	out-of-sample leg: +\$14 over 236 sessions, CI [−0.204, +0.246]	236-session OOS; live rider_w5 red in the same band
7	board_overnight — the same + 21:00-07:00Z	+\$4,142, +0.444R, LOO-month +\$2,388	held-out sample did not reproduce the separation; controls reverse sign	held-out 5s lake; placebo p=0.097; ER30 substitution flips it
8	board_book — far-side L2 depletion as the selector	—	clean null: r = −0.009, placebo p=0.62	311 fires, 264M book rows
9	rider_w5 — the arm shipped 2026-08-15	+\$4,841 / +\$19.29 as quoted	−\$1,471 over 51 live forward trades; −\$5.43/tr over 1,339 OOS trades; −\$174 on the census week	live shadow book + 236-session OOS + strip-3 (+ \$282 of +\$1,936)

10. DISPOSITION

LEAD	VERDICT	WHY / WHAT WOULD REVIVE IT
rider_w5 (live shadow arm)	REFUTED	Named tests: 51 live forward trades at −\$28.84/tr, red in all four ATR buckets; 1,339 OOS trades at −\$5.43/tr with strip-3 at −\$9,293; −\$174 on the census week it was built for; and its in-

		sample +\$1,936 collapses to +\$282 on strip-best-3. Mechanism stated in §8: at fire time the median run has 12–41pt left against a 33–48pt stop, and only ~1 fire in 20 is a run. Action: retire the arm.
board_atrband (the candidate)	PARKED	Passed every in-sample test including two placebos, then measured +\$14 over 236 sessions. Revive only with a SELECTOR — something that predicts which board becomes a run. Not more days: the power calculation says ~10,500 trades to separate it from zero. The census's own finding points the way (size threshold, ≥ 120 pt boarding), and that is a different study from this one.
board_overnight	SHADOW	Right sign in two independent samples (+0.444R and +0.224R), counter-trend leg negative in both, survives a full point of slippage. But its daytime and counter-trend controls reverse on the held-out sample and its placebo p is 0.097. Shadow it as an instrumented question, not as a candidate — the specific thing to buy is whether overnight boards really differ from daytime ones, which the two samples disagree about. Block and promotion bar below. Needs two small code changes first (§11).
board_wide / board_holdcap / board_blanket	PARKED	All failed as built on both-halves or the OOS sweep. Revive only if the selection problem is solved first — the exit was never the binding constraint here, which the 42-cell OOS surface establishes.
board_chandelier	PARKED, not refuted	It was never honestly tested — the trail was broken. With the corrected trail it is negative everywhere, but the shape (wide stop + armed trail) is what grind_long's deployed pure-6.0 chandelier runs on, so the mechanism is not disproven, only this application of it. Revive on a selected entry set, never on this unselected one.
board_book (L2 far-side depletion as selector)	REFUTED	Named test: 311 fires against 264M book rows, correlation -0.009 , no monotone band, placebo $p=0.62$. No reformulation of this measurement (far-side share, 3 deep, 30s pre-entry) survives; a different book statistic — resting-size decay rate, or the sweep-through print — would be a different lead, not a revival of this one.
Direction information itself	CONFIRMED but sub-scale	With-trend beats counter-trend in every one of three OOS exit configurations, by +0.06 to +0.11R. That is real and it is about the size of the round trip (0.060R). Keep as a prior, do not trade it alone.
The oracle ceiling	FINDING — carry it forward	The \$9,610 is a hindsight number. A perfect late boarder reaches \$5,094 (53%); a deployable one reaches \$1,910–\$2,286 (20–24%); 13% of runs are over before any honest trigger fires.

There is no LIVE candidate from this phase, and I am saying so plainly. The one stone I would call still unturned is **selection by expected SIZE** — the census's own greenfield work found that boarding only fires whose move is ≥ 120 pt pays +\$69.38/trade and passes the whole battery, and this study never tested a size predictor at the board moment. That is the next attempt, and it does not need a new data source; it needs a different question asked of the same tape.

11. THE SHADOW BLOCK — AND TWO THINGS THAT MUST BE BUILT FIRST

△ **This cannot be pasted in as-is.** I checked the shadow book's fields against the rule and found two genuine blockers. Both are small, but shipping around either one would record a different strategy under this name — which is exactly how `rider_w5` came to be quoted at +\$19.29 while making -\$28.84.

1. **gate_board cannot express a wrapping clock window.** It tests `hh_lo <= utc_hour < hh_hi`, which is never true for 21:00→07:00. Needs one line:
`inw = (hh_lo <= h < hh_hi) if hh_lo < hh_hi else (h >= hh_lo or h < hh_hi).`
2. **ShadowVariant has an `atr_max` but no `atr_min`.** The rule is a band, and both edges carry weight (\$3). An arm shipped with only the ceiling is a different arm. Needs one field, defaulted to 0.0 so nothing existing changes.
3. **Note on ER.** `ShadowVariant.er_min` reads **ER30**, not the entry window's ER. Substituting ER30 flips this rule's sign on the held-out sample (\$6), so the arm must gate on the **10-minute** ER. Either add `er_win_min` to the gate params or accept that an `er_min` arm is testing a different hypothesis and label it as such.

With those three in place:

```
ShadowVariant("board_overnight_band", "board",
              # ★ RIDER_ALL 2026-08-25. An INSTRUMENTED QUESTION, not a promotion
candidate.
              # Generated on the 236-session 1-min backfill (+$10.46/tr, +0.444R,
n=396,
              # strip-10 +$2,281, L00-month worst +$2,388); held-out on the 28-
session 5s lake
              # (+$5.34/tr, +0.224R, n=95) where the SIGN held but strip-3 fell to -
$51 and the
              # DAYTIME control came back POSITIVE (+$5.21/tr) — i.e. the overnight-
vs-daytime
              # SEPARATION that generated the idea did NOT reproduce. Random-time
placebo inside
              # the same band and session: p=0.097. Session split was 1 of 3 post-
hoc comparisons.
              # △ THE THING THIS ARM EXISTS TO ANSWER is whether overnight boards
differ from
              # daytime ones. It is NOT here because we think it makes money.
              # △ ER is the 10-MINUTE ER, not ER30. Substituting ER30>=0.30 FLIPS
the held-out
              # leg to -$2.25/tr and turns both controls positive. Do not use er_min
here.
```

```

# Δ adverse_cut_atr=99 ON PURPOSE. rider_w5's adverse_cut_atr=3.0
fired on 35% of
# its live trades for -$1,449 – a tight give-back cut is the exact
opposite of the
# wide-exit thesis this arm is testing. Do not "helpfully" restore it.
{"k": 2.5, "w": 10, "er_win_min": 0.50,
 "hh_lo": 21.0, "hh_hi": 7.0,          # NEEDS the wrapping-window fix
 "cooldown_min": 60},
symbol="MNQ", qty=1.0,
atr_min=11.0, atr_max=16.0,          # NEEDS the new atr_min field;
BOTH edges matter
stop_atr_mult=3.0, target_r=2.0,      # = target 6.0 x ATR
adverse_cut_atr=99.0, absorption_flow_min=1e9,
chandelier=False,                    # every chandelier variant
tested RED once the
# trail bug was fixed – do not
add one
time_cap_s=240 * 60)

```

Its router rule, in the router's own vocabulary

ARM board_overnight_band when, and only when, all four are true at the tick: 1. the clock is between **21:00 and 07:00 UTC**; 2. the router's own **ATR meter reads 11.0 to 16.0 points** (this is the mean 1-min range over 20 minutes — the number the router already prints; it is a **band**, so bench it above 16 as well as below 11); 3. **RUN STATE is RUN or RUN-FADING** — the arm is a boarding rule and has no business firing in declared chop; 4. the **untradeable meter is clear** and the tape is not inside a scheduled gateway window (04:28–05:00Z sits squarely inside this arm's session and hangs with the port open). **BENCH it immediately if the ATR meter leaves the band in either direction**, and treat a bench here as cheap: the arm's whole thesis is that both edges of the band carry weight. **Expectancy on the home regime only** (overnight, ATR 11–16, wide exit): **+\$10.46 a trade on the 236-session generating sample, +\$5.34 a trade on the 28-session held-out sample.** Take the lower of those as the working expectation, and note that its own strip-best-3 on the held-out sample is **–\$51** — three trades were the whole thing. **This is not a number to size on.**

Promotion bar — deliberately set high

Do **not** promote this to the paper tournament on P&L alone. It needs **all four**:

1. **200 shadow trades**, including at least one genuinely trending overnight week;
2. **≥ +0.15R** over those trades **after** stripping its best three;

3. the **daytime control arm** (identical rule, 13:00–20:00Z) must be **materially worse** — that is the claim under test, and it is the one that failed to replicate;
4. the **counter-trend control** must stay negative. It reversed on the held-out lake, and if it reverses again the arm is measuring noise and should be retired without argument.

I would also add the daytime control arm at the same time. Shadowing the treatment without the control is what produced this week's ambiguity in the first place.

VERDICT

NO RIDER SURVIVES. The pooled, direction-agnostic late-boarding run-catcher **DIES AT STEP 3 — THE CAUSAL FILTER** — killed by the **OUT-OF-SAMPLE LEG: 1,289 trades across 236 untouched sessions for a total of +\$14, a 95% confidence interval on expectancy of $[-0.204R, +0.246R]$, and a per-trade result that goes negative on one tick of slippage.** The verdict is independently corroborated by the live shadow arm rider_w5, which has taken 51 forward trades since 08-17 for $-\$1,471$ ($-\$28.84/\text{trade}$), red on six of seven days and red in every ATR bucket including the candidate's own — rider_w5 is **REFUTED** and should be retired. Step 1 **SUCCEEDED** and should be believed: a 5-minute boarding trigger fires inside 62 of 62 sat-out runs at a median 3 minutes late with a median 41 points still ahead, so detection is solved. Step 2's wide-exit thesis is **CONFIRMED IN DIRECTION** but **INSUFFICIENT IN SIZE** — a wide stop and a far target reliably make this rider less negative and never make it positive across 42 out-of-sample exit cells. The binding constraint is **SELECTION**: the same trigger fires 34 times a day and roughly 1 fire in 20 is a run, so the rider risks a 33-48 point stop to collect a median 12-41 points. The honest prize is **NOT \$9,610** — a perfect late boarder with a cheating exit reaches \$5,094 (53%), a deployable one reaches \$1,910–\$2,286 (20-24%), and 13% of the runs are finished before any honest trigger can see them. The causal result on the census week itself was $-\$245$ across 42 trades while participating in 32 of the 62 runs. **ONE lead goes forward:** board_overnight_band to **SHADOW** as an instrumented question ($+0.444R$ on 236 generating sessions, $+0.224R$ held-out, but its daytime and

counter-trend controls reverse and its placebo p is 0.097), gated behind two small code changes and a four-part promotion bar. The next attempt needs a SIZE PREDICTOR at the board moment, not more days, not a wider exit, and not a new data source — the L2 book was tested as the selector across 311 fires and 264M rows and returned a clean null ($r = -0.009$, placebo p = 0.62).

UNCLASS — the biggest bucket, 41 of the 68 runs

gf_full_UNCLASS.md · 9,219 words · written 2026-08-25 22:34 UTC

GREENFIELD HUNT — CLUSTER UNCLASS

MNQ · full parquet lake · 5s tape 2026-06-19 → 08-24 (51 sessions) + a 205-session 1-minute leg back to 2025-09-14 · \$2.00/point · \$1.50 per round trip

THE SHORT VERSION, BEFORE ANY OF THE WORKING

Garrath — I did the three steps in the order you set them. I built a direction-agnostic run-catcher and rode the UNCLASS runs without needing to know why they happened. I swept the exit far wider than the desk has ever run one. I filtered it with a causal rule and placebo-controlled every cut against throwing away the same number of fires at random. Then I routed it. It still lost.

It died on the PLACEBO/SHUFFLE TEST, corroborated by the OUT-OF-SAMPLE LEG. On the biggest independent sample I have — 205 sessions of tape that chose nothing — the real signal makes **+\$3.54 a trade** and the **fake signal, taken thirty minutes early with every other rule untouched, makes +\$4.44 a trade**. On the recent forward leg the real signal makes **−\$9.58 a trade** and the fake one makes **+\$1.48**. When a shifted copy of your signal outperforms your signal on two independent samples, you do not have a signal — and per your judging rule that is the named cause of death, not the 25% win rate, which I have deliberately ignored throughout.

Four things came out of the week that are worth more than the dead rider:

1. **UNCLASS is not a cluster and this is now measured twice, independently.** I evaluated the census's own waterfall on **every one of the 6,784 minutes** of its window. UNCLASS is true on **73.9% of the tape** and takes **60.3% of the runs** — a **lift of 0.82**. A label with a lift below 1.0 is not a footprint. It is the null hypothesis with a name tag, and "UNCLASS: 41 runs" reads as a discovery when it means "41 runs happened somewhere in the three-quarters of the week we have no test for."
2. **And it is worse than last week thought: 21 of the 68 labels flip on nothing but where a minute starts.** I changed one thing — the phase of the 60-second flow buckets — and 31% of the census's cluster assignments changed. Only 17 of 68 run timestamps are minute-aligned, so this is not a hypothetical.
3. **Last week's survivor was never shipped, and the tape since has buried it.** UNCL-RIDER-ASIA went to SHADOW on 2026-08-15 and no arm by that name exists in shadow.db. I replayed it myself. It is now **four consecutive red weeks**, and the newest week — pure forward tape it has never seen — is **-\$472 over 16 trades, -\$29.51 a trade, 18.8% wins**. The drawdown that made it a non-deploy last week did not recover; it deepened.
4. **The prize is not \$5,840.** A boarder that fires inside every UNCLASS run and then exits at the exact top of each move with no costs at all reaches **\$3,239 — 55%**. Put an honest stop on it and bank half the excursion and it is **\$1,592, 27%**. Two thirds of the headline is a close-to-close number that no late entry can ever collect.

The full ledger of every attempt — thirteen named candidates, none of them quietly dropped — is at the bottom, along with the two things I do want to carry forward, both of which are router instruments rather than gates.

△ **One housekeeping note on the brief.** The prompt I was handed says UNCLASS is "34 of the 60 sat-out runs and \$4,097 of the \$8,025 ceiling". That is last week's census. The frozen file for this week says **41 of the 62 sat-out runs and \$5,840 of the \$9,610 ceiling** — 61% of the money on the table, the biggest bucket again by a wide margin. Every number below is scored against the frozen file, not the brief.

0. THE ANCHOR — I REPRODUCED THE FROZEN CENSUS EXACTLY BEFORE TOUCHING ANYTHING

I did not re-run `run_census.py`. I re-implemented its algorithm against the `parquet lake` and checked it row for row against `census_stdout.txt`.

CHECK	RESULT
window	2026-08-17 00:00 → 08-21 20:59 UTC, 81,326 5s bars
typical 15-min range → threshold	37pt → 55pt — identical to the frozen file
runs found	68
rows matching frozen on time and direction and move	68 / 68
rows matching frozen on the cluster label	68 / 68
UNCLASS runs	41, all of them sat out, ceiling \$5,840

That is a full-strength anchor — every number that follows sits on the same run set the report's Movement 1 shows. (Last week's version of this phase reproduced 59 of 68 labels; the missing 9 were the bucket-phase problem, which I isolate deliberately in §1 instead of leaving it as noise.)

What the 41 runs are, before any hunting

SPLIT	BREAKDOWN
direction	24 DOWN / 17 UP
ASIA 00-07Z	11 runs · \$1,564
EUROPE 07-13Z	14 runs · \$2,048
US 13-20Z	12 runs · \$1,676
LATE 20-24Z	4 runs · \$552
median move	67 pt

1. ★ INTERROGATE THE LABEL FIRST. This outranks anything built on top of it.

Your standing rule, and it changes the whole question. `run_census.cluster()` is a four-branch waterfall and UNCLASS is the else clause:

```
|flow z| >= 1.0    (needs >= 30 one-minute tick buckets in the trailing 2h)
                    -> FLOW-LED if the flow sign agrees with the move, else
VACUUM
amp > 0.30 %      -> VOL-EXPANSION
13 <= hour < 15   -> OPEN/NEWS
otherwise         -> UNCLASS
```

I ran that waterfall on **every minute** of the census window, not just the 68 run minutes.

Base rate of every label, on all tape

LABEL	MINUTES	% OF ALL TAPE	SHARE OF THE 68 RUNS	LIFT
UNCLASS	5,013	73.9%	41 = 60.3%	0.82 ←
OPEN/NEWS	368	5.4%	13 = 19.1%	3.52
VACUUM	701	10.3%	9 = 13.2%	1.28
FLOW-LED	677	10.0%	4 = 5.9%	0.59
VOL-EXPANSION	25	0.4%	1 = 1.5%	3.99

Across the whole tick-covered lake — 49,064 minutes, ten times the sample — it is the same picture: **UNCLASS 75.4%**, OPEN/NEWS 4.9%, a flow event on 18.0%, VOL-EXPANSION 1.7%.

UNCLASS is true on three quarters of the tape and is UNDER-represented among the runs. That is the whole finding and it outranks every candidate in this document. Last week's number was 74.9% / lift 0.76; mine is 73.9% / lift 0.82 on a different week with a re-derived implementation. Two independent measurements a week apart landing in the same place is about as settled as this desk gets.

And the label is decided by where a minute starts

This is new, and it is worse than the "13% of assignments" figure last week reported, because I isolated the cause instead of estimating it. I changed **one thing** — the phase of the 60-second flow buckets, wall-clock instead of anchored to each run's own timestamp — and left everything else byte-identical.

TEST	RESULT
run timestamps that are actually minute-aligned	17 / 68
labels that CHANGE on bucket phase alone	21 / 68 = 31%

The census slides its 15-minute window by array index, not by clock, so where the 5s tape has holes the two drift apart and the run lands at :10, :25, :55 past. Nearly a third of this week's cluster assignments are decided by an arbitrary implementation choice nobody made on purpose.

The clock edge is doing real work too

WHERE THE UNCLASS RUNS SIT	COUNT
----------------------------	-------

in the hour on either side of the OPEN/NEWS window (12Z and 15Z)	10 / 41 (24%)
hour histogram	00:2 01:2 02:2 04:1 05:2 06:2 07:5 08:2 10:1 11:2 12:4 15:6 16:3 17:2 18:1 20:1 23:3

The 13:00–15:00Z fallback's right-hand edge is arbitrary — the hunt last week found the densest ignition minute is 13:25, not 13:30, so it is not a news-release footprint either. **One hour of slack in either direction moves 10 of these 41 runs out of this bucket without a single new measurement being taken.**

What "UNCLASS" actually means, minute by minute

REASON A MINUTE LANDS IN UNCLASS	CENSUS WINDOW	WHOLE LAKE
a flow verdict was available, and the flow was ordinary ($ z < 1$)	99.3%	85.6%
no flow verdict — fewer than 30 tick buckets in the trailing 2h	0.7%	14.4%

So on the report week UNCLASS means one thing almost exclusively: **the tape was not doing anything unusual with aggressor flow**. The thin-data path, which was a live hypothesis, is 0.7% of it.

And the members look like ordinary tape at their own start

This is the part that matters for the hunt. I measured the 41 members against the whole tape's distribution, at the run's first minute:

FEATURE AT THE RUN'S START	THE 41 UNCLASS RUNS	THE OTHER 27 RUNS	ALL-TAPE MEDIAN (THE NULL)
ATR (20-min, points)	11.65	24.15	12.31
5-min volume ÷ trailing-hour median	1.06	1.41	1.02
30-min efficiency ratio	0.153	0.178	0.163
pre-run amplitude %	0.08	0.19	—
$ flow\ z $	0.37	1.00	—
direction: continuation of the prior 30-min drift	22 / 41	—	coin flip

The UNCLASS runs are, at the moment they begin, statistically indistinguishable from a random minute of tape. Their ATR sits below the all-tape median. Their volume is on it. Their efficiency is below it. And you cannot even call the direction — 22 continuation, 19 reversal.

That is not a failure of the census; it is the definition. The other three branches are the tests that find footprints, so the residue is by construction the runs with no footprint. **Hunting for a footprint inside UNCLASS is a category error.**

Which is exactly why your instruction to RIDE first rather than classify first is the right one, and why I did not stop here.

2. STEP ONE — RIDE FIRST. Board it late, and measure what is left.

The rule is deliberately dumb and direction-agnostic: at each completed one-minute bar take the net move over the trailing **W** minutes; if it is at least **k × ATR20**, board it in whichever direction the move went. No prediction, no classification.

W	K	BOARDS INSIDE	MEDIAN MINUTES LATE	MEDIAN POINTS LEFT	MEDIAN ATRS LEFT	FIRES/DAY	
	3	1.5 41/41		2.2	36.0	3.10	145
	3	2.5 28/41		7.5	26.4	2.10	25
	5	2.0 40/41		3.3	31.2	2.57	137
	5	2.5 34/41		4.4	27.5	2.20	68
	5	3.0 25/41		5.5	20.2	1.79	33
	10	2.0 40/41		4.7	27.2	2.15	285
	15	2.5 41/41		5.8	25.5	1.77	277

STEP 1 PASSES, unambiguously. A five-minute boarding trigger gets inside **40 of the 41** runs at a median **3.3 minutes** late with a median **31 points / 2.57 ATRs** still ahead. Boarding is not the problem and never was. On gold it was 19 minutes in with 80% left; on MNQ it is three minutes in with two and a half ATRs left, which is better.

The trouble is in the right-hand column. **That same trigger fires 137 times a day.** After the cooldown of actually being in a trade it settles to about 27 entries a day at k=2.0, or 5 a day at k=2.5 with the filter of \$4 on. The gap has never been detection. It is selection.

And the excursion profile inside a run is superb

Boarding at the first W5/k2.0 thrust inside each run and then measuring the path:

HORIZON AFTER BOARDING	N	MEDIAN MFE	MEDIAN MAE	PERFECT-EXIT TOTAL
to the run's own 15-min end	37	44.2 pt	6.0 pt	\$3,239
+60 minutes	39	60.2 pt	13.8 pt	\$5,633
+240 minutes	38	99.0 pt	37.9 pt	\$8,660

Median MAE of 6.0 points against an ATR of 11.65 is **half an ATR** — and **0 of the 40 boarded runs touched a 2.5×ATR stop before reaching their best excursion inside the run window**. Once you are on one of these, a wide stop is never threatened. The entire problem is the fires that are not on one.

The prize, restated honestly

	\$	% OF THE CENSUS HEADLINE
census UNCLASS ceiling (close-to-close, hindsight, 1 lot)	\$5,840	100%
a cheating oracle — boards every run, exits at the exact top, no costs	\$3,239	55%
honest-ish — same boards, 2.5×ATR stop, bank half the excursion	\$1,592	27%

Two thirds of the number on the census page cannot be collected by anything that enters after the move has proven itself. When you see "\$5,840 on the table" in Movement 1, the rideable part of it is about sixteen hundred dollars.

3. STEP TWO — THE EXIT, SWEEP WIDE BEFORE ANY VERDICT ON THE ENTRY

Your standing finding is that a tight or trailing exit gets shaken out of a move that grinds. I swept it properly — stop 1.0–4.0×ATR against target 2–16×ATR, caps 30 minutes to 12 hours — and I did it on segments, not blanket.

Blanket, all tape, W5/k2.0, \$/trade — this is what a bug looks like

STOP \ TARG	2	3	4	6	8	10	12
1.5	−2.04	−0.27	−1.04	−1.21	1.12	1.51	1.52
2.0	−1.31	−0.37	−1.02	−0.52	−1.48	1.22	0.92
2.5	−0.85	1.29	−0.06	−0.93	−1.67	3.45	0.76
3.0	−0.89	−0.97	−2.14	−1.59	−3.06	−0.73	−8.44
4.0	−0.87	−4.57	−2.51	−5.89	−6.71	−6.16	−8.01

Blanket over 1,403 trades the rider makes **−\$1,308, −\$0.93 a trade**. Sprayed across every regime it is a coin. Exactly as your backtest discipline predicts, that number is not a verdict on anything — it is the average of the regimes it should be on with the ones it should be off.

Segmented, and the segments are not equal

BY SESSION (BLANKET RIDER)	N	NET\$	WIN%	\$/TR	
ASIA 00-07		433	+1,195	32.6%	+2.76
US 13-20		437	-588	33.2%	-1.34
LATE 20-24		139	-720	29.5%	-5.18
EUROPE 07-13		394	-1,196	29.7%	-3.03

BY 30-MIN EFFICIENCY AT N BOARD TIME	NET\$	WIN%	\$/TR	PF	
trend ER ≥ 0.40	196	+4,115	38.3%	+20.99	1.40
mid 0.20-0.40	454	-1,227	30.0%	-2.70	0.94
chop ER < 0.20	730	-4,117	30.7%	-5.64	0.89

△ AND HERE IS A TRAP I WALKED INTO, WHICH IS WORTH MORE THAN THE TABLE ABOVE

That +\$20.99 a trade is **not achievable and the table above is misleading**. It is a post-hoc slice of a trade list produced by an unfiltered rider. A rider that actually refuses the low-ER fires is not busy during them — it is free, so it takes entries the sliced version never had. Run the identical filter inside the simulation and:

THE SAME ER ≥ 0.40 FILTER	N	NET\$	WIN%	\$/TR	PF
as a post-hoc SLICE (unimplementable)	154	+4,363	40.3%	+28.33	1.53
run INSIDE the simulation (what you could trade)	317	+1,801	33.1%	+5.68	1.10

The number more than halves and n doubles. Every filtered figure in the rest of this document is the in-sim form. I am flagging it loudly because slicing a trade list is the natural way to test a filter, it is what the tables above do, and it is wrong by a factor of five.

The exit sweep on the home segment, in-sample, W5/k2.5/ER≥0.45 — \$/trade (n)

STOP \ TARG	3	4	6	8	10	12	16
1.0	7.05 (198)	8.21 (186)	8.98 (177)	10.77 (173)	9.32 (167)	6.96 (166)	10.41 (161)
1.5	13.27 (194)	14.53 (180)	14.91 (169)	26.59 (163)	31.97 (157)	29.01 (154)	25.84 (151)
2.0	15.73 (188)	14.62 (173)	17.38 (161)	29.10 (152)	38.04 (145)	27.62 (144)	25.32 (140)

2.5	14.57 (187)	14.90 (170)	17.94 (157)	30.05 (146)	37.05 (139)	23.37 (136)	18.05 (133)
3.0	8.69 (184)	8.95 (168)	10.33 (154)	25.32 (142)	38.14 (135)	22.71 (132)	17.58 (130)
4.0	10.44 (180)	13.95 (165)	20.29 (151)	35.10 (138)	42.55 (130)	35.54 (128)	28.12 (126)

Your wide-exit thesis is CONFIRMED on the target axis and REFUTED on the stop axis. There is a genuine ridge at a **10×ATR target** that holds across every stop width tested — that is a plateau, not a point, and it is the single most robust thing in this section. The stop width barely matters: 1.5, 2.0, 2.5 and 4.0 all land in the same place, which is exactly what you would expect given the median MAE inside a run is half an ATR. So the "wide stop" half of the finding does not apply here — the stop is not what is shaking us out, because we are not being shaken out.

Managed exits, on the same segment, all lost to the flat far target:

EXIT MECHANISM	N	NET\$	WIN%	\$/TR	PF
flat stop 3 / target 8	369	−2,467	27.1%	−6.69	0.90
chandelier trail 3.0×ATR	395	+1,454	34.9%	+3.68	1.10
chandelier trail 2.0×ATR	474	−1,777	30.8%	−3.75	0.84
breakeven stop at +2×ATR	400	−845	13.8%	−2.11	0.94
tight stop 1.5 / target 3	510	+926	33.9%	+1.82	1.06

Time cap sweep on the same cell: 30 min +\$3.80/tr, 60 min +\$3.92, 120 min −\$5.06, 240 min −\$6.69, 720 min −\$4.45. Nothing monotone, nothing to build on.

4. STEP THREE — THEN FILTER, WITH A PLACEBO ON EVERY SINGLE CUT

Every cut below is scored against **dropping the same NUMBER of fires at random**, 400–500 draws, exactly as you specified. Unfiltered baseline in-sample: n=688, +\$3,885, **+\$5.65/tr**, PF 1.11.

CAUSAL CUT	N	NET\$	\$/TR	KEEPS	RANDOM-CUT P95	P	VERDICT
ER30 ≥ 0.40 (trending tape)	196	+6,211	+31.69	28%	+20.99	0.006	BEATS placebo
ER30 ≥ 0.40 AND ATR ≥ 12pt	138	+5,278	+38.25	20%	+25.19	0.012	BEATS placebo
	120	+3,754	+31.28	17%	+27.87	0.030	BEATS placebo

ER30 $\geq$ 0.40 AND volume $\geq$ 1.5 $\times$						
ATR $\geq$ 12pt	478	+4,355	+9.11	69%	+12.72	0.240 ~placebo
ER30 $\geq$ 0.30	348	+2,728	+7.84	51%	+15.51	0.386 no better than random
volume $\geq$ 1.5 $\times$ (volume confirms)	420	+3,211	+7.64	61%	+13.67	0.338 no better than random
ATR $\geq$ 16pt (loud tape only)	298	+2,327	+7.81	43%	+16.20	0.402 no better than random
volume $\leq$ 0.9 $\times$ (thin, no aggression)	213	+732	+3.44	31%	+19.25	0.600 no better than random

The thin-volume cut deserves a note because it was my own best causal hypothesis and the data refused it. The members move on ordinary volume with no aggression, so "price is walking because nobody is there" was the obvious story — and boarding only the thin-volume fires is **worse than throwing away the same number of fires by coin flip**. That hypothesis is dead on its own evidence.

One cut passes with room to spare: **efficiency**. So the candidate is built on it.

5. THE CANDIDATE — UNCL-RIDE-ER

Chosen from the **interior of the in-sample shelf, not its peak**. The sweep's best cell was $ER \geq 0.45$ / $k=3.0$ at +\$42.68 a trade; I took the shelf's middle instead, because a boundary optimum is a curve-fit tell and I would rather report a smaller honest number.

board — the 5-minute net move is $\geq 2.5 \times ATR_{20}$ and the 30-minute efficiency ratio is ≥ 0.40 **side** — the sign of that move. No prediction, no classification, no direction view. **entry** — the OPEN of the next 1-minute bar (no same-bar fills, no look-ahead) **stop** — $2.0 \times ATR_{20}$ **target** — $10 \times ATR_{20}$ **cap** — 240 minutes **costs** — \$2.00/point, **\$1.50 per round trip**; a bar that touches stop and target in the same 5 seconds is scored as a STOP

Plateau checks — is this a shelf or a cliff?

ER30 GATE	0.00	0.10	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.70
\$/trade	6.05	12.22	4.49	0.32	7.84	22.88	31.69	38.04	32.75	5.02	-27.20	-67.57

n	687	586	495	429	348	270	196	145	105	66	38	6
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BOARDING WINDOW	2	3	4	5	6	8	10	15	20
\$/trade	18.39	33.07	38.65	38.04	19.95	8.55	2.40	−0.76	−2.32

THRUST K	1.5	1.75	2.0	2.25	2.5	2.75	3.0	3.5	4.0
\$/trade	9.57	14.62	11.93	25.22	38.04	39.33	42.68	27.43	35.15

Honest read: **W and k are clean shelves. ER is not.** It sags at 0.20–0.30 (4.49, 0.32, 7.84), rises to a peak at 0.45 and falls off a cliff after 0.50. A non-monotone parameter with a peak two cells wide is a warning I recorded before spending the out-of-sample leg — and the leg then confirmed it.

The full battery, in-sample (2026-06-19 → 08-08, 36 sessions)

Headline: n=196, +\$6,211, 25.0% wins, +\$31.69 a trade, PF 1.64. A 25% win rate is not a kill criterion here; the exit mix is 37 targets (+\$13,067), 12 time caps (+\$2,888), 147 stops (−\$9,744) — a textbook few-big-winners shape.

TEST	RESULT	PASS?
placebo — ER filter vs a random cut of the same size	actual +\$31.69 vs random mean +\$6.09, p95 + \$20.99 · p = 0.006	☐
placebo — signal shifted 5 min early	−\$4.33/tr	☐
placebo — signal shifted 10 min early	−\$5.16/tr	☐
placebo — signal shifted 30 min early	+\$19.67/tr	⚠ weak
placebo — signal shifted 60 / 120 min early	−\$8.94 / +\$2.00	☐
placebo — direction flipped	−\$3.97/tr	☐
strip the best 1 / 3 / 5 trades	+\$26.79 / +\$17.67 / +\$9.52	☐
strip the best 10 trades	−\$3.97/tr	⚠ ten trades of 196 are the whole result
leave-one-day-out, worst	+\$26.53/tr (dropping 2026-06-25)	☐
sessions green	18 / 36 = 50%	⚠
long / short symmetry	long +\$24.65 (n=91) · short +\$37.79 (n=105) — both green	☐
cost stress: fee ×3	+\$28.69/tr	☐
cost stress: 1 full point of slippage per side	+\$27.69/tr	☐
in-sample first half (06-19→07-14)	+\$36.04/tr (n=103)	☐
in-sample second half (07-15→08-08)	+\$26.88/tr (n=93)	☐

Both halves of the in-sample period are strongly green. Nothing here says stop.

In-sample segments

SESSION	N	NET\$	WIN%	\$/TR	PF
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US 13-20	59	+6,360	39.0%	+107.79	2.91
LATE 20-24	19	+1,231	42.1%	+64.81	2.27
ASIA 00-07	66	+81	18.2%	+1.23	1.03
EUROPE 07-13	52	-1,461	11.5%	-28.09	0.44

ATR LEVEL	N	NET\$	\$/TR	PF	
Q4 loud		70	+5,089	+72.70	2.02
Q1 quiet		30	+607	+20.23	1.79
Q2		44	+295	+6.70	1.19
Q3		52	+221	+4.25	1.09

6. SPENDING THE OUT-OF-SAMPLE LEG — and it goes red

2026-08-09 → 08-24, 11 sessions, untouched until this moment.

	N	NET\$	WIN%	\$/TR	PF
in-sample	196	+6,211	25.0%	+31.69	1.64
out of sample	71	-680	15.5%	-9.58	0.75
OOS strip-3	68	-1,558	11.8%	-22.92	0.42
OOS sessions green	4 / 11				
OOS placebo (signal 30 min early)	49	+73	20.4%	+1.48	1.05

The fake signal beats the real one out of sample. The OOS session split does not rescue it either — ASIA +\$11.91 (n=20), LATE -\$2.31 (n=4), US -\$18.62 (n=23), EUROPE -\$20.04 (n=24) — and note that **US 13-20 was the in-sample star at +\$107.79 and is the OOS loser.**

The second, much larger independent leg — 205 sessions back to 2025-09-14

Eleven sessions is thin, so I spent a second leg that is not thin: the 1-minute lake, every session before the 5s tape begins, none of which contributed to a single choice above. Path resolution here is 1-minute OHLC and a bar touching both levels is scored as a stop, so this leg is conservative, not optimistic.

	N	NET\$	WIN%	\$/TR	PF
UNCL-RIDE-ER on the untouched year	1,246	+4,417	18.7%	+3.54	1.09
strip the best 3				+1.25	
strip the best 10				-2.35	
sessions green	93 / 205 = 45%				
placebo — signal 30 min early	725	+3,222	17.2%	+4.44	1.13
	704	-5,373	14.2%	-7.63	0.78

placebo — signal 120
min early

placebo — direction flipped	1,405	−1,801	16.7%	−1.28	0.97
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On the largest honest sample I have, a signal taken thirty minutes early makes more money than the real signal. Ten trades out of 1,246 carry the entire result. Monthly it is noise: −8.14, +14.60, +0.16, +1.35, +6.42, +4.84, −0.05, +13.01, −4.62, +0.38.

That is the cause of death, and it is a robustness test, not a win rate.

The final arbiter — is ANY configuration green on both independent legs?

I swept 137 configurations ($W \times k \times ER \times \text{stop} \times \text{target}$) with $n \geq 150$ on the year leg and $n \geq 20$ on the recent OOS leg, and asked how many are positive on both.

	COUNT	SHARE
positive on the 205-session YEAR leg	77	56%
positive on the 11-session recent OOS leg	29	21%
positive on BOTH	18	13%
what pure chance predicts if the two legs are independent	—	12%

Thirteen per cent against a chance rate of twelve. There is no configuration of this rider that survives two independent samples any more often than a coin would put it there. That is the cleanest kill in this document and it is not about any one parameter choice — it retires the whole family, not just the cell I picked.

7. THE DIAGNOSIS — which half of the rule was ever doing anything?

The shifted placebo beating the real signal has a specific meaning worth chasing: efficiency is persistent, so a signal taken 30 minutes early still lands on trending tape. So I decomposed it.

ARM	YEAR LEG (205 SESSIONS)	RECENT 5S LAKE (51 SESSIONS)
A thrust trigger + $ER \geq 0.40$ (the candidate)	$n=1,246 \cdot +\$3.54/\text{tr}$	$n=267 \cdot +\$20.72/\text{tr}$
B same, signal shifted 30 min	$n=725 \cdot +\$4.44/\text{tr}$	$n=192 \cdot +\$15.03/\text{tr}$
C NO trigger at all — $ER \geq 0.40$, direction = sign of the 30-min move	$n=2,326 \cdot +\$3.05/\text{tr}$	$n=541 \cdot -\$6.13/\text{tr}$
D arm C with the direction flipped	$n=2,970 \cdot -\$2.55/\text{tr}$	$n=702 \cdot -\$0.62/\text{tr}$
E arm C without the ER gate (every minute)	$n=5,205 \cdot +\$0.05/\text{tr}$	$n=1,193 \cdot -\$1.31/\text{tr}$

The two legs give opposite answers. On the year leg the thrust trigger is worthless (F is negative) and the ER regime is everything (C ≈ A ≈ B, and D is its clean mirror). On the recent leg the ER regime alone loses money (C is negative) and the trigger is what earns (F positive). Neither component is stable, and a rule whose own decomposition inverts between two samples is measuring the sample, not the market.

8. THE ESCALATION — the size threshold, which is the finding you asked for

Per the brief I escalated: full sat-out set → top-25 → top-15 → top-10 → top-5. **The escalation runs flat, and on the one axis that matters it runs BACKWARDS.**

Does a footprint appear as the runs get bigger? (AUC vs the whole tape; 0.50 = no discrimination)

CUT	N	MEDIAN MOVE	ATR	VOLUME RATIO	ER30	5-MIN THRUST
all 41 UNCLASS	41	67 pt		0.444	0.517	0.424
top-25 by size	25	78 pt		0.476	0.518	0.428
top-15 by size	15	83 pt		0.419	0.572	0.332
top-10	10	90 pt		0.486	0.535	0.296
top-5	5	96 pt		0.515	0.633	0.313
≥ 80 pt	12	88 pt		0.428	0.571	0.280

Every AUC sits on 0.5, and the "5-min thrust" column falls to 0.28-0.33 as the runs get bigger — meaning **the biggest UNCLASS runs are LESS preceded by a thrust than an average minute of tape.** The top-5 shows a volume AUC of 0.633 on n=5, which is five observations and not a finding.

Where the money actually is, once you are on a run

Building the census's run algorithm across the whole 5s lake (608 runs on 51 sessions, 11.9/day, median 100pt) and tagging every rider fire by the run it boarded:

	N	NET\$	\$/TR
fires that landed inside an aligned run	116 (43%)	+\$6,157	+\$53.08

fires on tape that never became a run	151 (57%)	-\$626	-\$4.15
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RUN SIZE THE FIRE BOARDED	N	NET\$	WIN%	\$/TR	PF
55-70 pt	12	-194	16.7%	-16.12	0.57
70-85 pt	17	+8	23.5%	+0.47	1.01
85-100 pt	12	+668	41.7%	+55.63	2.57
100-130 pt	22	-82	22.7%	-3.75	0.93
≥ 130 pt	45	+5,695	42.2%	+126.56	3.12

Cumulatively: ≥55pt +\$56.43/tr, ≥70pt +\$65.50, ≥90pt +\$77.27, ≥120pt +\$109.81. **The size threshold is real and it is around 130 points on MNQ** — below that the runs pay for themselves and nothing more; above it they carry the entire book.

But that is the PRIZE, not a RULE, and here is why. The rider cannot know the run's eventual size at the moment it boards. The obvious counter-argument is that it can: ATR at board time correlates +0.719 with eventual run size in points, and the 5-minute move correlates +0.851. Those numbers are a scale artefact and I nearly filed them as a finding. Run size is measured in points and so is volatility, so a loud day produces big runs by arithmetic. Normalise the run size by the ATR at board time and the correlations collapse:

BOARD-TIME READ	VS RUN SIZE IN POINTS	VS RUN SIZE IN ATRS
ATR	+0.719	-0.221
5-min move	+0.851	+0.133
5-min move ÷ ATR	+0.239	+0.477
ER30	+0.336	+0.045

The only surviving read is the thrust size expressed in ATRs, at $r=+0.48$ — and that is already the k parameter, whose sweep in \$5 shows no usable gradient. **There is a size at which these runs are worth catching. There is no instrument on this desk that identifies it at board time.**

Big-moves-caught on the frozen census week: **UNCL-RIDE-ER boards 14 of the 41 UNCLASS runs, and its trades on that week are $n=33$, -\$209, -\$6.33 a trade.**

9. CLOSING THE LOOP — last week's survivor was never shipped, and it is now four weeks red

Last week this phase sent **UNCL-RIDER-ASIA** to SHADOW: thrust board $w=10$ / $k=2.0$, 00:00–06:00 UTC, $ATR \geq 8pt$, $ATR \text{ percentile} \leq 0.60$, stop $2.5 \times ATR$, target $6 \times ATR$, 2-hour cap, minus the 04:15–05:15Z gateway outage. Quoted at $n=157$, +\$5,006, +\$31.88 a trade.

There is no arm by that name in shadow.db. I listed all 82 strategies in the shadow book; the only rider shipped from last week's greenfield work is rider_w5 from the pooled phase, which the sibling RIDER_ALL section grades at $-\$1,471$. So the SHADOW recommendation from this phase was made and not executed. **A disposition that nobody actions is a disposition that cannot teach us anything, and this is the second week running that a "put it in shadow" line has gone nowhere.**

Since nothing recorded it, I replayed it myself across the whole 5s lake. (My absolute numbers are smaller than last week's — I have no V5 leg here and I had to re-derive the ATR-percentile definition — so read the shape, not the totals. The shape is unambiguous.)

ISO WEEK	DATES	N	NET\$	WIN%	\$/TR
26	06-23 .. 06-26	25	+233	36.0%	+9.32
27	06-29 .. 07-03	30	+751	46.7%	+25.02
28	07-07 .. 07-10	25	+510	40.0%	+20.38
29	07-13 .. 07-17	24	+213	37.5%	+8.88
30	07-20 .. 07-24	25	+368	40.0%	+14.73
31	07-27 .. 07-31	29	−85	27.6%	−2.93
32	08-03 .. 08-07	25	−398	24.0%	−15.90
33	08-10 .. 08-14	15	−547	6.7%	−36.46
34	08-19 .. 08-21	14	−533	14.3%	−38.08
35	08-24	2	+61	50.0%	+30.53

	N	NET\$	WIN%	\$/TR
up to 08-15 — everything last week could see	198	+1,045	33.8%	+5.28
since 08-17 — pure forward tape	16	−472	18.8%	−29.51

First five weeks +\$2,075 on 129 trades. Last five weeks **−\$1,502 on 85 trades**. The forward week is the worst of the lot. And it was never an answer to this bucket anyway: **it boards 3 of the week's 41 UNCLASS runs.**

UNCL-RIDER-ASIA is REFUTED by the forward out-of-sample leg — not parked, not thin-n. It had its shelf, the shelf kept going down for four straight weeks, and the ATR floor that was added specifically to explain the first two red weeks did not restore any of them.

10. THE RENAMING DELIVERABLE — four shapes the census could actually test for

You asked that if the bucket splits into real shapes I name them so the census can classify them next week. It does split, on axes the census already computes. Here are the four, with definitions that are one if statement each.

PROPOSED SHAPE	N	CEILING	MEDIAN MOVE	MEDIAN ATR	MEDIAN VOL RATIO	MEDIAN ER30	DN/UP
TURN	19	\$2,716	67 pt	12.5	1.06	0.142	11/8
STEP-REPRICE	10	\$1,465	66 pt	12.3	2.06	0.308	6/4
DRIFT-VOID	6	\$797	62 pt	9.9	0.75	0.091	4/2
GRIND-CONT	6	\$862	70 pt	11.4	0.82	0.267	3/3

- **STEP-REPRICE** — 5-minute volume $\geq 1.5\times$ the trailing-hour median but $|\text{flow } z| < 1$. Volume arrives and aggression does not: a two-sided repricing, not a sweep. This is the one shape with a genuinely distinctive signature and it is 10 runs / \$1,465.
- **GRIND-CONT** — the run continues the previous 30-minute drift and $\text{ER30} \geq 0.20$. On below-baseline volume, which is what makes it a grind rather than a thrust.
- **DRIFT-VOID** — volume below baseline ($< 0.95\times$) and $\text{ATR} < 12\text{pt}$. Price walks on nothing.
- **TURN** — everything else: a reversal on ordinary volume. Still the biggest half at 19 runs, and still genuinely featureless.

⚠ **Do not read the ride results per shape.** I ran them and they are $n = 8, 3, 3$ and 1 trades respectively. Quoting +\$224/trade off three trades would be exactly the kind of number this report exists to refuse. The shapes are offered as **census vocabulary**, not as a selector.

11. THE ROUTER — what I would arm, and what I would need to arm it

You are right that a gate only has to work in the regime it is armed for, so I am not allowed to kill this on the blanket average. Here is the router work, honestly.

The one bench rule that replicates on all three independent legs

EUROPE 07-13Z, ER-GATED RIDER	N	NET\$	\$/TR	PF
in-sample (36 sessions)	53	-1,557	-29.38	0.42
recent OOS (11 sessions)	24	-481	-20.04	0.44
year leg (205 sessions)	373	-905	-2.43	0.93

BENCH any late-boarding continuation rider between 07:00 and 13:00 UTC. Red on every leg, including the 373-trade one. It is the only piece of the session structure that replicates.

And the session rule that does NOT replicate — stated so nobody ships it

SESSION	IN-SAMPLE \$/TR	RECENT OOS \$/TR	YEAR-LEG \$/TR
ASIA 00-07	+4.61 (67)	+11.91 (20)	+1.67 (405)
EUROPE 07-13	-29.38 (53)	-20.04 (24)	-2.43 (373)
US 13-20	+107.79 (59)	-18.62 (23)	+12.76 (404)
LATE 20-24	+48.71 (22)	-11.62 (6)	-5.45 (128)

Spearman rank correlation of the session ordering: **IS vs OOS +0.20 · IS vs YEAR +0.40 · OOS vs YEAR +0.00**. In-sample this rider begs to be armed US-only at +\$107.79 a trade with a random-cut placebo p below 0.001. Out of sample that is its worst session. **This is precisely the router rule I would have shipped last Friday if I had stopped at the in-sample battery, and it would have been wrong.**

The ER meter as an arm/bench instrument — real, but it does not replicate forward either

SAME RIDER, ONLY THE ARMING DIFFERS	ARMED ALWAYS	ARMED ONLY WHEN ER30 ≥ 0.40
year leg, 205 sessions	n=3,238 · -\$3.40/tr	n=1,246 · +\$3.54/tr
recent OOS, 11 sessions	n=228 · -\$1.37/tr	n=71 · -\$9.58/tr

On 3,238 trades of untouched tape the efficiency meter turns a losing rider into a marginally positive one — that is a large sample and a real effect, and it is the only thing in this document with that kind of n behind it. On the most recent

eleven sessions it makes things worse. I am reporting both and recommending neither as a deploy.

The instrument we would actually need

The binding constraint is named precisely in §8: **a SIZE PREDICTOR at board time, in ATR units**. Every board-time read we own — ATR, volume, efficiency, thrust size, and (per last week's L2 work, which I did not re-dig) depth imbalance — correlates with a run's eventual size in ATRs at $r \leq 0.13$, except thrust-in-ATRs at +0.48, which is already a swept parameter with no gradient. Until something answers "given the volatility, is this going to be a 3-ATR move or a 12-ATR move", this rider risks a 2-ATR stop to collect a median 2.57 ATRs and the arithmetic never closes. **That is a missing instrument, not a failed strategy, and it is what the next attempt should go and build.**

12. EVERY ATTEMPT, BY NAME, INCLUDING THE GRAVES

Nothing here was quietly dropped. All scored identically: \$2.00/point, \$1.50/round-trip, 5s path resolution on the two recent legs, 1-minute on the year leg.

#	CANDIDATE	IS N	IS \$/TR	STRIP-3	OOS N	OOS \$/TR	YEAR N	YEAR \$/TR	CAUSE OF DEATH
1	UNCL-RIDE-ER (the candidate)	196	+31.69	+17.67	71	−9.58	1,246	+3.54	placebo — shifted signal beats it on both independent legs; OOS leg red
2	UNCL-RIDE-RAW (no filter)	688	+5.65	+1.86	228	−1.37	3,238	−3.40	negative on the 3,238-trade year leg
3	UNCL-RIDE-ER30 (looser gate)	348	+7.84	−0.21	122	−2.53	1,958	+0.97	strip-3 kills it in-sample; fails the random-cut placebo (p=0.386)
4	UNCL-FADE (flip the thrust — fade, don't ride)	262	−3.97	−13.89	86	+2.15	1,405	−1.28	red in-sample and on the year leg; the fade is not the answer
5		272	+11.18	+7.57	90	−3.52	1,683	−2.50	

	UNCL-TIGHT (stop 1.5 / targ 3 / cap 60)								year leg — confirms the tight exit is shaken out
6	UNCL- CHANDELIER (trail 3×ATR)	212	+17.02	+4.99	77	−5.67	1,350	−3.35	year leg; trailing loses to the flat far target
7	UNCL-THIN (volume ≤ 0.9× — the "price walks on nothing" hypothesis)	213	+3.44	−3.49	72	−1.91	n/a	n/a	random-cut placebo, p = 0.600 — worse than a coin flip
8	UNCL- VOLCONF (volume ≥ 1.5×)	420	+7.64	+1.81	149	+3.81	n/a	n/a	random-cut placebo, p = 0.338
9	UNCL-LOUD (ATR ≥ 16pt)	298	+7.81	−0.96	27	+27.06	n/a	n/a	random-cut placebo p = 0.402; strip-3 negative
10	UNCL-SLOW (W=15 boarding)	365	−0.74	−8.48	114	−18.03		1,876	+4.72 red in- sample AND out of sample
11	UNCL-FAST (W=3, k=3.0, stop 3 / targ 8)	62	−0.21	−27.64	26	+18.39		486	+8.45 strip-3 — three trades are the entire result
12	UNCL- NOEUROPE (bench 07- 13Z)	144	+53.28	+34.54	47	−4.24		893	+5.93 OOS leg
13	UNCL-US (arm 13-20Z only)	59	+107.79	+63.53	23	−18.62		404	+12.76 OOS leg — the in- sample star is the OOS loser
14	UNCL- RIDER-ASIA (last week's SHADOW pick)	198	+5.28	—	16	−29.51	—	—	forward OOS — four consecutive red weeks

⚠ **Row 14 uses different leg boundaries from rows 1-13.** Its "IS" column is everything up to 2026-08-15 (what last week could see) and its "OOS" column is the pure forward tape from 08-17, which is the only split that means anything for a candidate that was published last Friday. Rows 1-13 use the fixed 06-19→08-08 / 08-09→08-24 split chosen before any of them were scored.

Candidates green on all three legs with usable n: ZERO.

Two more nulls that did not earn a row because they are not candidates:

- **the post-hoc-slice method itself** — inflated the ER filter's expectancy from +\$5.68 to +\$28.33 a trade, a factor of five, and it is how a filter naturally gets tested. Recorded as a tooling finding, not a strategy.
- **VOL-EXPANSION as a census branch** — last week it fired on 0.03% of tape and was called dead code. This week it fires on 0.4% of the census window and 1.7% of the lake, so it is no longer dead, but at n=1 run out of 68 it is still not classifying anything. Recalibrate to a percentile rather than a fixed 0.30% amplitude.

13. DISPOSITION

ITEM	DISPOSITION	REQUIRED
UNCLASS as a cluster label	REFUTED as a footprint	Base rate 73.9% (census week) / 75.4% (whole lake), lift 0.82 , and its members sit ON the all-tape median for ATR, volume and efficiency. Rename it UNTESTED and report it as coverage, not as a finding. Second independent measurement in two weeks.
the census's flow-bucket phase	FIX — this is a real bug	21 of 68 labels change on bucket phase alone; only 17 of 68 run timestamps are minute-aligned. Snap the window slide to the clock, not the array index.
the OPEN/NEWS 15:00Z right edge	MEASURE IT	10 of the 41 UNCLASS runs sit in the hour either side of the window. One hour of slack reassigns a quarter of this bucket with no new evidence.
VOL - EXPANSION branch	RECALIBRATE	fires on 0.4-1.7% of tape at a fixed amp > 0.30%; make it a percentile
UNCL-RIDE-ER	PARKED	Revive only with a board-time SIZE PREDICTOR expressed in ATRs ($r \geq 0.4$ against eventual run size in ATR units on ≥ 300 runs). More days will not fix it — I already spent 205 untouched sessions and the placebo won.
UNCL-RIDER-ASIA (last week's SHADOW pick)	REFUTED	Forward out-of-sample: W31 —\$2.93, W32 —\$15.90, W33 —\$36.46, W34 —\$38.08 . Four consecutive red weeks and the ATR floor added to explain the first two restored none of them. Do not re-shadow.
the thin-volume / no-aggression mechanism	REFUTED	random-cut placebo $p = 0.600$ — the cut is worse than discarding the same number of fires by coin flip
the tight / trailing exit family	REFUTED for this shape	negative on the 1,350-1,683-trade year leg; the 10×ATR target ridge holds across every stop width
BENCH 07:00-13:00Z for any late-boarding rider	SHADOW → adopt as a router rule	red on all three legs (−\$29.38 / −\$20.04 / −\$2.43, the last on $n=373$). Cheap, fail-safe direction, no new instrument needed.
ER30 as a router ARM meter	SHADOW as an instrumented question	+\$3.54 vs −\$3.40 a trade on 3,238 untouched trades — the largest-n effect in this section — but it reverses on the recent 11 sessions.

Shadow the meter with its own control arm, do not arm a gate behind it.

the "arm US 13-20Z" rule	REFUTED before it shipped	+\$107.79/tr in-sample (placebo $p < 0.001$), – \$18.62/tr out of sample, session rank correlation IS-vs-OOS +0.20
the \$5,840 UNCLASS ceiling	RESTATE	perfect-exit oracle \$3,239 (55%); honest late board with a real stop \$1,592 (27%)
the four shapes (TURN / STEP-REPRICE / DRIFT-VOID / GRIND-CONT)	SHIP TO THE CENSUS	one if each, all computable at the run's start minute. Not a selector — vocabulary.

One stone still unturned: whether the ER meter's forward reversal is a genuine regime break or an eleven-session artefact. That is answerable for free by shadowing the meter with its control arm and waiting, and it is the single cheapest question left in this bucket.

VERDICT

VERDICT: THE UNCLASS RIDER DID NOT SURVIVE. It was BUILT (a direction-agnostic late boarder that gets inside 40 of the 41 runs at a median 3.3 minutes late with 2.57 ATRs still ahead), RIDDEN (exit swept stop $1.0\text{--}4.0\times\text{ATR}$ against target $2\text{--}16\times\text{ATR}$ and caps 30 min–12 h, finding a real $10\times\text{ATR}$ target ridge that holds across every stop width), FILTERED (eight causal cuts each placebo-controlled against discarding the same NUMBER of fires at random; $\text{ER}_{30} \geq 0.40$ beat its placebo at $p = 0.006$ and the thin-volume, volume-confirm, loud-tape and looser-ER cuts all failed theirs and are named in the ledger), and ROUTED (arm/bench rules written in the live router's ER / ATR / session vocabulary) — your order of work, in your order — and it STILL LOST. THE TEST THAT KILLED IT IS THE PLACEBO / SHUFFLE TEST, corroborated by the OUT-OF-SAMPLE LEG: on 205 untouched sessions the real signal makes +\$3.54 a trade and the SAME rule taken thirty minutes early makes +\$4.44, with ten trades out of 1,246 carrying the whole result and only 45% of sessions green; on the 11-session forward leg the real signal makes –\$9.58 a trade and the shifted fake makes +\$1.48. The family kill is stronger than the single cell: across 137 configurations with usable n on both independent legs, 13% are positive on both against a 12% chance rate, so no setting of this rider replicates any better than a coin. THE LABEL ITSELF IS THE PHASE'S PRIMARY FINDING AND IT IS REFUTED AS A FOOTPRINT — UNCLASS is true on 73.9% of the census window and 75.4% of the whole lake, takes

60.3% of the runs for a LIFT OF 0.82, its members sit ON the all-tape median for ATR (11.65 vs 12.31), volume (1.06 vs 1.02) and efficiency (0.153 vs 0.163), their direction is a 22/19 coin flip, and 21 of the 68 cluster assignments change on nothing but where a sixty-second bucket starts — rename it UNTESTED and report it as coverage. THE BANNED VERDICT IS NOT BEING USED: these runs ARE boardable (40/41) and I have NAMED four shapes (TURN 19, STEP-REPRICE 10, DRIFT-VOID 6, GRIND-CONT 6) for next week's census. The reason they cannot be ridden for money is priced, not asserted: the \$5,840 headline is worth \$3,239 to a cheating oracle and \$1,592 to an honest late board, 57% of fires never become a run, and the size threshold at which a boarded run becomes properly tradeable is ~130 POINTS on MNQ (+\$126.56/trade above it versus -\$16.12 in the 55-70pt band) — which is a PRIZE and not a RULE, because every board-time instrument we own correlates with eventual run size in ATR units at $r \leq 0.13$ once the volatility scale artefact is removed (the tempting +0.72 ATR and +0.85 thrust correlations are scale, and in ATR units they are -0.22 and +0.13). SEPARATELY AND IMPORTANTLY: last week's survivor UNCL-RIDER-ASIA was sent to SHADOW and NEVER SHIPPED — no arm of that name exists in shadow.db — and replayed forward it is REFUTED by four consecutive red ISO weeks (-\$2.93, -\$15.90, -\$36.46, -\$38.08), with the pure forward week since 08-17 at -\$472 over 16 trades (-\$29.51/trade, 18.8% wins); it also boards only 3 of this week's 41 runs and was never an answer to this bucket. TWO THINGS GO FORWARD, both ROUTER INSTRUMENTS rather than gates: BENCH 07:00-13:00Z for any late-boarding rider (red on all three legs including $n=373$), and SHADOW the ER30 arm-meter with its own control arm (+\$3.54 vs -\$3.40 a trade on 3,238 untouched trades, but reversing on the last eleven sessions — that reversal is the one cheap question left). THE NEXT ATTEMPT NEEDS A BOARD-TIME SIZE PREDICTOR IN ATR UNITS, not more days, not a wider exit, and not a new cluster label.

OPEN/NEWS — the 13:00-15:00Z cash-open window

gf_full_OPEN-NEWS.md · 10,349 words · written 2026-08-25 22:58 UTC

GREENFIELD HUNT — cluster OPEN/NEWS

Friday report 2026-08-21 week · Movement 3 · full working

(MNQ · \$2.00/point · **\$1.50 per round trip** · parquet lake *gazbot7.lake*
2026-07-16 → 2026-08-21 · frozen census *census_summary.json*)

0. THE SHORT VERSION, BEFORE ANY OF THE MONEY

Garrath — I was sent to invent a new entry that catches the runs we sat out in the OPEN/NEWS bucket. I built nine, killed all nine, and the thing that killed them is not any one of them.

The order below is the order of importance, not the order I did the work in.

#	FINDING	ONE LINE
1	The label was fixed on 08-15, and the fix worked — but only halfway	OPEN/NEWS is no longer if 13<=hour<15 checked first. It is now the residual : the 13-15Z clock minus a flow event minus a vol-expansion. It went from 100% → 58.7% of its own window. It is still true on the majority of the tape it covers.
2	★ Its non-clock content is ANTI-predictive	P(run) is 3.427% on the bare 13-15Z clock and 3.158% on the labelled subset. The two footprint tests strip out 51 of 111 in-window runs and keep the lower-density half. Label lift over the bare clock = 0.92x . Deleting the footprint tests would make the bucket a better predictor than leaving them in.
3	★★ The bucket is defined by the ABSENCE of evidence	A run is OPEN/NEWS because no flow footprint and no amplitude footprint was found. Hunting a footprint inside a bucket built from footprint-failures is close to self-defeating, and it dictated where I looked: only channels <code>cluster()</code> never reads — price structure, raw volume, and L2.
4	Nine gates, nine graves	ORB30, COILBRK, PULSE, PULSE-13h, FADE1, RNGX60, VWRC, OAB, MOMBRK, BOOKLEAN. Every one reported below by name with its stats and its cause of death. Best survivor-on-paper was MOMBRK at +\$32.87/trade over 52 trades .
5	★★★ THE KILL: a gate with NO TRIGGER AT ALL beats every gate I built	"Enter at 13:05Z every day , direction = sign of the last 15 minutes, stop 0.5xU, hold 30 min" — no signal, no arm, no condition — books +\$47.29/trade over 26 trades . That is double the best triggered candidate. Every candidate on this page sits inside the noise band of a control that contains no information.
6	The window cannot settle the question	PULSE's t-statistic is 1.30 . Day-block bootstrap 95% CI [-\$4.14, +\$26.27]. To get a ±\$10/ trade confidence interval needs 561 trades ≈ 79 trading days . We have 27.
7		Expectancy by realised 30-min follow-through: <+20pt → -\$38/tr · +60...+120pt → -\$4.10/

★ The size threshold IS the deliverable, and it is ~120pt	tr · ≥+120pt → +\$184.75/tr. Below ~120pt of follow-through the fee and the stop eat everything. 30.8% of armed events cleared it; break-even needed 19.2%.
8 Detection has a real footprint; direction does not — and the footprint's honest n is ELEVEN	Trade-count z-score separates ≥200pt runs at AUC 0.848, rising monotonically from 0.623 at 55pt. But on non-overlapping 30-min blocks the armed sample is n=11, 54.5% ±29.4pp. The AUC is real; the sample behind it is not big enough to trade.
9 PULSE and its exact mirror catch the same 6 of the 10 sat-out runs	Continuation catches 6/10 for +\$150; the direction-reversed twin catches the same 6/10 for +\$46. Catching a run is trivial. Getting the side right is the entire problem, and neither does.
10 A censoring recommendation, free	VOL - EXPANSION — the label the census barely uses — has 4.18× run lift on 1.95% of the tape, the best of any cluster. OPEN/NEWS has 2.85× on 5.20%. If a bucket is going to route a hunt, that is the one worth widening.

VERDICT (full statement in §12): NULL. Nothing ships, not even to shadow. Cause of death for the section: the trigger-free NULL-CLOCK control (§9) and the day-block bootstrap (§10). Cause of death for the two leaders individually: strip-the-best-trades and the OOS leg. The deliverable is the ~120pt size threshold (§8) and the label finding (§1-2).

1. ★ INTERROGATING THE LABEL FIRST

The brief says to check the cluster label's base rate before trusting anything built on top of it, and that a label true on most of the tape is noise, not a footprint — and that this finding outranks any signal built on top. I did it first. It changed where I was allowed to look.

1.1 What the label is now

Last year's version of this page found OPEN/NEWS was literally `if 13 <= hour < 15` checked **before** every footprint test. That has been fixed. From `scripts/run_census.py:78-84`:

```
def cluster(hour, flow, mv, amp, fz=None):
    if flow is not None and fz is not None and abs(fz) >= FLOW_Z:
        return "FLOW-LED" if (flow > 0) == (mv > 0) else "VACUUM"
    if amp is not None and amp > 0.30:
        return "VOL-EXPANSION"
    if 13 <= hour < 15:
```

```
return "OPEN/NEWS"           # no footprint, but the densest window – say so
return "UNCLASS"
```

The clock now goes **last**. OPEN/NEWS is a **residual**, and the full predicate is a conjunction:

```
13:00 ≤ hour < 15:00 UTC AND |flow z| < 1 (no aggressor-imbalance event
in the prior 60s) AND amp ≤ 0.30 (prior-5-min hi-lo ≤ 0.30% of price)
```

That is a materially different object from a clock, and the docstring's own claim — that the window is worth keeping as a fallback — is testable. So I tested it.

1.2 Base rate — measured on ALL bars, not on runs

I re-implemented the census's run detector and its three feature functions (flow_at, flow_z_at, amp) against the **full parquet lake** at 5s — 438,253 bars, 2026-07-16 → 2026-08-21 — and scored the label on **every** 60s-step 15-minute window, run or not. 36,507 windows.

CLUSTER	WINDOWS	% OF TAPE	RUNS (≥1.5×ATR)	P(RUN)	LIFT VS 1.109% BASE
UNCLASS	28,007	76.72%	240	0.857%	0.77×
FLOW-LED	2,988	8.18%	33	1.104%	1.00×
VACUUM	2,901	7.95%	39	1.344%	1.21×
OPEN/NEWS	1,900	5.20%	60	3.158%	2.85×
VOL-EXPANSION	711	1.95%	33	4.641%	4.18×

Two things fall straight out.

First: the label is true on the majority of its own window. 3,239 windows sit inside 13:00–15:00Z; **1,900 of them (58.66%) wear the label**. The 08-15 fix took it from 100% of its window to 59% of its window. That is progress, and it is not enough to call it a footprint — a coin that lands heads 59% of the time inside its own domain is still mostly describing the domain.

Second — and this is the one that matters:

SLICE OF 13-15Z	WINDOWS	RUNS	P(RUN)	LIFT VS BASE
the bare clock, everything in 13-15Z	3,239	111	3.427%	3.09×
OPEN/NEWS (clock ∧ flow-quiet ∧ amp-quiet)	1,900	60	3.158%	2.85×
what the label throws away (in-window, has a footprint)	1,339	51	3.809%	3.43×

★ **The label's lift OVER the bare clock is 0.921×**

The two footprint tests take a 3.427%-dense window, remove 41% of it, and keep the less dense half. The bit of the label that is not a clock is not merely uninformative — **it is mildly negatively informative**. Everything OPEN/NEWS knows about where runs happen, it knows because of the clock.

1.3 Is that an artefact of the pre-tick era?

The lake's ticks start 2026-07-24; before that `flow_z_at` returns `None` and the flow test cannot fire, so 07-16...07-23 windows fall through to the clock by default and could be inflating the label. Split it:

ERA	WINDOWS	13-15Z P(RUN)	OPEN/NEWS P(RUN)	LABEL LIFT OVER CLOCK	LABEL SHARE OF ITS WINDOW
ALL LAKE 07-16→08-21	36,507	3.427%	3.158%	0.921×	58.66%
TICK ERA 07-24→08-21	28,349	3.373%	3.190%	0.946×	53.49%

Same answer both ways. In the tick era the label covers 53% of its own window and still returns 0.946× the bare clock's lift. **Not an artefact.**

1.4 Which of the two footprint tests is doing it

`amp > 0.30` is true on only **3.04%** of all windows — it barely bites. `|fz| ≥ 1` is true on **16.13%**. So essentially all of the subtraction is the flow test, and essentially all of the anti-lift is the flow test removing in-window runs that did have a flow footprint and were, on the evidence, slightly more likely to be runs than the ones left behind.

That is not a bug in `cluster()` — it is doing exactly what it says, routing footprinted runs to FLOW-LED/VACUUM so they reach the hunts that can use them. It is a warning about this bucket: what is left is, by construction, **the runs for which the census's two instruments found nothing**.

1.5 ★★ What that forces on the hunt

This is the structural point of the whole section, and I want it stated plainly before any gate:

OPEN/NEWS is a bucket defined by the absence of evidence. A run lands here because aggressor flow was unremarkable in the 60 seconds before it and the

prior five minutes were not unusually wide. Any signal I build that uses aggressor flow or pre-run amplitude would, if it worked, have relabelled the run out of the bucket. The only channels that can legitimately catch an OPEN/NEWS run are the ones `cluster()` never reads: - **price structure** (ranges, levels, VWAP, momentum) — never read; - **raw volume / trade count** — never read (flow is the net signed aggressor sum; a two-sided auction has huge volume and ~zero net flow, and the label cannot tell it from a dead tape); - **L2 book** — the census measures `book_depletion()` and prints it, but `cluster()` never receives it. Free channel. - cross-asset — out of scope for this section.

Those three are exactly what §4-§7 test, in that order. Nothing here is a flow gate, deliberately.

1.6 A free recommendation for the census

VOL-EXPANSION covers **1.95% of the tape at 4.18× run lift** — the highest of any cluster, better than OPEN/NEWS's 2.85× on 5.20%, and it is the label the census currently produces least often (1 run in the frozen week, 33 across the lake). It is checked second, so a run that also has a flow footprint never reaches it. If a bucket is going to route next quarter's hunt, that is the one with the sharpest edge per unit of tape, and its `amp > 0.30` threshold — true on 3.04% of tape — is worth sweeping. Not my brief; recorded because I measured it.

2. THE TARGET, STATED EXACTLY

2.1 What the frozen census says

`census_summary.json` (frozen, not re-run): 68 runs $\geq 1.5 \times \text{ATR}$ over 2026-08-17 → 08-21, 62 sat out, sat-out ceiling **\$9,610**. OPEN/NEWS = **13 runs**, of which 3 were caught and **10 sat out**.

The three caught, for the record — all on 08-18, all inside 45 minutes:

TIME UTC	DIR	MOVE	GATE	REAL \$
08-18 13:40	DN		-137 exhaustion_short_A	-208
08-18 13:58	DN		-66 day_rider	+174
08-18 14:38	DN		-105 exhaustion_short_A	+250

The ten I was hired to catch (from census_stdout.txt, cluster column OPEN/NEWS, us = sat out):

#	TIME UTC	DIR	MOVE	CEILING \$	FLOW	AMP	BOOK
1	08-17 13:31	DN	-74	148	+147	0.19	0.57
2	08-17 14:53	UP	+70	140	-274	0.07	0.41
3	08-18 13:25	UP	+61	122	+89	0.09	0.52
4	08-18 14:14	UP	+91	182	-226	0.22	0.45
5	08-19 14:00	DN	-128	257	+661	0.22	0.51
6	08-19 14:25	UP	+150	300	-565	0.24	0.51
7	08-19 14:46	UP	+70	139	+281	0.13	0.49
8	08-20 13:22	UP	+120	240	+82	0.08	0.51
9	08-21 13:46	DN	-56	113	+18	0.18	0.47
10	08-21 14:41	UP	+64	127	-263	0.12	0.50

Ceiling on the table: \$1,768. Look at the flow column: five of ten have net aggressor flow pointing the wrong way (runs 2, 4, 6, 10 up on selling; run 5 down on +661 of buying). That is the \$1.5 point in raw numbers — this bucket's runs are the ones whose tape gave no directional tell.

2.2 Why I did not backtest on ten runs

Ten runs over five days cannot support a robustness battery, and capture.db — which is what the census reads — is a **5-trading-day rolling window**. Anything I fit on it would be fit on the sample. So the hunt runs on the **full parquet lake**:

```
[lake] MNQ 5s bars: 438,253 rows, 2026-07-16 00:00 → 2026-08-21 20:59 UTC
      27 trading days · typical 15-min range = 44.2pt · run threshold = 1.5× =
66.4pt
      405 runs ≥ threshold · 60 of them OPEN/NEWS
```

The lake's threshold (66.4pt) is stricter than the frozen week's (55pt) because the lake's typical 15-min range is wider (44.2 vs 37pt). I kept the census's own arithmetic rather than pinning the number, so the run definition is the census's definition on every day it is applied to.

60 OPEN/NEWS runs on 26 of 27 days — 32 UP, 28 DN, 32 in hour 13 and 28 in hour 14, sizes 68pt to 234pt. No minute-of-hour concentration at all (the busiest single minute has 3 of 60, and 13:30 — the cash open itself — has 2). **Whatever this bucket is, it is not a news-release footprint.**

2.3 The unit and the costs

- **U = 44.0pt** — median 15-min hi-lo range, computed causally (trailing 24h rolling median), so no config knows the future. This is the census's own "typical 15-min range" statistic.
- **\$2.00 / point** (MNQ; MGC is \$10 and is not in this section).
- **\$1.50 per ROUND TRIP.** Grepped the harness before reuse per the contract: engine.py: TICK, VPP, FEE = 0.25, 2.00, 1.50 and net = gross - FEE charged **once per trade**. Not \$5 (which would have killed every marginal candidate on a fee 3.3× too high), not per side.
- **1 tick (0.25pt) adverse slippage** on every stop-entry and every stop-exit.
- **Pessimistic intrabar:** within a 5s bar the stop is checked **before** the target. A bar that touches both is scored a loss.

2.4 Tick-honesty audit of the fills

The harness runs on 5s OHLC, so I audited the fills against the **tick tape** (memory: "a wrong-instrument tape PASSES the stop-width sweep" — consistency is not correctness).

CHECK	RESULT
entry price inside its own 5s fill-bar [min(price), max(price)] from ticks (40 sampled trades)	40/40
exit price inside its own 5s exit-bar tick range	40/40
exit mix — the wrong-tape signature is every leg at its exact stop	141 stop / 50 time, not 191/0
median hold	265s (max 1,830s)
instrument asserted	MNQ only, from data/tape/{bars,ticks,book}/MNQ/ — never a SELECT DISTINCT symbol

No off-tape fills, no all-stop degeneracy, right instrument.

3. WHAT THE TAPE ACTUALLY TELLS YOU BEFORE A RUN

Before inventing anything I measured what is knowable. Per the memory "census pre-run stats are a picker artefact", every discriminator below is measured on the **unselected** population — every 60s-step window whose forward 15-min move clears the threshold — **not** on the keep-strongest-deduped run list, which shifts these statistics badly.

3.1 Size is visible. Direction is not.

AUC for "this window is the start of a ≥ 66.4 pt move", 13:00–15:00Z, tick era, $n=2,520$ windows:

FEATURE	AUC
atr15 (trailing-day median 15m range)	0.658
rng_5m (prior 5-min range)	0.652
nt_z (trade-count z, 1-min vs trailing 120)	0.638
vol_z (volume z)	0.631
rng_30m	0.627
er_30m (efficiency ratio)	0.567
\	flow z
sq_5_30 (5m range ÷ 30m range — the "coil" test)	0.532

Direction, among the windows that did produce a run:

FEATURE	AUC FOR P(UP)
pos_60m (where in the 60-min range price sits)	0.554
pos_30m	0.548
er_30m	0.538
d_vwap	0.494
rng_5m	0.475
sq_5_30	0.423

and the crudest possible direction rule — does the run continue the last 15 minutes? — agrees on **569 of 1,049** in-window runs (54.2%), against a **53.4%** base rate on all in-window windows. **0.8 percentage points**.

Two consequences I obeyed for the rest of the page:

1. **sq_5_30 = 0.532 refutes the coil-break idea before it is built.** There is no compression footprint ahead of these runs. I built COILBRK anyway (\$4.2) because a refuted idea with an empty grave is worth less than a refuted idea with a measured one — and it produced one trade at the tight setting, which is itself the answer.
2. **A directional gate cannot be armed on a 0.55-AUC direction call.** The only honest shape is to let the tape choose the side — a two-sided stop-entry — or to accept a 54% momentum coin. Both are tested.

3.2 ★ The escalation: does a footprint appear at SIZE?

The brief says to escalate — full sat-out set, then top-25, then top-15 — because narrowing may reveal a footprint the marginal runs wash out. The clean way to do

that with enough n is to sweep the size threshold itself. 13-15Z, tick era, unselected windows:

Detection AUC vs run-size threshold

THRES HOLD	N WINDO WS ≥	VOL_Z	NT_Z	RNG_5M	RNG_30M	ATR15	FZ	SQ_5_30
55pt	988	0.618	0.623	0.648	0.629	0.650	0.558	0.524
66pt	805	0.631	0.638	0.652	0.627	0.658	0.557	0.532
80pt	597	0.655	0.662	0.656	0.620	0.651	0.565	0.543
100pt	393	0.671	0.678	0.661	0.617	0.671	0.569	0.546
120pt	259	0.674	0.681	0.694	0.626	0.703	0.576	0.573
150pt	137	0.711	0.718	0.710	0.628	0.728	0.598	0.571
200pt	43	0.842	0.848	0.791	0.618	0.718	0.668	0.666
250pt	8	0.895	0.901	0.932	0.867	0.799	0.800	0.681

The escalation DOES reveal a footprint, and it is the volume channel — the one the label cannot see. nt_z climbs monotonically 0.623 → 0.848. rng_30m, by contrast, is flat (0.629 → 0.618): this is not simply "wide days have wide runs". |fz| — the label's own instrument — climbs far more slowly (0.558 → 0.668), which is the same message as §1.2 from the other side.

Direction AUC vs size threshold (among windows whose |move| clears the threshold):

THRESHOLD	N	UP%	POS_60M	PREV15 (MOMENTUM)	D_VWAP	NETFLOW_Z
55pt	988	47.5	0.596	0.612	0.516	0.496
100pt	393	41.5	0.621	0.627	0.541	0.476
150pt	137	40.9	0.634	0.620	0.608	0.509
200pt	43	37.2	0.697	0.655	0.799	0.451

Direction improves too — but slowly, and from a much worse start.

3.3 ★★ ...and the honest n behind that footprint is ELEVEN

Those AUCs are computed on 60s-step windows carrying a 15-30 minute forward horizon. **They overlap by ~95%.** An n of 2,520 is not 2,520 observations; it is roughly 84 independent 30-minute blocks over 27 days. So I recomputed the headline claim on **non-overlapping** blocks:

P(30-min move in the momentum direction ≥ X), 13-15Z, one observation per non-overlapping 30-minute block, n=84 blocks:

X	ALL BLOCKS		NT_Z ≥ 1.5 BLOCKS	LIFT
80pt	19.0%	(n=84)	54.5% (n=11, ±29.4pp 95% CI)	2.86×
120pt	10.7%	(n=84)	54.5% (n=11, ±29.4pp)	5.09×
150pt	8.3%	(n=84)	45.5% (n=11, ±29.4pp)	5.45×

The volume footprint is real, it is large, and its independent sample size is eleven. A 95% confidence interval of ±29 percentage points around a 54% hit rate spans 25%–84%. That is not a footprint you can size a position on; it is a hypothesis that needs a quarter of tape.

For completeness, the same measurement on the raw overlapping 5s bars (n=30,169 in-window, 7,441 armed) gives 15.6% → 27.8% at X=120 — a 1.78× lift. **Both numbers are in this report because the gap between them is the finding.** Anyone quoting "n=7,441" off this study is quoting 5-second bars, not observations.

4. THE GRAVEYARD — every attempt, by name

Standing rule: every attempt gets reported, nulls and graves included, each with its stats and a one-line cause of death. Common config for the table below unless stated: stop **0.5×U = 22pt**, no target, 30-minute time cap, 1-tick slippage, \$1.50 round trip, 13:00–15:00Z, 15-minute cooldown.

GATE	N	NET \$	\$/TRADE	WIN%	STRIP-3	STRIP-5	TOP-3 % OF NET	LONG \$/ TR	SHORT \$/TR
ORB30	27	+418.5	+15.50	18.5	−26.98	−46.00	255%	+5.18	+26.62
COILBRK q0.60	9	+492.5	+54.72	22.2	−46.00	−46.00	156%	−46.00	+105.08
COILBRK q0.75	23	+693.5	+30.15	21.7	−25.38	−46.00	173%	−8.50	+65.58
PULSE	191	+2,167.0	+11.35	24.6	+2.63	−1.29	77%	+20.45	+1.13
PULSE-13h	95	+2,194.0	+23.09	26.3	+5.67	−1.89	76%	+30.80	+15.22
FADE1 (PULSE mirror)	211	−2,754.0	−13.05	13.7	−20.25	−24.19	—	−19.56	−7.30
RNGX60	81	−974.0	−12.02	14.8	−23.71	−30.47	—	−5.41	−18.80
VWRC	41	−352.0	−8.59	22.0	−24.46	−32.56	—	−6.69	−10.07
OAB	73	+1,330.5	+18.23	20.5	−2.59	−12.17	114%	+10.26	+26.42
MOMBRK	52	+1,709.0	+32.87	23.1	+4.02	−11.37	89%	+36.59	+29.91
BOOKLEAN	9	+108.0	+12.00	44.4	−29.58	−46.00	264%	+9.43	+21.00
NULL- CLOCK	26	+1,229.5	+47.29	34.6	+6.37	−15.76	88%	+49.38	+45.50

13:05 — no
trigger

NULL-CLOCK 13:30 — no trigger	27	−363.5	−13.46	11.1	−46.00	−46.00	—	−16.50	−11.03
NULL-CLOCK 14:05 — no trigger	27	+642.0	+23.78	29.6	−16.29	−29.30	161%	+52.79	+0.57

Read the last three rows first. They are in §9 and they are why nothing on this page ships.

Note on win%. Per the operator's 2026-08-01 judging rule, no candidate here is killed on win rate. PULSE-13h wins 26.3% of the time and I did not hold that against it — the live desk booked +\$625 this week at 39%. Every kill below names a robustness test.

4.1 ORB30 — pre-open range break — −\$26.98/tr after strip-3

Take the 13:00–13:30Z range (the half-hour into the US cash open), rest a buy-stop 1 tick above and a sell-stop 1 tick below, first touch wins, other cancelled. 27 events, 27 triggers — it fires every single day, which is at least a well-specified gate. Raw +\$15.50/tr looks alive. **Top 3 trades are 255% of the net** — strip them and it is **−\$26.98/tr**; strip 5 and every remaining trade is a full stop. 5 winners out of 27. **Cause of death: strip-the-best-trades.**

4.2 COILBRK — compression break — 1 trade at the setting the theory wanted

Arm when the 15-min range is $\leq q \times U$, then two-sided break. §3.1 predicted this would fail (sq_5_30 AUC 0.532) and it failed in the most literal way available:

Q	15M RANGE MUST BE UNDER	EVENTS	TRADES	\$/TRADE	
0.40	17.6pt	1	1	−46.00	−46.00
0.50	22.0pt	5	5	−46.00	−46.00
0.60	26.4pt	13	9	+54.72	+54.72
0.75	33.0pt	32	23	+30.15	+30.15

The 13:00–15:00Z window essentially **never coils**. To get a usable n you must relax the compression test until it stops meaning compression (q=0.75 is a 33pt 15-minute range — barely below the 44pt median). And at q=0.60 the whole \$492.50 is three trades (156% of net; strip-3 = −\$46.00, i.e. every survivor is a stop) and the **long side loses on every one of its trades** while the short side makes \$105/tr — n=9 split 2/7. **Cause of death: n, and parameter sweep —**

the edge exists only where the compression test has been relaxed into meaninglessness.

4.3 PULSE — 1-minute impulse continuation

Trigger: a 1-minute close-to-close move $\geq k \times U$ inside 13–15Z. Direction = the impulse's sign. Entry: stop order 1 tick beyond that minute's extreme. This is the largest-n candidate on the page and got the fullest battery — §5.

4.4 FADE1 — the exact mirror — $-\$13.05/\text{tr}$

Identical trigger, direction reversed, market entry. $-\$13.05/\text{tr}$ over 211. Not a failure so much as a **control that passed**: PULSE's $+\$11.35$ and FADE1's $-\$13.05$ are close to symmetric around $-\$1.50$ of fee, which says the two are genuine mirrors and the harness is not manufacturing money from the entry mechanics. That matters — it means PULSE's direction rule is doing something. §5.4 shows it is not doing enough.

4.5 RNGX60 — 60-minute structure break — $-\$12.02/\text{tr}$

Two-sided break of the trailing 60-minute high/low. 216 events, 81 triggers. Negative raw, negative at every strip, negative on both sides. **Cause of death: negative expectancy, unconditionally.** By 13:00Z the 60-minute extremes are the overnight range's edges, and MNQ spends the US open chopping across them.

4.6 VWRC — session-VWAP cross — $-\$8.59/\text{tr}$

Enter in the direction of a cross back through the running session VWAP. 42 events, 41 trades, $-\$8.59/\text{tr}$, negative both sides, $-\$24.46$ after strip-3. d_vwap had already scored **AUC 0.494** for direction in §3.1 — a coin. **Cause of death: negative expectancy; predicted by the direction AUC.**

4.7 OAB — volume-ignition two-sided break — no parameter plateau

The first gate built deliberately on the §1.5 orthogonal channel. Arm when raw 1-minute **volume** $z \geq V$ and $|\text{flow } z| < 1$ — a **two-sided auction**: heavy trade, balanced aggression, exactly the state the census's net-flow instrument is blind to. Then a two-sided break of the 5-minute range.

The arm itself is real (§3, and lift $1.66 \times$ at $\text{vol}_z > 2.5$ even restricted to flow-quiet minutes). The gate is not. **192-cell sweep** over $V \in \{1.0, 1.5, 2.0, 2.5\} \times \text{wait} \in \{1, 2, 3, 5\} \text{ min} \times \text{stop} \in \{0.375, 0.5, 0.75\}U \times \text{target} \in \{0.75, 1.0, 1.5, \text{None}\}U$:

overall mean \$/trade across all 192 cells: -5.15 fraction of cells positive: 0.234

mean \$/trade by arm strength × trigger window

vz \ wait	1min	2min	3min	5min
1.0	-2.26	-2.57	-6.50	-4.90
1.5	-10.32	-7.69	-7.82	-5.05
2.0	-11.89	-6.84	-10.75	-6.00
2.5	-5.23	+2.55	-1.47	+4.29

trades <- the only positive row, n=24..30

77% of the surface is negative, and the positive corner is the corner where n collapses to 24-30 trades. V=2.0 is negative in all four wait cells; V=2.5 flips positive. Adjacent parameter cells flip sign. **Cause of death: parameter sweep — no plateau; edge appears only as n collapses. That is the curve-fit tell named in the brief.**

4.8 MOMBRK — trade-count arm + momentum-direction break

The best-scoring gate I built. Full battery in §6.

4.9 BOOKLEAN — L2 imbalance — full L2 null in §7

9 trades. Reported in the table for completeness; the channel-level result in §7 is the finding.

5. PULSE — the largest-n candidate, killed

Spec (as it would ship):

```
GATE pulse_open
  SESSION      13:00 <= UTC hour < 15:00
  TRIGGER      |close[t] - close[t-60s]| >= 0.30 x U          (U = 44.0pt, causal
24h median 15m range)
  DIRECTION    d = sign(close[t] - close[t-60s])
  ENTRY        resting STOP at (high[t] + 1 tick) if d>0 else (low[t] - 1 tick);
valid 3 minutes
  STOP         entry - d x 0.5 x U    (22pt)
  EXIT         time cap 30 minutes; no target, no trail
  COOLDOWN     15 minutes
  COSTS        1 tick slippage in and out; $1.50 per round trip
```

Base: n=191, net **+\$2,167, +\$11.35/trade**, win 24.6%, median **−\$46.00**, best +\$695.50.

5.1 Parameter sweep — 144 cells

$k \in \{0.20, 0.25, 0.30, 0.35, 0.40, 0.50\} \times \text{stop} \in \{0.25, 0.375, 0.5, 0.75\}U \times \text{target} \in \{0.5, 0.75, 1.0, 1.5, 2.0, \text{None}\}U$, 15-min cap:

mean \$/trade by stop x target

stop\targ	0.50U	0.75U	1.00U	1.50U	2.00U	none
0.250	-3.30	-2.12	-2.33	-3.88	-4.66	-3.08
0.375	-1.57	-1.08	-0.78	-3.80	-4.82	-0.80
0.500	-1.51	-0.11	-0.76	-3.40	-3.18	-0.30
0.750	-1.37	+0.50	-0.39	-3.33	-4.01	-0.86

mean \$/trade by trigger size k

0.20	-0.85	0.25	-4.06	0.30	-0.39	0.35	-2.99	0.40	-1.65
0.50	-2.81								

23 of 24 exit cells are negative. And k is not monotone: $0.20 \rightarrow 0.25 \rightarrow 0.30$ gives $-0.85, -4.06, -0.39$. Adjacent trigger thresholds flip sign by \$3.7/trade. The best single cell ($k=0.20$, stop 0.75U, target 1.0U) prints +\$7.08/tr and is surrounded by cells averaging under zero.

5.2 The time cap is not an exit rule, it is a dial

The +\$11.35 headline comes from lengthening the hold, so I swept it (arm A=1.5 variant, same story for PULSE):

TIME CAP	5M	10M	15M	20M	30M	40M	60M	90M
\$/trade	+0.70	-7.91	+11.16	+22.19	+32.87	+41.26	+50.92	+51.51

It rises essentially monotonically with hold length (with a sign flip at 10m). **That is not "an exit rule works", that is "hold longer, capture more of whatever the tape did"**. A gate whose P&L is a monotone function of how long you stay in is not an entry edge; it is exposure.

5.3 Strip-the-best, day concentration, leave-one-out, OOS

TEST	RESULT	VERDICT
base	+\$11.35/tr, n=191	—
strip 3 best	+\$2.63/tr	top 3 trades = 77.2% of net
strip 5 best	-\$1.29/tr	FAIL
leave-one-day-out (27 folds)	all 27 folds positive, range +\$8.23 ... +\$13.53	pass
day balance	16 profitable days / 11 losing; top day 30.2% of net	pass
long / short symmetry	LONG +\$20.45 (n=101, +\$2,065) · SHORT + \$1.13 (n=90, +\$102)	FAIL — 95% of net is one side
OOS (IS 07-16...08-14 / OOS 08-17...08-21)	IS +\$10.69 (n=155) · OOS +\$14.18 (n=36)	pass

The long/short split is the one that should worry us most: MNQ fell **−318pt** over the sample and **−542pt of that accrued inside 13:00-15:00Z**, so the long side is not being paid by drift — it is being paid against drift, on 101 trades, while the short side in a falling tape returns \$1.13/trade. That asymmetry has no mechanism behind it. It is the signature of a handful of large long winners.

5.4 ★ The placebo — and a correction to how I first ran it

The brief's test: shift the signal series, keep every other rule; if a fake signal books most of the money, the real one is not the edge.

I initially ran this with shifts in both directions and I was wrong to. A negative shift feeds the gate the impulse that happens **5 minutes in the future** — that is not a placebo, it is a look-ahead oracle. Both directions are reported below, correctly labelled, because the look-ahead leg is a useful harness check (it must beat the real signal; if it did not, the harness would already be leaking).

	−30M	−15M	−5M	REAL	+5M	+10M	+15M	+30M	+60M
	look-ahead	look-ahead	look-ahead		stale	stale	stale	stale	stale
PULSE 13-15Z	+8.05	+4.57	+30.32	+11.35	−12.57	+8.14	−3.46	+6.45	+8.21
PULSE-13h	+10.16	+18.40	+48.45	+23.09	−9.51	+20.76	+18.81	+23.95	+18.47

- Harness check: the −5m look-ahead oracle books +\$30.32 vs the real + \$11.35 (and +\$48.45 vs +\$23.09 on the 13h variant). **The harness is not leaking** — knowing the future is worth $\sim 2.7\times$, as it should be.
- PULSE 13-15Z: stale placebos average **+\$1.35/tr = 12% of real**, 0 of 5 beat it. **Placebo passes.**
- PULSE-13h: stale placebos average **+\$14.50/tr = 63% of real**, and the +30m stale signal books **+\$23.95 — more than the real signal's +\$23.09. Placebo FAILS.**

A signal whose 30-minute-stale copy earns as much as it does is not a trigger. It is a slow-moving state variable being read at an arbitrary moment.

5.5 PULSE-13h — the home-segment variant, and its own battery

\$8's regime work shows PULSE's money is entirely in the **first hour**: 13:00-14:00Z **+\$23.09/tr (n=95, +\$2,194)** vs 14:00-15:00Z **−\$0.28/tr (n=96, −\$27)**. So I ran the battery on the home segment as its own candidate.

TEST	RESULT	VERDICT
------	--------	---------

base	+\$23.09/tr , n=95, win 26.3%	—
strip 3 best	+\$5.67/tr (top 3 = 76.2% of net)	weak
strip 5 best	−\$1.89/tr	FAIL
leave-one-day-out	27/27 folds positive, +\$14.74 ... +\$26.13	pass
day balance	17 up / 10 down, top day 38.2%	pass
long / short	+\$30.80 (n=48) / +\$15.22 (n=47) — both positive, both sides trade	pass
placebo (stale only)	mean +\$14.50 = 63% of real; 1 of 5 beats it	FAIL
OOS 08-17...08-21	IS +\$30.49 (n=78) · OOS −\$10.82 (n=17)	FAIL
day-block bootstrap 95% CI	[−\$2.79, +\$52.71] , P(≤0) = 0.044	inconclusive

PULSE-13h is the most attractive thing on the page — it is the only candidate that passes long/short symmetry with both sides positive. It still dies three ways.

Cause of death (PULSE and PULSE-13h): the placebo test (a 30-minute-stale copy of the signal books more money than the signal), confirmed by strip-5-best going negative and by the OOS leg (−\$10.82/tr).

6. MOMBRK — the best number on the page, killed hardest

Built directly on §3.2's finding: nt_z is the strongest size discriminator and it strengthens with run size. So: arm on trade-count z, take the momentum side, break-entry.

```
GATE mombrk_open
  SESSION      13:00 <= UTC hour < 15:00
  ARM          nt_z >= 1.5      (1-min trade COUNT vs trailing 120 non-overlapping
1-min buckets)
  DIRECTION    d = sign(close[t] - close[t-15min])
  ENTRY        resting STOP 1 tick beyond the 5-minute extreme on side d; valid 3
minutes
  STOP         0.5 x U (22pt)      EXIT   time cap 30 min      COOLDOWN 15 min
```

Base: n=52, net +\$1,709, +\$32.87/trade, win 23.1%.

6.1 The arm ladder — jumpy, not a plateau

NT_Z ≥	0.0	0.5	0.75	1.0	1.25	1.5	1.75	2.0	2.5	3.0
n	102	88	77	71	63	52	46	39	32	28
\$/trade	−0.73	+0.53	+4.50	+22.23	+20.53	+32.87	+8.55	+16.81	+6.97	−14.21

With the arm off entirely (A=0.0) the gate is −\$0.73/tr, so the arm is unambiguously doing the work. But the ladder is not a plateau: 1.5 → 1.75 → 2.0

reads +32.87, +8.55, +16.81, and 3.0 is −14.21. A quarter-of-a-standard-deviation move in the arm threshold changes expectancy by \$24/trade.

Swapping the arm column barely matters (nt_z +32.87, vol_z +24.02, netflow_z +30.98) — the three are near-collinear busy-tape meters, which is the \$5.4 point again: this is a state, not a trigger.

6.2 The battery

TEST	RESULT	VERDICT
base	+\$32.87/tr, n=52	—
strip 3 best	+\$4.02/tr — top 3 = 88.5% of net	FAIL
strip 5 best	−\$11.37/tr	FAIL
day balance	7 profitable days vs 14 losing. Top day = 50.1% of net. Two days (07-27 +\$801, 07-31 + \$857) = 97% of the +\$1,709	FAIL
leave-one-day-out	21/21 folds positive, +\$18.13 ... +\$37.69	passes — see below
leave-TWO-days-out	210/210 folds still positive, but worst fold + \$1.19/tr (drop 07-27 & 07-31)	fragile
long / short	+\$36.59 (n=23) / +\$29.91 (n=29) — symmetric	pass
OOS 08-17...08-21	IS +\$44.62 (n=42) · OOS −\$16.50 (n=10)	FAIL
placebo (stale +5/+15/+30/+60/+120m)	+23.91, +4.54, +25.20, −5.61, +30.13 → mean +\$15.63 = 48% of real , 0 of 5 beat it	pass
random-arm placebo (200 draws, same count, same window)	mean +\$2.61, sd \$14.87, p95 +\$27.96 → empirical p = 0.025	pass

6.3 ★ Why leave-one-day-out passed and should not have reassured anyone

All 21 LODO folds are positive. Two days carry 97% of the P&L. Both statements are true at once, and that is a property of the test, not of the gate: with 21 days and 2 money days, removing one day always leaves the other. **LODO with k money days cannot fail until you leave k days out.** I therefore ran leave-two-out — 210 folds — and the worst fold (drop 07-27 and 07-31 together) collapses to **+\$1.19/trade**, i.e. break-even.

That is worth carrying forward as a rule: **quote the number of profitable days alongside any LODO result.** 7-up/14-down with a passing LODO is a warning, not a clearance.

Cause of death (MOMBRK): strip-the-best-trades (+\$32.87 → +\$4.02 at strip-3, −\$11.37 at strip-5; top 3 trades are 88.5% of net) and the OOS leg (−\$16.50/trade). Confirmed by the day distribution: 7 winning days out of 21, two of which are 97% of the money.

7. THE L2 CHANNEL — a clean null, and one inverted surprise

The book is the one channel `cluster()` genuinely cannot see (§1.5): `run_census.py` computes `book_depletion()` and prints it in the book column, but never passes it to `cluster()`. Per the data contract, `capture.db.book` is MNQ-only and event-driven at 41ms — correct for this section — and I read it from the lake (`data/tape/book/MNQ/`, **2026-07-31 → 08-24**, so 16 trading days inside the study window, not 27).

Aggregated to per-minute 3-deep and 10-deep bid/ask size, restricted to 13:00–15:00Z, n=22,979 5s bars:

FEATURE	WHAT IT SHOULD PREDICT	AUC
$imb3 = (bid_3 - ask_3) / (bid_3 + ask_3)$	direction of the next 15 min	0.508
<code>imb10</code>	direction of the next 15 min	0.496
<code>imb3</code>	direction of the next 30 min	0.509
`	<code>imb3</code>	` (magnitude of the lean)
<code>depth_z</code> (3-deep size <code>z</code>)	size,	<code>fwd30</code>
<code>depth3</code> (raw 3-deep total size)	size,	<code>fwd30</code>

Book LEAN is a coin. 0.508 and 0.496 are as close to nothing as this measurement gets, and the gate built on it (BOOKLEAN, §4.9) produced **9 trades** — the threshold that gives a meaningful lean almost never fires inside the US open. On non-overlapping 30-minute blocks in the book era (n=64), $|imb3| \geq 0.1$ occurs **4 times** and ≥ 0.2 **once**. There is no sample here.

But depth3 at AUC 0.285 is not nothing — it is inverted. Read the other way, a THIN 3-deep book predicts a ≥ 120 pt move at AUC 0.715, which is better than volume-z (0.674) and better than any book lean measure by a mile. That is the same message as §3.2 from a third instrument: the tape tells you **how big**, never **which way**. And like `nt_z` it is a state variable — thin books cluster in time — so it inherits the same §5.4 problem and I did not build a gate on it.

Cause of death (BOOKLEAN): n=9. Cause of death (the L2 direction channel): AUC 0.508 — no directional information at any depth or horizon.

8. ★ THE SIZE THRESHOLD — this is the deliverable

The brief: report the size-threshold at which a footprint becomes tradeable, if any — that IS the finding. Here it is, three ways.

8.1 The money threshold: ~120pt of follow-through

MOMBRK's 52 trades, bucketed by the **realised** 30-minute move in the traded direction:

REALISED FOLLOW-THROUGH	TRADES	% OF EVENTS	NET \$	\$/TRADE
≤ -60pt (went hard against)	14	26.9%	-644.0	-46.00
-60 ... -20pt	6	11.5%	-276.0	-46.00
-20 ... +20pt (nothing happened)	9	17.3%	-346.0	-38.44
+20 ... +60pt	2	3.8%	+39.5	+19.75
+60 ... +120pt	5	9.6%	-20.5	-4.10
≥ +120pt	16	30.8%	+2,956.0	+184.75

A move of +60 to +120 points in your favour still loses money. -\$4.10/trade. The 22pt stop gets tagged on the way, or the 30-minute cap arrives before the move does, or the \$1.50 and the two ticks of slippage take the rest. Only follow-through beyond ~120pt pays, and when it does it pays +\$184.75.

The arithmetic that makes this a rule rather than an observation: with +\$184.75 on a hit and -\$44 on a miss, **break-even hit rate = $44/(184.75+44) = 19.2\%$** . MOMBRK's arm delivered **30.8%**. That 11.6-point margin is the entire edge — and §6 shows it does not survive strip-3 or OOS.

This lands on the same number as the desk's existing greenfield finding — "the size threshold is 120pt; ≥120pt boarding pays +\$69.38/tr" — from a completely independent cluster and a different harness. Two independent studies converging on ~120pt on MNQ is the most durable thing on this page.

8.2 The detection threshold: a footprint appears above ~120pt, sharpens above 200pt

From §3.2: nt_z AUC runs 0.623 (55pt) → 0.638 (66) → 0.662 (80) → 0.678 (100) → **0.681 (120)** → 0.718 (150) → **0.848 (200)**. The curve is gentle to 120 and steep after 150. **The escalation the brief asked for did reveal a footprint that the**

marginal runs wash out — and it is in the volume channel, not the price channel (rng_30m is flat at 0.62 across the whole range).

8.3 The threshold that matters most: ~11 independent observations

§3.3. On non-overlapping blocks, the ≥ 120 pt armed sample is **n=11 at 54.5% ± 29.4 pp**. So:

The tradeable size threshold on MNQ in this window is ≈ 120 pt of 30-minute follow-through. The footprint that predicts it is real (trade-count z, AUC 0.68 at 120pt rising to 0.85 at 200pt). The sample behind that footprint is eleven independent observations, and eleven is not enough to arm a gate on. The instrument we need is not a new measurement — it is more tape.

9. ★★★ THE CONTROL THAT KILLS THE WHOLE SECTION

Every candidate that scored well shared three ingredients: the 13:00–15:00Z window, a momentum-derived direction, and a ~30-minute hold. So I stripped out the only remaining ingredient — **the trigger** — and ran what is left.

NULL-CLOCK: no arm, no signal, no condition of any kind.

Enter at a FIXED clock time every trading day.

Direction = $\text{sign}(\text{close} - \text{close}[15 \text{ min ago}])$. Stop 0.5xU. Hold 30 min. Market entry.

ENTRY TIME	TRADES	NET \$	\$/TRADE	WIN%
13:05Z	26	+1,229.5	+47.29	34.6
13:15Z	27	+171.0	+6.33	14.8
13:30Z	27	-363.5	-13.46	11.1
13:45Z	27	-125.0	-4.63	14.8
14:05Z	27	+642.0	+23.78	29.6
14:30Z	27	-389.0	-14.41	14.8

A gate with no information in it books **+\$47.29/trade** — double the best triggered candidate on this page (PULSE-13h, **+\$23.09**) and 44% more than MOMBRK (**+\$32.87**).

And the same trigger-free rule swings from **+\$47.29 to −\$13.46 purely on which minute of the hour you pick it**, across 26–27 trades each. Same rule. Same direction logic. Same stop. Same hold. Twenty-five minutes apart.

For scale, the coin-flip floor: hold the 13:05Z entry times fixed and replace the momentum direction with a fair coin, 100 draws → **mean +\$2.05/tr, sd \$15.21, p95 +\$23.70**. So the momentum direction rule is worth something (+\$47 vs +\$2 is about 3 sd) — but **p95 of pure noise is +\$23.70, which is above PULSE-13h's +\$23.09 and within touching distance of MOMBRK's +\$32.87**.

Every candidate on this page lives inside the noise band of a control that contains no information. That is not a statement about any one gate. It is a statement about what 27 trading days of a two-hour window can and cannot resolve.

10. HOW MUCH TAPE WOULD IT TAKE

STATISTIC	PULSE (13-15Z)	PULSE-13H
n trades	191	95
mean \$/trade	+11.35	+23.09
sd of \$/trade	120.78	143.26
standard error	8.74	14.70
t-statistic	1.30	1.57
day-block bootstrap (5,000, blocks = trading days) mean	+11.08	+23.33
day-block bootstrap 95% CI	[−\$4.14, +\$26.27]	[−\$2.79, +\$52.71]
P(true expectancy ≤ 0)	0.075	0.044

Neither clears $t=2$. Both bootstrap intervals contain zero. And to settle it:

TARGET 95% CI HALF-WIDTH	TRADES NEEDED	AT 7.1 TRADES/DAY
±\$30	63	9 trading days
±\$20	141	20 trading days
±\$10	561	79 trading days

Roughly sixteen weeks of tape to distinguish a \$10/trade edge from nothing in this window. We have five and a half. That is the honest reason nothing here can be armed, and it is not fixable by a cleverer gate.

11. DID ANYTHING CATCH THE TEN RUNS?

Against the ten sat-out OPEN/NEWS runs of the frozen census (\$2.1), ceiling **\$1,768**. Counted as caught if the gate opened a **correctly-signed** position between 5 minutes before and 15 minutes after the run's start:

GATE	CAUGHT	NET \$ ON THOSE RUNS	CEILING CONVERSION
PULSE	6 / 10	+\$150	8.5%
FADE1 (PULSE's exact mirror)	6 / 10	+\$46	2.6%
VWRC	2 / 10	+\$50	2.8%
MOMBRK	1 / 10	-\$46 —	
ORB30	1 / 10	-\$46 —	
RNGX60	1 / 10	-\$46 —	

★ **PULSE and the gate that does the exact opposite of PULSE catch the same six of ten runs.**

That single line is the cluster's whole character. Something fires into these runs reliably — a 1-minute impulse of $\geq 0.30 \times U$ happens near 6 of the 10 — so **detection is not the problem**. But a gate and its perfect mirror both "catch" 60% of them, and the better of the two converts **8.5% of the ceiling**. The run is there; the side is a coin; the money is in neither.

Across the whole lake, MOMBRK caught 17 of the 60 OPEN/NEWS runs (28%) for + \$884, and 7 of the 23 that were $\geq 120pt$ for +\$921 — **essentially all of its run-money is in the $\geq 120pt$ runs**, which is \$8.1 restated on the run set rather than the trade set.

12. REGIME SEGMENTATION AND THE ROUTER

The backtest discipline forbids scoring a static config across the whole tape. Regimes are tagged **causally** per bar: ATR percentile-rank (trailing 2 days) $\times$ efficiency ratio over 30 minutes, in the router's own vocabulary.

A bug I fixed mid-study, reported because it would have silently deleted two regimes. My first ER summed 1-minute absolute changes at 5-second spacing — 360 overlapping legs — inflating the path $\sim 12\times$ and driving every ER to ~ 0.03 . Under that ER, clean-trend and building **did not exist anywhere in 13-15Z**, and the regime table showed only the three chop buckets. Fixed to

sum non-overlapping legs (median ER-30m now 0.167). Every regime number below is post-fix.

13-15Z bar counts: building 12,987 · violent-whipsaw 8,679 · dead-chop 8,505 · normal-chop 3,927 · clean-trend 2,575 · trend-thin 2,201.

PULSE by regime (n=191):

REGIME	N	NET \$	\$/TRADE	WIN%
normal-chop	16	+563.0	+35.19	25.0
clean-trend	11	+374.5	+34.05	36.4
dead-chop	36	+866.5	+24.07	27.8
violent-whipsaw	46	+333.5	+7.25	21.7
building	67	+150.0	+2.24	23.9
trend-thin	15	-120.5	-8.03	20.0

PULSE by regime × sub-session (\$/trade, n in brackets):

REGIME	13:00-14:00Z	14:00-15:00Z
normal-chop	+157.12 (4)	-5.46 (12)
clean-trend	+72.25 (6)	-11.80 (5)
dead-chop	+63.53 (17)	-11.24 (19)
violent-whipsaw	+4.22 (18)	+9.20 (28)
building	+0.12 (37)	+4.85 (30)
trend-thin	-2.19 (13)	-46.00 (2)

MOMBRK by regime (n=52): dead-chop +\$73.12 (25) · violent-whipsaw +\$5.24 (21) · normal-chop -\$38.17 (6). By slot: 13h +\$51.14 (42) · 14h -\$43.90 (10).

12.1 What the regime table actually says

The **time-of-day split is far stronger than the regime split**. PULSE is + \$23.09/tr in 13:00-14:00Z and -\$0.28/tr in 14:00-15:00Z; MOMBRK is +\$51.14 and -\$43.90. Within the first hour, every regime except trend-thin is positive; within the second, four of six are negative. Whatever this is, it is a **first-hour-of-the-US-cash-session** effect, not a volatility-regime effect.

And the regime cells that look best are the ones with no sample: normal-chop 13h is **+\$157/trade on 4 trades**; clean-trend 13h is **6 trades**. Those are not home segments, they are anecdotes. The only regime cell with both a decent n and a decent number is dead-chop 13h, +\$63.53 on 17 trades — and §9 has already shown that a no-trigger rule entering at 13:05Z makes +\$47.29 on 26 trades, so even that cell is not clear of the null.

12.2 ★ The router rule — stated for something that did not survive

A gate that is only viable behind a router condition we cannot yet measure is a finding, not a failure. Nothing here is viable, but the honest statement of what a survivor would need is:

```
# WOULD-BE ROUTER RULE — NOT FOR DEPLOYMENT. No candidate in this section earned it.
ARM  pulse_open  WHEN  session == US_OPEN
                        AND  13:00 <= utc_hour < 14:00          # the SECOND hour is
where it dies
                        AND  regime IN (dead-chop, normal-chop, clean-trend)  # n = 17 /
4 / 6 — NOT a sample
                        AND  nt_z(1min, 120-bucket) >= 1.5        # the size footprint, AUC
0.68 at 120pt
BENCH pulse_open  WHEN  utc_hour >= 14:00
                        OR   regime == trend-thin
                        OR   untradeable_meter == TRUE
```

Two of those four arm conditions rest on 4 and 6 trades respectively. **The router condition we cannot yet measure is not an instrument — it is sample size.** We have every input this rule needs (nt_z is computable from ticks in one query, ER and ATR-rank already exist in the router's vocabulary, and the session split is free). What we do not have is 79 trading days of it.

If the desk wants this line of enquiry alive, the cheapest useful thing is **not** another gate: it is to start logging nt_z, depth3 and the realised 30-min follow-through at 13:00–14:00Z every day, so that by December the ≥ 120 pt question has n in the hundreds instead of eleven.

13. WHAT I DID NOT DO, AND WHY

Stated so the absences are not read as findings.

- **No cross-asset (MGC) leg.** MGC is \$10/point and gets its own invented-fresh gates in gf_MGC; the contract forbids porting an MNQ gate onto gold and this section is MNQ.
- **No V5 archive extension.** gaz:v5archive would add MNQ history before 07-16, but the V5→V7 seam has a **real tick hole 07-17 → 07-24** (lake.py warns about it explicitly), and every candidate here needs nt_z/vol_z, which need ticks. Extending bars-only would have widened the sample for the price features while leaving the volume features — the only ones that showed a

footprint — on the short sample. I judged a clean 27-day tick-complete sample better than a contaminated 40-day one. **This is the single biggest thing that could change the answer**, and if the seam is ever bridged, §8's eleven observations is the number to re-run first.

- **No walk-forward re-optimisation.** With 27 days and a 2-hour window there is not enough tape to split into fit/select/test three ways; the IS/OOS 2-way split in §5–6 is already thin (OOS n=10–36).
- **No book-pressure gate beyond BOOKLEAN.** The L2 direction channel measured AUC 0.508 (§7). I did not build further on a coin.
- **The census was not re-run.** census_summary.json and census_stdout.txt are read frozen, as instructed. Everything in §1–3 re-implements the census's functions against the lake; it does not re-cut the census's run list.

14. ONE-LINE CAUSES OF DEATH

GATE	\$/TRADE	CAUSE OF DEATH
ORB30	+15.50	strip-the-best — top 3 trades are 255% of net; strip-3 → −\$26.98
COILBRK q0.40	−46.00	n = 1. The US open does not coil; sq_5_30 AUC 0.532 predicted it
COILBRK q0.60	+54.72	parameter sweep + n — edge only where compression is relaxed to 26pt; strip-3 → −\$46.00, long side loses every trade
COILBRK q0.75	+30.15	strip-the-best — strip-3 → −\$25.38, strip-5 → −\$46.00
PULSE	+11.35	placebo (30m-stale copy earns 63–100% of real on the home segment) + strip-5 → −\$1.29 + long/short asymmetry (95% of net on one side)
PULSE-13h	+23.09	placebo (+30m stale books +\$23.95 > real + \$23.09) + strip-5 → −\$1.89 + OOS −\$10.82
FADE1	−13.05	negative expectancy — reported as PULSE's mirror control, which it passed
RNGX60	−12.02	negative expectancy unconditionally, both sides, every strip
VWRC	−8.59	negative expectancy; d_vwap direction AUC 0.494 predicted it
OAB	+18.23	parameter sweep — 77% of a 192-cell surface negative, sign flips between adjacent arm levels, positives only where n falls to 24–30
MOMBRK	+32.87	strip-the-best (top 3 = 88.5% of net; strip-3 → +\$4.02, strip-5 → −\$11.37) + OOS −\$16.50 + 7 winning days of 21, two of them 97% of the money
BOOKLEAN	+12.00	n = 9. L2 lean carries no directional information (AUC 0.508)

15. VERDICT

NOTHING SURVIVED. Nothing ships — not to live, not to shadow.

What killed the two leaders, individually and by name:

- **MOMBRK (+\$32.87/trade, n=52)** — killed by **strip-the-best-trades** and the **OOS leg**. The top 3 of 52 trades are 88.5% of the net; removing them takes +\$32.87 to +\$4.02, and removing 5 takes it to −\$11.37. Out of sample (08-17 → 08-21) it made **−\$16.50/trade**. Seven of twenty-one days were profitable and two of them are 97% of the money. Its leave-one-day-out passed all 21 folds, which is a property of running LODO with two money days, not a clearance.
- **PULSE-13h (+\$23.09/trade, n=95)** — killed by the **placebo/shuffle test**, confirmed by **strip-5-best** and the **OOS leg**. A copy of the signal delayed by 30 minutes — same window, same direction rule, same stop, same hold, just a stale trigger — books **+\$23.95/trade, more than the real signal's + \$23.09**. Stale placebos average 63% of the real signal's money. Strip 5 trades and it is −\$1.89. Out of sample it made −\$10.82/trade.

What killed the section: §9. A gate containing **no signal at all** — "enter at 13:05Z, direction = sign of the last 15 minutes, stop $0.5 \times U$, hold 30 minutes" — books **+\$47.29/trade over 26 trades**, double the best gate I built, and the identical rule 25 minutes later books −\$13.46. Everything on this page sits inside the noise band of that control. §10 says the same thing in confidence intervals: $t = 1.30$ and 1.57 , bootstrap 95% CIs of $[-\$4.14, +\$26.27]$ and $[-\$2.79, +\$52.71]$, and **561 trades $\approx$ 79 trading days** needed to resolve a \$10/trade edge in this window. We have 27.

The deliverables, which are not gates:

1. ★ **The cluster label is a clock with an anti-predictive filter bolted on.** The 08-15 fix correctly demoted OPEN/NEWS from checked-first to checked-last, taking it from 100% to **58.7%** of its own window. But its lift **over the**

bare clock is 0.921× — the flow and amplitude tests remove 41% of the window and keep the less run-dense half. Same answer in the tick era (0.946×). Per the brief's own instruction, this finding outranks every signal on the page. **Recommendation:** either drop the two footprint tests from the OPEN/NEWS fallback and call it what it is (US-OPEN-WINDOW, 3.09× lift on 8.87% of tape), or route the hunt through VOL-EXPANSION instead — **4.18× lift on 1.95% of tape**, the sharpest cluster the census produces and the one it uses least.

1. ★ **The tradeable size threshold is ≈120 points of 30-minute follow-through.** Below it the gate loses even when right: +60...+120pt of follow-through still books **−\$4.10/trade**. Above it, **+\$184.75/trade**. Break-even hit rate is 19.2%; the best arm delivered 30.8%. This is the same ~120pt number the desk's independent greenfield work reached from a different cluster with a different harness — the most durable result on this page.
1. ★ **Detection has a real footprint and it is VOLUME, not price; direction has none; and the footprint's honest sample is eleven.** Trade-count z separates ≥ 200 pt runs at AUC **0.848**, rising monotonically from 0.623 at 55pt, while 30-minute range is flat at 0.62 throughout — so this is not "wide days have wide runs". Direction tops out at AUC 0.55–0.65 and the crude momentum rule agrees with runs 54.2% of the time against a 53.4% base. On **non-overlapping** 30-minute blocks the armed sample for the ≥ 120 pt claim is **$n=11$ at 54.5% ± 29.4 pp**. And the cluster's own character, in one line: **PULSE and its exact mirror catch the same 6 of the 10 sat-out runs**. Detection is solved. Selection is not.
1. **A test-design finding worth keeping.** Leave-one-day-out cannot fail when k days carry the money and you only remove one. MOMBRK passed 21/21 LODO folds while two of twenty-one days held 97% of its P&L. **Always quote profitable-days-out-of-total next to a LODO result**, and run leave-two-out when the day distribution is that skewed (MOMBRK's worst leave-two fold: +\$1.19/tr).
1. **A harness finding.** A signal shifted backwards is a look-ahead oracle, not a placebo. Only forward (stale) shifts are legitimate placebos; the backward shifts belong in the report as a harness leak-check, which this harness passed (look-ahead earns 2.7× the real signal, as it should). Reported because I ran it the wrong way first and it would have produced a false kill.

Full working, every sweep and every null: this file. Tick-honesty audit \$2.4 — 40/40 entries and 40/40 exits inside their own bar's tick range, MNQ only, \$1.50 per round trip, \$2.00 per point.

FLOW-LED — aggressor-driven continuation

gf_full_FLOW-LED.md · 4,186 words · written 2026-08-25 23:12 UTC

GREENFIELD HUNT — cluster FLOW-LED

Full working, all attempts, all graves

MNQ · tick-honest over the full Parquet lake, 2026-07-05 → 2026-08-25 (35 tape days, 45,466,604 ticks) Costs: MNQ \$2.00/point, fee **\$1.50 per round trip**, 1 tick (0.25pt) adverse slippage on entry and on every stop.

0. The headline, before any signal

The cluster label is not a footprint. FLOW-LED and VACUUM are a coin flip, and the coin is flipped by the future.

The census assigns FLOW-LED when the prior-60s net aggressor flow is unusual ($|flow-z| \geq 1.0$) **and its sign agrees with the FORWARD 15-minute move**; VACUUM is the identical test with the signs disagreeing. The second half of that definition is only knowable after the run has happened. At the moment a gate would have to fire, both labels look exactly the same.

Measured on 43,982 minute bars of lake tape that carry a valid flow-z and a resolvable +15min close:

TEST	RESULT
base rate of the flow-event half ($ fz \geq 1.0$) on all bars	20.58 %
given a flow event, flow agrees with the next 15 min	49.62 % (n = 9,022)
control — bars with $ fz < 1.0$, same question	49.29 % (n = 34,786)
⇒ information carried by the flow event about direction	0.33 percentage points
FLOW-LED base rate on all bars	10.21 %
VACUUM base rate on all bars	10.37 %

The flow event does **not** meet the "true on most of the tape" failure mode — 20.6 % is a real filter, and the 2026-08-01 z-score fix that replaced `abs(flow) > 50` (66.5 % of bars) did its job. The label fails for a different and worse reason: **the filter fires on a fifth of the tape and then splits that fifth into two almost exactly equal halves on a coin toss.** FLOW-LED vs VACUUM is not two tape behaviours, it is one population cut by its own outcome.

Directly:

E[fwd15 in the flow's direction]				
fz >=1.0	-0.299 pt	n=9,053	t = -0.61	
fz >=1.5	+0.408 pt	n=4,594	t = +0.54	
fz >=2.0	+0.067 pt	n=2,655	t = +0.06	
fz >=2.5	-0.029 pt	n=1,660	t = -0.02	

Zero at every threshold, with the sign wandering. The census's own 4 FLOW-LED / 13 flow-event runs (30.8 % agreement) is **not** evidence of a contrarian tape: against the lake's 49.62 % null, $P(\leq 4 \text{ of } 13) = 0.139$. Expected FLOW-LED among 68 census runs = 9.0, observed 4; expected VACUUM = 9.6, observed 9. Both inside noise. **The cluster has n=4 because 68 runs is too few to populate a 10 %-base-rate label, not because the tape did something.**

0b. Flow does predict magnitude — but that is a volume channel wearing a flow costume

CONDITION	E FWD15	VS BASE
base (all bars)	26.70 pt	1.000×
fz ≥ 1.5	33.64 pt	1.260×
fz ≥ 3.0	41.73 pt	1.563×
volume-z ≥ 2.0	41.75 pt	1.564×
flow unusual, volume ORDINARY (fz ≥1.5 & volz <1.0)	25.70 pt	0.963×
volume unusual, flow BALANCED (volz ≥1.5 & fz <0.5)	32.56 pt	1.220×

Strip the volume out of the flow event and the magnitude lift vanishes entirely (0.963× — below base). Strip the flow out of the volume event and the lift survives (1.220×). **The census's flow column is a volume proxy.** Net aggressor imbalance adds nothing over "a lot traded".

That is the strongest statement this section can make, and it outranks everything below.

1. Method

Everything derives from ticks in `gazbot7.lake` (2026-07-05 → 08-25; the documented 07-18→07-23 V5/V7 seam is absent, weekends absent — 35 tape days, 44,847 minute buckets). `capture.db` was never touched: it holds 5 trading days and would have shrunk the sample 7×.

- **No lookahead.** A minute row at `ts` covers ticks `[ts, ts+60)`, so it is only known at `ts+60`. Every entry is stamped `ets = ts+60`. ATR is `.shift(1)`-ed off closed 5-min bars; every z-score uses `closed='left'` on a trailing 120-min window with ≥ 30 buckets, exactly the census recipe.
- **Contiguity guard.** `er30/brk30` use `.shift(30)`; across a weekend or the V5/V7 seam that silently compares two different tapes. 97.5 % of rows pass the guard; the rest are dropped, not used. (The two largest "minute moves" in the tape, 408.5pt and 295.75pt, are both 00:00:00 Monday weekend gaps — they are excluded by this guard, not traded.)
- **Tick-honest exits.** Entry at the first print $\geq ets$, filled 1 tick adverse. Stops fill 1 tick adverse to the stop price; targets fill at the limit; the time stop fills at the print. **Stop is tested before target on every print** (the pessimistic ordering). One position at a time.
- **Regimes** keyed on ATR level + ER + range-break, not the clock: DEAD-CHOP (`atr_rel<0.70 & er30<0.30`), VIOLENT-WHIPSAW (`atr_rel $\geq$ 1.6 & er30<0.35`), CLEAN-TREND (`er30 $\geq$ 0.55`), BUILDING (`0.35 $\leq$ er30<0.55`), NORMAL-CHOP otherwise. Sessions ASIA / LONDON / US / LATE.

Engine integrity check (run before believing any positive number)

Random-direction entries on the **identical** row set, five seeds:

seed 0	n=221	\$ -5.27/tr	seed 3	n=222	\$ -13.59/tr
seed 1	n=219	\$ -11.05/tr	seed 4	n=224	\$ -2.24/tr
seed 2	n=220	\$ -1.30/tr			

All negative, as a healthy engine must be — a coin flip pays the fee and the slippage. There is no drift bug manufacturing the numbers below.

2. Every candidate, blanket screen

Stop $1.0 \times \text{ATR}$, target $1.5 \times \text{ATR}$, 15-min time stop (the census horizon). **The blanket number is context, never the verdict** — per-regime follows.

#	CANDIDATE	SPEC	N	NET	\$/TR	WIN%
FL-1	flow_impulse_cont	$ fz \geq 1.5 \rightarrow$ trade in flow's direction	2,306	−\$4,280.30	−1.86	44.5 %
FL-2	flow_impulse_fade	$ fz \geq 1.5 \rightarrow$ trade AGAINST flow (the VACUUM mirror)	2,323	−\$8,244.89	−3.55	43.7 %
FL-3	flow_persist_cont	3 consecutive same-sign flow minutes, $ 3\text{-min } z \geq 1.5$	1,411	−\$3,080.93	−2.18	44.6 %
FL-4	flow_ignition	$ fz \geq 1.5$ and price already broke $0.5 \times \text{ATR}$ in flow's direction	1,496	+\$1,409.59	+0.94	45.6 %
FL-5	flow_absorb_fade	$ fz \geq 1.5$ and $ 1\text{-min move} \leq 0.15 \times \text{ATR}$ (absorption) $\rightarrow$ fade	832	+\$452.23	+0.54	46.2 %
FL-6	imb_cont	z of the imbalance ratio (flow/volume), $\geq 1.5 \rightarrow$ continuation	2,343	−\$7,451.77	−3.18	44.4 %
FL-7	vol_ignition_dir	volume- $z \geq 1.5$, direction from the bar's own sign (the volume control)	1,774	−\$3,850.50	−2.17	42.6 %
FL-8	flow_x_vol_cont	$ fz \geq 1.5$ and volume- $z \geq 1.5$	1,287	−\$1,637.89	−1.27	43.3 %
FL-9	flow_lot	net aggressor flow restricted to prints $\geq L$ lots ($L = 2/5/10/25/50$)	see §5	—	−1.63 ... −2.85	~44 %

Six of nine are outright negative blanket. The two that clear zero (FL-4 +\$0.94, FL-5 +\$0.54) clear it by less than a tick. **Per the standing judging rule these are not killed on win-rate** — 42–46 % is normal for this desk, which booked +\$625 this week at 39 %. They are killed, below, on the robustness battery.

3. Per-regime — the cells that had a pulse

40 candidate×regime cells were scored. Four came back with a live-looking expectancy:

CELL	N	NET	\$/TR	WIN%
FL-1 flow_impulse_cont @ BUILDING	451	+\$2,495.07	+5.53	47.9 %

FL-4 flow_ignition @ VIOLENT-WHIPSAW	170	+\$2,070.12	+12.18 52.4 %
FL-8 flow_x_vol_cont @ CLEAN-TREND	50	+\$630.91	+12.62 52.0 %
FL-5 flow_absorb_fade @ CLEAN-TREND	28	+\$706.32	+25.23 57.1 %

△ These four were selected out of 40 cells. At a 50/50 null you would expect several positive cells for free. The selection is why the battery below is the deliverable and the table above is not.

4. The battery — every grave, by name and cause of death

💀 FL-1 flow_impulse_cont @ BUILDING — KILLED BY THE OOS LEG

home (full tape) n=451 net=+\$2,495.07 \$/tr= +5.53
 IS (first 20d) n=243 \$/tr=+11.72 OOS (last 15d) n=208 \$/tr= -1.69
 strip 3 best \$/tr= +3.57 strip 5 best \$/tr= +2.48
 LONG n=210 \$/tr=+0.76 SHORT n=241 \$/tr=+9.69 <- asymmetric, all of it is short-side
 leave-one-day-out \$/tr min +3.70 / med +5.59 / max +6.69 (20/33 days profitable)
 PLACEBO (flow shifted ±390..1560 min) \$/tr -6.65..+6.23, 1/6 placebos beat the real signal

Cause of death: OOS sign flip (+\$11.72 → −\$1.69/tr), with a failed long/short symmetry test as the corroborating wound — 94 % of the net is short-only, which is a directional accident of a 35-day tape that fell, not a footprint. A placebo also beat it 1-in-6.

💀 FL-8 flow_x_vol_cont @ CLEAN-TREND — KILLED BY THE PLACEBO AND STRIP-THE-BEST

n=50 net=+\$630.91 \$/tr=+12.62
 top-3 trades = 80.5% of net ; strip 5 best -> \$/tr = -3.14 (NEGATIVE)
 IS \$/tr=+25.27 (n=31) OOS \$/tr= -8.03 (n=19)
 PLACEBO: the shifted-flow control scored +\$20.06/tr, BEATING the real signal

Cause of death: the placebo out-earned the real signal, and stripping 5 trades turns it negative. Triple-killed — OOS is negative too.

💀 FL-5 flow_absorb_fade @ CLEAN-TREND — KILLED BY STRIP-THE-BEST

n=28 net=+\$706.32 \$/tr=+25.23
 top-3 trades = 96.1% of net ; strip 5 best -> \$/tr = -7.63 (NEGATIVE)

Cause of death: strip-the-3-best. 96.1 % of the money is three trades; the remaining 25 trades are worth +\$27.84 in total, i.e. +\$1.11/tr, which the fee eats. n=28 across 17 days is below the threshold at which a placebo test even means anything, so this never reached one.

☠ **FL-2, FL-3, FL-6, FL-7 — KILLED AT THE BLANKET SCREEN, no regime cell rescued them**

- **FL-2 flow_impulse_fade** −\$3.55/tr blanket. The direct test of "the census saw 9 VACUUM vs 4 FLOW-LED, so fade the flow". It is the worst candidate in the suite. Its only positive cell, CLEAN-TREND, is n=55 at +\$5.77/tr, i.e. inside the same 40-cell selection noise. **Cause of death: negative expectancy blanket and in 4 of its 5 regimes.**
- **FL-3 flow_persist_cont** −\$2.18/tr. Requiring three consecutive same-sign flow minutes (a working order, not one print) reduced n from 2,306 to 1,411 and left expectancy unchanged. **Cause of death: negative expectancy; persistence adds no information.**
- **FL-6 imb_cont** −\$3.18/tr. Volume-normalising the flow (imbalance ratio instead of net contracts) makes it worse, not better. **Cause of death: negative expectancy.**
- **FL-7 vol_ignition_dir** −\$2.17/tr. The volume control from \$0b. Confirms that volume's magnitude lift does not convert into a directional edge either. **Cause of death: negative expectancy.**

☠ **FL-4 flow_ignition @ VIOLENT-WHIPSAW — the one that got all the way to the end**

This was the only candidate to survive strip-the-best, long/short symmetry, leave-one-day-out and a 6-shift placebo. It took the full battery to kill it, so it is documented in full.

Spec (as it would have shipped): at each minute close, if $|\text{flow-z}| \geq 1.5$ **and** the 30-minute signed range-break $(\text{close} - \text{close}_{-30})/\text{ATR}$ is $\geq +0.5$ in the flow's direction, enter at market on the next print in the flow's direction. Stop $1.0 \times \text{ATR}$, target $1.5 \times \text{ATR}$, flat at 15 minutes. Armed only when $\text{atr_rel} \geq 1.6$ and $\text{er30} < 0.35$.

Survived:

home regime	n=170	net=+\$2,070.12	\$/tr=+12.18	win=52.4%	median +\$3.75
strip 3 best		\$/tr= +8.08	strip 5 best	\$/tr= +5.49	(top-3 = 34.8% of

```

net)
LONG  n=70  $/tr= +8.51    SHORT n=100 $/tr=+14.75    <- symmetric
leave-one-day-out  $/tr min +8.68 / med +12.23 / max +15.04  (22/33 days profitable)
net without the best day  +$1,415.11  (of +$2,070.12)
6-shift placebo      0/6 beat it
parameter plateaus stop 1.0-2.0xATR all +$10..+16 ; target 0.75-3.0xATR all
positive ;
                                time stop 5-45 min all positive ; break threshold 0.0-1.5 all
positive

```

Died on:

(1) The 40-shift placebo — empirical $p = 0.150$. Shifting only the flow series $\pm 120 \dots 2400$ minutes and keeping every other rule identical:

```

real FL-4  $/tr = +12.18
placebo null (n=40)  mean +2.60  sd 8.37  min -13.81  MAX +16.25
placebos >= real: 6/40  ->  p = 0.150

```

Six fake flow series matched or beat it, and the best fake beat it outright. That is the operator's own stated kill criterion: if a FAKE signal books most of the money, the real one is not the edge.

(2) The paired ablation — the flow term contributes nothing out of sample. Holding the breakout population fixed and splitting it by what flow was doing:

BREAKOUT POPULATION, SPLIT BY FLOW	N	\$/TR	IS	OOS
flow AGREES ($ fz \geq 1.5$) — this is FL-4	72		+13.15	+39.43 -11.71
flow QUIET ($ fz < 1.0$)	338		+4.92	+8.13 +1.44
flow OPPOSES	30		-1.88	-4.28 +1.72

In-sample the ordering is beautiful and monotone (agrees $\gg$ quiet $>$ opposes) — which is exactly what a fitted term looks like. Out of sample it **inverts**: the flow-agreement bucket is the worst of the three, and the only stable positive bucket is the one where flow is doing nothing at all.

(3) Five-fold walk-forward — 3/5, and worse than the flow-free version.

fold		FL-4 (flow+break)	break only, NO flow term
F1 07-05..07-12	n= 25	+\$26.19/tr +\$654.82	n= 65 \$+19.30/tr +\$1,254.50
F2 07-13..07-27	n= 42	\$ -4.02/tr -\$168.71	n=123 \$ +1.20/tr +\$147.59
F3 07-28..08-05	n= 25	+\$62.54/tr +\$1,563.38	n= 91 \$+10.16/tr +\$924.70
F4 08-06..08-14	n= 38	\$ -9.13/tr -\$347.00	n=112 \$ +0.49/tr +\$54.38

F5 08-17..08-25	n= 40	\$ +9.19/tr	+\$367.64	n=100	\$ +5.49/tr	+\$548.52
folds profitable: 3/5				5/5		

Adding the flow term costs two folds and 65 % of the sample.

(4) Block bootstrap CI straddles zero. 10,000 day-block resamples: \$/tr = +12.18, **95 % CI [-2.71, +27.70]**, $P(\$/tr > 0) = 0.945$. It cannot clear a two-sided bar even before the $p=0.150$ placebo is accounted for.

(5) The regime it was armed for is half fiction. Sweeping the home definition:

```
atr_rel >= 1.2 +$7.92 | 1.4 +$13.02 | 1.6 +$12.18 | 1.8 +$10.08 | 2.0 -$0.62
er30 < 0.25 +$16.35 | 0.35 +$12.18 | 0.45 +$13.05 | 0.55 +$12.62 | 1.01 (no ER
filter at all) +$12.66
```

The ATR half is a genuine plateau. **The ER half does nothing** — removing the ER condition entirely changes \$/tr by 4 cents. So "VIOLENT-WHIPSAW" was never the home regime; "high ATR" was, and half the router condition was decoration.

(6) It is a US-session-only signal whose US-session OOS is negative. 192 of 212 trades sit in 13:30–20:00 UTC: US $n=192$, \$/tr +\$13.44, **IS +\$32.07 → OOS −\$8.13**.

CAUSE OF DEATH — FL-4: the PLACEBO test ($p = 0.150$, 6/40 shifted-flow controls beat it, best fake +\$16.25 vs real +\$12.18), confirmed by the paired OOS ablation in which the flow term's contribution flips from +\$39.43/tr in-sample to −\$11.71/tr out-of-sample.

5. The escalation the brief ordered — and the size-threshold answer

5a. Hunt the full sat-out set → top-25 → top-15

Against the frozen census's 68 runs (62 sat out), asking how many FL-4 caught in the correct direction inside a $\pm 5/+10$ min window:

	CENSUS RUNS CAUGHT	ON THOSE TRADES
FL-4 flow_ignition	10 / 68	+\$254.14
the flow-free breakout	18 / 68	+\$1,323.80
FL-4 on the 4 FLOW-LED runs (its own cluster)	1 / 4	—
FL-4 on the top-15 biggest runs	3 / 15	—

Narrowing did **not** reveal a footprint the marginal runs washed out — it went the other way. FL-4 catches its own cluster 1-in-4 and the biggest runs 3-in-15, worse than its 10/68 overall rate.

5b. The size threshold, asked on both sides

Predictor side — raise the flow bar and read expectancy (home = atr_rel≥1.6, break≥0.5):

FLOW-Z BAR	N	\$/TR	IS	OOS
≥1.0	296	+2.32	+11.81	−8.99
≥1.5	212	+12.66	+26.37	−3.29
≥2.0	157	+14.46	+40.61	−12.71
≥2.5	122	+4.26	+27.40	−18.89
≥3.0	96	+3.96	+25.07	−17.14
≥3.5	69	−5.39	+12.69	−22.95
≥4.0	43	+1.90	+26.90	−21.96

Every single threshold is positive in-sample and negative out-of-sample. The OOS column does not merely decay with size, it gets worse. That is the signature of a fitted term, not a footprint that sharpens.

Outcome side — flow agreement as runs get bigger, over all 35 days:

RUN SIZE	N	FLOW-LED	VACUUM
fwd15 ≥ 55pt	1,367	48.4 %	51.6 %
≥ 80pt	677	49.5 %	50.5 %
≥ 100pt	401	48.6 %	51.4 %
≥ 120pt	249	50.2 %	49.8 %
≥ 150pt	147	53.7 %	46.3 %
≥ 200pt	57	50.9 %	49.1 %
≥ 250pt	14	too few	

The only departure from 50/50 is 53.7 % at ≥150pt on n=147 — **0.9 sigma**, and it does not survive to the ≥200pt bucket. There is no size at which the tape starts telling you which way it is going.

💀 5c. FL-9 flow_lot — KILLED BY THE PARAMETER SWEEP (no plateau)

The census counts every contract equally: one 250-lot sweep scores the same as 250 one-lots. If a footprint hides anywhere it should hide in the blocks.

directional accuracy of large-lot flow, z >=1.5, all bars					
lot>=2	n=4,622	agree 48.55%	E[fwd15 in dir]	+0.74pt	t=+0.98
lot>=5	n=4,732	agree 48.77%		+0.74pt	t=+1.02

lot>=10	n=4,571	agree	49.60%	+0.81pt	t=+1.11
lot>=25	n=4,028	agree	50.15%	-0.13pt	t=-0.16
lot>=50	n=1,580	agree	50.70%	+0.89pt	t=+0.62

tick-honest replay, continuation

lot>=2	blanket	-\$2.49/tr	home(atr_rel>=1.6)	-\$0.85/tr	
lot>=5	blanket	-\$2.85/tr	home	-\$2.58/tr	
lot>=10	blanket	-\$2.30/tr	home	-\$2.45/tr	
lot>=25	blanket	-\$1.71/tr	home	+\$6.05/tr	<- the only positive
cell					
lot>=50	blanket	-\$1.63/tr	home	-\$1.75/tr	

Cause of death: the parameter sweep. Every lot threshold is negative blanket, agreement never leaves 48.5–50.7 %, and the single positive cell (lot≥25 on the high-ATR home, +\$6.05/tr) is an isolated spike with −\$2.45 immediately below it and −\$1.75 immediately above. An edge that exists at 25 lots but not at 10 or 50 is a bucket, not a plateau.

6. The residue — and why it is not this section's survivor

The flow-free half of FL-4 is the only thing on this page with a pulse after the battery:

plain 30-min range-break, $\geq 0.5 \times \text{ATR}$, in high ATR ($\text{atr_rel} \geq 1.6$). No flow term whatsoever. n = 491, net +\$2,929.68, \$/tr +\$5.97, win 50.5 %, 5/5 walk-forward folds profitable, 18/68 census runs caught for +\$1,323.80. Exit mix: TIME 285 (+\$20.6/tr) · TGT 102 (+\$150.8/tr) · STOP 158 (−\$112.9/tr).

I am **not** reporting this as a survivor, for three reasons, and the operator should treat it as a lead for the structure-break phase rather than a gate:

1. Its block-bootstrap 95 % CI is **[−2.12, +14.36]** — it straddles zero exactly as FL-4 did.
2. Its own IS/OOS split is +\$10.38 → **+\$1.04/tr**. Positive, but \$1.04 is a third of the fee.
3. Its US-session leg, which is 483 of its 491 trades, goes **IS +\$13.64 → OOS −\$2.15**.

More importantly: **it is not a flow gate, and it is not this cluster**. Reporting a breakout gate as the FLOW-LED cluster's finding would be the exact error the census made — attaching a flow name to something the flow had no part in.

Router rule, if it is ever picked up by another phase (stated in live-router vocabulary so it is falsifiable): ARM when `atr_rel` ≥ 1.4 (ATR-14 on 5-min bars vs its trailing 24h median) **and** `session` = US; BENCH otherwise; no ER condition — the ER term was measured to be worth 4 cents. Note that `atr_rel` is **not currently an instrument the live router reads** — the router's LAST HR ATR is a mean range, not the true-range ATR-14 this is keyed to, so arming on the router's existing number would arm on the wrong measurement. That is the instrument we would need to build first.

7. What would have to be true for this cluster to be worth revisiting

The FLOW-LED hypothesis is not refuted in principle — it is refuted with the instrument we have. The census's flow is a 60-second net-aggressor count from `ticks.aggressor`, and `$0b` shows it is a volume proxy. Three measurements would make it a fresh question rather than a repeat of this one:

- 1. **Resting-liquidity flow, not print flow.** `capture.db.book` is 41 ms event-driven and MNQ-only. The question worth asking is not "who crossed the spread" but "did the far side get pulled" — that is a book measurement, and it is the one the census's book column already gestures at.
 - 2. **Flow at a level, not flow in a clock minute.** A fixed 60-second bucket splits a single institutional order across two buckets and merges two unrelated ones. Volume-clock or order-imbalance-at-price buckets are the standard fix, and the tick data supports both.
 - 3. **A run population large enough to populate a 10 %-base-rate label.** 68 runs cannot; the lake's 5,066 bars at $\geq 55pt$ can. Any future cluster work should be labelled on the lake and only illustrated on the week's census.
-

8. Summary table

CANDIDATE	BEST CELL	\$/TR	FATE	CAUSE OF DEATH
FL-1 <code>flow_impulse_cont</code>	BUILDING	+5.53	👻	OOS leg (+11.72 $\rightarrow$ -1.69), long/short asymmetry
FL-2 <code>flow_impulse_fade</code>	—	-3.55	👻	negative expectancy blanket + 4/5 regimes
FL-3 <code>flow_persist_cont</code>	—	-2.18	👻	negative expectancy; persistence adds nothing

FL-4 flow_ignition	VIOLENT-WHIPSAW	+12.18 	placebo p=0.150 ; paired OOS ablation flips +39.43 → −11.71
FL-5 flow_absorb_fade	CLEAN-TREND	+25.23 	strip-the-best (top-3 = 96.1 % of net; strip-5 → −7.63)
FL-6 imb_cont	—	−3.18 	negative expectancy
FL-7 vol_ignition_dir	—	−2.17 	negative expectancy (volume control)
FL-8 flow_x_vol_cont	CLEAN-TREND	+12.62 	placebo beat it (+20.06 vs +12.62); strip-5 → −3.14; OOS −8.03
FL-9 flow_lot (5 lot sizes)	atr_rel≥1.6, lot≥25	+6.05 	parameter sweep — isolated cell, no plateau
(residue, not a survivor) break-only, high ATR	US session	+5.97 —	bootstrap CI [−2.12, +14.36]; OOS +1.04; not a flow gate

VERDICT

NOT SURVIVED. No FLOW-LED entry signal reached a tradeable expectancy. Nine candidates were built and all nine are dead: FL-2/3/6/7 on negative expectancy at the blanket screen, FL-1 on the **OOS leg**, FL-5 and FL-8 on **strip-the-3-best** (with FL-8 also beaten outright by its own **placebo**), FL-9 on the **parameter sweep**, and the last survivor FL-4 flow_ignition on the **PLACEBO/SHUFFLE TEST — empirical p = 0.150, with 6 of 40 time-shifted fake flow series matching or beating it and the best fake (+\$16.25/tr) beating the real signal (+\$12.18/tr) outright** — corroborated by a paired ablation in which the flow term's contribution inverts from +\$39.43/tr in-sample to −\$11.71/tr out-of-sample, and by 3/5 walk-forward folds against 5/5 for the same gate with the flow term deleted.

There is no size threshold at which a FLOW-LED footprint becomes tradeable. On the predictor side every flow-z bar from 1.0 to 4.0 is positive in-sample and negative out-of-sample. On the outcome side, flow agreement with the run is 48.4 %–53.7 % from 55pt to 200pt and never departs from a coin flip by as much as one sigma.

★ **The finding that outranks all of the above: the cluster label itself is not a footprint.** The flow-event filter fires on 20.58 % of all bars and then splits that fifth into FLOW-LED (10.21 % of tape) and VACUUM (10.37 %) on a 49.62 % coin toss — 0.33 percentage points of directional information versus the $|fz| < 1$ control. Whatever lift the flow event carries is a **magnitude** lift, and that lift is entirely a

volume proxy: flow unusual with volume ordinary gives $0.963 \times$ base $E|fwd15|$ (no lift at all), while volume unusual with flow balanced gives $1.220 \times$. The census's flow column measures how much traded, not who was buying.

Recommendation: stop splitting the run census into FLOW-LED/VACUUM. The two labels are one population cut by its own outcome, and the week's 4-vs-9 split is noise (expected 9.0 vs 9.6; $P(\leq 4 \text{ of } 13) = 0.139$ against the lake null).

VACUUM — the snap-back / stop-run

gf_full_VACUUM.md · 5,224 words · written 2026-08-22 00:18 UTC

★★ REV3 — THIS DOSSIER IS ON THE OLD CENSUS FREEZE

Written 2026-08-22 00:18 UTC, three days before the census was re-frozen on 08-25, and anchored throughout to the superseded list: **11 VACUUM runs of 70, of which 9 sat out, a \$9,986 week-ceiling, and every catch figure scored “x/65”**. Movement 1’s frozen census for the same week says **VACUUM = 9 runs of 68, 62 sat out, \$9,610. Do not line up per-run numbers between this page and Movement 1** — they count different lists, and neither is wrong on its own terms. Nothing here depends on the count: four candidates, four graves, and the cluster is killed by its own LABEL rather than by any run list, so the verdicts stand as written and this is banner-marked rather than re-run. Movement 2 carries the same warning for the same reason. Movement 1’s banner used to say “all seven greenfield labs” reproduce the new freeze; it is six, and this is the seventh.

GREENFIELD HUNT — cluster VACUUM (census 2026-08-16 → 2026-08-21)

The job. Ignore every gate we own. Invent a brand-new entry signal that shows up for the runs the census filed under **VACUUM**, spec it mechanically, backtest it tick-honest over the **full Parquet lake** at the **true \$1.50 round-trip fee**, escalate to the biggest runs if it fails, and try hard to kill it. Working below; every attempt is named, including the graves.

THE THREE ANSWERS, IN ORDER OF HOW MUCH THEY MATTER

1. ★ The VACUUM label is not a footprint — and this week I found where it comes from. Last Friday's section established that the label is a coin flip. This week's larger sample (33 trading days, 43.2M ticks, 42,143 tape-minutes) confirms that and goes one step further. The flow event the label is built on fires on **20.41% of every minute of tape we own**. Given that event, the direction test comes up "VACUUM" **49.90%** of the time — against **49.91%** on minutes where no flow event happened at all. That is a **−0.01pp** difference. But the census still reports VACUUM on 15.7% of its runs, well above the 10.2% base rate, and I traced the whole of that excess to a **selection artefact in run_census.py's own run-picker**. The picker takes every 60s-slid 15-min window over threshold and dedupes "keep-strongest", so the surviving minute is by construction the local extremum of the move. Rebuilding that picker on the lake: the prior-5-minute price direction agrees with the run **48.66% of the time on unselected big minutes and 22.55% after the dedup — a −26.11pp shift created purely by the selection rule**. The census's own 65 sat-out runs read 24.6%, matching the rebuilt picker almost exactly. Push the VACUUM condition through the same three populations and it goes **10.45% → 14.01% → 21.87%**. **VACUUM is the name the census gives to the fact that it picked a local bottom.**

2. Detection is real, it is size-graded, and it is the answer to the size-threshold question — but direction is a coin flip and I could not break it. A volume-z / ATR-expansion detector separates the biggest runs from ordinary tape at **AUC 0.69 (p<0.0001)** on the 113 lake-wide runs of 120pt+, rising monotonically from 0.557 at 55–70pt. On the census's own top-15 the volume-z AUC is **0.730, p=0.0004**. Turned into a bracket gate (**VAC-IGN-BRK**) it catches **9 of the 65 sat-out runs and 5 of the top-15 on only 48 fires in the census week — 3.6× the random-fire baseline**. And it still cannot be traded, because **the sign-flip control earns exactly as much as the signal** (+\$3.08 vs +\$3.21 per trade). We can see the runs coming. We cannot tell which way they go.

3. Last week's shadow candidate failed its forward test, and last week's one open stone is now closed. VAC-ABS reproduced to the dollar on the in-sample leg (+\$24.04/tr vs the +\$24.91 reported) and then lost **−\$529.50 over 27 trades (−\$19.61/tr)** on the five fresh days. Its strip-5 is **−\$427**. It is dead, not shadowed. And the L2 far-side-depletion discriminator that last week's report

nominated as "the one stone still unturned" is **refuted**: on 21 days and 246.9M book rows it predicts the next 15 minutes' direction at **49.44% ±2.52pp**.

△ **A method note that changed every headline in this section.** My first pass let overlapping entries run concurrently. Five separate shorts booked the same 2026-07-31 impulse and were worth **\$3.9k of a \$10.3k headline**. A live gate takes one lot and cannot re-enter until flat. Everything below is scored **one position at a time**; where the unconstrained number is shown it is labelled as such and is never the verdict. Under the honest accounting VAC-IGN-BRK falls from **+\$10,316 / +\$15.49 per trade to +\$911.55 / +\$3.21 — 91% of the apparent edge was one impulse counted five times**.

1. What the census labelled, and what the label is made of

scripts/run_census.py::cluster() calls a run **VACUUM** when, in the 60 seconds before it starts, net aggressor flow was unusual for that moment ($|z| \geq 1.0$ against the trailing two hours of the same statistic) **and pointed the opposite way to the move that followed**. Plain English: the tape was full of buyers and then price fell.

This week: **11 VACUUM runs of 70**, of which **9 sat out**, carrying **\$1,595** of 1-lot hindsight ceiling out of the week's **\$9,986** total sat-out ceiling.

#	RUN START (UTC)	DIR	MOVE	\$ CEILING	60S FLOW	AMP%	BOOK	US
1	08-17 07:27	UP	+57	\$114	−216	0.05	0.42	sat out
2	08-18 01:38	DN	−101	\$202	+458	0.07	0.49	sat out
3	08-18 13:40	DN	−138	\$276	+485	0.29	0.49	caught (exhaustion_sh ort_A, −\$208)
4	08-18 13:56	DN	−70	\$140	+1850	0.21	0.34	caught (day_rider, + \$32)
5	08-18 14:11	UP	+100	\$200	−1738	0.32	0.52	sat out
6	08-19 12:32	UP	+150	\$300	−131	0.12	0.56	sat out
7	08-19 13:59	DN	−108	\$215	+741	0.23	0.52	sat out
8	08-19 18:03	DN	−89	\$178	+552	0.19	0.54	sat out
9	08-20 05:54	UP	+56	\$112	−128	0.05	0.44	sat out
10	08-20 15:02	DN	−80	\$161	+1542	0.37	0.49	sat out
11	08-21 14:42	UP	+56	\$113	−819	0.14	0.49	sat out

Read the definition again. **The label uses the forward move to decide the label.** You cannot know a minute is VACUUM until you already know where price went for the next fifteen minutes. That is fine for a taxonomy and fatal for a trigger. The only thing knowable in advance is the **flow z-score**, so that is what I tested — and then I tested the label itself.

2. ★ THE LABEL INTERROGATION — the finding that outranks everything else here

reports/friday_v7/vacuum/label_test.py, run over the whole lake: every one of 42,143 tape-minutes scored with exactly the census's own arithmetic (60s prior net aggressor flow, z-scored against the trailing 2h of the same statistic, ≥ 30 non-empty buckets required).

	VALUE
tape-minutes scored	41,182
the flow event ($ z \geq 1.0$) fires on	20.41% of all minutes
VACUUM (event and direction disagrees) on all minutes	10.18%
P(direction disagrees a flow event happened)	49.90% (95% CI 48.84-50.97)
P(direction disagrees no flow event)	49.91% (95% CI 49.37-50.46)
information the flow event adds to the direction test	-0.01pp

One minute in five is a "real flow event", and knowing that tells you nothing whatsoever about which way the next fifteen minutes go. **VACUUM is a normal flow minute with a coin landed on it.**

2.1 So why does the census see 15.7% and not 10.2%?

Because the run-picker selects local extrema. stage5_artifact.py rebuilds the picker on the lake — every minute with $|fwd15| \geq 55pt$, then "keep-strongest" dedup at 15-min spacing — and compares three populations:

FEATURE (SIGN AGREEMENT WITH THE RUN)	ALL-BIG (NO SELECTION)	PICKED (CENSUS PICKER)	SHIFT
prior 5-min price change	48.66% (n=4,930)	22.55% (n=683)	-26.11pp
prior 15-min price change	50.39% (n=4,932)	31.38% (n=682)	-19.01pp
flow z (the VACUUM leg)	48.92% (n=4,928)	41.38% (n=679)	-7.54pp
5-min cumulative-delta z	48.95% (n=4,887)	40.75% (n=670)	-8.20pp

Verification: the census's own 65 sat-out runs read **24.6%** on prior-5-min agreement — the rebuilt picker says 22.55%, the unselected tape says 48.66%. The picker, not the tape, is doing the work.

And the label rate itself, decomposed:

POPULATION	VACUUM RATE	= FLOW-EVENT RATE	× DISAGREEMENT RATE
all scored minutes	10.45%	20.72%	50.42%
all-big minutes (fwd15 ≥ 55pt, unselected)	14.01%	27.05%	51.79%
PICKED (the census's own population)	21.87%	28.43%	76.92%
the census's reported week	15.71% (11/70)		

The disagreement rate goes 50.4% → 51.8% → **76.9%** purely by selecting local extrema. **The VACUUM cluster does not describe a state of the tape. It describes the census's dedup rule.** ★ This finding outranks every signal in this document and is the reason none of them was ever going to work as specified.

What this does not say. The census is not broken as a taxonomy — "where was the money and did we show up" is answered correctly, and the sat-out ceiling of \$9,986 stands. What is unsafe is treating a cluster name as a hunting ground, and any pre-run price-path statistic quoted off the run list.

3. Every candidate, tick-honest, one position at a time

Harness (reports/friday_v7/vacuum/vaclab.py). 1-minute bars built **from the trade ticks**, so price and the aggressor split are the same object. Entry is the **first tick after the trigger bar closes** — no same-bar fill, no lookahead. Stops and targets are walked **tick by tick** over all 43,186,167 MNQ lake ticks, stop checked **before** target on every tick. Costs charged in every number: **\$1.50 per round trip** (the true venue fee) **plus 0.25pt slippage each way**, MNQ at **\$2.00/point**. Verified by hand-replaying the largest trade against the raw tick array — entry, stop price, exit and net all reconcile to the cent.

Serial accounting. One position at a time. See the ⚠ note in the summary.

CANDIDATE	NET	\$/TR	WIN%	T	SAT-OUT CAUGHT	TOP-15	CAUSE OF DEATH
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VAC-Z1 — trade the label literally	1,768	−\$4,150.79	−2.35	35.7%	—	16/65	3/15	Refuted outright. The label has no edge; unconstrained it loses — \$32,184 over 8,589 trades.
VAC-ABS — 08-15 spec, forward-tested	97	+\$1,319.21	+13.60	38.1%	1.39	0/65	0/15	OOS leg — \$529.50 / — \$19.61 per trade on the 5 fresh days, strip-5 — \$427 , t=1.39. Dead.
VAC-ABSGO — its sign-flip control	96	−\$2,290.43	−23.86	22.9%	—	—	—	Working as intended — confirms the mechanism is directional even though the edge is gone.
VAC-SNAP — absorption → structural break	410	+\$370.64	+0.90	34.9%	0.17	6/65	2/15	Strip-3 — \$1,436 (top-5 trades = 738% of net), OOS — \$18.02/tr, LOO goes negative, 12/33 days green.
VAC-SNAP/GO — its sign- flip control	428	−\$625.71	−1.46	39.0%	—	—	—	Control behaves; the corpse is still a corpse.
VAC-CUMDIV z1.5 / z2.0	506 / 264	−\$1,814.64 / −\$1,668.71	−3.59 / −6.32	32%	—	4/65	2/15	Refuted outright, both parameterisati ons, both directions of the sweep.
VAC-IGN-BRK — the detection bracket	284	+\$911.55	+3.21	32.4%	0.45	9/65	5/15	SIGN-FLIP: the fade earns + \$3.08/tr against the signal's + \$3.21. Both directions win = volatility, not edge. Also strip-3 −\$915, no plateau.
CTRL-UNCOND — same break, arm every minute	2,404	−\$4,657.54	−1.94	29.7%	—	30/65	7/15	Control. Proves the arm is load-bearing for the detection , and that the break rule alone loses.
VAC-ABS placebo +240m	92	+\$239.00	+2.60	37.0%	—	—	—	A fake trigger still earns — further

								evidence VAC-ABS's remaining number is noise.
L2 far-side depletion (direction discriminator)	—	—	—	—	—	—	—	Refuted. 49.44% ±2.52pp on unselected big moves. \$6.

Nulls that never reached a spec, recorded so nobody re-walks them: **flow-sign continuation** (z5 agreement 48.95% on unselected big moves), **near-30-min-extreme position** (49.05% on all minutes), **coil ratio** (AUC 0.501 on the sat-out set — no detection at all).

3.1 The three mechanical specs, for the record

VAC-SNAP — invented here, refuted here. On completed 1-min bars: $\text{cumflow5} = \Sigma \text{ netflow over the last 5 bars}$; z5 = its z-score against the trailing 2h of the same statistic. **ARM** when $\text{z5} \geq +1.5$ (buyers) and the 5-bar price change is $\leq 0.35 \times \text{ATR}(1\text{m}, 14)$ — heavy one-sided aggression that failed to move price. **TRIGGER** within 10 bars when a bar closes **below the arming window's low** → **SHORT** (mirror for sellers absorbed → **LONG** on a close above the window high). **ENTRY** first tick after that bar closes. **STOP** $2.0 \times \text{ATR}(1\text{m}, 14)$. **EXIT** stop or 30-min time-stop.

VAC-IGN-BRK — invented here, the one that detects. volz = z-score of bar volume against the trailing 120 bars; $\text{atr_ratio} = \text{ATR}(1\text{m}, 14)$ averaged over 5 bars ÷ over 60 bars. **ARM** when $\text{volz} \geq 2.0$ **and** $\text{atr_ratio} \geq 1.15$. **TRIGGER** within 10 bars when a bar closes outside the 5-bar range — **above the high** → **LONG, below the low** → **SHORT**. Non-directional by construction: the tape picks the side, the gate only picks the moment. **ENTRY** first tick after that bar closes. **STOP** $1.5 \times \text{ATR}(1\text{m}, 14)$. **EXIT** stop or 30-min time-stop.

VAC-ABS — last week's spec, unaltered, forward-tested. $|z(\text{netflow})| \geq 3.0$ **and** $|\text{close} - \text{prev close}| \leq 2.75\text{pt}$ → fade the aggressors. Stop $2.0 \times \text{ATR}$, 30-min time-stop.

4. VAC-ABS — the forward test, in full

This is the honest part of last Friday's deliverable coming due. The spec was frozen on 2026-08-15; the lake now holds five trading days it has never seen.

LEG	N	NET	\$/TR	WIN%
v5 archive, ≤ 07-17		17	+\$439.14	+25.83 —
in-sample, 07-24 ... 08-15		53	+\$1,409.57	+26.60 —
OUT OF SAMPLE, 08-17 ... 08-21		27	-\$529.50	-19.61 28.6%
all		97	+\$1,319.21	+13.60 38.1%

The in-sample leg reproduces last week's +\$24.91/tr almost exactly (+\$26.60 here; the small gap is the serial constraint dropping 6 overlapping fires). **The forward leg is negative and it is negative in the home regime too** — the ATR ≥ 15pt cell that last week's router rule was written around books 4 OOS trades for **-\$154.64**. Add **strip-5 = -\$427** and **t = 1.39** and there is nothing to defend.

Cause of death: the out-of-sample leg, confirmed by strip-5. Not the win rate — 38.1% would have been perfectly acceptable per the operator's judging rule if the expectancy had held up. It did not.

5. ★ THE SIZE THRESHOLD — the escalation, and the one place it pays off

The mandate asks for the escalation and for the size at which a footprint becomes tradeable. Running the census picker across the **whole lake** gives **686 runs** instead of 70, so the bands have real n. AUC is against a random lake minute; p is a 4,000-draw permutation of the same n.

BAND (MOVE)	N	VOLUME-Z AUC	ATR AUC	FLOW Z AUC	PRIOR-5MIN DIRECTION
55-70 pt	257	0.557 (p=0.0003)	0.616 (p<0.0001)	0.558 (p=0.0005)	19.8%
70-90 pt	188	0.542 (p=0.026)	0.670 (p<0.0001)	0.488 (p=0.73)	26.9%
90-120 pt	128	0.567 (p=0.004)	0.801 (p<0.0001)	0.552 (p=0.021)	20.5%
120 pt +	113	0.690 (p<0.0001)	0.867 (p<0.0001)	0.634 (p<0.0001)	23.9%

And on the census's own week, permutation-tested at n=15/25/65:

FEATURE	SAT-OUT 65	TOP-25	TOP-15
volume-z	0.597 (p=0.0029)	0.637 (p=0.0089)	0.730 (p=0.0004)

ATR(1m,14)	0.553 (p=0.071)	0.583 (p=0.072)	0.676 (p=0.0081)
flow z	0.522 (p=0.27)	0.525 (p=0.34)	0.679 (p=0.0088)
ATR expansion ratio	0.529 (p=0.21)	0.508 (p=0.44)	0.674 (p=0.0093)
ER15	0.442	0.442	0.403 — big runs start from chop, replicating last week

The answer to the size-threshold question: yes, there is one, and it is real — but it is a DETECTION threshold, not a tradeability threshold. Three of the four detectors are indistinguishable from noise on the full sat-out set and all four become significant at the **top-15 band, |move| ≥ 87pt**; lake-wide the effect keeps climbing to **AUC 0.867 on ATR at 120pt+**. Narrowing revealed a footprint exactly as the mandate hoped it might.

And it converts into catches. VAC-IGN-BRK fired **48 times in the census week** and caught **9 of the 65 sat-out runs and 5 of the top-15**, against a random-fire baseline of **2.5** — a **3.6× lift**. That is the first thing in two Fridays of greenfield work that demonstrably shows up for the big runs.

It just cannot be traded, because the direction column above is the picker artefact and nothing else. The prior-5-min "20-27% agreement" looks like a screaming fade signal until you notice it reads **48.66% on unselected big moves**. Strip the selection and there is no direction anywhere:

DISCRIMINATOR, ON UNSELECTED FWD15 ≥ 55PT MINUTES		AGREEMENT
prior 5-min price change		48.66%
prior 15-min price change		50.39%
flow z / 5-min cumulative-delta z	48.92% / 48.95%	
position in the 30-min range		49.89%
L2 far-side depletion		49.44%

6. ★ THE STONE LAST WEEK LEFT UNTURNED — L2, and it is now closed

Last Friday's section ended: "revive ATR-IGNITION with an L2 far-side-depletion discriminator once MNQ book coverage ≥ 30 days". Coverage is now **21 days (2026-07-31 ... 2026-08-21), 246,934,712 book rows, 22,119 book-minutes** — short of 30 days but enough for a verdict at this effect size. Ask-share = top-3 ask depth ÷ (ask + bid), z-scored against its own trailing 2h so "thin" means thin for now. Convention: asks thin → should telegraph UP.

POPULATION	N	DIRECTION AGREEMENT
all book minutes	21,249	49.72% ($\pm 0.67\text{pp}$)
fwd15 $\geq$ 30pt	5,009	50.37% ($\pm 1.38\text{pp}$)
fwd15 $\geq$ 55pt (unselected big moves)	1,515	49.44% ($\pm 2.52\text{pp}$)
fwd15 $\geq$ 90pt	418	49.76% ($\pm 4.79\text{pp}$)
thin book only (ask-share z $\geq$ 1)	344	47.38% ($\pm 5.28\text{pp}$)
census sat-out 65 / top-15 / VACUUM	65 / 15 / 9	40.0% / 26.7% / 22.2%

The census-population rows look like strong **inverse** information — until you note they are the same -26pp local-extremum artefact as every other pre-run price statistic. On the honest unselected population the book reads **49.72% on 21,249 minutes with a $\pm 0.67\text{pp}$ error bar**. That is as close to a definitive null as this desk can measure.

△ Also worth recording: the census's own book column **cannot be read as a forecast**. It is far/(far+near) where far is chosen using the realised direction. It is a description, not a prediction, and quoting it as one would be the third instance of this same class of error in this document.

Verdict on the stone: REFUTED. MNQ L2 top-3 depth does not supply direction. Do not revive ATR-IGNITION on it at 30 days either — the null is not n-limited.

7. ★ THE ROUTER — what would be armed, and the honest answer that nothing is

The mandate is explicit that a blanket cross-tape number is a bug and that a surviving gate must come with its home regime and an arm rule in the live router's vocabulary. **Nothing survived**, so what follows is the segmentation evidence and the rule that would have been written, stated so it is falsifiable rather than aspirational.

The regime split is consistent across all three candidates and is the one genuine policy finding here — **every one of them is an ATR-high, US-session animal and every one of them bleeds in low ATR:**

	VAC-ABS	VAC-SNAP	VAC-IGN-BRK
ATR low tercile	-\$5.26/tr (n=33)	-\$8.10/tr (n=137)	-\$11.64/tr (n=95)
ATR mid tercile	-\$5.99/tr (n=31)	-\$10.09/tr (n=136)	+\$3.24/tr (n=94)
ATR high tercile	+\$50.86/tr (n=33)	+\$20.82/tr (n=137)	+\$18.03/tr (n=95)

US session 13:00-21:00Z	+\$51.24/tr (n=26)	+\$22.89/tr (n=104)	+\$16.33/tr (n=110)
overnight / pre-open	-\$0.18/tr (n=71)	-\$6.57/tr (n=306)	-\$5.08/tr (n=174)

ER is not a discriminator here, in either direction. Serially, VAC-IGN-BRK books $-\$0.49/\text{tr}$ in dead chop ($\text{ER}_{15} < 0.15$, $n=83$), **+\$17.85/tr in the 0.15-0.35 middle** ($n=93$) and $-\$6.55/\text{tr}$ in clean trend ($n=108$) — a middle bucket carrying the whole of a \$911 net across three buckets is noise, not a regime. $\triangle$ Recorded because it matters: the unconstrained run said $+\$22.28$ in chop and $+\$13.79$ in trend, and that was the overlap-inflated ledger — exactly the number this section warns against quoting.

What **is** a router finding, and it comes from the detection work in \$5 rather than from any backtest: the biggest runs **start from chop** — ER_{15} is the only feature measured whose AUC sits below 0.5, and it falls further as the runs get bigger (**0.442** on the sat-out 65, **0.403** on the top-15). Every ER-floor arm condition the live router owns is standing gates down at the moment the largest moves begin. That belongs to the router section, not to a gate.

The rule that would have been written, had the sign-flip control not killed it:

ARM `vac_ign_brk` when $\text{ATR}(1\text{m}, 14) \geq 21.0\text{pt}$ on MNQ **and** $13:00 \leq t < 21:00\text{Z}$. **BENCH** below 19.0pt (2-point hysteresis, matching the existing 2-mark convention) with a **≥ 5 -minute dwell on the arm side** — ATR crossing a bare threshold on a partial bar is exactly the oscillation that produced 20 switch changes for 4 trades on 08-07. **No ER condition**, for the reason above. Expectancy on that home regime only: **+\$18.03/trade over 95 trades**. Blanket, as context and never as the verdict: **+\$3.21 over 284**.

Do not arm it. The home-regime number is the same number the fade control earns; conditioning on ATR-high selects trades with more range in both directions, and a gate that makes money whichever way you point it is a volatility harvester with a stop, not an edge.

What we would need to measure to make this tradeable. A **direction discriminator that is not price, not aggressor flow, and not top-3 book depth** — all three are ruled out above at tight error bars. The concrete candidates, in order of how cheaply we could instrument them: (i) **full-depth book imbalance beyond level 3**, including queue-position and cancel/replace rate — `depth.db` carries the wide snapshot but at 250ms it cannot see the fleeting

quotes capture.db.book does at 41ms, so this needs the 41ms feed widened past level 3, which is an IBKR subscription question, not a code question; (ii) **cross-instrument lead-lag** — we already capture MGC L1+L2 and MES/M2K depth, and none of this week's work looked at whether gold or the breadth complex turns first; (iii) **a trade-size distribution split** (large-lot vs small-lot aggressor), which the tick stream already carries in size and which no census or gate on this desk currently uses. (iii) is free and is the one I would run first next Friday.

MGC: deliberately not attempted. The scope is explicit that gold gates are **invented fresh, never ported or re-tuned from an MNQ gate**, and the VACUUM cluster is a property of the MNQ census. Porting VAC-IGN-BRK to gold would have been the wrong kind of easy number — and at \$10.00/point a mis-priced transfer would have looked five times more exciting than it was. Gold is gf_MGC's work.

8. Robustness, laid out test by test

Every candidate went through the full battery. The table records what each test did, not just whether it passed.

TEST	VAC-ABS	VAC-SNAP	VAC-IGN-BRK
sign-flip	−\$23.86/tr □ passes	−\$1.46/tr □ passes	+\$3.08/tr vs +\$3.21 ☹ KILL
time-shifted placebo (+30/+90/+240/−90m)	+240m earns +\$2.60 △	—	+90m +\$1.06, +240m +\$0.06 △
random-arm bootstrap (40 draws, same arm count)	—	—	mean −\$2.92, max +\$1.98, base + \$3.21 — beaten 100% □
unconditional control (arm every minute)	—	—	−\$1.94/tr □ the arm is load-bearing
parameter sweep / plateau	—	edge only as n collapses (+\$0.90 @ n=410 → +\$7.02 @ n=63) ☹	4 of 15 cells negative; −\$1.92 @ n=813 → +\$13.26 @ n=143 — step, not plateau ☹
strip-the-3-best	+\$118 (from +\$1,319) △	−\$1,436 ☹	−\$915 ☹
strip-the-5-best	−\$427 ☹	−\$2,364 ☹	−\$1,685 ☹
leave-one-day-out	min +\$9.53 □	min −\$2.55 ☹	min −\$0.26 ☹ (drop 07-31)
out-of-sample leg (08-17...08-21)	−\$529.50 / −\$19.61 ☹	−\$1,189.64 / −\$18.02 ☹	+\$184.57 / +\$3.85 □
v5 archive leg (07-05...07-17)	+\$439.14 □	−\$758.14 ☹	−\$1,904.73 / −\$19.44 ☹
long/short symmetry	+\$27.60 vs −\$2.58 ☹ 10:1	+\$8.76 vs −\$6.57 ☹	+\$6.74 vs −\$0.47 △
big-moves-caught	0/65 sat-out, 0/15 top-15 ☹	6/65, 2/15	9/65, 5/15 — 3.6× baseline □
t-statistic on \$/trade		1.39	0.17
days green	15/28	12/33	15/32

Two things in that table deserve to be read rather than skimmed. **VAC-IGN-BRK is the only thing in two Fridays of VACUUM work that passes the placebo family, the unconditional control and the bootstrap, and catches the big runs at 3.6× baseline** — and the sign-flip control still kills it stone dead. **And VAC-ABS is the only candidate whose sign-flip control is convincingly negative** — a real directional mechanism, forward-tested, and it lost money anyway.

9. Files

FILE	WHAT
reports/friday_v7/vacuum/vaclab.py	shared tick-honest harness (walk, walk_serial , costs)
reports/friday_v7/vacuum/label_test.py	★ the label base-rate interrogation
reports/friday_v7/vacuum/stage5_artifact.py	★ the run-picker selection artefact + permutation AUCs
reports/friday_v7/vacuum/stage6_ignbrk.py	detection by size band (686 lake runs), IGN-BRK surface
reports/friday_v7/vacuum/stage7_kill.py	controls, unconstrained (kept to document the overlap trap)
reports/friday_v7/vacuum/stage8_serial.py	everything re-scored one position at a time
reports/friday_v7/vacuum/stage9_l2.py	★ the L2 direction test that closes last week's open stone
reports/friday_v7/vacuum/stage10_final.py	strip / LOO / OOS / symmetry battery
reports/friday_v7/vacuum/ser_*.csv, stage*.json	per-trade ledgers and machine-readable results

VERDICT

The VACUUM cluster did NOT survive, and for the second Friday running it was killed by the LABEL itself rather than by a backtest — but this week the mechanism is identified, not just the symptom. The census's flow event fires on **20.41% of every tape-minute we own**, and conditional on it firing the VACUUM direction test comes up VACUUM **49.90%** of the time against **49.91%** on non-event minutes — **−0.01pp**, inside a ± 1.1 pp error bar. The apparent excess in the census (15.7% of runs vs a 10.2% base rate) is **entirely a selection artefact of run_census.py's keep-strongest run-picker**: prior 5-min price direction agrees with the run **48.66% of the time on unselected big moves and 22.55% after the dedup**, and the VACUUM rate climbs **10.45% → 14.01% → 21.87%** across the same three populations. VACUUM is the name we give to having picked a local bottom. **Traded literally (VAC-Z1) it loses −\$4,150.79 over 1,768 trades serial, −\$32,184 over 8,589 unconstrained.**

Four candidates were invented and all four are graves, each with a named cause of death. VAC-SNAP (absorption → structural break, +\$370.64 / n=410) — killed by **strip-the-3-best** (−\$1,436) and the **OOS leg** (−\$18.02/tr). **VAC-CUMDIV — refuted outright**, negative at every threshold. **VAC-ABS** (last week's shadow candidate, forward-tested unaltered, +\$1,319.21 / n=97 / +\$13.60 per trade net of \$1.50 per round trip) — killed by its **OUT-OF-SAMPLE LEG**, −\$529.50 over 27 trades on the five days it had never seen, confirmed by **strip-5 = −\$427**; it is now **RETIRED, not shadowed**. **VAC-IGN-BRK** (the detection bracket, +\$911.55 / n=284) — killed by the **SIGN-FLIP CONTROL**: fading the break earns **+\$3.08/trade against the signal's +\$3.21**, so both directions win and it is volatility harvesting, not edge; the parameter surface is a step rather than a plateau and 4 of 15 cells are negative.

The size-threshold question has a real answer this week, and it is a detection threshold, not a tradeability one. Below $|\text{move}| \approx 87\text{pt}$ (the top-15 band) three of four detectors are indistinguishable from noise; at and above it all four go significant — volume-z **AUC 0.730, p=0.0004** on the census's top-15, rising lake-wide to **AUC 0.867 on ATR for the 113 runs of 120pt+**. **VAC-IGN-BRK converts that into 9 of 65 sat-out runs and 5 of the top-15 caught on 48 fires — 3.6× the random-fire baseline, the first thing in this workstream that demonstrably shows up for the big runs.** It is still not a gate, because **detection without direction is a coin flip with a stop**, and the sign-flip control proves it.

Last week's one open stone is closed: the L2 far-side-depletion discriminator is REFUTED — 49.44% ±2.52pp on unselected 55pt+ moves over 21 days and 246.9M book rows, **49.72% ±0.67pp** on all 21,249 book-minutes. Do not revive it at 30 days; the null is not n-limited.

△ **And one methodological finding that should outlive this section: the first pass of this backtest let overlapping entries run concurrently, and five shorts into the single 2026-07-31 impulse were worth \$3.9k of a \$10.3k headline. Scored one position at a time — the only way a live gate can trade — VAC-IGN-BRK falls from +\$15.49 to +\$3.21 per trade. 91% of the apparent edge was one impulse counted five times. Every greenfield harness on this desk should be checked for it.**

Dispositions: NOTHING to LIVE. NOTHING to SHADOW. VAC-ABS retired from shadow. Two findings handed onward — (1) the census run-picker's

local-extremum artefact, which invalidates every pre-run price-path statistic quoted off the run list and should be fixed or flagged in `run_census.py`; (2) the biggest runs start from CHOP (ER15 AUC 0.403), so every ER-floor arm condition the router owns is standing gates down at the moment the largest moves begin. The single cheapest next experiment is a large-lot vs small-lot aggressor split — the size field is already in the tick stream and no gate or census on this desk uses it.

THE CHOP-DAY TURN SCALP — the operator's own commission, L2-led

`gf_chopscalp.md` · 5,750 words · written 2026-08-25 23:39 UTC

GREENFIELD — THE CHOP-DAY TURN SCALP (L2-LED)

The short version

Garrath, you asked for a purpose-built chop-turn scalper: catch the oscillation turns on the days the desk currently donates, bank tiny, and **lead with the order book** — same lineage as `exhaustion_short`, which was built out of L2 footprints and is one of the best things we own.

I built it and raced it tick by tick on **17 full-stack trading days** — 20.7 million trade ticks, **261.9 million 41-millisecond ten-deep book rows**, 4.6 million 250ms depth snapshots and 26.2 million quote updates. Sixteen named candidates, 768 costed parameter cells in the main sweep and 1,080 more in the finalist search, every fill at \$1.50 the round trip.

The answer is no, and this time the cause of death is not the one I expected.

1. Your lead hypothesis is refuted, and it points the wrong way. Requiring the far side to be ABSORBING — a big resting wall on the side price has to eat to continue — does not help the fade. It **hurts** it, consistently, in two independent tests. Unfiltered chop turns drift **+0.054 pt** in the fade's favour over the next minute. Add "wall $\geq 1.5\times$ " and that becomes **-0.117 pt**. Add "five-deep wall $\geq 1.3\times$ " and it becomes **-0.374 pt** ($t = -1.72$). In the costed sweep the wall filter is the **worst** candidate on the board: CT2_WALL medians **-\$3.09/trade** against the naive fade's **-\$2.54**. The wall is not absorption at a range edge. It is a target.

2. The one thing that did survive to the battery was killed by handing the null my own search. CT15 — close at a 30-minute extreme, on a ≥ 4 pt 20-second thrust, with the 41ms refill count above median — made **+\$158.00 on 26 trades, +\$6.08 a trade, 77% win, positive in-sample AND out-of-sample, positive on 13 of 13 leave-one-day-out folds, still +\$4.07/trade after stripping its best five.** On a single pre-registered cell its shuffle p-value is **0.017**. But that cell was found by a 1,080-cell search, so I gave the null the same search: shuffle the entry features within each day and let it hunt. **The null's own best cell averages +\$8.42/trade and beats the real one 85% of the time.** Matched fairly, at cells at least as wide as the real winner, **p = 0.367**. It is a search artefact.

3. And even if it were real, the arithmetic does not reach your bar. CT15 fires **1.53 times a day**. To clear +\$250 on a chop day at that rate it must earn **\$163.50 a trade — 81.7 MNQ points**. The median chop 30-minute block's entire range is **51.75 points**. Selectivity and the +\$200-300 target are in direct conflict: at the measured \$6.08/trade you need 41 trades a day, and the filter loose enough to fire 41 times a day is the naive fade at **−\$2.54**.

4. ★ The premise needs one correction, and it is the finding I would act on. The desk did not donate on this week's chop days. **It did not trade them at all.** On 2026-08-20 and 2026-08-21 the tournament took **zero** trades. The week's **−\$2,323** is day_rider's book, not the gates'. Across the whole 18-day window the tournament's loss is concentrated in **violent-whipsaw** blocks (**−\$1,383.50** on 95 trades) — high ATR going nowhere — **not** in plain chop (**−\$403.00** on 58). That is a router bench to price, not a gate for greenfield to build.

1. Method, data and the cost model — stated before any number

ITEM	VALUE
Symbol	MNQ , \$2.00 per point, 0.25pt tick
Fee	\$1.50 per ROUND TRIP (not \$5, not \$2, not per side)
Entry fill	first tick at or after the decision, 1 tick adverse (we cross)
Stop fill	stop-market, 1 tick beyond the trigger
Full-stack days	17 — 07-31, 08-03...07, 08-10...14, 08-17...21, 08-24
Trade ticks	20,672,140
Book rows (41ms, 10-deep, event-driven)	261,887,502
Depth snapshots (250ms, 10-deep)	4,647,174

Quote updates	26,188,882
Source	Parquet lake data/tape/ via gazbot7.lake — not capture.db, which is a 5-day rolling window
Turn candidates built	6,762 — of which 3,439 on chop-classified tape
Report week	2026-08-17 → 2026-08-21

Regime segmentation, per the standing backtest discipline. Nothing is scored blanket across the tape. Every 5s decision carries the regime of the **last completed minute**, keyed on the 30-minute efficiency ratio and on ATR-14 **relative to its own trailing one-session median** — never an absolute ATR floor, because MNQ ran ~20pt ATR in late July and ~7pt in mid-August and an absolute threshold would label all of July violent and all of August dead. The block counts below are the full 28-day bar window (07-16 → 08-24, 1,264 blocks); the backtests use only the 17 full-stack days.

REGIME	RULE	30-MIN BLOCKS	SHARE
DEAD_CHOP	ER30 < 0.20, ATR < 0.70× session median	105	8.4%
CHOP	ER30 < 0.20, normal ATR	586	46.6%
WHIPSAW	ER30 < 0.20, ATR ≥ 1.50× session median	109	8.7%
BUILD	0.20 ≤ ER30 < 0.40	387	30.8%
TREND	ER30 ≥ 0.40	70	5.6%

Day types come off the US-session (13:30–20:00Z) efficiency: TREND ≥ 0.45, CHOP ≤ 0.25. The four chop days inside the full-stack window are **07-31, 08-07, 08-10 and 08-20**.

⚠ **Two harness bugs found and fixed in this section — both would have flattered or destroyed the result, and both are worth carrying forward.**

1. **bar_ts is the bar's START, not its close.** The first race entered on the first tick after bar_ts, i.e. before the high or low that defined the event. That is anti-look-ahead: the trade is entered knowing the bar will reach an extreme it has not reached yet, guaranteeing the stop. It read **19% target-first on BOTH sides** against a 47.8% random-time control. Fixed to bar_ts + 5; both sides moved to ~45%.
2. **The extreme definition was not held constant across the parameter sweep.** The 30-minute test required the bar to **CLOSE** at the extreme; the 15- and 20-minute tests only required the **wick** to touch it. That made a definition change look like a hard step in the lookback parameter, and I very

nearly reported "the plateau is a cliff at L=30". It is not. \$6 re-runs it with the definition held constant and the surface is monotone in both parameters.

2. The prize — where does the desk ACTUALLY donate?

Every live MNQ trade attributed to the regime of the 30-minute block it **opened** in. day_rider is a separate book taking its own size on the same IB account and is reported on its own line.

Tournament gates, 07-31 → 08-24, 281 trades, −\$1,756.00

REGIME OF THE OPENING BLOCK	N	NET	\$/TRADE	WIN%
WHIPSAW	95	−\$1,383.50	−\$14.56	34%
CHOP	58	−\$403.00	−\$6.95	33%
BUILD	103	+\$128.00	+\$1.24	43%
TREND	23	+\$83.00	+\$3.61	39%

−\$1,383.50 of the tournament's −\$1,756.00 net is lost in violent-whipsaw blocks — high ATR, no direction — and it is lost at twice the per-trade rate of plain chop. A chop scalper does not address this. A router bench keyed on $ATR \geq 1.5 \times \text{session median}$ AND $ER30 < 0.20$ does.

And the report week itself:

DAY	DAY TYPE	TOURNAMENT N	TOURNAMENT NET	DAY_RIDER NET
08-17	TREND	8	+\$17.00	+\$456.00
08-18	TREND	20	+\$7.50	+\$188.00
08-19	TREND	8	−\$29.00	+\$544.00
08-20	CHOP	0	\$0.00	−\$1,288.50
08-21	MIXED	0	\$0.00	−\$2,218.44

The gates did not bleed the chop days this week. They were silent on them. That does not make your ask wrong — it sharpens it. A scalp clearing +\$200–300 on 08-20 would have been **new money**, not recovered money. It also means there is no donation on chop days for this section to claim back; the week's −\$2,323 is day_rider's, and **\$2,149.44 of it is the 4-lot carry opened 08-21 13:13Z and closed by the operator by hand on 08-24 (MANUAL_CLAIM).**

3. Does the book separate a turn from a break? The model-free test

No exits, no fees, no fitting. From every chop-tape turn candidate, the signed move in the fade's favour at fixed horizons. **The fee alone is 0.75 points round trip** — that is the line to beat.

POPULATION	N	+60S	+120S	+300S	T(60S)
all chop turns	3,439	+0.054	+0.088	+0.425	0.44
+ closed at 30m extreme	104	+1.113	+1.486	+0.853	1.89
+ closed at 30m ext & 20s thrust ≥ 4pt	49	+1.883	+2.592	+3.107	1.85
+ wall ≥ 1.5× at the touch (absorption)	1,328	−0.117	−0.168	−0.171	−0.60
+ five-deep wall ≥ 1.3×	934	−0.374	−0.264	+0.066	−1.72
+ 41ms refill ≥ median	1,765	+0.164	+0.226	+0.695	0.82
+ VWAP-flat ≤ 0.5 ATR	3,293	+0.079	+0.127	+0.505	0.63
ALL TAPE (any regime)	6,762	+0.353	+0.451	+0.717	3.14

Three things fall out of that table, and they set up everything after it.

The unfiltered chop turn earns 0.054 points against a 0.75-point cost. That is the known coin-flip, measured. It is not a cost problem you can fix with a cheaper broker; it is 7% of the fee.

The wall has the wrong sign in both L2 sources. The quintile test agrees independently — on a symmetric 4pt/4pt first-touch race over 3,434 resolved chop turns (base 45.02%), the top quintile of the five-deep wall ratio reverts **39.62%** against the bottom quintile's **45.93%**, $z = -2.37$; the level-1 wall gives $z = -2.03$. Every other book term I built is inside noise:

FEATURE (TOP VS BOTTOM QUINTILE, CHOP TAPE, 4/4 RACE, BASE 45.02%)	TOP	BOTTOM	Z
60s price move into the fade	49.23%	42.63%	+2.52
VWAP steepness	48.33%	41.63%	+2.49
aggressor fraction, 60s	46.14%	41.05%	+1.90
41ms support-pull count	45.49%	41.48%	+1.52
41ms refill count	45.05%	42.19%	+1.09
aggressor delta, 20s	48.48%	45.57%	+1.08
stall ratio (exhaustion_short's own term)	45.64%	42.79%	+1.06
ten-deep imbalance	46.81%	43.96%	+1.06
resting-book depletion, 20s	44.89%	43.61%	+0.53

wall building over 20s	43.56%	44.57%	−0.43
level-1 wall ratio	40.08%	45.15%	−2.03
five-deep wall ratio	39.62%	45.93%	−2.37

The extreme-fade works BETTER off chop tape than on it — +0.353 pt (t = 3.14) across all tape against +0.054 (t = 0.44) in chop. That is the first sign that whatever mechanism is here is not a chop specialist, and §9 confirms it.

★ **And your VWAP-flat filter is inert.** +0.079 pt filtered against +0.054 unfiltered, and in the costed sweep CT1_VWAPFLAT medians −\$2.53 against the naive fade's −\$2.54 — a one-cent difference across 48 cells. It is not wrong. It is doing no work. Last week's section reached the same verdict by a different route; two independent passes now say the same thing.

4. Is it a cost problem or a sign problem? — fees set to zero

Before blaming the fee, I ran every candidate **frictionless**: no commission, no stop slippage.

CANDIDATE	COSTED MEDIAN \$/TR	COSTED CELLS +	FRICITIONLESS MEDIAN \$/TR	FRICITIONLESS CELLS +
CT0_NAIVE fade every chop-tape 10m extreme	−\$2.541	0/48	−\$0.772	0/48
CT1_VWAPFLAT + your VWAP filter	−\$2.533	0/48	−\$0.780	0/48
CT2_WALL + wall ≥ 1.5×	−\$3.086	0/48	−\$1.335	0/48
CT2b_WALL5 + five-deep wall ≥ 1.3×	−\$3.190	0/48	−\$1.402	0/48
CT5_STALL + top-quintile stall ratio	−\$2.575	0/48	−\$0.825	0/48
CT6_ANTIWALL + wall ≤ 0.8× (the inverse)	−\$2.289	0/48	−\$0.541	0/48
CT7_SPIKE + 20s thrust ≥ 4pt	−\$2.308	0/48	−\$0.543	0/48
AT0_ALLTAPE any regime	−\$2.228	0/48	−\$0.478	1/48
CT10_DEEPEXT closed at a 30m extreme	−\$1.500	5/48	+\$0.216	29/48
CT13_DEEP_SPIKE that, on a ≥4pt thrust	−\$0.566	16/48	+\$1.102	39/48

Fading a turn on chop-classified MNQ is a SIGN problem, not a cost problem. The naive fade is negative in all 48 cells before a cent of fee. Adding the absorption wall makes the frictionless number nearly twice as negative. Only the

two deep-extreme candidates cross zero without costs, and only one of them survives contact with the fee.

★ **The exhaustion_short detector is essentially never satisfied on chop tape.** Running its literal published thresholds — $|20s \text{ signed aggressor net}| \geq 400$, $|20s \text{ price move}| \leq 2pt$, $hit\text{-}side \text{ wall} \geq 1.5\times$ — over 3,439 chop turns: the net floor alone passes **2.94%**; net **and** stall passes **8 events (0.23%)**; all three pass **exactly 1 event in 17 trading days**. The median $|20s \text{ net}|$ on chop tape is **44 contracts** against a floor of 400. That gate's detector is a US-session-thrust instrument. It cannot be ported to chop tape by tuning, and this is why CT4_EXH_LITERAL never reached the grid at all.

5. The full candidate board — every attempt, including the graves

Sixteen candidates × a 48-cell exit grid (target 2–12pt × stop 3–10pt), taken **sequentially and non-overlapping** on one slot, which is the only way a real desk can trade them.

#	CANDIDATE	EVENTS/DAY	CELLS +	MEDIAN \$/TR	BEST \$/TR	CAUSE OF DEATH
CT0	NAIVE fade every 10m extreme in chop	202.3	0/48	−\$2.54	−\$2.25	frictionless sign test — negative in 48/48 before fees
CT1	VWAPFLAT CT0 + $ VWAP \text{ slope} \leq 0.5 \text{ ATR}$	193.7	0/48	−\$2.53	−\$2.24	inert — moves CT0 by \$0.01/trade
CT2	WALL CT0 + level-1 wall $\geq 1.5\times$ (your lead hypothesis)	78.1	0/48	−\$3.09	−\$2.34	wrong sign — worse than CT0 costed AND frictionless
CT2b	WALL5 CT0 + five-deep wall $\geq 1.3\times$	54.9	0/48	−\$3.19	−\$2.37	wrong sign, $z = -2.37$ in the model-free race
CT3	FOOTPRINT CT0 + aggressor delta ≥ 150	24.4	0/48	−\$2.46	−\$1.80	frictionless sign test
CT4	EXH_LITERAL the published exhaustion_signal	0.06	—	—	—	never fires — 1 event in 17 days
CT4b	EXH_SOFT loosened to net ≥ 200 / move ≤ 3 / wall ≥ 1.2	0.7	1/48	−\$4.48	+\$1.21	$n = 12$; IS +\$7.90/tr → OOS −\$3.57/tr
CT5	STALL top-quintile stall ratio	40.5	0/48	−\$2.58	−\$1.89	frictionless sign test

CT6	ANTIWALL the inversion — wall ≤ 0.8×	32.1 0/48	−\$2.29	−\$1.73	the inverse is not tradeable either
CT7	SPIKE 20s thrust ≥ 4pt	100.2 0/48	−\$2.31	−\$1.92	frictionless sign test
CT8	PULL top-quintile 41ms support-pull count	40.5 0/48	−\$2.24	−\$1.26	frictionless sign test
CT10	DEEPEXT bar CLOSES at a 30m extreme	6.1 5/48	−\$1.50	+\$0.53	survives frictionless, dies on the fee
CT11	SPIKE_ANTIWALL	43.2 0/48	−\$2.34	−\$1.64	frictionless sign test
CT12	SPIKE_PULL	52.2 0/48	−\$2.19	−\$1.85	frictionless sign test
CT13	DEEP_SPIKE close@30m extreme + ≥4pt thrust	2.9 16/48	−\$0.57	+\$2.54	→ the battery, \$8
CT14	US_ONLY CT0 restricted to 13:30-20:00Z	45.1 0/48	−\$2.37	−\$1.67	frictionless sign test
CT21	RETEST touch of the 60m edge, no new 30m extreme	29.6 0/24	−\$2.54	−\$1.91	frictionless sign test
CT22	RETEST + thrust ≥ 4pt	16.6 0/24	−\$2.19	−\$1.34	frictionless sign test

Thirteen of sixteen are negative in every cell of their own grid. The exit sweep cannot rescue an entry whose sign is wrong; that is what §4 established and this table is what it looks like.

6. The plateau — with the extreme definition held constant

This is the table I nearly got wrong. H = the **wick** touched the trailing L-minute extreme; C = the bar **CLOSED** at it. 15 exit cells per row.

DEFN	L	THRUST K	EVENTS	CELLS +	MEDIAN \$/TR	MAX \$/TR
H	10	0 / 3 / 4 / 5	3351 / 2169 / 1704 / 1327	0 / 0 / 0 / 0	−2.55 / −2.32 / −2.35 / −2.15	all negative
H	15	0 / 3 / 4 / 5	2384 / 1525 / 1194 / 933	0 / 0 / 0 / 0	−2.64 / −2.43 / −2.68 / −2.42	all negative
H	20	0 / 3 / 4 / 5	1850 / 1207 / 951 / 750	0 / 0 / 0 / 0	−2.49 / −2.11 / −2.29 / −2.01	all negative
H	30	0 / 3 / 4 / 5	1205 / 788 / 618 / 496	0 / 0 / 0 / 0	−2.56 / −2.06 / −2.21 / −2.12	all negative
C	10	0 / 3 / 4 / 5	348 / 218 / 151 / 112	0 / 0 / 0 / 2	−2.40 / −2.19 / −2.18 / −0.54	+0.25
C	15	0 / 3 / 4 / 5	232 / 145 / 104 / 75	0 / 0 / 1 / 5	−2.05 / −1.65 / −1.73 / −0.28	+1.41
C	20	0 / 3 / 4 / 5	174 / 111 / 80 / 60	1 / 4 / 7 / 10		+3.03

				-1.56 / -0.50 / -0.10 / +0.84	
C	30 0 / 3 / 4 / 5	103 / 66 / 49 / 34	1 / 5 / 9 / 13	-1.88 / -0.63 / +0.39 / +1.71	+4.27

The lookback is not the discriminator — the wick-versus-close distinction is. Under H the family is negative in **0 of 240** cells at every lookback from 10 to 30 minutes. Under C it improves monotonically in both parameters. That monotone surface is a genuine gradient, not a spike, and it is the reason I took the survivor all the way through the battery instead of discarding it on a plateau failure. **The plateau test did not kill this candidate. Something else did.**

7. The book ablation — the operator's actual question

CT13 base (chop tape, bar closes at a 30-minute extreme, 20s thrust ≥ 4 pt), then one book term at a time, 35 exit cells each. The honest number is the **median** across the grid, not the max.

TERM ADDED	EVENTS	CELLS +	MEDIAN \$/TR	BEST CELL \$/TR
(base, no book term)		49 15/35	-\$0.29	+\$2.54
+ 41ms refill count $\geq$ median		26 32/35	+\$2.25	+\$6.08
+ 41ms support-pull $\geq$ median		27 24/35	+\$0.43	+\$2.61
+ level-1 wall $\geq 1.5\times$ (absorption)		26 16/35	-\$0.17	+\$4.25
+ support draining over 20s		19 14/35	-\$0.29	+\$2.34
+ VWAP-flat ≤ 0.5 ATR		45 14/35	-\$0.50	+\$2.56
+ five-deep wall $\geq 1.3\times$		30 9/35	-\$0.65	+\$1.38
+ wall BUILDING over 20s		26 5/35	-\$1.75	+\$0.50
+ aggressor delta ≥ 100		2 —	—	too few
+ ten-deep imbalance > 0		10 —	—	too few
+ five-deep wall $\leq 1.0\times$ (inverse)		11 —	—	too few

Exactly one L2 term earns its keep on this base — **the 41-millisecond refill count**, which is only visible in the event-driven book stream and cannot be computed from the 250ms depth sample. Nothing keyed on **resting size** helps: the wall at the touch is inert, the five-deep wall hurts, and the wall building over 20 seconds is the worst term on the board. **What travels from the exhaustion_short lineage is the quote-level activity, not the wall.**

That gives the finalist: **CT15 = chop tape + bar closes at a 30-minute extreme + ≥4pt 20-second thrust + 41ms refill count above median, target 8pt / stop 10pt.**

8. The robustness battery — three finalists, and they all pass it

TEST	CT13 (6/10)	CT16 K≥5 (8/10)	CT15 +REFILL (8/10)
headline	n=49, +\$124.50, +\$2.54/tr, 75.5%	n=34, +\$145.00, +\$4.27/tr, 70.6%	n=26, +\$158.00, +\$6.08/tr, 76.9%
days traded	16 of 17	15 of 17	13 of 17
in-sample 07-31→08-14	n=31, +\$33.00, +\$1.07/tr	n=20, +\$88.00, +\$4.40/tr	n=13, +\$79.00, +\$6.08/tr
out-of-sample 08-17→08-24	n=18, +\$91.50, +\$5.08/tr	n=14, +\$57.00, +\$4.07/tr	n=13, +\$79.00, +\$6.08/tr
report week only	n=14, +\$82.00	n=11, +\$50.00	n=10, +\$72.00
strip the best 3	+\$93.00 (+\$2.02/tr)	+\$101.50 (+\$3.27/tr)	+\$114.50 (+\$4.98/tr)
strip the best 5	+\$72.00 (+\$1.64/tr)	+\$72.50 (+\$2.50/tr)	+\$85.50 (+\$4.07/tr)
leave-one-day-out	16/16 positive, \$1.64-\$4.14	15/15 positive, \$3.27-\$5.63	13/13 positive, \$4.98-\$8.15
unresolved legs at the 15-min horizon		0	1
single-cell shuffle placebo	—	—	p = 0.017

Every one of them passes every test in the standard battery. Positive out-of-sample, positive on every leave-one-out fold, robust to stripping the best five trades, and a pre-registered shuffle p-value of 0.017. On the criteria we normally apply, CT15 is a promotion candidate.

It is not, and \$9 is why.

9. ★ THE KILL — give the null the same search that found the survivor

CT15 was not pre-registered. It is the winner of a search over **6 thrust thresholds × 12 book terms × 15 exit cells = 1,080 costed cells**, on a pool of 104 events. A 1.7% p-value on the best of 1,080 cells is not a 1.7% p-value on anything.

So I built the null tape and ran the **entire search on it**, 300 times.

- **Null B — SHUFFLE.** Permute the entry features across events within each day. Every marginal distribution and each day's outcome mix is preserved exactly; only the feature→outcome link dies.

- **Null A — DIRECTION.** Flip each event's side at random, so the fade becomes a chase.

	REAL	NULL B (SHUFFLE) BEST CELL	NULL A (DIRECTION) BEST CELL
best \$/trade found by the search	+\$6.077	mean +\$8.422 , median +\$8.396, p95 +\$12.418	mean +\$8.122, median +\$8.375
median n of the winning cell	26	16	14
p(null ≥ real), unrestricted	—	0.853	0.790
p(null ≥ real), n-matched (cells n ≥ 26)	—	0.367	0.150

The null, handed the same grid, finds a better cell than the real one 85% of the time. That comparison is unfair to the null in one direction — it wins by picking 14- and 16-trade corners where the variance is enormous — so I matched it: restrict the null to cells at least as wide as the real winner. It still lands at **p = 0.367**. There is no reading of this test on which CT15 is significant.

This is the same failure mode last week's section found by a different route (a placebo beating the best cell 77% of the time). Two independent weeks, two independent harnesses, the same answer: **the deep-extreme chop fade is search noise.**

10. And the two tests it fails on its own terms

It must hold on ALL the chop days, not one. The four chop days inside the full-stack window are 07-31, 08-07, 08-10 and 08-20.

CANDIDATE	CHOP-DAY N	CHOP-DAY NET	PER CHOP DAY	DAYS GREEN	07-31	08-07	08-10	08-20
CT13	15	+\$27.50	+\$6.88	1 of 4	-\$1.00	-\$12.50	+\$42.00	-\$1.00
CT16	12	+\$8.50	+\$2.12	3 of 4	+\$7.00	+\$7.00	+\$24.00	-\$29.50
CT15	8	-\$30.00	-\$7.50	1 of 3 traded	+\$7.00	-\$7.50	no trade	-\$29.50

The best candidate is negative on chop days. Its entire edge is on days that are not the days you asked it to rescue. CT13's whole chop-day P&L is one good day (08-10, +\$42.00) against three flat-to-negative ones.

Where the money actually is, and it is not chop. The same rule, armed off chop tape:

CANDIDATE	ON NON-CHOP BLOCKS	ON TREND DAYS
CT13	n=73, −\$24.00, −\$0.33/tr	n=50, +\$59.50, +\$1.19/tr
CT16	n=57, +\$133.00, +\$2.33/tr	n=36, +\$230.00, +\$6.39/tr
CT15	n=50, +\$141.00, +\$2.82/tr	n=31, +\$230.50, +\$7.44/tr

It does not bleed the trend days — it earns more on them than it does in chop. Combined with §3's model-free +0.353 pt across all tape against +0.054 in chop, the mechanism, to whatever extent it exists at all, is not a chop specialist. Restricting it to chop tape is throwing away the two thirds of it that reads better.

11. The arithmetic against your +\$200-300 bar

This is the structural argument, and it holds even if you disbelieve every p-value above.

	CT13	CT15
fires per day		2.88
measured \$/trade		+\$2.54
produces per chop day		+\$6.88
\$/trade needed to make +\$250/day at this rate		\$86.70
MNQ points needed per trade		43.4
median chop 30-min block total range	51.75 pt	51.75 pt
median chop 30-min block net travel	17.75 pt	17.75 pt
trades/day needed at the measured \$/trade		98
		41

You cannot take 82 points out of a block whose entire 30-minute range is 52. And the other direction closes just as hard: to fire 41 times a day the filter has to be loosened back toward CT0_NAIVE, which is −\$2.54 a trade and would lose −**\$104 a day** at that rate. The selectivity you correctly asked for and the +\$200-300 target are not reachable at the same time by this mechanism. The gap is about **30×**.

12. Honest caveats — including three that cut against my own numbers

- **The survivors' P&L is overnight and one-sided.** CT13 made **+\$110.50 of its +\$124.50 in ASIA (00:00-07:00Z)** and −\$2.00 in the US session on 6 trades; CT15 made **+\$123.00 of +\$158.00 in ASIA**. CT13 is also effectively

long-only: **LONG +\$142.00, SHORT −\$17.50**. A two-sided mechanism that earns on one side, in the thinnest hours, is showing you its sample size.

- **My cost model is optimistic and every \$/trade above is up to \$0.50 too high.** I cross one tick on entry; the median MNQ spread is **two ticks (0.50 pt) in every session**, mean 0.55pt in Asia against 0.40pt in the US. A full-spread cross removes another **\$0.50 a trade** from everything — which does not change any verdict's sign, but it turns CT13's +\$2.54 into +\$2.04 and CT0's −\$2.54 into −\$3.04, and it bites hardest exactly where the survivors trade.
 - **The 15-minute race horizon.** Legs unresolved at the horizon are marked OPEN and carried at their mark, never predicted. There were 0, 1 and 0 of them in the three finalists — the verdicts do not rest on an unresolved leg.
 - **The regime attribution in §2 is hindsight labelling.** It answers "where did the money go", not "what should have been armed at 09:14". Every gate feature in §3 onward is strictly trailing and lagged by one completed minute.
 - **Two harness bugs were found and fixed (§1).** I cannot rule out a third. What I can say is that a leak manufactures edge rather than destroys it, so an undiscovered leak would make this section's verdict more negative, not less.
 - **Six chop days exist in the wider tape; only four are inside the full-stack window** (07-17 and 07-28 pre-date the 41ms book capture). The all-chop-days test in §10 is four days, not six.
-

13. What I would do with this instead

1. **Bench the whipsaw blocks, do not scalp the chop ones.** the tournament's 18-day loss is concentrated (−\$1,383.50 of a −\$1,756.00 net) opened in blocks with $ATR \geq 1.5 \times$ the session median and $ER30 < 0.20$, at −\$14.56 a trade against plain chop's −\$6.95. That is a two-term router rule against a measured number, and it is worth roughly ten times what any scalper in this section produced.
2. **Stop treating a resting wall as absorption at a range extreme.** Two independent tests over 3,439 chop turns and 261.9M book rows put the wall's coefficient on the wrong side of zero. If any live or shadow gate keys on hit-side resting size at an extreme, it is keying on a term whose point estimate is negative. `exhaustion_short`'s wall term should be re-examined in its own home regime — this section only refutes it on chop tape.

3. **The 41ms refill count is the one L2 term worth keeping on the shelf.** It was the only book feature to improve a base across 32 of 35 exit cells, and it is invisible to the 250ms depth sample — so any study that reaches for `depth.db` for book dynamics is looking through the wrong instrument. It did not survive here; it earned the right to be tried somewhere with more n .
 4. **Do not re-run this hunt at 25 days.** The null is not n -limited. The naive fade is negative in 48 of 48 cells frictionless, and the wall is negative in every test at $n = 1,328$. More tape makes those tighter, not positive.
 5. **If you still want chop-day money, the next experiment is a different instrument, not a different threshold.** MGC is \$10 a point with its own 10-deep book in `depth.db`, and the points-per-trade arithmetic in §11 is five times kinder there. That is the `gf_MGC` phase's ground, and this section's result argues for spending the next chop-day effort on it.
-

VERDICT — the chop-day turn scalp DID NOT SURVIVE. The lead candidate CT15 (chop tape, bar closes at a 30-minute extreme, ≥ 4 pt 20-second thrust, 41ms refill count above median, 8pt/10pt) was killed by the **placebo/shuffle test run with the search included** — the null, handed the same 1,080-cell grid, finds a better cell than the real +\$6.077/trade **85.3% of the time** unrestricted and **36.7% of the time** when fairly matched at $n \geq 26$ — and, independently, by the **all-chop-days requirement**, on which it is **−\$7.50 per chop day and green on 1 of the 3 chop days it traded**. It passed everything else: out-of-sample (+\$79.00 = in-sample exactly), leave-one-day-out (13/13), strip-the-best-5 (+\$4.07/trade). The operator's core L2 hypothesis — that at a price extreme the far side ABSORBING separates a turn from a break — is **REFUTED with the sign inverted**: a hit-side wall $\geq 1.5\times$ turns the chop turn's +0.054pt one-minute drift into **−0.117pt** (five-deep $\geq 1.3\times \rightarrow$ **−0.374pt**, $t = -1.72$; quintile $z = -2.37$), and CT2_WALL is the **worst** of sixteen candidates at −\$3.09/trade. The VWAP-flat filter is **inert** (\$0.01/trade across 48 cells). The naive fade is a **sign problem, not a cost problem** — negative in 48 of 48 cells before a cent of fee. Projected chop-day P&L is **+\$6.88 (CT13) to −\$7.50 (CT15) per chop day against the +\$200-300 bar**, and the gap is structural: at 1.53 fires a day the rule must earn **81.7 points a trade** out of a chop block whose whole range is 51.75. The section's transferable results are elsewhere — **(1) the desk does not donate in chop, it donated −\$1,383.50 on 95 trades in violent-whipsaw blocks** (and

took **zero** tournament trades on this week's chop days, 08-20 and 08-21), which is a two-term router bench worth $\sim 10\times$ this scalper; **(2)** the resting wall is a target, not absorption; **(3)** the only L2 term that ever helped is the **41ms refill count**, which the 250ms depth sample cannot see.

GOLD — MGC, and the shadow arm that went live and lost

gf_MGC.md · 10,446 words · written 2026-08-25 21:36 UTC

THE GOLD GATE HUNT — MGC, 2026-08-25

Garrath — read this paragraph and you have the week. Last Friday we shadowed a gold gate called `mgc_hole_break_fade`. It has now been running live in the shadow book since Monday, and **it is losing money: −\$1,196 across 80 trades in seven days, every one of its three arms red**. When I replayed the same seven days in the lab it made **+\$512**. That is not a small disagreement, and this section is mostly about running that contradiction to ground — because the answer turned out to be a single, fixable thing.

The gate is being fed the wrong tape. The research built its one-minute bars from the depth mid — the midpoint of the order book, which updates every 250 milliseconds whether or not anything trades. The live service builds its bars from trades. Gold prints sparsely, so those are not the same series, and a sixty-minute high taken from one is not the sixty-minute high of the other. Hold absolutely everything else constant — same break rule, same book filter, same exit, same spread-honest race — and change only the input tape:

THE SAME RULE, THE SAME EXIT, THE SAME RACE, ON THE SIX FORWARD DAYS BOTH TAPES COVER			
	N	NET	\$/TRADE
depth-mid bars (what the research measured)	56	+\$172	+\$3.06
trade bars (what the live service actually reads)	53	−\$475	−\$8.96

The sign flips on the bar source alone, and the trade-bar number is the one the live shadow independently produced. So this is not a mystery any more, and it is not a research failure — it is a plumbing mismatch between the lab and

production, and the fix is mechanical: the service already holds a DepthFeed connection to gold's order book (it needs it for the liquidity filter anyway), so it already has the bid and the ask. It just needs to build its minute bar from their midpoint instead of from trades.

Until that is done, **I would not promote a single gold cell, and I am not proposing one.** Everything below stays in shadow. But three other things came out of the week that are worth your time:

- **A year of gold arrived.** The lake gained 320 trading days of MGC minute bars (2025-07-28 → 2026-08-19) — thirteen months against last week's twenty-nine days. It kills last week's router rule outright, closes last week's one loose thread, and puts a much colder light on the trigger.
- **The unturned stone paid.** Gold against the Nasdaq on the same clock produced a real, well-powered **veto**: over 278 paired days, fading a gold break the Nasdaq agrees with loses \$10.52 a trade and sits at the 2.5th percentile of its own placebo. That is a genuinely new input.
- **I caught a look-ahead in my own harness before it reached you.** A volatility cell read +\$71,608 until I noticed it was entering at the open of the bar whose high it had already used. Corrected, it reads −\$12,462. The working is in §7 because that number would otherwise have been the biggest in this report.

1. THE RULER — what changed about the measuring stick this week

1.1 The tape is longer in two different ways

STREAM	LAST FRIDAY	TODAY	WHAT IS NEW
MGC depth-quote minute bars (the lab tape)	22 days, 07-16 ... 08-14	29 days, 07-16 ... 08-25	7 genuine forward days the rules have never seen
MGC 10-level L2 (depth.db + lake)	22 days	29 days	no gap
MGC trade bars (5s)	18 days	24 days	production's input
MGC 1-minute bars (backfill_1m in.parquet)	—	320 days, 2025-07-28 ... 2026-08-19	thirteen months, brand new
MNQ 1-minute bars, same file	—	2025-09-14 ... 2026-08-19	278 days paired with gold

The seven forward days are the important ones for judging last week's gate. The 320-day file is the important one for judging the mechanism, because last week's

single largest caveat was stated plainly in its own §9.1: "twenty-two days is twenty-two days ... one macro regime." That is now answerable.

1.2 A landmine came with the backfill, and it will bite something

Those `backfill_*.parquet` files were added to the lake since last Friday, and they sit in the same directory as the daily 5-second files. So a query that reads gold's bars without filtering the timeframe now silently unions 1-minute, 5-minute, hourly and **daily** bars into the same series. Take a high across that union and the 10:00 minute inherits the high of the day it is in.

The tape module's own audit caught it as a side effect: its "August control, expected 0" line, which read **0** last week, now reads **9,469 contaminated minutes**. `gf_mgc_tape.load_ll()` filters `timeframe='5s'` and is safe. Anything that does not — `mgc_session_anchor.load()` is the one I know of — is now reading four-minute-to-one-day look-ahead on every day rather than only the ten V5 archive days it was doing before. Nothing in this section reads it unfiltered.

1.3 The \$7.50 round trip is not the cost, and that is already fixed

Last week's section led with "a round trip in gold costs \$7.50, not \$1.50" — 0.30pt of spread $\times$ 2 legs $\times$ \$10, plus the fee. That figure double-counts. Crossing the spread costs you **half** the spread against the mid on each leg, so a round trip crosses it **once**, not twice: \$3.00 of spread plus \$1.50 of commission = **\$4.50 all-in from the mid**.

I measured this independently before finding it had already been caught: across the 288 fades in this section the mean gap between a mid-to-mid mark and the spread-honest race is **\$0.38 a trade**, nothing like \$6.00. The 08-15 audit had reached the same conclusion the same day and shipped the correction into `shadow_mgc.py` (`MGC_FEE_RT` \$7.50 $\rightarrow$ \$4.50). Worth knowing why it matters: that error is what made the coil bouncer read $-\$454$ and therefore dead; at the true cost it is $+\$8.42$, which is breakeven rather than a grave.

So the honest cost lever is not a fixed fee — the racer fills at real bid and ask already. It is **slippage**, and every candidate below is swept against it in ticks (MGC tick = 0.10pt = \$1.00).

1.4 The harness reproduces last week exactly, before it is allowed to disagree with it

No comparison means anything if the machinery moved. Re-running the identical spec on the identical 22 days:

ARM	LAST WEEK REPORTED	THIS HARNESS	
fade_primary (obstacle == 0)	n=219, +\$2,730	n=219, +\$2,730	✓ exact
fade_double (obstacle == 0 & support == 0)	n=157, +\$2,648	n=157, +\$2,648	✓ exact
mgc_wall_break_go	n=63, +\$746	n=116, +\$814	✗ see below

1.5 ★ The momentum arm's shipped line is not the momentum arm that was measured

That third row is a real defect and it is worth naming precisely. Last week's number (n=63) came from a cut written ratio > 1, where ratio = obstacle / support if support > 0 else nan — so it **silently required support > 0**. Last week's shadow-ready line said require_obstacle_gt_support: True, which is obstacle > support and includes every row where nothing at all was resting behind the level. Those are different populations:

DEFINITION	N (22D)	NET	\$/TRADE	N (FORWARD 7D)	NET	\$/TRADE
ratio > 1 — the number that was reported	63	+\$746	+\$11.84	38	+\$320	+\$8.43
obstacle > support — the line that was shipped	116	+\$814	+\$7.02	73	+\$693	+\$9.50

Shipped as written it would trade **1.8× as often** as the cell that was measured, at 59% of the per-trade edge. Both are positive and both forward-walked positive, so nothing was lost — but a spec that does not express the thing it was measured on is exactly how a shadow book records the wrong experiment. Both are carried separately below.

2. ★★★ THE LIVE SHADOW — last week's recommendation, and what it actually did

Last week's disposition table ended with **MGC shadow plumbing — BUILD REQUIRED**, four blockers listed. **All four were built.** There is now a `level_break` gate branch, a `chand_arm_k` field (added specifically because last week's \$10.2 asked for it), a `DepthFeed` that reads gold's L2, and a completely separate service and store so that gold can never be repriced at MNQ's \$2 a point. `gazbot7-shadow-mgc.service` has been active since **2026-08-18 17:49 UTC**.

△ Its store is **`data/shadow_mgc.db`**, deliberately not `shadow.db`. I looked in the wrong database first and it reported a confident zero. Worth writing down, because "the gold shadow has recorded nothing" is exactly what a broken service also looks like.

Seven days of live, observe-only gold, scored on tick-repriced `real_pnl`:

ARM	N	DAYS	NET (REAL_PNL) \$/TRADE	WIN%
<code>mgc_break_fade_nobo</code> ok — the control , no book filter	66	7	-\$940.00	43.9%
<code>mgc_holebreak_fade_</code> short	7	5	-\$153.50	42.9%
<code>mgc_holebreak_fade_</code> long	7	4	-\$102.50	42.9%
all three	80	7	-\$1,196.00	-\$14.95

Every arm red, and the two book-filtered arms are worse per trade than the unfiltered control they exist to beat. That is the one comparison last week said would decide the thesis: "if the two hole arms do not beat this one, the book is decorative and the whole thesis is wrong." On seven days they do not.

Two honest qualifications before that lands too hard. **n=7 per side is nothing** — the two hole arms are 14 trades between them and I would not conclude anything from 14 trades in either direction. And the exit reasons are 42 chandelier / 38 stop with all 80 marked filled, so the machinery is working correctly; this is not a plumbing failure recording garbage.

One thing in the ledger is worth seeing, because it explains why `ceiling_pnl` must never be scored. Two of the fourteen:

ENTRY	SIDE	SIM ENTRY → EXIT	SIM'S OWN NUMBER	TICK-REPRICED REAL_PNL
08-24 17:20	LONG		+\$413.50	-\$1.50

		4656.50 → 4698.00 (+41.5pt)		
08-24 15:45	LONG	4675.60 → 4662.80 (−12.8pt)	−\$129.50	+\$9.50

The sim believed it held a 41-point winner overnight. Raced honestly on the real tape, the 2-ATR chandelier gave it all back and it came out flat. The second row is the same thing inverted. The repricer is right and the sim's own marks are fantasy — which is the desk's standing rule working as intended, and a reminder that a gold "winner" measured off bar closes can be worth nothing.

3. ★★★ WHY THE LAB AND THE LIVE DESK DISAGREE — and it is one thing

My lab replay of those same seven days says `fade_primary` made **+\$512** and `fade_double` made **+\$656**. The live desk says both lost. Only one of them can be describing the gate we own.

This was already flagged as an **open** item on 08-15 (`MGC_SHADOW_SCOPE.md` §8: "the lab and production do not read the same tape"), measured then at 41% fire overlap on the older window, and explicitly left unresolved: "the shadow's job is to produce the trade-bar number honestly over weeks. Until it has, treat the gold cells as unquantified in the direction that matters."

It has now produced that number, and I can also produce it in the lab. Restricting to 2026-08-17 ... 08-24, the six forward days both tapes cover, and changing **only** the series the sixty-minute level is measured on — same break rule, same book gate, same 250ms bid/ask race, same 3-ATR stop and 2/2 chandelier, same costs:

ARM	DEPTH-MID BARS (THE LAB)		TRADE BARS (PRODUCTION)	
	n / net	\$/trade	n / net	\$/trade
nobook control	129 / −\$317		−\$2.46 124 / −\$904	−\$7.29
hole PRIMARY	56 / +\$172		+\$3.06 53 / −\$475	−\$8.96
hole DOUBLE	35 / +\$605		+\$17.28 28 / −\$183	−\$6.55

★★ REV3 — IS THE “HOLE” A HOLE, OR IS IT AN ABSENT FEED? THE STONE THIS DOSSIER RAISES AND DOES NOT TURN

The cell selects for an empty book, and an empty book is what a quiet tape and a dropped feed look like. If the money is only there when nothing is trading, the bar-source fix cannot save it — a gate that needs a dead tape cannot be filled on a live one. So: the PRODUCTION fire set, the hole cell exactly as shipped, over the whole L2 era (2026-07-16 → 08-26, n=250), split by how busy the tape actually was — MGC trade prints per minute over the 15 minutes before each fire, counted off the lake's own tick table.

TAPE BUSYNESS AT THE FIRE	FIRES	NET	\$ / TRADE	WIN %	MEDIAN PRINTS/MIN
Q1 — quietest	63	-\$673.50	-\$10.69	47.6%	9
Q2	62	-\$90.00	-\$1.45	58.1%	15
Q3	62	-\$662.00	-\$10.68	48.4%	24
Q4 — busiest	63	+\$1,827.50	+\$29.01	69.8%	65
All 250 fires	250	+\$402.00	+\$1.61	56.0%	18

The answer is the reassuring one, and it is emphatic: the hole pays when the tape is **BUSY** and loses when it is **quiet**. Every dollar of the cell's positive expectancy is in the top quartile of trade activity — a median of 65 prints a minute, four times the overall median — and the other three quartiles are all negative. So the signal is a genuine liquidity hole inside an active tape, not an absent feed.

And it explains the bar-source finding from a second angle. Run the same split on the LAB (depth-mid) fire set and the quietest quartile has a median of **ZERO trade prints per minute** — 229 of 461 lab fires occur with essentially nothing trading, which is exactly what a mid-price series can do and a trade series cannot. The lab tape can fire when there is no tape. That is a second, independent reason to distrust the depth-mid numbers, and it points the same way as §3.

★ The caveat, and it is the one this dossier itself keeps catching. A quartile cut chosen after seeing the result is a sample percentile — the same error the regime-routing rule died of in §10. It is not a rule and it is not a promotion. It survives what can be thrown at it (n=63 over 16 days, strip-best-1 +\$1,574, strip-best-3 +\$1,080, strip-best-5 +\$655, worst leave-one-day-out +\$1,316, 12 of 16 days green), which is more than most cells here manage — but it must ship as a SHADOW arm with the threshold pinned in

ABSOLUTE terms (> **33 trade prints per minute over the prior 15 minutes**), never as a percentile, and it must be graded forward before it is believed. Filed as an addition to the gold shadow set, behind the bar-source fix.

And the fires themselves are only two-thirds the same event: **129 depth-mid fires, 124 trade-bar fires, 86 in both** — 67% of the lab's exist in production, 69% of production's exist in the lab.

That reconciles everything. The trade-bar column has the same sign and roughly the same magnitude as what the live shadow actually recorded (control −\$7.29 in replay vs −\$14.24 live; hole arms −\$8.96 and −\$6.55 vs −\$14.64 and −\$21.93 — live worse, but n=7 a side). **There is no mystery and no bug. The lab and the desk are running different gates, and the desk is running the worse one.**

Why the mid is the better input, in plain English. A "sixty-minute high" taken from trades is the highest price at which somebody happened to transact; taken from the depth mid it is the highest price the market was showing. Gold's book updates four times a second whether or not anyone trades, so its extremes are a cleaner statement of where the level actually was. The gate is a statement about a **level**, and the mid knows where the level is.

△ **I have to correct my own first draft on the mechanism here.** I initially wrote that the trade tape is sparser — fewer minutes, so staler extremes. On the sixteen days both of my tapes cover, that is simply false: **21,699 depth-mid minutes against 21,600 trade-bar minutes, a dead heat.** The two tapes do not differ in how many minutes they have; they differ in **where the high and the low of each minute sit**, and that is enough to move a third of the fires. Note also that my trade-bar proxy is the 5-second lake bars resampled, which is close to but not identical to what md actually folds — the 08-15 audit measured a much larger minute gap (29,565 vs 13,440) against production's real bar stream. So the direction of this finding is solid and triple-confirmed, but the exact mechanism by which the two tapes diverge is not fully pinned down, and I am not going to pretend otherwise.

△ **And the caveat that stops me claiming more than that.** The lab's forward result leans hard on one day. `fade_primary` made +\$512 across the seven forward days, of which **+\$341 is 2026-08-25 alone** — and 08-25 has no trade-bar file, so it is the one day this comparison cannot check. Strip it and the lab's six

checkable days are +\$171 against production's −\$475. The direction of the finding is unchanged, but "the mid tape wins" currently rests on six days and a fire-overlap argument, not on a fortress.

The recommendation, and it is a build not a study: point the MGC shadow's bar builder at the depth mid, keep the existing trade-bar arms running alongside as the control, and let the two accumulate side by side. The service already opens `DepthFeed("MGC")` and `book_at()` already returns `bid1p/ask1p`, so the mid is sitting in a variable the process already holds. **That is the single highest-value change on the gold line and nothing else here should be acted on before it.**

4. THE FORWARD WALK — last week's gate on seven days it had never seen

This is the lab tape (depth mid), which §3 has just shown is not what production reads. Read it as "is the research result real on its own terms", not as an expectation.

Nothing was refitted. The spec is copied from last week's §6.1. **The regime labels are frozen too** — the ATR and efficiency percentile cuts are computed on the 22 in-sample days only and then applied to the new week, so the forward days cannot help define their own regime boundaries.

ARM	N	DAYS	NET	\$/TRADE	WIN%	LONG \$/TR	SHORT \$/TR
fade_primary (obstacle == 0)	69	7	+\$512	+\$7.43	63.8%	+\$15.73	+\$0.65
fade_double (obst == 0 & sup == 0)	43	7	+\$656	+\$15.26	65.1%	+\$25.23	+\$4.81
wall_break_go (obstacle > support)	73	7	+\$693	+\$9.50	60.3%	+\$9.44	+\$9.56

Per day, fade_primary: 08-17 +\$169 · 08-18 +\$40 · 08-19 −\$78 · 08-20 +\$81 · **08-21 −\$319** · 08-24 +\$279 · 08-25 +\$341.

The controls on the same forward stamps — this is what stops it being the exit or the drift:

	FADE_PRIMARY	FADE_DOUBLE
the gate	+\$512 (+\$7.43)	+\$656 (+\$15.26)
the MIRROR (follow instead)	−\$1,183 (−\$17.15)	−\$848 (−\$19.72)
always LONG at the same stamps	−\$236 (−\$3.42)	+\$211 (+\$4.92)
always SHORT at the same stamps	−\$435 (−\$6.31)	−\$403 (−\$9.38)
random-minute placebo, 30 draws	beaten 0/30 , mean −\$1,173	beaten 0/30 , mean −\$409

wall_break_go fails the same placebo **23/30**, exactly as it did last week.

4.1 ★ The refinement held better than the arm it was hedging

Last week's §9.3 named the double-hole cut as "the most likely thing here to shrink", and the simpler obstacle == 0 arm was shipped as primary specifically to hedge that. It went the other way: forward, the refinement made **+\$15.26** a trade against the primary's **+\$7.43**, and it is the only arm with every week green. I am not going to over-read one week of it, but the hedge was pointed at the wrong risk.

4.2 ★ The short side is where the decay is

IN-SAMPLE 22D			FORWARD 7D		
fade_primary	n	\$/trade	n	\$/trade	
LONG	103		31		+\$15.73
SHORT	116		38		+\$0.65

The short side went from earning to earning nothing. Before calling that a mechanism failure: it is two days. Forward shorts by day are 08-24 −\$316, 08-21 −\$229, 08-19 −\$73, then +\$65, +\$128, +\$175, +\$274. **Drop the single worst day and the short side is +\$341 over 33 trades.** So it is two bad sessions inside a seven-day sample, not a side that stopped working — but with the long side intact and the short side flat, it is the thing to watch and the reason the two sides stay as separate shadow arms.

4.3 The 29-day ledger, both blocks together

ARM	N	DAYS	NET	\$/TRADE	STRIP BEST 3 / 5 / 10	DROP BEST DAY	DAYS GREEN	WEEKS GREEN
fade_primary	288	29	+\$3,243	+\$11.26	+\$2,216 / + \$1,616 / + \$336	+\$2,858	72.4%	6/7
fade_double	200	29	+\$3,304	+\$16.52	+\$2,281 / + \$1,689 / + \$704	+\$2,848	75.9%	7/7
wall_break_g o	189	23	+\$1,508	+\$7.98	+\$786 / + \$472 / − \$192	+\$1,126	60.9%	4/7

Leave-one-day-out is green on every one of the 29 days for all three. **ISO week 34 is the first red week fade_primary has ever had** (−\$107) — last week's "five weeks, five green" is now six of seven. fade_double is seven of seven.

Slippage, in extra adverse ticks charged to each leg:

	+1 TICK	+2	+3	+4
fade_primary, all 29 days	+\$2,667	+\$2,091	+\$1,515	+\$939
fade_double, all 29 days	+\$2,904	+\$2,504	+\$2,104	+\$1,704
fade_primary, forward only	+\$374	+\$236	+\$98	−\$40
wall_break_go, all 29 days	+\$1,130	+\$752	+\$374	−\$4

The reversion arms carry three to four ticks. The momentum arm breaks even at four. Since the gate deliberately selects for moments when nobody is resting within a point of the level, this is not an academic column — it is the single most likely way it degrades, and it was last week's \$9.8 too.

5. ★★★ THIRTEEN MONTHS OF GOLD — the caveat that could not be answered, answered

320 trading days, 2025-07-28 → 2026-08-19, one 14-day gap in January. **What this can and cannot say, stated before the numbers:** there is no L2 before 2026-07-16, so the liquidity-hole filter — which is where most of the edge lives — **cannot be applied**. This tests the bare trigger and its direction. A pass promotes nothing; a fail would retire the family.

ON 320 DAYS, 60-MIN BREAK, 45-MIN COOLDOWN, 3-ATR STOP + 2/2 CHANDELIER							
	N	NET	\$/TRADE	LONG	SHORT	DRIFT-NEUTRAL	
FADE the break	4,610	+\$5,155	+\$1.12	+\$3.98	−\$1.58	+\$1.20	
FOLLOW the break (the mirror)	4,610	−\$6,425	−\$1.39	−\$4.40	+\$1.79	−\$1.31	
always LONG at the same stamps	4,610	−\$1,513	−\$0.33				
always SHORT at the same stamps	4,610	+\$243	+\$0.05				

The direction call survives thirteen months. Fade beats follow by \$2.51 a trade across 4,610 trades, and beats both constants. That is the mechanism holding up on a sample nineteen times the one it was found on, and it is the strongest thing in this section.

And then the robustness battery takes it apart:

TEST	RESULT
green months	5 of 14
best / worst month	2025-10 +\$3,636 / 2025-12 -\$2,532
strip the best 10 trades (of 4,610)	-\$4,392
strip the best 1%	-\$21,580
first half / second half	-\$5,847 / +\$11,002
days green	46.4%
survives extra slippage of	0.05pt/leg (+\$914); dead at 0.10pt (−\$3,250)

Read plainly: **the bare trigger is not a standalone edge.** Its entire thirteen-month profit is a few dozen trades, the first half loses, nine months of fourteen lose, and it dies on a tick of extra slippage. It is a real but very thin directional tilt that needs a filter to be worth anything — and the filter it needs is precisely the one that cannot be tested back here.

This is the right cold light on last week's headline. fade_primary earned +\$12.47 a trade on 22 days; the same trigger unfiltered earns **+\$1.12** across 320. The book filter is doing enormous work, or 22 days was a favourable stretch, and this test cannot separate those two.

5.1 ★ The loose thread from last week, closed — and it closed the good way

Last week's \$6.5 flagged one unfinished item: the trigger's lookback × margin grid was **0 of 15 positive**, but it had been run on the tp1.0/sl1.0 scalp — the exit the section itself had shown cannot pay gold's spread — and "until I re-run it on the chandelier, 'the trigger works at other lookbacks' is an untested claim." Re-run on the chandelier, on 320 days:

LOOKBACK	MARGIN 0.00 ATR	MARGIN 0.10	MARGIN 0.25
30 min	+\$0.15	−\$0.07	−\$0.09
45 min	+\$0.39	+\$0.27	−\$0.19
60 min	+\$1.51	+\$1.12	+\$1.95
90 min	+\$3.35	+\$2.60	+\$4.75
120 min	+\$4.16	+\$4.20	+\$5.36

12 of 15 positive, against 0 of 15 on the scalp — the trigger is not fragile across lookbacks, it was the exit that was killing it, and last week's own diagnosis was right. It also says something new: the gradient is **monotone in lookback**, and 60 minutes is nowhere near the top.

5.2 ...but the lookback gradient does not survive the book gate

Tested on the 29-day sample where the liquidity filter can be applied:

LOOKBACK	IN-SAMPLE 22D		FORWARD 7D		ALL 29D	
	n	\$/trade	n	\$/trade	n	\$/trade
30 min	280	+\$2.58	101	−\$9.19	381	−\$0.54
45 min	252	+\$7.97	83	+\$1.65	335	+\$6.40
60 min	219	+\$12.47	69	+\$7.43	288	+\$11.26
90 min	181	+\$10.75	57	+\$5.38	238	+\$9.46
120 min	164	+\$3.92	47	+\$10.34	211	+\$5.35
180 min	138	+\$10.34	35	+\$14.44	173	+\$11.17

The year's monotone climb does **not** reproduce once the book filter is on: 60 minutes is at or near the top in-sample, and everything from 45 to 180 minutes is positive in both blocks while 30 minutes fails forward outright. So **60 minutes sits inside a broad plateau rather than on a peak**, which is what you want, and there is no case for changing it. The year's gradient is a statement about the unfiltered trigger, and the book gate evidently substitutes for the longer lookback.

5.3 ★★ The year kills last week's router rule

Last week's §6.3 said: ARM on CLEAN_TREND or DEAD_CHOP, **BENCH on NORMAL_CHOP** — it was the only losing bucket in-sample, at −\$9.93 a trade on the short side, and 75 of 157 trades.

NORMAL_CHOP, THE BUCKET LAST WEEK BENCHED	VERDICT
in-sample, 22 days	SHORT −\$9.93/tr — the only losing bucket
forward 7 days, fade_primary	LONG +\$9.39 , SHORT +\$4.13
forward 7 days, fade_double	LONG +\$16.58 , SHORT +\$20.83
the year, 320 days	LONG +\$7,276 (+\$6.72/tr) — the single biggest earner of any bucket

And CLEAN_TREND, the bucket last week armed on, is **negative** on the year (LONG −\$1.92/tr). **Two independent samples contradict the rule, one of them nineteen times larger than the sample it was fitted on.** Tested as an actual policy:

POLICY		IN-SAMPLE 22D		FORWARD 7D	
fade_primary	n	\$/trade	n	\$/trade	
blanket, no router at all	219	+\$12.47	69	+\$7.43	
last week's rule (ARM trend/dead only)	76	+\$35.49	28	+\$9.89	
bench NORMAL_CHOP only	115	+\$24.54	38	+\$8.31	
fade_double					
blanket, no router at all	157	+\$16.87	43	+\$15.26	
last week's rule	57	+\$44.65	15	+\$12.53 — worse than blanket	

Last week's rule looks spectacular in-sample (+\$35 and +\$44 a trade) and buys almost nothing forward — on fade_double it is actively worse than not routing at all, while discarding two thirds of the trades. **That +\$44.65 was the sample memorising itself.** The regime cut points were whole-sample percentiles, which last week's \$9.2 honestly flagged as a mild in-sample peek; frozen properly and walked forward, the peek was most of the number.

Recommendation: run the gold fade UNROUTED for now. That is an unusual thing for this desk to say, and I want to be clear it is not "regimes don't matter" — it is that the specific ARM/BENCH rule we wrote does not replicate, and shadowing it would bench the gate in the regime that is currently paying best.

5.4 The session read — the one conditioning that triangulates

	ASIA	LONDON	US
the year, 320 days, unfiltered trigger	−\$2.74/tr		+\$1.60
			+\$5.70
29 days, fade_primary, book-gated, LONG		+\$10.76	+\$10.28
			+\$20.14
29 days, fade_primary, book-gated, SHORT		+\$9.92	+\$2.35
			+\$16.01
forward 7d, fade_primary, US only			+\$13.45 (n=20)
forward 7d, fade_double, US only			+\$31.26 (n=12)

The US session is the best bucket in all three samples, which is the only conditioning statement in this section that survives every tape I own — and it happens to be the session you actually watch.

But note the interesting part: on the year's unfiltered trigger Asia **loses**, while on the book-gated 29 days Asia is solidly positive on both sides. **The liquidity filter does its best work exactly where the raw trigger is worst** — which is what you would expect if the mechanism really is about empty books, because Asia is when gold's book is thinnest. That is the most encouraging mechanistic evidence in the section and it is not something I went looking for.

So: no Asia bench. A US-only arm is worth shadowing as a second arm because it is the strongest and best-triangulated cut, but it costs two thirds of the sample and the full-clock version is positive.

6. ★★ THE UNTURNED STONE — gold against the Nasdaq

Last week's §13 named exactly one thing never tried: "the relationship between MGC and MNQ on the same clock ... a break in gold that coincides with a matching break in the Nasdaq is a different event from one that does not, and nothing in this section can tell them apart."

We hold 278 paired trading days of both symbols' minute bars. Everything is strictly causal — MNQ's state is read at the gold signal bar, and MNQ's own 60-minute extreme is shifted so the signal bar is excluded. Every cut is placebo-controlled by **keeping the same number of trades at random, 200 times**, because a filter that keeps 10% of trades looks brilliant whenever the 90% it dropped lost.

6.1 It works, and it is a veto rather than a confirm

FADE THE GOLD BREAK, 278 PAIRED DAYS	N	NET	\$/TRADE	FILTER PLACEBO
blanket (control)	3,800	+\$6,522	+\$1.72	—
MNQ is CONFIRMING the break (breaking the same way)	389	−\$4,093	−\$10.52	beaten 195/200 — 2.5th pctile
MNQ's 15-min drift AGREES with the break	2,417	−\$543	−\$0.22	beaten 188/200 — 6.0th pctile
MNQ is not breaking at all	3,324	+\$8,926	+\$2.69	beaten 18/200 — 91.0th pctile
MNQ's drift OPPOSES the break	1,345	+\$7,299	+\$5.43	beaten 6/200 — 97.0th pctile

This is the strongest form the test can take: **both sides move, in opposite directions, well past random**. The bad cut is significantly worse than a random keep and the good cut significantly better. Fading a gold break the Nasdaq agrees with turns a +\$1.72 blanket into −\$10.52; fading one the Nasdaq is leaning against turns it into +\$5.43, more than tripling the edge on 35% of the trades.

The mechanism is plausible and was predicted before pricing: a gold break that coincides with a Nasdaq break is a macro risk event with real flow behind it,

so it keeps going. A gold break with a quiet Nasdaq is gold's own idiosyncratic noise, and noise snaps back.

6.2 ...and it does not show up on the 29 days where the book gate lives

FADE_PRIMARY, 29 DAYS	N	NET	\$/TRADE	FORWARD 7D
no veto	288	+\$3,243	+\$11.26	+\$512 (+\$7.43)
veto when MNQ confirms	244	+\$2,566	+\$10.52	+\$445 (+\$7.29)
veto when MNQ's drift agrees	95	+\$851	+\$8.96	+\$336 (+\$12.46)
filter placebo of the confirm-veto				beaten 132/200 — 34th pctl

On 288 trades the veto is indistinguishable from random and reduces the total. On 3,800 it is a strong two-sided result. The year sample is thirteen times larger and the book sample has 44 trades in the vetoed bucket, so I lean towards the year — but I am not going to fold an unconfirmed cut into the primary arm on that reasoning.

Disposition: a SEPARATE shadow arm. `mgc_holebreak_fade_{long,short}_xveto` alongside the existing pair, identical in every other respect. That is the only way to attribute it, it costs nothing, and in a few weeks the book will simply say which sample was right.

6.3 The cross-asset momentum attack — a null, and a clean one

Rule 2 says hunt momentum deliberately. Every previous gold momentum gate has been built on gold's own tape, and that family is refuted six ways. A different instrument as the confirm is genuinely new. It does not work:

FOLLOW THE GOLD BREAK	N	NET	\$/TRADE	PLACEBO
blanket (control), 29 days	600	-\$3,213	-\$5.36	—
when MNQ confirms the break, 29 days	76	-\$695	-\$9.15	30th pctl
when MNQ's drift agrees, 29 days	381	-\$3,572	-\$9.37	1.0 pctl
when MNQ confirms, the year	389	-\$890	-\$2.29	36th pctl
when MNQ confirms AND the book shows a wall	19	+\$334	+\$17.58	92nd pctl

Following a gold break because the Nasdaq agrees makes things **worse**, not better, on both tapes — the drift-agrees cut sits at the 1st percentile of its own

placebo, which is a real anti-signal. The last row is the only positive and it is **19 trades**; at that n the control is uninformative and so is the cell. Noted, not claimed.

7. THE VOLATILITY CELL — right verdict last week, wrong reason, and a look-ahead of my own

Rule 2 asks explicitly: if direction is unpredictable but size is predictable, does a breakout-either-way beat a directional gate? Last week refuted it on 22 days — the tightest 20-minute range quintile went on to move less than average. That deserves the 320-day tape as much as any survivor does, and it turns out the reasoning was wrong even though the verdict was right.

The two measures disagree completely, on identical rows, 296,475 minutes:

SORTED ON THE RAW 20-MIN RANGE (LAST WEEK'S MEASURE)					
N	NEXT-60MIN MOVE, \$	VS ALL	IN ATR UNITS	VS ALL	
Q1 tightest	59,295	\$49.05	−\$50.92	10.60	+4.75
Q5 widest	59,295	\$186.33	+\$86.36	3.83	−2.02

SORTED ON 20-MIN RANGE					
÷ ATR (RELATIVE QUIET)	N	NEXT-60MIN MOVE, \$	VS ALL	IN ATR UNITS	VS ALL
Q1 tightest	59,295	\$110.52	+\$10.56	4.62	−1.23
Q5 widest	59,295	\$74.15	−\$25.81	10.48	+4.63

Rank correlation with the next 60 minutes' size: raw range **+0.371**, range÷ATR **−0.134**.

Why they disagree. Rank corr(ATR, next-60-min size) is **+0.412** — big begets big. So sorting on the raw range mostly sorts on the volatility level, and last week's table was measuring volatility clustering, not compression. Dividing by ATR asks the straddle's actual question — is this quiet relative to its own recent behaviour — and the sign flips: relatively quiet gold does go on to move **more** in dollars. **Last week's stated reason for the refutation was an artefact.**

So I traded it, because a gradient in dollars can still be worth nothing once the stop scales to the same small ATR. Either-way entry: stop orders at $\pm k \cdot \text{ATR}$ around a compressed minute, 60-minute arming window, take whichever fills, no direction called anywhere.

COMPRESSION CUT	TRIGGER	STOP	CHANDELIER N	NET	\$/TRADE	
tightest 20%	1.0 ATR		3.0 2.0 / 2.0	3,396	−\$13,340	−\$3.93
tightest 33%	1.0 ATR		3.0 2.0 / 2.0	3,945	−\$1,248	−\$0.32
tightest 50%	1.0 ATR		3.0 2.0 / 2.0	4,371	−\$10,776	−\$2.47
tightest 33%	0.75 ATR		2.0 1.5 / 1.5	3,945	−\$13,162	−\$3.34
no compression filter (control)	1.0 ATR		3.0 2.0 / 2.0	5,025	−\$14,001	−\$2.79

All eight compression cells negative; the control negative too. The best cell is −\$0.32 a trade, 4 green months of 13, strip-best-3 takes it to −\$8,591, drop-best-day −\$6,258, and one tick of extra slippage per leg takes it to −\$2.32. **REFUTED again — this time on 320 days and by an actual trade rather than a gradient, and for the correct reason:** compressed gold does expand more in dollars, but less in ATR units, and once the stop is scaled to the same ATR there is nothing left.

7.1 **The look-ahead I put in it, and how it read before I found it**

The first version of that harness reported the control at **+\$71,608, +\$14.25 a trade, 62% win.** That number is a bug and it is worth showing you exactly which one, because it is a class this desk keeps meeting.

The scan found the bar whose high or low breached the trigger level, then entered at **that same bar's open**. But a bar's extremes are only known once it has closed — so it was buying at a price it had only chosen because it had already seen the rest of that bar. A guaranteed favourable fill on every single trade. Entering at the open of the next bar, which is the first price genuinely available after the signal, turned **+\$71,608 into −\$12,462.**

It is the same family as the resample left-edge trap that cost the coil gate +\$906 → +\$470 on 08-14, and it produced by far the biggest number in this report. Anything in a gold study that looks like \$14 a trade over five thousand trades is a bug until proven otherwise.

8. RULE 3 — THE FULL EXIT MATRIX, on both surviving entries

The entry and the exit are separate questions. Both grids are on the 29-day book-gated sample.

REVERSION — mgc_hole_break_fade primary, n=288:

FAMILY	CONFIG	NET	\$/TRADE	LONG	SHORT	DRIFT-NEUTRAL
(a) tight scalp	tp 0.5 / sl 0.5	−\$1,025	−\$3.56	−\$3.97	−\$3.21	−\$3.59
(a) tight scalp	tp 1.0 / sl 1.0	−\$269	−\$0.93	−\$0.18	−\$1.59	−\$0.89
(a) tight scalp	tp 1.5 / sl 2.0	+\$429	+\$1.49	+\$1.50	+\$1.48	+\$1.49
(b) chandelier	stop 2.0 / arm 1.5 / trail 1.5	+\$2,115	+\$7.34	+\$8.60	+\$6.24	+\$7.42
(b) chandelier	stop 2.5 / arm 2 / trail 2	+\$3,190	+\$11.08	+\$9.05	+\$12.84	+\$10.95
(b) chandelier	stop 3.0 / arm 2 / trail 2 — shipped	+\$3,243	+\$11.26	+\$13.83	+\$9.03	+\$11.43
(b) chandelier	stop 4.0 / arm 2.5 / trail 1.5	+\$3,691	+\$12.82	+\$18.05	+\$8.26	+\$13.16
(b) chandelier	stop 5.0 / arm 3 / trail 3	+\$4,379	+\$15.21	+\$27.86	+\$4.20	+\$16.03
(c) dual slot	Lot A 1R + Lot B chandelier	+\$2,974	\$10.33/signal on TWO lots		vs \$11.26 on one	
(d) time cap	stop 3.0, flat at 480min	+\$5,090	+\$17.67	+\$40.89	−\$2.53	+\$19.18

MOMENTUM — mgc_wall_break_go, n=189:

FAMILY	CONFIG	NET	\$/TRADE	LONG	SHORT
(a) tight scalp	tp 1.0 / sl 1.0	−\$769	−\$4.07	−\$4.16	−\$3.96
(a) tight scalp	tp 2.0 / sl 2.0	+\$204	+\$1.08	+\$0.79	+\$1.42
(b) chandelier	stop 3.0 / arm 2 / trail 2	+\$1,508	+\$7.98	+\$8.30	+\$7.61
(b) chandelier	stop 5.0 / arm 3 / trail 3	+\$2,140	+\$11.32	+\$22.27	−\$1.24
(c) dual slot	Lot A + Lot B	+\$739	\$3.91/signal on TWO lots		vs \$7.98 on one
(d) time cap	stop 3.0, flat at 480min	+\$5,779	+\$30.58	+\$78.93	−\$24.92

Every one of last week's exit conclusions replicates on the larger sample, and I checked rather than assumed:

- **The tight scalp is dead on gold.** Negative at 1R on both entries. Gold's snap-back is a ten-to-twenty-minute grind, and a 1-ATR target clips it before it has paid for the spread.
- **The dual slot — the live MNQ shape — is worse per lot than running Lot B alone,** on both entries (\$10.33 vs \$11.26; \$3.91 vs \$7.98). Lot A is negative on gold and cancels Lot B. This remains the most actionable

structural finding: **gold wants one lot and a wide chandelier, which is the opposite of the live MNQ slot configuration.**

- **The two biggest numbers in both grids are discarded on their side-split.** The 480-minute time cap makes +\$40.89 long and −\$2.53 short on the reversion entry, +\$78.93 and −\$24.92 on the momentum entry. On a tape that rose most of the sample, that is a long position with extra steps, not a gate. The same applies to the 5.0/3/3 chandelier (+\$27.86 / +\$4.20).

Shipped 3.0/2/2 is the most side-balanced cell in the grid, which is why it stays even though two cells beat it on the headline.

Which exit does each cell NEED? All four want **(b), a wide chandelier, on a single lot.** Not the scalp, not the dual slot, not a time cap.

9. RULE 4 — HOME REGIME AND THE ROUTER RULE, per cell

Given §5.3, the honest answer for the reversion cells is **no router rule yet**, and that is a finding rather than an evasion.

REVERSION-LONG / REVERSION-SHORT — `mgc_hole_break_fade`

HOME REGIME	none that replicates. Last week's ARM CLEAN_TREND DEAD_CHOP / BENCH
NORMAL_CHOP	is REFUTED by the forward walk AND by 320 days: NORMAL_CHOP is the biggest earner
ROUTER RULE	on the year (+\$6.72/tr) and CLEAN_TREND LONG is negative (-\$1.92/tr). ARM ALWAYS, all sessions, all regimes. The only conditioning that triangulates across all three tapes is that the US session is the strongest bucket - but ASIA and LONDON are both positive under the book gate, so there is nothing to bench. Shadow a US-only arm alongside, not instead.
EXPECTANCY	home regime == blanket: +\$11.26/tr (primary, 29d) and +\$16.52/tr (double, 29d)
△ MEASURABLE?	on the LAB tape. On PRODUCTION's tape both are NEGATIVE (section 3). The regime labels here are whole-sample percentile cuts. Live they must be computed on a trailing window. Since no regime rule survived anyway, this no longer bites.

MOMENTUM-LONG / MOMENTUM-SHORT — `mgc_wall_break_go` (29 days, chandelier 3/2/2)

REGIME

LONG N / \$PER

SHORT N / \$PER

CLEAN_TREND	21 / +\$33.98	6 / -\$12.60
DEAD_CHOP	26 / +\$13.52	15 / +\$3.95
NORMAL_CHOP	37 / -\$3.49	46 / +\$12.88
BUILDING	14 / -\$1.64	18 / -\$1.13

HOME REGIME LONG -> CLEAN_TREND and DEAD_CHOP (+\$33.98 and +\$13.52/tr)
 SHORT -> NORMAL_CHOP (+\$12.88/tr)

ROUTER RULE BENCH the BUILDING band on both sides (mid ATR, high ER) - a wash, -
\$1.64/- \$1.13.
 BENCH momentum-SHORT in CLEAN_TREND (n=6, -\$12.60 - thin, so a soft
bench).

△ MEASURABLE? Yes, all of it is in the router's existing ATR/ER vocabulary. BUT this
split has
 NOT been walked forward - it is a 29-day in-sample read, which is
exactly the shape
 of the reversion rule that just failed. Do not ship it; record it and
re-test.

SESSION ASIA +\$12.23/+\$14.54, LONDON +\$13.53/+\$4.20, US -\$0.67/+\$3.30. The
momentum arm is
 the mirror image of the reversion arm on the clock, and it is WEAKEST
in the session
 you watch. Worth knowing before anyone gets attached to it.

10. EVERYTHING THAT DIED THIS WEEK, BY NAME

The honest failures are the deliverable.

#	CANDIDATE	SHAPE	N	RESULT	CAUSE OF DEATH	DISPOSITION
1	last week's ROUTER RULE (ARM CLEAN_TREND DEAD_CHOP, BENCH NORMAL_CHOP)	filter	219 → 76	in-sample +\$35.49/tr → forward + \$9.89/tr; on fade_double worse than blanket (+\$12.53 vs +\$15.26)	Forward walk + 320-day tape, independently. NORMAL_CHOP is the year's biggest earner (+\$6.72/tr) and CLEAN_TREND LONG is negative. The in-sample number was the whole-sample percentile cuts memorising the sample.	REFUTED as a rule
2	mgc_straddle / breakout-either-way	VOL	3,945	best cell -\$1,248 (-\$0.32/tr); 4/13 green months; strip-best-3 - \$8,591	Traded, not just measured. All 8 compression cells negative, control negative, dies on 1 tick of slippage. Right verdict last week, wrong reason — raw range sorts on ATR,	REFUTED (upgraded: 320 days, and by a trade)

						not on compression.
	cross-asset momentum				year –\$2.29/tr, 29d –\$9.15/tr; drift-agrees at the 1.0 percentile of its own placebo	Following a gold break because the Nasdaq agrees makes it worse on both tapes. A genuinely new input, cleanly negative.
3	confirm (follow a gold break MNQ confirms)	MOM ×2	389 / 76			REFUTED
4	the lookback gradient (60min → 90/120min)	param	3,184–4,610		year says +\$1.12 → +\$4.20/tr going 60→120min	Does not reproduce under the book gate: 60min is at the in-sample top and 30min fails forward. The year's gradient describes the unfiltered trigger only.
						PARKED — revive only if the book gate is ever shown to substitute for lookback; 45–180min is a plateau, do not move 60
5	the bare 60-min break trigger as a standalone	REV ×2		4,610	+\$5,155 (+\$1.12/tr) over 320 days	5/14 green months, strip-best-10 = –\$4,392 , first half –\$5,847, dead at 0.10pt extra slippage. Beats its mirror and both constants, so the direction is real — the standalone P&L is not.
						PARKED — it is a filter's raw material, never a gate on its own
6	mgc_break_fade_n obook (the live control arm)	REV		66	–\$940 live (–\$14.24/tr) in 7 days	The live shadow, on production's trade-bar tape.
						PARKED pending the bar-source fix (\$3) — it is the control, it must keep running
7	cross-asset veto applied to the book window	filter		244	+\$2,566 vs +\$3,243 unvetoes; filter placebo beaten 132/200, 34th pctl	Indistinguishable from random on 29 days, while strongly two-sided on 278. Not refuted — unresolved , and the two samples disagree.
						SHADOW as a separate arm (\$6.2)
8	mgc_wall_break_g o random-minute placebo	MOM ×2		189	+\$1,508 (+\$7.98/tr) but beaten 29/30	Fails the same placebo it failed last week, now on 189 trades instead of 63. Strip-best-10 is –\$192. Its filter placebo passes, its random-minute placebo does not — the cut is a good selector inside a bad population.
						SHADOW, on probation (thin n is never a kill)
9	my own straddle harness	bug		5,026	read +\$71,608 (+\$14.25/tr)	Entered at the open of the bar whose high/low triggered it. Caught pre-publication; corrected to –\$12,462.
						FIXED — recorded so the trap is on the record
10	mgc_wall_break_g o's shipped spec	fidelity	63 vs 116		last week's number was ratio>1 (n=63); last week's	Different populations — 1.8× the trades at 59%
						FIXED in the blocks below —

ShadowVariant line	of the per-trade	
was obstacle >	edge. Both positive,	both arms carried
support (n=116)	both forward-	separately
	walked positive, so	
	nothing was lost.	

Not re-derived, deliberately: the six attacks closed on 2026-08-05 (momentum/direction, arm-then-confirm, router-filtered, L2 book direction, fader-on-extension, thrust-continuation), and last week's twelve. Two of last week's twelve now have new evidence and are updated above (#2, #5); the rest stand unchanged. The London-open break fade (n=16) remains PARKED, untouched, at n>40.

11. WHAT COULD STILL BE WRONG

1. **The whole reversion result may be a bar-construction artefact.** §3 shows the sign flips on the input tape. My reading is that the depth mid is genuinely better information about where a level is — but the alternative reading, that the mid tape's break and the mid-derived quote race share a construction and flatter each other, is not excluded by anything I ran. **A re-pointed shadow is the only clean way to settle it**, which is why §3's recommendation is a build and not a claim.
2. **The forward walk leans on one day.** +\$341 of fade_primary's +\$512 is 08-25, and 08-25 is the one day with no trade-bar file to cross-check.
3. **Fourteen live trades.** The two hole arms have 7 fires each. I would not conclude from that in either direction, and I have tried not to.
4. **The book is a 250ms sample, not an event stream.** capture.db.book is 41ms and event-driven but MNQ-only. The live gate reads a slightly different book than any backtest here, in the direction of missing brief refills.
5. **The fill model assumes the touch is where the last quote said it was** — in a genuine liquidity hole, which is precisely what this gate selects for, it may not be. This is why §4.3's slippage ladder matters more than any single headline, and why 3-4 ticks of headroom is the number to hold on to rather than +\$11.26.
6. **The year tape has no book, and the book is where the edge is.** §5 can validate the direction and nothing else. The two things I most want tested — the hole filter and the double-hole refinement — are exactly the two things 291 of those 320 days cannot speak to.

7. **The momentum cells still have no passing random-minute placebo,**
two weeks running, on n=63 and now n=189.

12. SHADOW-READY BLOCKS

The plumbing exists — `mgc_slate()`, the `level_break` gate, `chand_arm_k`, `DepthFeed`, a separate service and store. **The change needed is to the bar source, not to the slate.**

12.1 The build item, which gates everything else

```
★ POINT THE MGC SHADOW'S BAR BUILDER AT THE DEPTH MID.  
  shadow_mgc.py currently folds md's TRADE bars. Build the 1-minute OHLC from the  
  depth-quote  
  MID ((bidlp+asklp)/2) instead - the tape every number in the research was measured  
  on.  
  The service ALREADY holds DepthFeed("MGC") and book_at() already returns bidlp/  
  asklp, so the  
  input is in a variable the process holds. This is plumbing, not research.  
  △ KEEP the existing trade-bar arms running unchanged alongside, renamed  
  *_tradebar. Without  
  them the bar-source question can never be attributed, and it is the live  
  question.
```

12.2 The slate

```
# — MGC GOLD, 2026-08-25 revision. Runs in the MGC service only (cfg.symbol="MGC",  
#   cfg.value_per_point=10.0, its own store). Never in the MNQ slate:  
reprice_pending()  
#   takes ONE value_per_point and would price gold at $2/pt with no error.  
MGC_EXIT = dict(stop_atr_mult=3.0, chandelier=True, chand_lock=True,  
                chand_start_k=2.0, chand_arm_k=2.0, lock_r=99.0, lock_k=2.0,  
                time_cap_s=480 * 60)      # unchanged - the exit grid re-confirms it  
on 29 days  
  
# — CELLS 1+2, REVERSION. Unchanged rules; the CHANGE is which bars feed them  
(12.1).  
ShadowVariant("mgc_holebreak_fade_long", "level_break", symbol="MGC", side="LONG",  
              params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,  
                    "fade": True, "book_band_pt": 1.0, "obstacle_max": 0},  
**MGC_EXIT),  
ShadowVariant("mgc_holebreak_fade_short", "level_break", symbol="MGC", side="SHORT",  
              params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,  
                    "fade": True, "book_band_pt": 1.0, "obstacle_max": 0},  
**MGC_EXIT),  
  
# — The DOUBLE-HOLE refinement. PROMOTED to a full arm: it was shipped as the  
hedged-against  
#   case and it forward-walked BETTER than the primary (+$15.26/tr vs +$7.43), 7/7
```

```

weeks green.
ShadowVariant("mgc_holebreak_fade_long_dbl", "level_break", symbol="MGC",
side="LONG",
    params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,
"fade": True,
    "book_band_pt": 1.0, "obstacle_max": 0, "support_max": 0},
**MGC_EXIT),
ShadowVariant("mgc_holebreak_fade_short_dbl", "level_break", symbol="MGC",
side="SHORT",
    params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,
"fade": True,
    "book_band_pt": 1.0, "obstacle_max": 0, "support_max": 0},
**MGC_EXIT),

# — ★NEW: the CROSS-ASSET VETO arm (section 6). Do not fold this into the primary -
the year
#    tape (n=3,800) and the book window (n=288) disagree, and a separate arm is the
only way to
#    find out which was right. REQUIRES A NEW BRANCH: ShadowSim._entry cannot
express it, because
#    the sim sees ONE symbol's bars and this needs MNQ's 60-min extreme on the same
clock.
#    Cheapest honest route: a small MNQ state feed alongside DepthFeed, read at the
signal bar.
ShadowVariant("mgc_holebreak_fade_long_xveto", "level_break", symbol="MGC",
side="LONG",
    params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,
"fade": True,
    "book_band_pt": 1.0, "obstacle_max": 0,
    "veto_ref_symbol": "MNQ", "veto_if_ref_confirms": True},
**MGC_EXIT),
ShadowVariant("mgc_holebreak_fade_short_xveto", "level_break", symbol="MGC",
side="SHORT",
    params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,
"fade": True,
    "book_band_pt": 1.0, "obstacle_max": 0,
    "veto_ref_symbol": "MNQ", "veto_if_ref_confirms": True},
**MGC_EXIT),

# — CELLS 3+4, MOMENTUM. △ TWO arms, because last week's NUMBER and last week's
LINE were
#    different populations (section 1.5). Ship both or the wrong one gets
attributed.
ShadowVariant("mgc_wallbreak_go_long", "level_break", symbol="MGC", side="LONG",
    params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,
"fade": False,
    "book_band_pt": 1.0, "require_obstacle_gt_support": True},
**MGC_EXIT),
ShadowVariant("mgc_wallbreak_go_short", "level_break", symbol="MGC", side="SHORT",
    params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,
"fade": False,
    "book_band_pt": 1.0, "require_obstacle_gt_support": True},
**MGC_EXIT),
# the n=63 cell that was actually MEASURED: ratio>1 requires support>0.
ShadowVariant("mgc_wallbreak_go_long_ratio", "level_break", symbol="MGC",
side="LONG",
    params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,
"fade": False,

```

```

        "book_band_pt": 1.0, "require_obstacle_gt_support": True,
        "support_min": 1}, **MGC_EXIT),
ShadowVariant("mgc_wallbreak_go_short_ratio", "level_break", symbol="MGC",
side="SHORT",
        params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,
"fade": False,
        "book_band_pt": 1.0, "require_obstacle_gt_support": True,
        "support_min": 1}, **MGC_EXIT),

# — THE CONTROL. Keep it. It is the arm that produced this week's most important
number.
ShadowVariant("mgc_break_fade_nobook", "level_break", symbol="MGC", side="",
        params={"look_min": 60, "margin_atr": 0.10, "cooldown_min": 45,
        "fade": True, "book_band_pt": 1.0, "require_book": False},
**MGC_EXIT),

```

Can ShadowSim._entry express these? The `level_break` branch handles everything except the cross-asset veto, which needs a second symbol's state and therefore **a new branch plus a small MNQ state feed**. `support_max` and `support_min` need to be honoured in `gate_level_break` if they are not already. Everything else is a params change.

The router rule to attach: for the reversion arms, **none** — arm always, all sessions, all regimes (§5.3, §9). For the momentum arms, bench the BUILDING band, and treat that as recorded rather than shipped until it has been walked forward.

13. ★ PER-CELL VERDICT TABLE

CELL	GATE	N (29D)	NET	\$/TRADE	FORWARD 7D	PLACEBO	LIVE SHADOW	VERDICT
REVERSION-LONG	mgc_hole_break_fade — 60-min break DOWN into a liquidity hole → buy	134	+\$1,853	+\$13.83	+\$488 (+ \$15.73)	0/30 ✓	−\$102.50 (n=7)	SHADOW — lead candidate, blocked on the bar-source fix
REVERSION-SHORT	mgc_hole_break_fade — 60-min break UP into a liquidity hole → sell	154	+\$1,390	+\$9.03	+\$25 (+ \$0.65)	0/30 ✓	−\$153.50 (n=7)	SHADOW — same, plus a side-decay watch
MOMENTUM-LONG	mgc_wall_break_go — break UP through a defended level → buy	101	+\$838	+\$8.30	+\$378 (+ \$9.44)	29/30 ✗	not deployed	SHADOW, on probation (2nd week failing the placebo)
MOMENTUM-SHORT	mgc_wall_break_go — break DOWN through	88	+\$669	+\$7.61	+\$315 (+ \$9.56)	29/30 ✗	not deployed	SHADOW, on probation

a defended level → sell								
(both reversion cells, double- hole arm)	..._dbl	200	+\$3,304	+\$16.52	+\$656 (+ \$15.26)	0/30 ✓	not deployed	SHADOW — promote to a full arm
(volatility, the 5th cell you asked about)	mgc_straddle	3,945	−\$1,248	−\$0.32	—	control also negative	—	REFUTED on 320 days

Exit family every cell needs: the wide chandelier on a single lot (stop 3.0 ATR, arm 2.0, trail 2.0, 480-min backstop). Not the tight scalp — negative at 1R on both entries. Not the dual slot — Lot A is negative on gold and cancels Lot B, on both entries. Not a time cap — its headline numbers are a long-drift artefact on both entries. **This is the opposite of the live MNQ slot configuration and it replicates on 29 days.**

14. DISPOSITION TABLE

LEAD	VERDICT	REQUIRED NOTE
MGC shadow bar source	BUILD REQUIRED — the gating item	Point the shadow's bar builder at the depth mid; keep the trade-bar arms as *_tradebar controls. Everything else on this line waits behind it. The service already holds the feed.
mgc_hole_break_fade LONG + SHORT (primary + double-hole)	SHADOW	Survived the forward walk on the lab tape (+ \$512 / +\$656, placebo 0/30, mirror and both constants negative, 3–4 ticks of slippage headroom). Losing in live shadow on production's tape (−\$256 over 14 trades) — \$3 says that is the bar source. Promote the double-hole arm to full status.
Cross-asset MNQ veto	SHADOW — new arm	+\$5.43/tr vs +\$1.72 blanket on 278 paired days, two-sided, placebo 97th/2.5th percentile. Does not reproduce on the 29-day book window (34th pctl). Needs a new _entry branch and an MNQ state feed. Revive/settle by running it alongside.
mgc_wall_break_go LONG + SHORT	SHADOW, on probation	Forward-walked +\$693, mirror negative, filter placebo 0/200 — but fails the random-minute placebo 29/30 for the second week, and strip-best-10 is −\$192. Ship both definitions (\$1.5). Promote no further until n ≥ 150 with a passing random-minute placebo.
Last week's router rule for the fade	REFUTED as a rule	Killed by the forward walk and the 320-day tape independently. Do not shadow it — it would bench the gate in NORMAL_CHOP, the regime currently paying best. Run the fade unrouted.
Momentum router split (bench BUILDING)	PARKED	A 29-day in-sample read only — the same shape as the rule that just failed. Revive after a forward walk.
Bare 60-min break trigger, standalone	PARKED	Direction real on 320 days (beats mirror and both constants), P&L is not (strip-best-10 − \$4,392, 5/14 green months, dies at 0.10pt slippage). Raw material for a filter, never a gate.
Lookback 90/120 min	PARKED	

The year's monotone gradient does not survive the book gate. 45-180 min is a plateau; leave 60. Revive only with evidence that the book gate and the lookback are substitutes.

mgc_straddle / breakout-either-way	REFUTED	Upgraded from 22 days to 320 and from a gradient to an actual trade. All 8 compression cells negative, control negative, dies on one tick. Last week's reason was wrong (raw range sorts on ATR); the verdict was right.
Cross-asset momentum confirm	REFUTED	-\$2.29/tr on 278 paired days, -\$9.15/tr on the book window, drift-agrees at the 1st percentile of its own placebo. A new input, cleanly negative.
mgc_coil_bounce	PARKED (updated)	Last week killed it partly on the \$7.50 cost error ; at the corrected \$4.50 the 08-15 audit puts it at +\$8.42, i.e. breakeven rather than dead. Not re-derived this week. Revive by re-running it on the 29-day quote tape at the corrected cost.
London-open break FADE	PARKED (unchanged)	n=16. Revive at n > 40. Not re-derived.
mgc_session_drift_rider, one-sided vacuum, book-imbalance, obstacle-size, depletion, dual-slot, time-cap exits	unchanged	Nothing new this week. The dual-slot and time-cap refutations were re-confirmed on 29 days (\$8).
⚠ Lake bars timeframe landmine	FIX REQUIRED	New backfill_{1min,5mins,1hour,1day}.parquet files sit beside the daily 5s files. Any query without a timeframe filter now inherits up to a full day of look-ahead on every day. gf_mgc_tape.load_ll() is safe; mgc_session_anchor.load() is not.

15. THE ONE STONE STILL UNTURNED

Last week's stone — cross-asset — is turned, and it paid (\$6). The next one is narrower and it is the direct consequence of §3.

We have never checked whether gold's book is as sparse as gold's trades.

The whole bar-source finding says the depth mid is a better price series than the trade tape because gold prints sporadically. But the liquidity filter reads that same book, and if the book is also thin or stale in exactly the minutes where trades are absent, then the "empty far side" the gate selects for may sometimes be **an absent feed rather than an absent bid**. Last week checked one version of this — 72% of hole rows had size in front 60 seconds earlier — but that was on the depth-mid tape and it did not ask whether the quiet-tape holes and the busy-tape holes pay differently.

The concrete next three runs, in order:

1. **Build the depth-mid bar source into the MGC shadow** (§12.1) and let both arms run. Nothing else on this line is worth doing first.

2. **Split the hole cell by tape activity** — trades-per-minute in the minute before the break — and check the empty-book cell still pays in the busiest quartile. That is the cheapest possible test of the concern above.
3. **Re-run mgc_coil_bounce at the corrected \$4.50 cost** on the 29-day quote tape. It was killed at \$7.50 and the audit already says it is breakeven at the true cost; it deserves the honest look it did not get.

Files

ARTEFACT	PATH
the shared honest gold tape (unchanged, now 29 days)	scripts/gf_mgc_tape.py
the forward walk + frozen regime calibration	scripts/gf3_mgc_forward.py
the 320-day year tape: mechanism, plateau, cost sweep	scripts/gf3_mgc_year.py
the router / lookback / exit-matrix / slippage re-derivation	scripts/gf3_mgc_router.py
cross-asset MGC × MNQ, both tapes	scripts/gf3_mgc_cross.py
the volatility cell and the look-ahead it hid	scripts/gf3_mgc_vol.py
the full battery and the per-cell verdicts	scripts/gf3_mgc_verdict.py
results	reports/friday_v7/sections/ gf3_mgc_{forward,year,router,cross,vol,verdict,barsource,exit_ mom}.json
the live shadow's own ledger	data/shadow_mgc.db